SEISMIC FRAGILITY ASSESSMENT OF MASONRY BUILDINGS IN THE KURDISTAN REGION

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The thesis is submitted in partial fulfilment of the requirements for the award of the degree of Doctor of Philosophy of the University of Portsmouth

October 2015

Abstract

The collapse of building structures during recent earthquakes, particularly in the countries around the Kurdistan region (KR), including Turkey (2011 Van earthquake) and Iran (2003 Bam earthquake), has raised many questions about the safety of existing buildings in the region and structures that are going to be constructed in the future. The KR, which is located in northern and northeastern Iraq, is also considered to be the most hazardous region of Iraq. However, many buildings in the region, especially unreinforced masonry (URM) buildings, were not engineered to withstand seismic loads. The seismic vulnerability assessment of these types of buildings in this region is a necessary step towards the development of regional seismic retrofitting and pre-disaster mitigation plans. Fragility analysis is often used for this purpose and to graphically represent a structure's seismic vulnerability in terms of fragility curves.

Considering that there are several important uncertainties involved in such an analysis and after developing and proposing seismic zonation maps, response spectra and the seismic zone factor Z for the KR, the results of analytical fragility analyses of URM buildings in the KR indicated that the correct selection of a ground-motion intensity measure (IM) is the most important variability involved in assessing the fragility of URM buildings. The results suggest that the variability in the mechanical parameters of materials can be neglected because the variability of ground motion is considerably more substantial. Furthermore, the use of incremental dynamic analysis (IDA) and a well-selected IM allows fragility curves to be derived with only a few records (i.e., a minimum of 7 records) with the same performance as for with numerous records (i.e., 60 records). Moreover, the pronounced difference in the results when using only one IM vs. using two IMs cannot be ignored; hence, fragility surfaces are preferred over the more commonly used fragility curves.

Furthermore, a framework based on using machine learning models (i.e., a wrapper-based approach) for the optimal selection of an IM for developing fragility curves is proposed in this study. The feasibility of the wrapper method for selecting the best IM is compared to a statistical regression (log-logistic regression) used to develop fragility curves and the results were encouraging.

Finally, the outcomes of the study indicate that the seismic safety of the investigated low-rise buildings in the region is questionable and that these structures must be strengthened to prevent failure.

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Declaration

Whilst registered as a candidate for the above degree, I have not been registered for any other research award. The results and conclusions embodied in this thesis are the work of the named candidate, and they have not been submitted for any other academic award.

Word count: 53,000

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Chapter 8

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Symbols and Abbreviations

ANN	Artificial neural network
ASI	Acceleration spectrum intensity (m/s)
A95	Maximum acceleration that corresponds to 95% of the IA value (m/s^2)
C	Numerical factor that depends on the on-site soil characteristics and the fundamental period of the structure
CAV	Cumulative absolute velocity (cm/s)
CART	Classification and regression tree
C_i	Spectral amplitude at frequency f_i
CMS	Conditional mean spectrum
D	Damage measure
E	Modulus of elasticity (MPa)
EDA	Effective design acceleration (m/s^2)
EpiD	Epicentral distance (km)
F	Resistance function
G	Shear modulus (MPa)
GMSM	Ground motion selection and modification
I	Occupancy importance factor
IA	Arias Intensity (m/s)
IC	Characteristic intensity
IDA	Incremental dynamic analysis
ID3	Iterative dichotomiser 3 tree algorithm
IH	Housner Intensity (cm)
IM	Ground-motion intensity measure
IndCART	Inverse CART
K	User-defined constant in the nearest neighbour algorithm
KR	Kurdistan region
L(M)	Conditional probability of the dependent variable given the independent variables
M	Bending moment
M*	Inelastic component of bending moment
M_c	Minimum magnitude of complete reporting data for M_S

MDOF	Multiple-degree-of-freedom system
ML	Maximum likelihood
MM	Modified Mercalli intensity scale
M_S	Surface wave magnitude
M_u	Ultimate bending moment
M_W	Moment magnitude scale
Mx	Cumulative percentage of the effective mass in the x-direction
M_x	Size or magnitude of earthquake events
My	Cumulative percentage of the effective mass in the y-direction
N	Axial compressive force
N*	Inelastic component of the axial force
N’	Number of observations in a dataset
N_m	Cumulative number of events with a magnitude $\geq M_x$
P	Annual exceedance probability of an earthquake event
PEER	Pacific earthquake engineering research center
P_F	Probability of failure
PGA	Peak ground acceleration (m/s^2)
PGV	Peak ground velocity (cm/s)
PGD	Peak ground displacement (cm)
P_n	Probability of failure at the n-th level of the IM
Po	Probability of occurrence of earthquake events
PSV	Peak pseudo-velocity (cm/s)
R	Return period or recurrence interval (years)
RMSA	Root-mean-square acceleration (m/s^2)
RMSD	Root-mean-square displacement (cm)
RMSV	Root-mean-square velocity (cm/s)
Rw	Structural system performance factor
S	Internal stress vector
S_a	Spectral acceleration (m/s^2)
$S_a(T1)$	First-mode spectral acceleration (m/s^2)
SDOF	Single-degree-of-freedom system
S_d	Spectral displacement (cm)
SED	Specific energy density (cm^2/s)

$S_e(T_x)$	Elastic response spectrum (m/s^2)
SMA	Sustained maximum acceleration (m/s^2)
SMV	Sustained maximum velocity (cm/s)
S_s	Soil factor
S_v	Spectral velocity (cm/s)
T’	Period of interest for conditioning the response spectrum (second)
T_B	Period corresponding to the constant acceleration branch of the smoothed response spectrum (second)
T_C	Period corresponding to the constant velocity branch of the smoothed response spectrum (second)
T_D	Period corresponding to the constant displacement branch of the smoothed response spectrum (second)
Tm	Mean period (second)
T_{min} and T_{max}	Smaller and larger of the two periods of interest (second)
T_n	Significant fundamental period range of the structure (second)
Tp	Predominant Period (second)
T_x	Time period (second)
UHS	Uniform hazard spectrum
URM	Unreinforced masonry
V	Shear force (Newton)
V*	Inelastic component of shear force (Newton)
VARCLUS	Variable clustering
VSI	Velocity spectrum intensity (cm)
V_u	Ultimate shear strength (Newton)
W	Total seismic dead load (Ton)
Z	Seismic zone factor
a	Constant in the Gutenberg-Richter equation that reflects the total seismicity rate in the region
a_g	Design ground acceleration on type A ground (m/s^2)
a(t)	Acceleration time history (m/s^2)
b	Constant that represents the slope of the magnitude frequency law (Gutenberg-Richter equation)
b’	Shape factor (ratio of height to width)

c	Non-dimensional coefficient that controls the inelastic deformation
d	Displacement (cm)
d_f	Focal depth (km)
d_i	The i-th limit state
d(t)	Displacement time history (cm)
d_u	Ultimate displacement (cm)
d_y	Yield displacement (cm)
f	Global friction coefficient
f_b	Compressive strength of masonry units (MPa)
f_k	Characteristic compressive strength of masonry (MPa)
f_m	Compressive strength of mortar (MPa)
f'_m	Average resistance in compression of the masonry (MPa)
f_{vo}	Shear resistance of the masonry without compression (MPa)
g	Gravitational acceleration (m/s^2)
h, b and s	Height (cm), width (cm), and thickness (cm) dimensions of the macro element (Page 156)
k	A constant value that ranges from 0.4 to 0.6 depending on the type of masonry and mortar
l	Width of the masonry panel (cm)
l'	Actual width of the masonry panel subjected to compression (cm)
m_b	Body wave magnitude
q	Mass density of the material (kg/m^3)
t	Thickness of the panel (cm)
t_d	Total duration of the seismic motion (second)
t_s	Significant duration (time intervals between 5-95% or 5-75% of IA) (second)
t_1	Initial time of the seismic motion (second)
t_2	End time of the seismic motion (second)
u	Horizontal displacement of the macro element at nodes i and j (cm)
v	Wave speed in the material (m/s)
v(t)	Velocity time history (cm/s)
w	Axial displacement of macro element at nodes i and j (cm)
Δ	Thickness of the inferior and superior layers of macro elements (cm)
$\Phi(\ )$	Normal cumulative distribution function

Φ_d	Failure function
α	A damage parameter
β	Standard deviation
β_1, β_2 and β_3	Regression coefficients
δ	Axial displacement of macro element at the central node (cm)
ε	Number of standard deviations between the logarithms of observed S_a and predicted S_a
$\varepsilon(T_i)$	Conditional mean epsilon for the i-th period
η	Damping correction factor
$\boldsymbol{\theta}$	Mean
λ	Average rate of occurrence or mean seismic activity rate
μ	Mohr–Coulomb friction coefficient
ξ	Damping coefficient
ρ	Correlation coefficient
σ_o	Normal compressive stress over the entire area (MPa)
τ_o	Shear strength at the first diagonal crack (MPa)
φ	Rotation of the macro element at nodes i and j
ϕ	Rotation of the macro element at the central node

Acknowledgements

I would like to express my sincere gratitude and appreciation to my PhD supervisors Dr. David Begg and Dr. Nikos Nanos for their continuous guidance, support and encouragement throughout the course of my studies. I am very grateful to them for the education and direction they have provided.

This PhD study was supported by the Kurdistan Regional Government through the Human Capacity Development Program (HCDP) in Higher Education. This financial support is gratefully acknowledged.

I am also grateful to the TREMURI staff for providing the academic and commercial versions of the TREMURI software, which were used in the example analyses. I would like to especially thank Professor Cattari, University of Genova, for her advice regarding the correct usage the academic version of the software.

I would like to thank Jayaram et al. for providing the MATLAB implementation of the proposed ground-motion selection algorithm, which is available at http://www.stanford.edu/~bakerjw/gm_selection.html.

Great thanks go to my family and friends, whose constant presence, help, and intuition mark the cornerstone of my being. I would like to thank my wife, who was with me throughout my studies, even during the hardest times when we both endured the loss of our babies. I would have never finished this thesis without the great support of my father, mother, and brothers. I really love you. Thank you very much.

Dissemination

2015

International Journal of Structural Analysis & Design– IJSAD, 2(1), pp.1-9

Seismic fragility assessment of low-rise unreinforced masonry buildings in the Kurdistan region of Iraq

2014

Second International Conference on Advances in Civil, Structural and Mechanical Engineering (CSM 2014), Birmingham, UK

Seismic fragility assessment of low-rise unreinforced masonry buildings in the Kurdistan region of Iraq

First International Engineering Conference (IEC2014), Erbil, Kurdistan Region of Iraq

Time History Analysis of Existing Buildings in the Kurdistan Region of Iraq

First International Engineering Conference (IEC2014), Erbil, Kurdistan Region of Iraq

Selection of the Optimal Ground-motion Intensity Measure in Building Fragility Using the Wrapper Method

International Journal of Advanced Research in Computer Science and Software Engineering, 4(8), pp. 8-16

Wrapper Subset Evaluation Facilitates the Automated Detection of Ground-Motion Intensity Measures and Derivation of the Seismic Fragility Curves

CHAPTER 1. INTRODUCTION

1.1 Background

Earthquakes have been shown to be the cause of significant disasters when they strike crowded areas in both rural and urban environments. The majority of hazards and dangers that people must confront originate from man-made structures because these structures can be affected by earthquake-induced shaking.

Devastating earthquakes have taken the lives of millions of people and have resulted in economic losses totalling billions of dollars over the years. Most of these human losses have occurred in developing countries as a result of poorly constructed structures (Khalfan, 2013). For example, in 2003, Bam, which is a city of Iran, was struck by an earthquake measuring 6.6 on the Richter magnitude scale (Richter, 1935), resulting in the deaths of more than 43,000 people and leaving more than 60,000 people homeless due to the collapse of many (approximately 60%) unreinforced masonry (URM) buildings in that area (Nasrabadi et al., 2007). Therefore, learning from past earthquakes and increasing our understanding of earthquakes, including their impact, are fundamental requirements for developing procedures that must be performed to mitigate these devastating effects and to protect communities in the future. However, rapid development and the growth in urban populations, especially in developing countries, present a new challenge to governments, engineers, and architects in terms of increasing urban safety with respect to seismic vulnerability. Rapid urbanisation has resulted in a variety of common building types that are extremely exposed to earthquakes. Notably, 75% of casualties related to earthquakes are due to the collapse of buildings (Coburn & Spence, 2002); unreinforced masonry buildings represent the greatest vulnerability (Grünthal, 1998). This type of building (i.e., URM buildings) represents the majority of buildings in the area under study, i.e., the Kurdistan region (KR) of Iraq.

The KR has a population of approximately 4.5 million people. The region is located in northern and north-eastern Iraq and is considered to be the most seismically hazardous region of the country. Because of its location in a relatively active seismic zone (it borders the Zagros-Tauros Belt), where the Arabian, Eurasian, and Anatolian tectonic plates collide (Gok et al., 2006), Ameer et al. (2005) asserted that frequent destructive earthquakes can be expected in this part of Iraq. The maximum expected earthquake magnitude is 7.68 ± 0.52 for this region. The most recently recorded and moderately devastating earthquake

was 5.1 on the surface wave magnitude scale M_S (Gutenberg, 1945a,b,c) and 5.5 on the moment magnitude scale M_W (Hanks & Kanamori, 1979) and occurred at a depth of 26 km and a distance of 37.2 km from the regional capital city of Erbil on July 24, 1991. Approximately twenty people were killed, and several houses either collapsed or were damaged (Utsu, 2002). Additionally, Aziz et al. (2001) stated that seismic activity has increased in the region over the last decade. Although the KR is considered to be the most hazardous region in Iraq based on the aforementioned facts, many buildings in the region, especially URM structures, were not originally engineered to withstand seismic loads. Furthermore, the collapse of building structures during recent earthquakes, particularly in countries surrounding the KR, such as Turkey (2011 Van earthquake) and Iran (2003 Bam earthquake), has raised many questions regarding the safety of the existing buildings in the region and structures that will be constructed in the future. Thus, assessing the seismic vulnerability of the building environment in the KR is essential for loss assessments and for identifying the most vulnerable structures in the region. Such an assessment will provide preparatory measures for enabling proper emergency management, awareness, improvement, and a recovery plan to reduce the seismic vulnerability of structures.

Living in a catastrophe-free environment is technically impossible, although it is possible to reduce the impact of disasters with the assistance of suitable risk-reducing strategies. Over the last decade, researchers have attempted to develop reliable vulnerability assessment procedures to determine the safety of seismically deficient buildings and evaluate retrofit or demolition alternatives, especially following several destructive earthquakes, including the 1999 İzmit and Düzce earthquakes (Turkey), the 1999 Chi Chi (Taiwan) earthquake, the 2001 Bhuj (India) earthquake, the 2003 Bam (Iran) earthquake, the 2011 Tōhoku (Japan) earthquake, and the 2011 Van (Turkey) earthquake. The damage caused by these natural disasters, in addition to their direct relation to the number of casualties, highlights the poor performance of the buildings located in these regions subjected to large earthquakes. The inability to withstand the forces produced by earthquakes increases the volatility of buildings. Thus, without a proper risk reduction plan, estimating losses in post-earthquake events will be difficult and inefficient in most cases. Fragility analysis, which is an effective tool in the seismic assessment of different structure types, such as buildings (Erberik, 2008), bridges (Mackie & Nielson, 2009), and nuclear power plants (Zentner, 2010), has become increasingly popular in recent years. Fragility curves, which are produced by this type of analysis, are usually used to represent the probability of reaching or exceeding a structure's specific damage level with regard to

a seismic intensity measure. It is a probabilistic procedure aimed at treating all uncertainties involved in a seismic safety assessment.

To perform a fragility assessment and derive fragility curves, the following information must be utilised: 1) building stock data and corresponding construction materials associated with the region under study to select typical buildings that properly represent the building stock; 2) seismic data relevant to the region for selecting an appropriate number of ground motions and parameters that can represent the region's seismicity; 3) structural performance levels and damage state limits; and 4) analytical simulation methods to estimate the response of the selected building/buildings. All of these areas are formally uncertain and are considered to be the primary sources of uncertainty in determining fragility curves (Choun & Elnashai, 2010). The uncertainties (e.g., earthquake hazards, intensity of earthquakes, ground-motion features, soil effects, structural properties, and the nonlinear dynamic behaviour of structures) can be divided into two categories: inherently random (aleatoric), such as variability in the ground motions or mechanical properties of materials, and epistemic uncertainty, which is caused by a lack of knowledge, model error, a lack of objectivity in damage state definition, ground-motion intensity measures (IMs), and the selection of a ground-motion suite (Mackie & Nielson, 2009; Gehl et al., 2013b). Therefore, the accuracy of information used in the process of buildings' fragility assessment is critical for reducing these uncertainties.

Generating fragility curves usually requires considering a single ground-motion parameter, such as the peak ground acceleration, to represent the seismic action in the area of interest, neglecting other important characteristics such as the frequency or energy content. This assumption is a source of epistemic uncertainty that can be included in seismic risk analysis via the construction of fragility curves. It has been recently shown that using two parameters results in a considerable reduction in the scatter of the final results and provides better and more accurate information about a building's vulnerability (e.g., Seyedi et al., 2010; Gehl et al., 2011; Gehl et al., 2013b). As a result, fragility curve is converted to a fragility surface, which expresses damage probability as a function of two parameters. This task, i.e., determining the feasibility of using two ground-motion intensity measures as an efficient alternative to the widely used single parameter approach during the construction of fragility curves, is the primary objective of this study. However, the choice of appropriate ground-motion parameters requires additional effort to facilitate the fragility analysis and achieve such an objective.

To investigate which ground-motion characteristics are most important, several accelerograms that are appropriate for the seismic hazard analysis of the selected region must be selected. Subsequently, the correlation between different parameters of each accelerogram and the inducted damage corresponding to the related accelerogram on the tested structure should also be tested and contrasted to identify the most appropriate parameters. Because there is an unlimited range of possible earthquake ground motions, it is useful to typify them and utilise several important motion parameters that approximately represent the different properties of the motions such as the amplitude, frequency content, duration, and energy. Expressing the ground motion in terms of various intensity parameters, such as peak ground acceleration (PGA), peak ground velocity (PGV), spectral acceleration (S_a), and others, is stipulated by the fact that they offer simplicity in exhibiting a complex seismic event through a single number; and this number can be easily derived from direct measurements or simple mathematical models.

As mentioned above, one of the most important issues in deriving fragility curves is the large number of IMs that have been proposed by researchers to represent seismic action in a specific region. Because a single ground-motion parameter is used to derive fragility curves, all other IMs are ignored, thereby making it difficult to select the optimal parameter. The techniques currently used to select IMs often consider many of the available IMs, therein classifying them in terms of efficiency and sufficiency via a predefined regression model, and ultimately select the optimal IM. However, performing numerous regression analyses may require a considerable amount of time, and selecting a different regression function may also impact the obtained results. In machine learning, this issue can be addressed by applying a process known as feature selection (which is also called attribute selection or variable selection) to identify and remove irrelevant and redundant information to the greatest extent possible. For that purpose, this study attempts to propose a framework to select the best IM or subsets of IMs using machine learning techniques.

Furthermore, due to the nonlinear behaviour of masonry units in progressive collapse cases (Colliat et al. 2002), it is of great importance to concentrate on the nonlinear dynamical behaviour of this type of structures for developing precise fragility curves. Nonlinear dynamic analysis is one of those methods that can be used successfully for that purpose. The selection of seismic inputs that are compatible with the seismic and site conditions of the area under investigation is critical for this type of analytical method (i.e., nonlinear dynamic analysis), specifically when using incremental dynamic analysis

(Vamvatsikos & Cornell, 2002), which is an approach that has become very prevalent among researchers. For that purpose, defining seismic characteristics of the KR is an essential step towards correctly selecting seismic inputs for the nonlinear dynamic analysis of target buildings in the region. Furthermore, because aleatoric uncertainty in the form of variability in the mechanical properties of URM structures (the focus in this study) is more prevalent than for other structures, such as reinforced concrete and steel frame constructions, this variability should also be considered for target buildings in the region to enable a comparison between it and the ground-motion variability.

Finally, the number of ground-motion records that should be used, the method for selecting records, and the effects of them on the derived fragility curves are various challenges that are addressed in this study because only a few topical studies have been published to date, especially regarding URM buildings.

1.2 Objectives

As mentioned above, the main goal of the present study is to investigate the fragility of existing URM buildings in the KR using an analytical approach. The accuracy of the information used in such an approach is crucial for reducing uncertainties involved in the process. Therefore, understanding the impact of the following three main sources of uncertainty can also be considered as an important objective of this study. The identified uncertainties are as follows: the variability in the mechanical properties of the target buildings versus ground motions, the variability in the ground-motion intensity measures, and the variability in the number of records and the method used to select such records. Examining the efficiency of using two parameters (fragility surfaces) instead of a single parameter (fragility curve) is another goal of the current study.

The difficulty in developing fragility curves is associated with identifying a suitable ground-motion intensity measure (IM) that relates seismic hazards to structural response. Current techniques require considerable computational efforts because many regression analyses must be performed to select an optimal IM from the numerous available IMs. This study addresses the problem of IM selection through machine learning models. A wrapper method is used to classify a set of IMs into different subsets containing various IMs. The performance of each IM is evaluated with respect to the known response of the tested structures. Reducing the size of IM sets could minimise the time required for selecting the optimal IM and may facilitate the use of different regression models to fit the data and

derive fragility curves with an increased accuracy. Therefore, another main objective of this study is to investigate the feasibility of machine learning models (namely, the wrapper feature selection method) for selecting the best IM to develop fragility curves. The results of the wrapper method are compared with the results from a statistical regression method. A framework for selecting the best IM or subsets of IMs is eventually proposed.

The study performs a seismic analysis on existing buildings in the Kurdistan Region (KR) of Iraq as most hazardous part in the whole of Iraq in order to achieve the above mentioned objectives. Having a clearly defined building classification for the area and developing the corresponding fragility curves or surfaces for each class is an essential step towards evaluating the seismic vulnerability of a building stock and performing a loss estimation study. Statistical data reveal that the majority of the buildings in the study area contain one and/or two storeys and are primarily URM buildings, which are represented by low-rise unreinforced masonry bearing walls (URML) in the Hazards United States Multi-Hazard (HAZUS-MH) model (FEMA, 2003). Thus, the focus of this study is on two URM buildings (one- and two-storey buildings) located in this region. The damage level is evaluated based on hundreds of nonlinear time history analyses using an incremental dynamic analysis approach. The maximum displacement at the top of the structure is defined as the damage index, and the structure is modelled by specific macro-elements signifying the masonry's seismic conduct.

To perform this study, the following information and data are necessary:

- Building stock data and corresponding construction materials relevant to Kurdistan.
- Classified building information for characterising several typical buildings that can properly represent the building stock.
- Seismic data relevant to Kurdistan for selecting an appropriate number of ground motions and parameters that are indicative of the region's seismicity.
- Generated data from simulation analyses.

The choices made for the analytical method, structural idealisation, seismic hazard characterisation, and damage models strongly impact the derived fragility curves/surfaces, which may cause significant discrepancies in seismic risk assessments made by different groups for the same location, structure type, and seismicity.

1.3 Organisation of the Dissertation

Various seismic vulnerability evaluation methods are reviewed in Chapter 2 following the introduction of a brief definition of the terminology related to the process of

vulnerability assessment. Based on the literature review, the research strategy is discussed in Chapter 3 by identifying the implemented tools needed to achieve the final objective, referring to either the seismic record selection, structural type classification or the nonlinear dynamic analysis and data post processing approaches. All data extracted from the aforementioned procedures are organised in the remainder of this dissertation as follows. In Chapter 4, the characteristics of the building stock in the region are summarised and graphically visualised to make such information on the region publicly available; a description of the selected building models is then provided. Sources of earthquake hazards are reviewed in Chapter 5; based on the area's tectonic configuration and seismic history, for the first time, a seismic hazard zone map is defined and constructed for the region. Smoothed response spectra are then proposed for each seismic zone, after which different sets of strong ground motions are chosen. Analytical modelling and an application of the nonlinear dynamic analysis are discussed in Chapter 6. To study the correlation between ground-motion characteristics and building response, the incremental dynamic analysis results are used in Chapter 7. Furthermore, the fragility curves and surfaces of the studied buildings are produced. Chapter 8 presents a discussion of the results obtained in Chapter 7. The roles of different sources of uncertainty (i.e., the variability in the material properties of the target buildings versus ground motions, the variability in the ground motion intensity measures, and the variability in the number of ground motions and in the method of their selection) in accurately deriving fragility curves is discussed more thoroughly in Chapter 8. Furthermore, the feasibility of the machine learning tool (in terms of the wrapper feature selection method) is investigated and demonstrated. Finally, conclusions, application norms, and implications for future work are presented in Chapter 9.

CHAPTER 2. LITERATURE REVIEW

The first earthquake risk and loss estimation studies date back to 1968 with the development of a new seismic risk analysis field, known as probabilistic seismic hazard analysis (PSHA), introduced by Allin Cornell (Cornell, 1968). PSHA attempts to measure uncertainties pertaining to the location, size, and intensity of future earthquakes and combines them to produce a clear description of the probability distribution of a site's future shaking potential. According to potential seismic hazard or earthquake scenarios, the probability of structure damage and economic losses can be predicted (Baker, 2008). This is a key aspect of risk management, especially in terms of risk mitigation. Decisions to mitigate seismic risk require a rational plan for emergencies, raising public awareness, the structural enhancement of buildings, and damage reduction for infrastructure such as water, roads, highways, and electric power systems. To make such decisions, the contributions of five main fields are to be considered: (1) hazard analysis, (2) local site effects, (3) exposure information, (4) vulnerability analysis, and (5) the estimation of risk and loss (Coburn et al., 1994; Chandler & Nelson, 2001; Bendimerad, 2001). Accordingly, the main goal of the abovementioned analyses is to assess the risk facing a structure from earthquake shaking and to establish processes that can be used to enable more informed and competent decisions regarding seismic safety.

As discussed above, seismic risk in an urban environment is closely associated with the structures, buildings, and their distribution in the area of their location. Therefore, an understanding of building characteristics and an area's seismicity is essential for assessing their seismic vulnerability and planning for future disaster mitigation. Finding the most vulnerable building types in a given area allows engineers to distinguish the necessary steps that must be taken when designing new buildings or retrofitting existing structures. Fragility analysis, which is an effective tool for assessing the seismic vulnerability of structures (Park et al., 2009), is often implemented to represent the 'damageability' of a structure in terms of fragility curves. These curves allow engineers to monitor the damage probability of a structure with respect to a seismic intensity measure (IM). Thus, this method can be used to advise policy makers regarding the seismic resistance required for a region and how to prepare and integrate proper emergency management, awareness, rehabilitation, and strengthening plans.

In this chapter, a literature review of fragility analysis as an effective tool for the seismic vulnerability assessment of buildings is discussed. However, prior to this review, it

is important to distinguish among several terms related to such kind of studies, including hazard, vulnerability, risk and loss. This not only contributes to a comprehensive understanding of the necessary elements utilised throughout the project but also results in the development of the most reliable procedure among available methods for the seismic vulnerability assessment of buildings.

2.1 Earthquake Hazard

Earthquake hazard identification and seismic vulnerability assessment are two main components that are applied when developing plans and policies to mitigate earthquake risk in a region. Risk has different meanings in different fields (e.g., insurance and engineering) (Wang, 2008). In general, any likelihood of harm is considered to be a risk. However, in earthquake engineering, risk can be defined as the possibility of ground motion at a certain site exceeding a specific level and causing damage and loss to a given element within a certain period of time (Coburn et al., 1994). Furthermore, the potential to cause harm is defined as a hazard, which can be represented by three modes: inactive, pre-active, and active (MacCollum, 2007). Vulnerability is the probability that a hazard capability of an element exceeds its ability to resist the hazard; as a result, the element suffers a loss (Oliveira et al., 2008). Consequently, loss can be determined as the socio-economic impacts of damage or harm such as injuries, deaths, and repair costs. As noted by Coburn et al. (1994), a combination of estimated losses obtained from all levels of hazard intensity represents the risk, while the vulnerability is an expression of losses for a given hazard intensity level.

In the case of earthquake loss assessment, the hazard is best described as strong ground shaking (Dissen et al., 1992; Bird & Bommer, 2004), which means that seismic vulnerability assessments are generally conducted with that in mind. Seismic events include hazards other than ground shaking such as landslides, liquefaction, surface fault ruptures, and tsunamis. Nevertheless, in terms of the larger-scale losses that may be induced in an affected area, ground shaking is considered to be the only significant impact (Bird & Bommer, 2004), which is why seismic vulnerability assessments are generally related to this hazard.

To provide quantitative estimates of the expected levels of seismic ground motion as the basic input to seismic risk assessments, it is essential to characterise the complex nature of strong motion accelerograms using simple parameters and to develop predictive relationships for these parameters. Since the first recording of strong motion in terms of

accelerograms, the Long Beach earthquake of 1933, many parameters have been proposed to define the main features of ground motion. However, a comprehensive study of the literature demonstrates that there is often substantial vagueness or disagreement regarding the definition of even the simplest ground-motion parameters (Bommer & Martinez-Pereira, 2000). In the next section, some of the most commonly used parameters are reviewed.

2.2 Ground-motion Intensity Measures

For the purposes of seismic vulnerability assessment and to characterise the probability of damage caused by seismic ground motion in terms of fragility curves, a ground-motion parameter called an intensity measure (IM) is typically used. A good IM should be capable of capturing all intensity, frequency content, and duration data that are needed to accurately evaluate the structural response without additional ground-motion information. Such a measure would satisfy the requirements of efficiency and sufficiency for the selected ground-motion parameters (or IMs) that are necessary to reduce the variability in the predicted structural response, to minimise the number of input records, and to decrease the complexity of the record selection procedure (Luco, 2002; Krawinkler et al., 2003). Moreover, Padgett et al. (2008) summarised five requirements of an IM used in the development of fragility curves: (1) efficiency, which can be measured using the standard deviation of the relation between the IM and the structural response (lower standard deviations are better); (2) practicality, which is a measure of the strength of the correlation between the IM and the demand (steeper slopes are better); (3) proficiency, which is a combination of the first two requirements; (4) sufficiency, which implies no need for other earthquake characteristics, such as magnitude and source-to-site distance; and (5) hazard computability, which refers to the availability and accessibility of the IM.

Various studies (e.g., Braga et al., 1982; Barbat et al., 1996; Kappos et al., 1998; Okada & Takai, 2000; Masi, 2003; Di Pasquale et al., 2005) have examined the role of seismic intensity measure using a variety of macro-seismic parameters such as measures of seismic damage potential. Furthermore, several studies (e.g., Elenas, 2000; Elenas & Meskouris, 2001; Riddell, 2006; Nanos et al., 2008; Alvanitopoulos et al., 2010; Nanos, 2011; Buratti, 2012; Elenas, 2013) have also showed that different IMs may have different capabilities in predicting structural responses when being used as a damage descriptor. In a broad sense, it is a common procedure to use a single parameter because of its simplicity;

meanwhile, the large dispersion of the predicted structural response is usually neglected. Krawinkler et al. (2003) indicated that the use of a single parameter is truly questionable for that purpose, and this is only one of several basic challenges related to performance-based earthquake engineering. Numerous studies (e.g., Shome, et al., 1998; Song & Ellingwood, 1999; Ellingwood, 2001; Kafali & Grigoriu, 2004; Schotanus et al., 2004; Seyedi et al., 2010; Gehl et al. 2011; Xu & Wen, 2012; Gehl et al., 2013b) have also claimed that a single parameter cannot comprehensively represent the seismic action in a particular region and describe its complex nature, implying that the use of a single parameter results in great uncertainty during the vulnerability assessment. Furthermore, it has been recently demonstrated that using two ground-motion parameters instead of one may reduce this uncertainty and offer a better understanding of the vulnerability of a building's typology (Seyedi et al., 2010; Gehl et al. 2011; Gehl et al., 2013b).

Over the years, numerous ground-motion parameters have been proposed by researchers for use as indices of a ground motion's damage potential (e.g., Housner, 1952; Housner & Jennings, 1964; Arias, 1970; Shome et al., 1998; Fajfar et al., 1990; Bojórquez & Iervolino, 2011). One of the most important objectives put forth in these studies was to identify the ground-motion parameter that was best correlated with the damage index, which represents a measure of structural performance. It is desirable to minimise the variability in this correlation to determine the expected damage for increased accuracy. For that purpose, a good awareness of the ground-motion parameters and the factors affecting them are critically important.

Ground motion is usually described through a time history. A time history can be classified into three basic types: acceleration, velocity, and displacement time histories. Usually, only one of these time histories is directly recorded, and the others can be calculated through integration or differentiation (Maniyar & Khare, 2011). These time histories are naturally designed to describe seismic action and provide general information (Nanos, 2011). However, the relevant important characteristics (i.e., amplitude, frequency content, and duration) of strong ground motion must be described using a proper parameter for the accurate evaluation of structural performance. For that purpose, several important parameters have been extracted and derived by researchers (as mentioned above) using appropriate mathematical methods applied to time histories. These parameters can be classified according to the time histories that they are derived from and are known as acceleration-, velocity-, or displacement-based intensity measures (Riddell, 2006; Buratti, 2012). Most of these parameters measure one of the ground-motion characteristics: the

amplitude, duration or frequency content of ground motion. However, there are some other parameters that are based on a combination of the aforementioned characteristics; these parameters are usually known as energy-based parameters. The duration is another important characteristic of ground motion that may affect the level of damage experienced by a structure. Long-duration motion can cause substantial damage, even if it has a low or moderate amplitude. On the other hand, a motion with high amplitude but short duration may not result in any structural damage. The parameters that are defined and based on the duration of the ground motion are recognised as duration-based parameters. In addition to the aforementioned IMs, several other parameters, which are known as hybrid IMs, have also been developed. These parameters are generally defined using different combinations of IMs.

The following is a brief description of some of the parameters considered in this study.

2.2.1 Acceleration-based intensity measures

Acceleration-based IMs are parameters that are derived from acceleration time histories using proper mathematical methods. The acceleration time history displays more high-frequency content (Kramer, 1996; Moratto, 2008). Thus, the IMs derived from this history are associated with the high-frequency components of an earthquake's ground motion. The following is a description of several acceleration-based parameters that are used in this study. Figure 2.1 shows an example of the acceleration time history recorded during the 1979 Imperial Valley-06 earthquake (M_W=6.53 and epicentral distance EpiD=27.47 km).

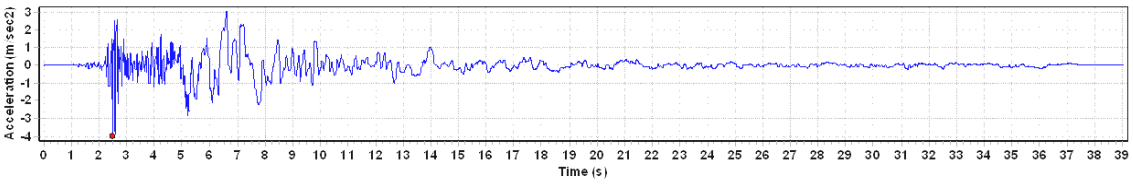


Figure 2.1: Acceleration time history for the 1979 Imperial Valley-06 earthquake (M_W=6.53 and EpiD=27.47 km).

Peak ground acceleration (PGA)

The maximum absolute value of the earthquake acceleration (rate of change in speed) on the ground recorded by a strong motion instrument during an earthquake is known as the PGA. To obtain this peak value, an accelerogram, which is a graphical representation of the recorded history of acceleration versus time (shown in Figure 2.1), is implemented (Maniyar & Khare, 2011). For example, Figure 2.1 indicates a PGA of 4 m/s^2.

Because of its computational simplicity and relatively good accuracy for potential damage prediction for low-rise buildings, the PGA has been one of the most commonly used and widely accepted parameters to describe seismic intensity and remains relevant today (Ye et al., 2011). However, due to the complex conditions (e.g., magnitude, earthquake depth, fault length, distance from the epicentre, duration, and the ground's geology) affecting the PGA, apart from the fact that it ignores duration-related information, frequency content, and amplitude effects and consequently produces variability in the predicted results, several other IMs have been suggested by researchers (e.g., Housner & Jennings, 1977; Fajfar et al., 1990).

Ground motions with high PGAs are typically more damaging than those with lower PGAs. However, this concept depends on several factors. For example, high peak acceleration over a very short time period may cause minimal damage to many types of structures despite the low peak acceleration that persists for a longer time period (Maniyar & Khare, 2011). The PGA is directly related to seismic loads that could damage short-period structures (one and two-storey buildings), which has been mentioned by several researchers, including Fajfar et al. (1990), Riddell and Garcia (2001), Allen (2007), and Ye et al. (2011). Moratto (2008) mentioned that the PGA is indicative of the maximum response for rigid structures with a natural period of less than (or equal to) 3 s. Maniyar and Khare (2011) argued that the PGA alone is unable to provide true information about an earthquake's damage potential. Furthermore, with increasing building height, evidence has suggested that the PGA is a poor measure of an earthquake ground motion's damage potential (Fajfar et al., 1990). The PGA is usually expressed in terms of g (gravitational acceleration); its mathematical expression is as follows:

$$PGA = max|a(t)|, \qquad \text{Equation 2.1}$$

where a(t) is the acceleration time history.

Root-mean-square acceleration (RMSA)

Many researchers started to apply theories of random vibration to earthquake engineering, especially after earthquake motion was first considered as a random process by Housner (1947). In mechanical engineering, random vibration is a motion in which its future behaviour cannot be precisely predicted. To specify the motion, the acceleration spectral density, which is the square root of the power spectral density, is usually determined (Norton & Karczub, 2003). Because the power is essentially represented by the integral of the square of the acceleration over a specified duration, the root-mean-square acceleration can be expressed mathematically as follows (Housner & Jennings, 1964):

$$RMSA = \sqrt{\frac{1}{t_d}\int_{t_1}^{t_2}[a(t)]^2 dt}\,, \qquad \text{Equation 2.2}$$

where t_1 and t_2 are the initial and end times of the motion, respectively, and t_d is the total duration, i.e., (t_2-t_1). Thus, the RMSA is a measure of the average energy rate (power multiplied by time) imparted by the ground motion. The RMSA is a useful intensity index for engineering purposes because it incorporates the effect of duration and is not seriously influenced by very-high-frequency accelerations (over a very short period of time). However, the RMSA can be impacted by methods used to define strong motion duration; in addition, the RMSA does not provide any information regarding the frequency content because it represents the sum of the input energies for all frequencies (Danciu & Tselentis, 2007).

Arias Intensity (IA)

The Arias Intensity, which is a ground-motion parameter that was defined by Arias (1970), can predict an earthquake's potential damage using the integral of the square of the acceleration time history. The Arias Intensity is also a measure of ground-motion strength, which represents the total energy content of seismic ground motion (Arias, 1970). The IA is one of the most widely accepted seismic parameters and is well correlated with commonly used index measures of structural performance because it considers the intensity of the PGA over a specified time period, which was shown by Urban (2007). However, Riddell (2006) indicated that the IA is only appropriate when the response of rigid systems is of interest because acceleration-related indices are best for rigid systems. The IA can be defined as follows:

$$IA = \frac{\pi}{2g}\int_0^{t_d}[a(t)]^2 dt. \qquad \text{Equation 2.3}$$

The Arias Intensity has units of velocity and is often expressed in metres per second.

Characteristic intensity (IC)

Because of its reliable correlation with structural damage based on the damage index proposed by Park and Ang (1985), Park et al. (1985) proposed the characteristic intensity index as a combination of the root-mean-square acceleration value and a record's significant duration. The significant duration is defined as the time interval (usually between 5 and 95%) in which a specified amount of the total IA is accumulated. The IC is linearly related to an index of structural damage due to maximum deformations and absorbed hysteretic energy (Sperbeck, 2009; Valamanesh et al., 2010). The IC can be calculated as follows:

$$IC = (RMSA)^{1.5}\sqrt{t_s} \ , \qquad \text{Equation 2.4}$$

where t_s is the significant duration defined by time intervals between 5 and 95% (or between 5 and 75%) of the IA. Here, t_s can also be substituted by the record length of t_d.

Acceleration spectrum intensity (ASI)

For analysing structures with fundamental periods of less than 0.5 s, Von Thun et al. (1988) introduced the ASI as a strong ground-motion parameter defined by calculating the area under the acceleration response spectrum between periods of 0.1 and 0.5 s:

$$ASI = \int_{0.1}^{0.5} S_a(\xi = 0.05, T_n)dT_n \ , \qquad \text{Equation 2.5}$$

where ξ is the damping ratio and T_n is the significant fundamental period range of the structure. The ASI was originally proposed as a ground-motion intensity measure (IM) applicable to the seismic analysis of concrete dams. The response spectrum is viewed as a graphical representation of the maximum response experienced by a series of elastic single-degree-of-freedom oscillators due to the applied acceleration at the bases (Danciu, 2006).

Based on a range of spectral periods, as argued by Bradley (2010), the ASI is considered to be a useful ground-motion parameter for simultaneously predicting

acceleration and displacement demands in individual structures and regions where short-period structures are typically prevalent. Bradley (2010) asserted that this ground-motion intensity measure also has a smaller uncertainty compared with pseudo-spectral accelerations. Kadaş and Yakut (2013) recommended this parameter as a basis for collecting ground-motion records because of its reduced dispersion and good correlation with the structural response, especially over short periods.

Cumulative absolute velocity (CAV)

To indicate the onset of structural damage in engineered structures, Benjamin (1988) developed an IM by accounting for the cumulative effects of ground-motion duration through the integration of the absolute acceleration time series:

$$CAV = \int_0^{t_d} |a(t)| dt \, . \qquad \text{Equation 2.6}$$

Because the velocity can be defined as $v(t) = \int a(t)dt$, the proposed IM by Benjamin (1988) was called the cumulative absolute velocity (CAV; units of velocity). Notably, this IM is not directly related to the ground-motion velocity. The CAV has been used to represent the damage potential of nuclear power plant facilities (USNRC, 1997) and potentially liquefiable soils (Kramer & Mitchell, 2006). The relation between the CAV and damage is indirect due to its strong relationships with other proposed IMs (for example, the Housner intensity and local macroseismic intensities, such as magnitude and distance), as mentioned by Koliopoulos et al. (1998), Kostov (2005), Danciu and Tselentis (2007), and Martinez-Rueda et al. (2008). As Kadaş and Yakut (2013) outlined, the CAV exhibits a stable performance at intermediate to long periods, meaning that it can be used in the seismic analysis of structural systems that may potentially experience various degrees of inelasticity.

Spectral acceleration (S_a)

The spectral acceleration describes the maximum acceleration caused by ground motion in a single-degree-of-freedom harmonic oscillator within a specified spectral period (usually the oscillator's natural period) and spectral damping level (usually 5%). A plot of S_a against a given natural vibration period and the fixed damping ratio is called the acceleration response spectrum. S_a is measured in units of g and is the most commonly used intensity measure in earthquake engineering (Song, 2014). Spectral parameters

convey a variety of information, including data provided by peak parameters and frequency contents. Generally, the first-mode spectral acceleration $S_a(T1)$ is applied as an intensity index because of a seismic response's sensitivity of a first-mode-dominated structure to the strength of the frequency content near its first-mode frequency, which was mentioned by Bazzurro et al. (1998). Vamvatsikos and Cornell (2002) stated two major advantages of this index: its methodological simplicity and ability to produce a smaller dispersion when evaluating seismic responses. Formally, $S_a(T1)$ is well correlated with the structural top displacement, the maximum storey drift, and the total input energy of structures. However, for tall and long-period structures that have higher vibration modes, the spectral acceleration of the first-mode period may not be a good solution (Ye et al., 2011). Therefore, other spectral response ordinates at various periods (e.g., 0.3, 0.5, 1.0, and 2.0 s) can be used (Allen, 2007). Kadaş and Yakut (2013) recommended using $S_a(T1)$ as a basis for compiling ground-motion records because it introduces less dispersion and is appropriately correlated with major engineering demand parameters, including the maximum top drift, the maximum inter-storey drift ratio, and the maximum base shear. Nevertheless, Ye et al. (2011) indicated that $S_a(T1)$ is not suitable for multi-degree-of-freedom (MDOF) systems (nonlinear systems) due to its poor correlation with the maximum base shear applied to these types of systems. The spectral shape is ignored when $S_a(T1)$ is used; thus, the spectral ordinates for other periods of shaking, which are important for MDOF inelastic structures, are also neglected in processing. It is widely accepted that $S_a(T1)$ is the most efficient IM for elastic SDOF systems (Bojórquez et al., 2010; Buratti, 2012). However, Ye et al. (2011) insisted that for structures (SDOF systems) with a period of less than 0.4 s, the PGA represents a better alternative. Furthermore, Fontara et al. (2012) showed that the spectral acceleration at a structure's fundamental period is only effective when it is used to predict the seismic response of regular structures and structures with low nonlinearity. Moreover, Yahyaabadi and Tehranizadeh (2011) stated that in terms of near-fault pulse-like ground motions, the estimated seismic demands may have substantial scatter when only $S_a(T1)$ is used as an intensity index.

A95 parameter

Sarma and Yang (1987) proposed a new parameter, i.e., A95, defined as the maximum value of acceleration corresponding to 95% of the Arias Intensity. More specifically, A95 is the acceleration required to generate 95% of the energy represented in

an earthquake record. Despite its low popularity, Sarma and Yang (1987) argued that it is a good measure of the damage potential of a ground-motion record.

Sustained maximum acceleration (SMA)

Nuttli (1979) proposed an intensity index known as the sustained maximum acceleration (SMA) by determining the 3rd- or 5th-largest peaks in an acceleration time history. This IM was rationalised by assuming that damage often requires repeated cycles of high-amplitude motions. Similarly to A95, this parameter is also less prevalent in the engineering sciences.

Effective design acceleration (EDA)

The effective design acceleration, which was proposed by Benjamin (1988), corresponds to the peak acceleration that remains after filtering out accelerations exceeding 8–9 Hz. Benjamin (1988) suggested this parameter after recognising that pulses of high-acceleration via high frequencies do not impact a structure's seismic response. Before Benjamin (1988), Kennedy (1980) defined the EDA as an acceleration exceeding 25% of the third-highest (absolute) peak acceleration obtained from the filtered time history. Ye et al. (2010) showed that the EDA is among the parameters (e.g., PGA, IA, IC, and RMSA) that have a high correlation for short periods and a poor correlation within the intermediate and long periods. Moreover, the EDA, similar to some of the aforementioned parameters, such as the SMA, IC, and A95, is considered less often in studies and is less popular due to its low correlation performance and its high cross-correlation with prominent IMs (Kadaş & Yakut, 2013).

2.2.2 Velocity-based intensity measures

Velocity-based intensity measures can be extracted from the velocity time history, which can also be derived from the integral of the acceleration time history. The velocity time history exhibits more intermediate frequency content (relatively), which was mentioned by Kramer (1996) and can describe intermediate period motion in a period range between 1 and 2 s (Moratto, 2008). The following are some of the most important velocity-based indices used in this study. For example, Figure 2.2 shows the velocity time history corresponding to the 1979 Imperial Valley-06 earthquake (M_W=6.53 and EpiD=27.47 km).

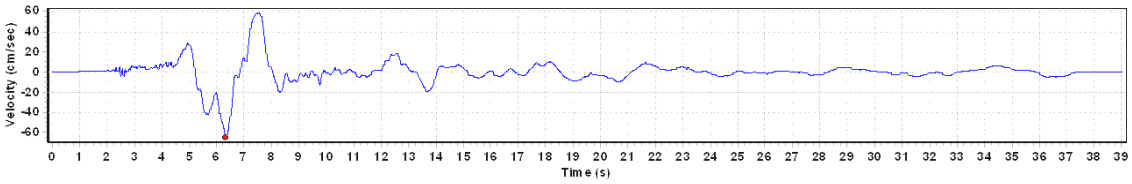


Figure 2.2: Velocity time history of the 1979 Imperial Valley-06 earthquake (M_W=6.53 and EpiD=27.47 km).

Peak ground velocity (PGV)

The peak ground velocity (PGV) is another useful parameter for characterising the ground-motion amplitude. The PGV is the maximum absolute value of the ground velocity time history and can be mathematically expressed as the maximum of the integral of an acceleration record (Equation 2.7):

$$PGV = max|v(t)| = max|\int a(t)|. \qquad \text{Equation 2.7}$$

For example, the PGV is approximately 65 cm/s for the velocity time history shown in Figure 2.2. For structures that are more sensitive to loading in the intermediate frequency range (e.g., tall or flexible buildings and bridges), the PGV is thought to be a better indicator of the damage potential compared with the PGA because the velocity is less sensitive to the higher-frequency components of the ground motion (Allen, 2007; Maniyar & Khare, 2011). Similarly, Hao et al. (2005) clarified that the PGV is more efficient than the PGA for representing the damage intensity level. Ye et al. (2011) also recommended using the PGV as an intensity index for performance-based seismic analysis and design (except for very rigid systems with high frequencies) due to its (relatively) high correlation with seismic responses over intermediate and certain long period regions. This observation is also compatible with the findings of Akkar and Özen (2005) and Kadaş and Yakut (2013), who showed that the PGV presents a better correlation beyond short periods. Akkar and Özen (2005) proposed the PGV as a stable candidate for a ground-motion IM in simplified seismic assessment approaches used to evaluate the structural performance in ground-motion hazard analysis. The PGV has been used as an index in seismic analysis in the Japanese Building Code (Japanese Ministry of Construction, 2000). Akkar and Özen (2005) also noted that the PGV is better correlated with the SDOF deformation demands compared with other ground-motion intensity measures. Moreover, the PGV can also provide useful information regarding the ground-motion frequency content and the strong

motion duration. However, as Kadaş and Yakut (2013) claimed, both the PGV and the PGA present notable dispersion when evaluating the seismic response of MDOF systems.

Root-mean-square velocity (RMSV)

In geophysics, the square root of the sum of the squares of the velocities divided by the number of values is called the root-mean-square velocity (RMSV). This parameter was proposed by Housner and Jennings (1964) as the square root of the integral of the square of the velocity over a duration t_d, which is similar to what was mentioned in the RMSA section:

$$RMSV = \sqrt{\frac{1}{t_d}\int_{t_1}^{t_2}[v(t)]^2 dt}, \qquad \text{Equation 2.8}$$

where v(t) is the velocity time history. Because the RMSV is a velocity-related index, it is highly correlated with the maximum seismic responses at intermediate periods, which is similar to the PGV. However, the RMSV is considered to be an insignificant IM due to its low correlation performance and/or its high cross-correlation with other prominent IMs, as mentioned by Kadaş and Yakut (2013).

Housner intensity (IH)

Housner (1959) defined the response spectrum intensity as the area under the pseudo-velocity response spectrum between periods of 0.1 and 2.5 s (Equation 2.9) and used it to provide an indication of a structure's potential response:

$$IH = \int_{0.1}^{2.5} PSV(\xi = 0.05, T_n)dT_n \qquad \text{Equation 2.9}$$

As a single parameter, the Housner intensity can capture important aspects of the amplitude and frequency content. The velocity response spectrum intensity can be computed with any structural damping coefficient, although 5% is commonly used. The term *pseudo* is used to avoid confusion with the true peak relative velocity. The peak pseudo-velocity (PSV) for an SDOF system with a natural period T_n is related to its peak deformation (Danciu, 2006) and can be determined as follows:

$$PSV = \left(\frac{2\pi}{T_n}\right) S_d \qquad \text{Equation 2.10}$$

where S_d is the spectral displacement. A pseudo-velocity response spectrum example related to the 1979 Imperial Valley-06 earthquake (M_W=6.53 and EpiD=27.47 km) is shown in Figure 2.3.

Martines-Rueda (1998) indicated that the impact of important response parameters, such as the energy distribution in the frequency domain, the yield period, and the period elongation related to the structural damage, is not incorporated in the Housner intensity. However, Riddell (2006) recognised that the Housner intensity is the best index in the velocity region because of its good correlation with both spectral ordinates and energy responses. As indicated by the author, the Housner intensity is not a widely used intensity indicator because it is a spectral response quantity itself.

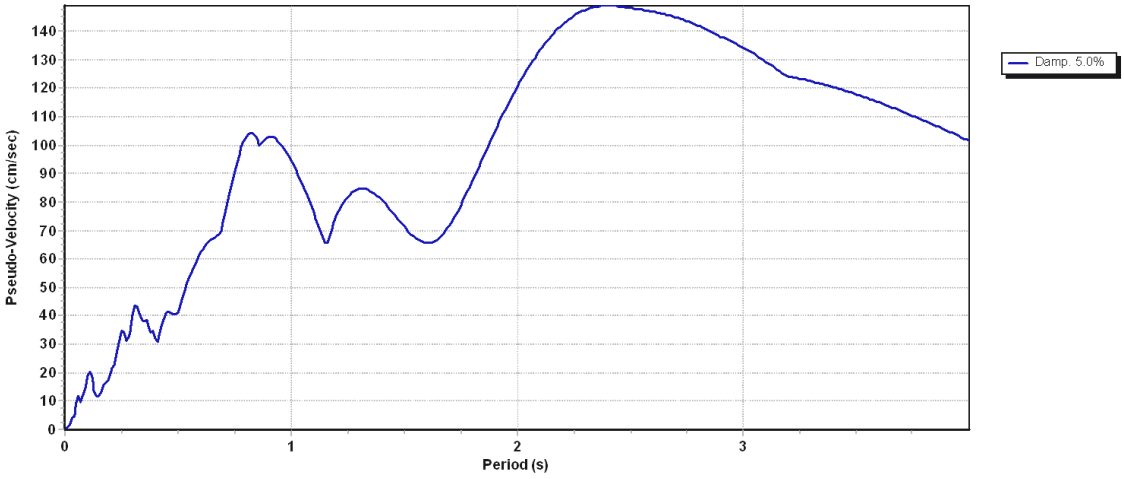


Figure 2.3: Pseudo-velocity response spectrum for the 1979 Imperial Valley-06 earthquake (M_W=6.53 and EpiD=27.47 km).

Velocity spectrum intensity (VSI)

For the seismic evaluation of earthen and rock-fill dams, Makdisi and Seed (1978) proposed the VSI by considering the fundamental periods between 0.6 and 2 s. However, because many structures have fundamental periods between 0.1 and 2.5 s, the VSI can be obtained by calculating the area under the velocity response spectrum between periods of 0.1 and 2.5 s:

$$VSI = \int_{0.1}^{2.5} S_v(\xi = 0.05, T_n) dT_n, \qquad \text{Equation 2.11}$$

Although the VSI is often interpreted as the Housner intensity, they are used individually in this study. An example of a velocity response spectrum is shown in Figure 2.4 and is related to the 1979 Imperial Valley-06 (M_W=6.53 and EpiD=27.47km) earthquake. Similar to what was discussed regarding the ASI, Kadaş and Yakut (2013) also recommended this index for selecting ground-motion records due to its reduced dispersion and appropriate correlation with major structural responses. Furthermore, other researchers (e.g. Riddell, 2006; Ye et al., 2011) have noted that the VSI tends to exhibit similar performance to the velocity-based indices at intermediate to long periods. Furthermore, Bradley et al. (2009) showed that the VSI is a sufficient intensity measure pertaining to the earthquake magnitude and source-to-site distance. They used the VSI in the seismic response analysis of pile foundations becouse of its good predictability.

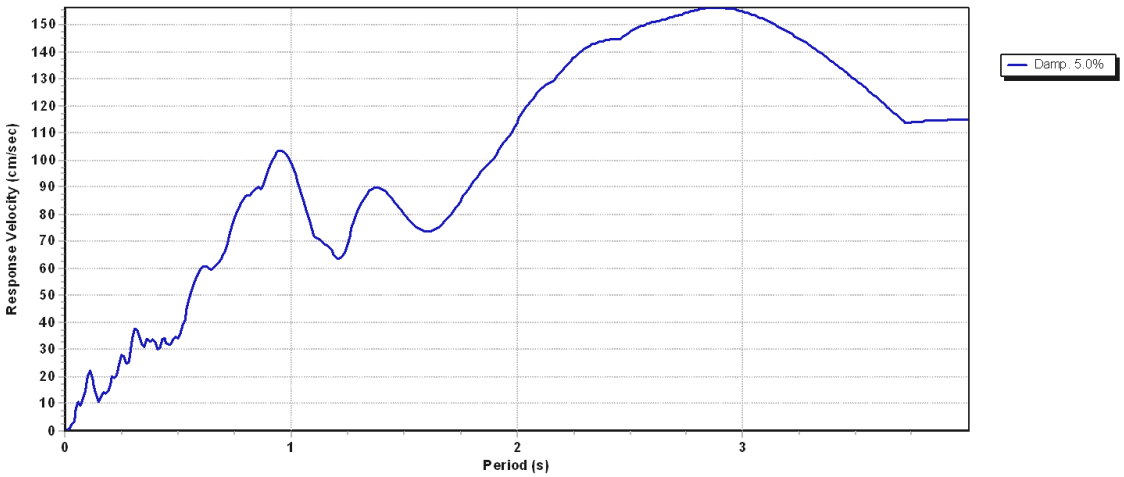


Figure 2.4: Velocity response spectrum for the 1979 Imperial Valley-06 earthquake (M_W=6.53 and EpiD=27.47 km).

Sustained maximum velocity (SMV)

Similar to the SMA, the sustained maximum velocity can be defined according to the 3rd- or 5th-largest peaks in a velocity time history (Nuttli, 1979). As mentioned earlier, for structures that require several repeated cycles of strong motion to induce damage, the sustained maximum values are a better indicator of the damage potential compared with the peak values.

Specific energy density (SED)

The specific energy density is obtained by integrating the square of the velocity over the duration of an earthquake:

$$SED = \int_0^{t_d} [v(t)]^2 dt \ . \qquad \text{Equation 2.12}$$

This measure is also known as the normalised energy density, which was initially proposed by Sarma (1971) and has units of m^2/s. The variation in kinetic energy input to the structure during an earthquake is captured by this parameter. The SED is not the true energy density; to be the true energy density, the SED must be multiplied by (vq/4), where v is the wave speed in the material and q is the mass density of the material (Sarma, 1971). Kadaş and Yakut (2013) showed that this parameter is poorly correlated with the structural response of MDOF systems, although it has been found to exhibit good correlation in SDOF systems with intermediate to long periods.

2.2.3 Displacement-based intensity measures

Parameters calculated and extracted from a displacement time history are known as displacement-based IMs. For example, the displacement time history of the 1979 Imperial Valley-06 earthquake (M_W=6.53 and EpiD=27.47 km) is shown in Figure 2.5. Several studies, such as Riddell and Garcia (2001), Riddell (2006), and Ye et al. (2011), have indicated that displacement-based IMs are suitable for the displacement region, which is characterised by long periods. Moratto (2008) stated that the displacement best represents long periods on the order of several seconds. Here, some of the IMs used in this study are described.

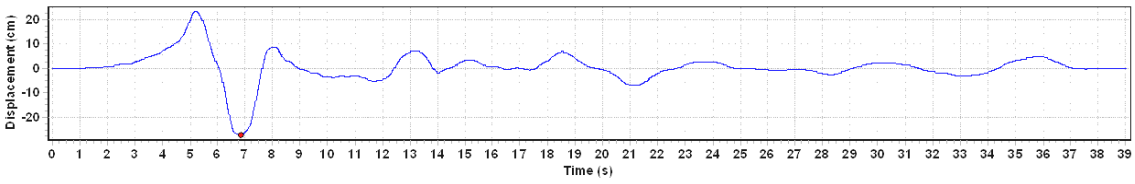


Figure 2.5: Displacement time history for the 1979 Imperial Valley-06 earthquake (M_W=6.53 and EpiD=27.47 km).

Peak ground displacement (PGD)

Because the behaviour of long-period structures is governed by the peak displacement, the peak ground displacement (PGD) and other displacement-related indices, such as the root-mean-square displacement (RMSD) and the spectral displacement (S_d), appear to be better if used for flexible systems and long periods (Ye et al., 2011; Riddell, 2006). The PGD is the simplest index in this case and can be defined as the peak value of a ground displacement history. The peak displacement for the displacement time history of the 1979 Imperial Valley-06 earthquake (M_W=6.53 and EpiD=27.47 km) is shown in Figure 2.5, which suggests a peak displacement of approximately 28 cm.

Peak ground displacements are associated with low-frequency components of earthquake ground motions and are less commonly used because of the difficulty in accurately determining correct values due to "signal processing errors in filtering, integration of accelerograms and long period noise" (Kramer, 1996). Mathematically, the PGD can be described as follows:

$$PGD = max|d(t)| \qquad \text{Equation 2.13}$$

where d(t) is the displacement time history.

Root-mean-square displacement (RMSD)

The RMSD is the root mean square of the displacement and is determined using (Housner & Jennings, 1964):

$$RMSD = \sqrt{\frac{1}{t_d}\int_{t_1}^{t_2}[d(t)]^2 dt}\ . \qquad \text{Equation 2.14}$$

As an index related to displacement, the RMSD is also suitable for long-period structures because its performance decreases with decreased structural period (Ye et al., 2011).

Spectral displacement (S_d)

Similar to S_a, the spectral displacement can be defined using the displacement instead of the acceleration and can be determined using the following relation (Karbassi, 2010):

$$S_d = S_a \left(\frac{T_n}{2\pi}\right)^2 . \qquad \text{Equation 2.15}$$

As mentioned above and based on numerical analyses, Riddell and Garcia (2001) stated that specific intensity measures are suitable for certain spectral regions. Hence, S_d as a displacement based intensity index is most suitable for the displacement region, which is characterised by long periods.

2.2.4 Duration parameters

The duration of ground motions is a quantity that is required by many fields related to earthquakes, including engineering analysis, the earthquake resistant design of structures, and any probabilistic analysis. The duration of an earthquake can greatly influence the subsequent damage to structures, especially in the case of deteriorated masonry and reinforced concrete structures. Strength of these structures degrades due to repairable cracks in masonry or concrete, yielding reinforced steel as a result of repeated cycles of seismic action. The stiffness decreases as the strength degrades; thus, the natural period increases. This implicitly assumes that "the longer period content of the ground motion occurs after the shorter period motions that initiate stiffness degradation" (Snaebjornsson & Sigbjornsson, 2008). Subsequently, the ultimate effects are major failures as mentioned by Danciu (2006). Thus, the damaging effects of an earthquake are strongly related to the duration of strong shaking. A high-amplitude but short-duration motion may not produce substantial damage to a structure, especially when compared with moderate-amplitude but long-duration motions.

The length of an earthquake is directly proportional to the length or area of the fault rupture. The time needed for rupture increases as the length or the area of the fault increase. Therefore, the magnitude and duration of strong motion both increase (Kramer, 1996). In this study, the period at which the maximum spectral acceleration occurs in an acceleration response spectrum calculated at 5% damping, which is known as the predominant period (Tp), is used in addition to the mean period (Tm), which is defined by Rathje et al. (1998) as follows:

$$Tm = \frac{\sum \frac{{C_i}^2}{f_i}}{\sum {C_i}^2}, \qquad \text{Equation 2.16}$$

where C_i is the Fourier amplitude of the accelerogram for each frequency f_i within the range 0.25–20 Hz.

The predominant period of earthquake ground motion is also a useful tool for mitigating possible damage to a structure if it is known because the structure can be designed such that its natural period does not coincide with the predominant period of earthquake ground motion. The findings of various studies that have considered the effect of strong motion duration on the structural response have been mixed, suggesting that the "effect of duration depends on the response parameter" (Foschaar et al., 2012). Foschaar et al. (2012) suggested a significant duration range of 5-95% for evaluating the structural response. However, in Kadaş and Yakut's (2013) study (nonlinear time history analyses of seven reinforced concrete moment-resisting frames with different structural properties), Tp, Tm, and the significant duration (Trifunac & Brady, 1975) were considered to be insignificant IMs because of their poor performance. Danciu (2006) indicated that a new definition of duration is needed to obtain a better relation because of the poor performance of the total and significant duration parameters studied in his work.

2.2.5 Hybrid intensity measures

Due to the complex nature of seismic loading, no single parameter is ideally suited for capturing all intensity, frequency content, and duration information that significantly influences the structural response (elastic and inelastic). For that purpose, researchers have attempted to define another set of parameters that can predict seismic demand with a smaller dispersion. These studies have shown that certain parameters that have little effect on the structural response when considered alone may positively influence the performance when combined with a second IM. These sets of parameters are known as hybrid intensity measures because they combine two or more IMs. The most commonly used parameters (also used in this study) of this type are those generated from combining two major IMs, e.g., PGA and PGV as in PGV/PGA, PGA/PGV, and PGV^2/PGA.

PGV/PGA is related to the frequency content of the ground motion because both parameters include peak values that are usually associated with motions of different frequencies (Kramer, 1996). Sucuoglu and Nurtug (1995) noted that the PGV/PGA ratio is closely correlated with the energy dissipated by an SDOF system. Furthermore, they indicated that this ratio is generally used to emphasise the effect of local soil conditions on ground-motion properties.

Another ratio derived from considering both the PGA and the PGV is PGA/PGV. Because a low PGA/PGV ratio is usually indicative of high energy contents and low frequencies, Kwon and Elnashai (2007) noted that this ratio can be used as an indicator of

the dynamic characteristics of ground motion. This concept agrees with what was indicated by Gehl et al. (2013b), who combined the PGA/PGV ratio with the PGV as earthquake descriptors. Similar to the PGV/PGA ratio, the PGA/PGV ratio combines ground-motion characteristics for high frequencies (represented by accelerations) and low frequencies (represented by velocities); hence, this ratio can reflect the effects of soil media (Elnashai & McClure, 1996). Zhu et al. (1988) showed that the PGA/PGV ratio has a significant effect on both the peak inelastic response and hysteretic energy dissipation of stiffness degrading systems. Another parameter that can be used as a hybrid IM, especially in fields related to the prediction of damage caused to pipelines by earthquakes, is PGV^2/PGA. Pineda-Porras and Ordaz (2007) showed that this parameter is related to displacement. They used the parameter for pipeline damage estimation in soft soils.

To this end and as discussed above, it could be mentioned that none of the existing indices with a single parameter or in simply combined forms performs the best for the entire range of natural frequencies. Therefore, the best solution is to individually select the most appropriate IMs for each case. According to the review above, acceleration-based IMs are most suitable for the acceleration (short period) region, IMs in terms of the velocity are most appropriate for the velocity (intermediate period) region, and displacement-based parameters are primarily applicable to the displacement region (long period). However, to avoid the bias introduced in the prediction model when using only a single parameter, a function of more than one IM (for example, two parameters) can also be applied if used for the seismic risk evaluation of buildings. This study focuses on this aspect of IMs.

2.3 Fragility Analysis

Many tools and methodologies have been proposed to mitigate the risk of an earthquake and understand its economic impacts, especially in terms of assessing the seismic vulnerability of the building stock. Fragility analysis is an effective tool that is often used for this purpose (Park et al., 2009). The fragility of a structure can be defined as its ability to resist damage caused by an earthquake (damageability); the result of this damageability is the structure's vulnerability (Porter, 2003). Both of these factors can be used to graphically represent the seismic vulnerability of a structure in terms of fragility or vulnerability curves. Although these terms appear similar and are often used

interchangeably, they are different in their explanation and description of information (Crowley et al., 2010).

Fragility curves represent the probability of reaching or exceeding specific damage levels with respect to seismic intensity measures, whereas vulnerability curves describe the probability of losses as a function of seismic intensity measures (Porter, 2003). Fragility curves can be used to develop vulnerability functions using consequence functions. Based on the availability of damage data, fragility curves can be derived using the following four methods: empirical (observed post-seismic statistics), analytical (lack of observed damage data), judgmental (expert judgment in both cases, i.e., available or unavailable damage data), or hybrid (combinations of these three methods) techniques (Rossetto & Elnashai, 2003; Calvi et al., 2006). In the absence of such observed data, analytical methods are often used to assess the vulnerability of structures and derive fragility curves (Nielson & DesRoches, 2007). Porter et al. (2007) indicated that in addition to their use in seismic risk assessments to evaluate overall structural performance, fragility curves can also be used within performance-based design through an evaluation of the fragility of building components (structural and non-structural). However, this study only accounts for seismic risk assessments; hence, only the development of overall structural fragility functions is addressed.

As mentioned above, fragility curve provide a graphical representation of exceeding a damage state with respect to one seismic ground-motion intensity measure (IM). A typical fragility curve is shown in Figure 2.6. In this figure, the probability of the structural response (damage measure), D, exceeding a specified threshold (the i-th limit state), d_i, as a function of the ground-motion IM is represented by the shaded area. It is clear from Figure 2.6 that this probability increases as the IM level increases.

The most common IMs for use in building loss assessment and fragility curve derivation were defined in previous sections. In this section, various noteworthy works that highlight methods and strategies used by researchers to develop fragility curves within the four aforementioned methods and their limitations are discussed. Afterward, the structural damage limit states that are commonly used as damage indices required by the fragility curve derivation process and those defined for URM buildings are discussed in the following section.

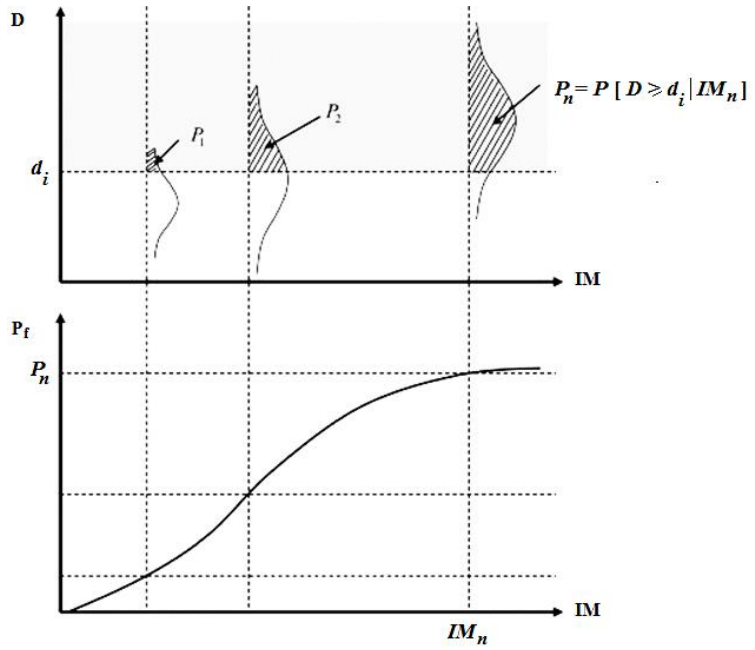


Figure 2.6: Fragility curve concept; modified from Park et al. (2009).

2.3.1 Fragility analysis based on empirical methods

The data collected through post-earthquake damage surveys can be used to derive empirical fragility curves. Sabetta et al. (1998) used the damage data collected after the 1980 Irpinia and 1984 Abruzzo earthquakes in Italy to derive fragility curves for three structural classes using the PGA, IA, and effective peak acceleration (average spectral acceleration exceeding 2.5 for periods of 0.1-0.5 s) as IMs. Murao and Yamazaki (2000) and Yamaguchi and Yamazaki (2000) constructed fragility curves in terms of the PGV for the Japanese building stock using damage data obtained after the 1995 Kobe earthquake. A cumulative lognormal distribution function was used in their study for constructing the fragility curves.

Damage statistics for 19 earthquakes (European and non-European seismic events) were used by Rossetto and Elnashai (2003) to develop fragility curves of reinforced concrete buildings. The PGA, spectral acceleration and displacement were used in their study as ground-motion measures. King et al. (2004) and Sarabandi et al. (2004) created empirical fragility curves for a variety of building types, including RC frame, rehabilitated URM, steel moment frame, RC shear wall, and wood frame buildings, using data from the 1994 Northridge, California, and 1999 Chi-Chi, Taiwan, earthquakes. In these studies, damage states defined by several guidelines, such as ATC-13 (Applied Technology Council [ACT], 1985) and the HAZUS (National Institute of Building Science [NIBS],

1999), were related to the best correlated parameters selected from 22 ground-motion intensities. Cumulative lognormal distribution functions were used to describe the fragility curves in these studies.

Fragility curves for several building types (e.g., RC and masonry construction types) characterising the Italian building stock were generated by Rota et al. (2008). Seismic severity was represented by the PGA and cumulative lognormal distribution was used for fitting the data. Spectral displacement as an indicator of seismic demand was used by Colombi et al. (2008) to derive empirical fragility curves for RC, masonry, and hybrid (RC and masonry) buildings. Four damage levels, i.e., none, slight, significant, and collapse, were used in their work. Khalfan (2013) developed empirical fragility curves for single-storey URM houses as a function of the PGA, PGV, and pseudo-spectral acceleration using damage data obtained from the 2006 Yogyakarta earthquake in Indonesia. Empirical fragility curves were also derived in several studies for other structure types such as bridges. For example, Basoz and Kiremidjian (1998) studied damage to bridges during the 1989 Loma Prieta and 1994 Northridge, CA, earthquakes. They used logistic regression analysis to obtain empirical fragility curves.

Despite its ability to capture many of the characteristics of the inventory, including topography, source characteristics, soil-structure interaction and site effects, a post-earthquake survey is unable to provide correct information regarding the materials used in buildings and the seismic design provisions (Orsini, 1999). Furthermore, the availability of observed damage data is also a main limitation related to these empirical fragility curves. To overcome this limitation, multiple data sets for various earthquakes with similar ground conditions are often used (Calvi et al., 2006). However, as Rossetto and Elnashai (2003) mentioned, this process is also not free from uncertainties that can be introduced by different construction practices applied for similar structural types in different regions. The other notable limitations of empirical fragility curves that should be highlighted here are the variety of available damage scales and the lack of systematic post-earthquake surveys.

2.3.2 Judgemental fragility curves

Judgemental fragility curves, which are also known as expert-opinion fragility curves, are obtained by relating the damage states to ground-motion IMs using the opinions and estimates of earthquake engineering experts.

Expert opinion values of damage (i.e., damage states described as none, slight, light, moderate, heavy, major, and destroyed) at each intensity level represented by the

Modified Mercalli intensity (MM) (Wood & Neumann, 1931) were used by Anagnos et al. (1995) in a joint undertaking by the National Center for Earthquake Engineering Research (NCEER) and the ATC to develop fragility curves for 40 building classes based on the information provided in ATC-13 (ATC, 1985). Lognormal functions were used to fit data in that study. Similarly, Jaiswal et al. (2011) derived fragility functions by fitting the expert estimates (from experts in over 30 countries) provided by the World Housing Encyclopedia (WHE) - Prompt Assessment of Global Earthquakes for Response (PAGER) project using the MM and PGA as the seismic intensities indices.

Jaiswal et al. (2011) and Rossetto and Elnashai (2003) noted that judgmental, or expert-opinion, fragility curves are greatly influenced by the experience of the experts. Additionally, due to the impact of expert opinions on estimating the response of different structural types at a regional scale, the level of conservatism and error present in their judgment cannot be assessed. Hence, this approach can only be applied to similar structures. Moreover, the inability to access or obtain expert opinions in regions where there is limited or a lack of experience is another issue involved within these types of fragility curves.

2.3.3 Analytical fragility assessment

Analytical procedures are usually used in cases where there is a lack of observed damage data. In these situations, data for fragility curves are collected via analytical simulations of structures using different structural representations and for increasing seismic loads. The analytical methods used in deriving fragility curves may vary in complexity from linear static to non-linear dynamic analyses. Usually, fragility analysis is used to construct fragility curves as a function of a single ground-motion intensity measure. However, more recently, deriving fragility surfaces as a function of two IMs has been recommended in several studies. Therefore, some of the major studies related to these topics (fragility curves and fragility surfaces) are presented herein.

Fragility curves

Fragility analysis was used by Korkmaz et al. (2013) to investigate the seismic risk of three different hall structures in Turkey by considering different modulus of elasticity values. In their study, nonlinear time history analyses were applied using data on 25 different real earthquakes. Using non-linear dynamic analysis, fragility curves for RC

frames (low-, mid-, and high-rise buildings) were developed by Singhal and Kiremidjian (1996) using the spectral acceleration as the IM and lognormal distribution to fit the damage distributions. Erberik (2008) developed a set of fragility curves to assess the seismic vulnerability of typical mid- and low-rise buildings in Turkey. Nonlinear time history analyses were conducted using random variables of structural input parameters with a set of 100 natural strong motion records and considering the PGV as a reference hazard parameter. The estimated damage data from these curves were then compared with the observed damage data of Duzce buildings after the 1999 earthquake. Furthermore, a probabilistic assessment study was undertaken by Korkmaz (2006) by employing a fragility analysis for 5- and 7-storey reinforced concrete buildings (bare frame), which is representative of typical buildings in Turkey. The PGA and PGV were used as the two seismic intensity parameters for a set of 25 earthquakes related to the 1999 Kocaeli earthquake. For different numbers of storeys (3, 5 and 7 storeys), Kirçil and Polat (2006) developed fragility curves in terms of S_a, PGA, and S_d to evaluate the vulnerability of mid-rise RC frame buildings in Istanbul to 12 artificial ground motions. Similarly, fragility analysis has been employed in many other regions and countries.

For example, Dumova-Jovanoska (2000) developed fragility curves for two RC structures in Macedonia using ground-motion data for 240 synthetic earthquakes. Kumar (2010) constructed fragility curves for mid-rise commercial reinforced buildings in India. Marefat et al. (2008) investigated the vulnerability of 6 groups of reinforced concrete buildings in Tehran (Iran) using fragility curves that were produced based on incremental dynamic analyses (IDAs) and 17 natural earthquake records. Similarly, Kalantari et al. (2010) performed a fragility assessment for 6 steel and RC school frame buildings in Shahre-Kurd, Iran. They used incremental dynamic analysis to study the effect of 20 strong motion records and considered the PGA as a measurement of earthquake intensity. Likewise, Bakhshi and Karimi (2006) proposed fragility curves and used them to examine the seismic vulnerability of URM buildings commonly used in Iran. Uncertainties in the structural materials and seismic inputs were treated using Monte Carlo simulation and 12 real earthquake records. Nonlinear dynamic analyses and CAV were used to construct capacity curves for buildings and to represent earthquake intensities, respectively. In another work, the fragility functions of URM buildings constructed of solid brick were derived by Ahmad et al. (2010) using a nonlinear static procedure. Additionally, fragility curves for a gravity-designed two-storey reinforced concrete frame building were produced by Ramamoorthy et al. (2006) using nonlinear time history analyses and a suite of

synthetic ground motions in the Memphis region. In that work, the spectral acceleration at the fundamental period of the building was selected as a representative parameter for the earthquake intensity. Mwafy (2012) developed fragility curves for high-rise buildings (10 to 60 storeys) in the UAE using 20 natural and synthetic ground-motion records. He used inelastic pushover analyses and incremental dynamic analyses and considered the PGA as the earthquake intensity measure.

In the study by Frankie (2010), a set of simulation-based fragility relationships for URM buildings in the central and eastern United States were compared to those generated using opinion-based methods (e.g., HAZUS curves). For that purpose, pushover analysis, artificial accelerograms, and the PGA were selected. Similarly, Park et al. (2009) derived fragility curves for a typical low-rise URM building (2 stories) in the central and southern US; the authors used nonlinear time history analysis and the spectral acceleration in their study. More recently, Mitropoulou and Papadrakakis (2011) developed fragility curves for RC buildings based on neural network (NN) incremental dynamic analysis predictions. The fundamental period spectral acceleration was used as an intensity measurement parameter in addition to 100 natural earthquake records in that study.

To investigate the nature of seismic parameters that may be used as reliable damage potential descriptors rather than the currently utilised peak ground parameters, Nanos (2011) developed a methodology based on measuring the response characteristics of a given RC frame structure (detailed in accordance to the Greek anti-seismic engineering practice), by implementing a non-linear dynamic analysis based on ground-motion data for 450 earthquakes derived from spectrum compatible artificial accelerograms. The extensive fragility curves showed that "energy parameters were more efficient in seismic damage potential derivation compared to peak ground and seismic duration parameters", and particularly, the Arias intensity (IA) was diagnosed as a better descriptor of the seismic damage potential compared with other peak and energy parameters. Moreover, Nanos (2011) asserted that this conclusion might be different for other structural types and/or systems (i.e., masonry or steel structures); therefore, extensive studies on this topic must be conducted.

Fragility curves relate a single ground-motion parameter (scalar-valued IM) to structural damage. Nevertheless, according to the literature, there are many types of ground-motion parameters that have been applied in different studies, in different regions, and for different types of structural typology. Therefore, the use of a single parameter may introduce a large scatter in the fragility function, which has been discussed in numerous

studies, including Shome et al. (1998), Song and Ellingwood (1999), Ellingwood (2001), and Kafali and Grigoriu (2004). To this end, certain researchers (Cordova et al., 2001; Baker & Cornell, 2005; Luco et al., 2005; Kafali & Grigoriu, 2007; Rajeev et al., 2008; Sei'ichiro Fukushima, 2010; Seyedi et al., 2010; Gehl et al. 2011; Gehl et al. 2013b) have encouraged the use of vector-valued IM functions or fragility surfaces. A review of some of these works is provided below.

Fragility surfaces

Several studies have shown that the use of more than one IM results in improved damage state prediction for buildings compared to the use of a single IM. Some of these works are discussed below.

Baker and Cornell (2005) showed that considering two IMs (which is called a vector-based IM), namely, the spectral acceleration (S_a) and epsilon (ε), instead of one IM can improve the accuracy of the collapse probability for a MDOF structure. They defined epsilon as the difference between the logarithms of the observed S_a and the predicted S_a for a given period. Rajeev et al. (2007) used S_a at the first and second natural periods of the structure as the IMs in the seismic risk analysis of a RC frame structure. Kafali and Grigoriu (2007) presented a method for calculating system fragility as a function of the moment magnitude and the source-to-site distance and derived fragility surfaces for that system (linear and nonlinear systems). Omine et al. (2008) used the PGA and PGV to develop fragility surfaces. Seyedi et al. (2010) asserted that a significant reduction in the scatter of the fragility function may result from an increase from one to two ground-motion parameters. They developed a set of fragility surfaces represented by two parameters (S_d at periods representing the first and second mode) for RC buildings using nonlinear time history analyses. In total, 169 natural and 571 artificial records were used in their study. Koutsourelakis (2010) illustrated a general procedure based on logistic regression to derive fragility surfaces and indicated that more accurate predictions can be attained when the probability of exceeding a damage level is expressed as function of several IMs. Gehl et al. (2011) proposed a methodology to develop fragility surfaces for a two-storey URM building using a non-parametric probability density via kernel density estimators. They showed that fragility surfaces can provide a more complete description of the seismic aggression by considering the epistemic uncertainty related to the second IM parameter. The PGA and PGD were used as ground-motion intensity indices.

A complete framework for developing vector-valued fragility functions for evaluating the damage level of an URM building (two stories) was formulated by Ghel et al. (2013b) using several hundred nonlinear time history analyses. Different IMs (e.g., the PGA, PGV, response spectral accelerations at various periods, IA and various duration parameters) were considered for characterising earthquake shaking. Fragility surfaces were derived using two pairs of IMs [PGV-PGA/PGV and S_a(0.15 s)-S_a(0.5 s)] and compared to scalar-based fragility curves using only a single IM.

With respect to the analytical fragility analysis approach, simulation of complex models has become easier and more computationally efficient than before due to modern advances in computational abilities. Consequently, analytical fragility functions are becoming more commonly used. However, soil-structure interaction, highly nonlinear behaviour, and post-peak behaviour represent additional noticeable limitations to this method (Rossetto & Elnashai, 2003).

2.3.4 Hybrid methods for fragility analysis

To overcome the various above-mentioned deficiencies involved in each method, hybrid fragility relationships based on a combination of any or all of the three methods may be used. Kappos et al. (2006) indicated that a hybrid approach is useful to reduce "the computational effort of analytical modelling and compensate for the subjective bias of expert judgment method". Various works related to the hybrid method are summarised below.

Akkar et al. (2005) used a hybrid method to determine the fragility functions for low- and mid-rise reinforced concrete buildings in Turkey. Dynamic responses and capacity of buildings were obtained from response history analyses of 32 two- to five-storey buildings. Variability in the fundamental period vibration, lateral strength, and damage limit state were considered as uncertainties. A set of 82 natural strong ground motions recorded worldwide was used in their study and PGV was selected as seismic intensity measure. Singhal and Kiremidjian (1998) applied a method to update analytical fragility curves using damage data from the January 17, 1994, Northridge earthquake in California through a Bayesian statistical analysis method. Kappos et al. (2006) and Kappos & Panagopoulos (2010) developed a hybrid method to derive fragility curves (in terms of the PGA and S_d) for RC and URM building types by combining statistical data and analytical procedures.

Despite its advantages, the hybrid method also has various drawbacks. For example, variations in materials and construction practices that are inherently reflected within the empirical data are ignored when simulating and modelling buildings within the analytical approach. Furthermore, uncertainties involved in the ground-motion data (variability of earthquakes), empirical damage data (lack of sufficient information or knowledge), and analytical models should be considered when using this method.

After defining ground-motion parameters to represent seismic action in a region and reviewing different fragility analysis methods, the next section presents and defines the various indices used in such studies (i.e., the seismic fragility analysis of structures) as indicators of the structural damage level due to earthquakes; only the damage indices related to URM buildings are reviewed herein.

2.4 Damage Measure and Performance Levels

Defining an indicator to represent the structural damage level induced by an earthquake is another main step that must be taken toward developing fragility curves or surfaces. The damage level (also known as the performance level or limit state) is a limit or point at which a structure can no longer carry the load and perform a desired function. These limits and levels can be obtained through post-earthquake investigations. These observations are expressed either in verbal (qualitative manner) or mathematical terms via the correlation between the observed damage and the system response (Erbay, 2007). The latter (quantitative manner) is widely accepted because the former (qualitative manner) is only applicable to similar building configurations and ground-motion conditions.

Several damage measures, such as displacement-based measures (Rodriguez & Aristizabal, 1999), energy-based criteria (Wong & Wang, 2001), and hybrid measures (Rodriguez & Aristizabal, 1999; Park & Ang, 1985), have been proposed by researchers to assess buildings subject to earthquake loadings. However, damage levels or limit states are typically specified in terms of displacement-based measures by defining the drift as a system response parameter (Pinto, 2000). Tomazevic (2007) hypothesised that structural damage in masonry buildings is independent of the masonry type and can be directly associated with storey drift. These damage values can be easily evaluated, and many correlations between them and structural damage limit states have been discussed in the literature.

Tomazevic (2007) defined four limit states on the capacity curve for masonry buildings and proposed threshold storey rotation values corresponding to each of these states (see Table 2.1). The response of a building is generally defined through capacity curves, which describe the relationship between the base shear and roof displacement; these characteristics are obtained using pushover analysis. FEMA-356 (ASCE, 2000) proposes three performance levels for URM buildings: immediate occupancy (IO), life safety (LS), and collapse prevention (CP), which are described in Figure 2.7 and Table 2.2. These limit states correspond to the stiffness, strength, and ductility requirements in structural design. Through an analytical study, Erbay (2007) proposed four threshold drift values, namely, 0.1, 0.6, 1, and 2%, for four limit states, namely, immediate occupancy (IO), life safety (LS), collapse prevention (CP), and total collapse (TP), respectively. HAZUS (NIBS, 1999) defines four threshold values based on the maximum drift ratio for four damage limit states (given in Table 2.3): slight damage, moderate damage, extensive damage, and complete damage. Furthermore, for URM buildings, four performance levels (i.e., LS1, LS2, LS3, and LS4) were suggested by Calvi (1999) (see Table 2.4). He proposed a relationship between the inter-storey drift ratio and the defined damage levels. All of the aforementioned performance levels are summarised in Table 2.5 in terms of IO, LS, and CP and with respect to each study.

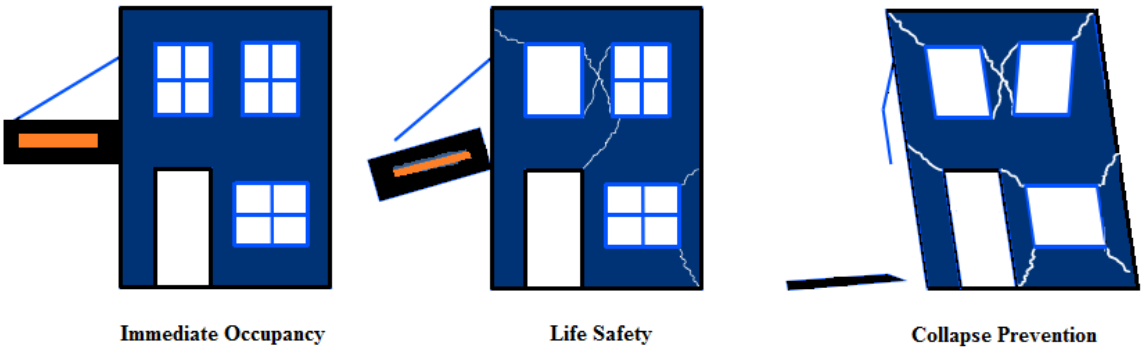


Figure 2.7: Performance levels for URM buildings provided by FEMA-356 (ASCE, 2000).

Table 2.1: Structural performance levels for URM structures according to Tomazevic (2007).

Structural performance levels	Overall damage descriptions	Drift ratio
Crack limit state	Defined by the formation of the first cracks in the walls. This limit is associated to the serviceability limit state of the structure.	0.2 - 0.4%
Maximum resistance	Defined by the maximum resistance of the system.	0.3 - 0.6%
Design ultimate limit state	The resistance of the system degrades below an acceptable level corresponding to 20% degradation of the maximum resistance.	
Collapse limit state	Defined by the partial or total collapse of the structure.	1.0 - 2.0%

Table 2.2: Structural performance levels for URM structures based on FEMA-356 (ASCE, 2000).

Damage levels	Structural performance levels		
	Collapse prevention (CP)	Life safety (LS)	Immediate occupancy (IO)
	Severe	Moderate	Light
Overall damage descriptions	Extensive cracking. Face course and veneer may peel off. Noticeable in-plane and out-of-plane offsets.	Extensive cracking. Noticeable in-plane offsets in the masonry and minor out-of-plane offsets.	Minor cracking of veneers. Minor spalling in veneers at a few corner openings. No observable out-of-plane offsets.
Drift ratio	1.0%	0.6%	0.3%

Table 2.3: HAZUS-defined damage limit states for URM structures (NIBS, 1999).

Structural performance levels	Overall damage descriptions	Drift ratio (Low code)	Drift ratio (Pre-code)
Complete damage	Structure has collapsed or is in imminent danger of collapse due to in-plane or out-of-plane failure of the walls. Approximately 15% of the total area of URM buildings with complete damage is expected to collapse.	3.5%	2.8%
Extensive damage	In buildings with relatively large wall openings, most walls have suffered extensive cracking. Some parapets and gable end walls have fallen. Beams or trusses may have moved relative to their supports.	1.5%	1.2%
Moderate damage	Most wall surfaces exhibit diagonal cracks; some of the walls exhibit larger diagonal cracks. Masonry walls may have visible separation from diaphragms. Significant cracking of parapets. Some masonry may fall from walls or parapets.	0.6%	0.5%
Slight damage	Diagonal, stair-step hairline cracks on masonry wall surfaces. Larger cracks around door and window openings in walls with large openings. Movements of lintels. Cracks at the bases of parapets.	0.3%	0.2%

Table 2.4: Damage limit states for URM buildings as suggested by Calvi (1999).

Structural performance levels	Overall damage descriptions	Drift ratio
LS1	No damage: expected response is linear elastic.	-
LS2	Minor structural damage and moderate non-structural damage: the building can immediately be placed in service after the earthquake without significant strengthening and repair.	0.1%
LS3	Significant structural damage and extensive non-structural damage exist: the building cannot be utilised after the earthquake without significant repair, although repair and strengthening are feasible.	0.3%
LS4	Collapse: repairs are neither possible nor economically reasonable. The structure must be demolished after the earthquake. Beyond this limiting state, global collapse with danger to human life is expected.	0.5%

Table 2.5: Performance levels for URM buildings according to different studies.

Author		Drift ratio %		
		IO	LS	CP
FEMA-356 (ASCE,2000)		0.3	0.6	1.0
HAZUS (NIBS, 1999)	Low code	0.3	0.6-1.5	3.5
	Pre-code	0.2	0.5-1.2	2.8
Tomazevic (2007)		0.2-0.4	0.3-0.6	1.0-2.0
Erbay (2007)		0.1	0.6	1.0
Calvi (1999)		0.1	0.3	0.5

Furthermore, various other studies have provided mathematical relationships between limit states and building responses. For example, Milutinovic and Trendafiloski (2003) related top displacements (in terms of the yield point d_y and ultimate displacement d_u) to the EMS-98 (Grunthal, 1998) damage scale through the equations shown in Table 2.6. The classification of damage to masonry buildings according to EMS-98 (Grunthal, 1998) is shown in Figure 2.8. Figure 2.9 shows a bilinear approximation of force-deformation capacity curves used to obtain yield and ultimate points based on the equations proposed by Milutinovic and Trendafiloski (2003) for defining damage limit states. Lagomarsino and Penna (2003) proposed a criterion to split the capacity curve into five parts after identifying the yield point and ultimate displacement of the structure. Following the suggested criterion, they defined the spectral displacements for the five damage limit states (see Table 2.7).

Based on Table 2.5, significant differences between the threshold drift values proposed in several studies for the same class of structures (URM buildings) are observed due to the use of different experimental results in each reference. Therefore, as Gehl et al. (2011) indicated, the definitions and criteria that consider fixed drift ratios for a generic building typology and do not differ for specific cases may not be as accurate as those that are defined based on the strength and ductility in each model such as the Milutinovic and Trendafiloski (2003) criterion. Therefore, the criterion recommended by Milutinovic and Trendafiloski (2003) has been used in several studies, including Gehl et al. (2011) and Gehl et al. (2013b), to develop fragility curves and surfaces for URM buildings. Similarly, the criterion proposed by Lagomarsino and Penna (2003) was used by Bilgin and Korini (2012) to evaluate the seismic capacity of URM buildings in Albania.

Table 2.6: Correlation between the EMS-98 (Grunthal, 1998) damage states and the transient top displacement for URM buildings according to Milutinovic and Trendafiloski (2003).

Damage state	Displacement limit equation
Slight	$d = 0.7d_y$
Moderate	$d = 0.7d_y + 0.05(0.9d_u - 0.7d_y)$
Extensive	$d = 0.7d_y + 0.2(0.9d_u - 0.7d_y)$
Very heavy	$d = 0.7d_y + 0.5(0.9d_u - 0.7d_y)$
Collapse	$d = 0.9d_u$

Table 2.7: Performance levels and criteria proposed by Lagomarsino and Penna (2003).

Damage state	Spectral displacement, S_d
No damage	$S_d < 0.7d_y$
Slight	$0.7d_y < S_d < d_y$
Moderate	$d_y < S_d < d_y + 0.25(d_u - d_y)$
Extensive	$d_y + 0.25(d_u - d_y) < S_d < d_u$
Complete	$S_d > d_u$

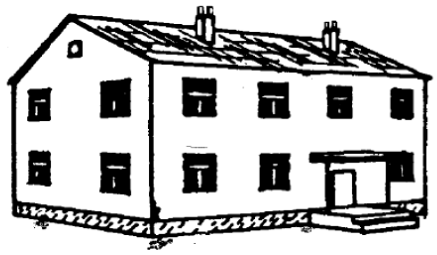
Classification of damage to masonry buildings	
	**Grade 1: Negligible to slight damage** **(no structural damage,** **slight non-structural damage)** Hair-line cracks in very few walls. Fall of small pieces of plaster only. Fall of loose stones from upper parts of buildings in very few cases.
	**Grade 2: Moderate damage** **(slight structural damage, moderate non-structural damage)** Cracks in many walls. Fall of fairly large pieces of plaster. Partial collapse of chimneys.
	**Grade 3: Substantial to heavy damage** **(moderate structural damage,** **heavy non-structural damage)** Large and extensive cracks in most walls. Roof tiles detach. Chimneys fracture at the roof line; failure of individual non-structural elements (partitions, gable walls).
	**Grade 4: Very heavy damage** **(heavy structural damage,** **very heavy non-structural damage)** Serious failure of walls; partial structural failure of roofs and floors.
	**Grade 5: Destruction** **(very heavy structural damage)** Total or near total collapse.

Figure 2.8: Classification of damage to masonry buildings from EMS-98

(Grunthal, 1998).

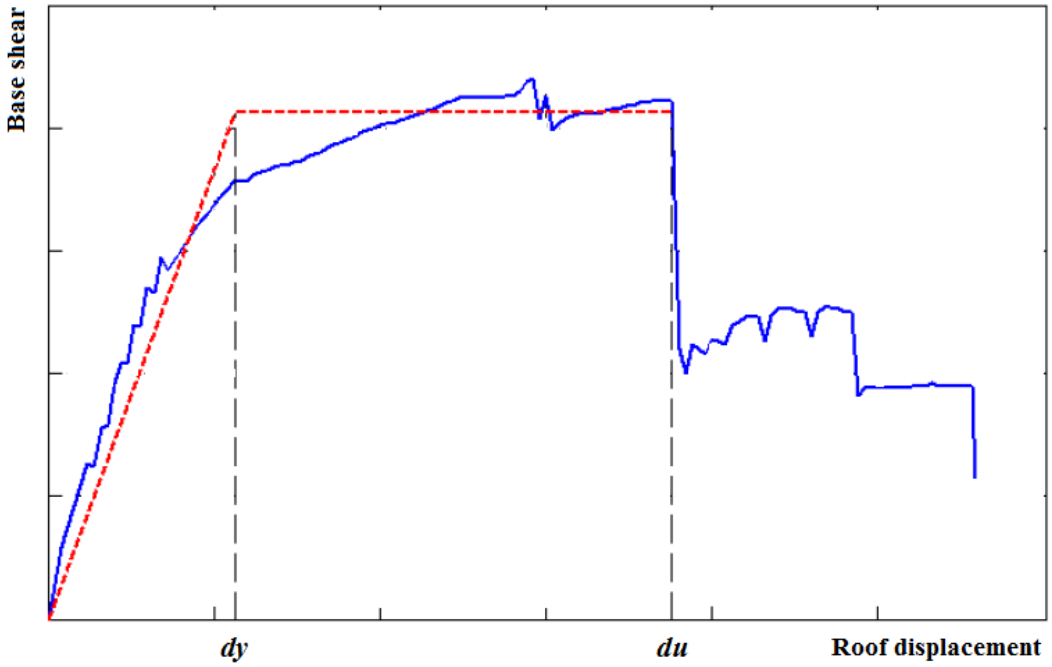


Figure 2.9: Bilinear approximation of force-deformation capacity curves used to obtain the yield and ultimate points used in the equations proposed by Milutinovic and Trendafiloski (2003) for defining the damage limit states.

CHAPTER 3. PROPOSED METHODOLOGY FOR ASSESSING THE EARTHQUAKE FRAGILITY OF EXISTING BUILDINGS

This study attempts to examine the fragility of existing buildings in the Kurdistan region of Iraq subject to earthquakes using a strictly analytical approach. As mentioned before, fragility analysis is an effective assessment tool that is often used to evaluate the vulnerability of structures and derive fragility curves (Park et al., 2009; Nielson & DesRoches, 2007). The literature review presented in Chapter 2 provided a comprehensive picture demonstrating the lack of observed post-earthquake damage data; an analytical method is a suitable choice for such a context. Generally, to perform an analytical fragility analysis and derive fragility curves in any region, knowledge acquisition concerning the following three main subjects must be performed:

1) Building stock data and corresponding construction materials associated with the region of study for the purpose of selecting typical structures that can represent the proper extent of the building stock;

2) Seismic activity data relevant to that region for selecting an appropriate number of ground motions and other important parameters that are indicative of the seismicity in the region; and

3) Analytical simulation methods used to estimate the structural response of the selected building or buildings.

Formally, all of these areas are vague and are considered to be main sources of technical uncertainty when producing fragility curves (Choun & Elnashai, 2010). These uncertainties can be categorised into two main groups. The first is inherently random (aleatoric) uncertainty such as indistinctive variability in the ground motions or mechanical properties of materials. The second is epistemic uncertainty caused by a lack of knowledge or competency, model error, objectivity in damage state definitions, ground-motion intensity measures (IMs) and the selection of ground-motion parameters (Mackie & Nielson, 2009; Gehl et al., 2013b). Thus, the accuracy of information used in the analytical process of a building's fragility assessment is fundamental to decreasing the effects of these uncertainties. Three main sources of uncertainty – namely, the variability in the mechanical properties of the target buildings versus ground motions, the variability in ground-motion intensity measures, and the variability in the selected ground motions - are of great scientific interest to this study.

This chapter identifies an appropriate approach to structural assessments by considering the aforementioned variabilities. Each stage of the methodology is thoroughly discussed, focusing on its pragmatic application and further potential contribution to the entire structural assessment. The methodology includes the following stages:

- Structure type classification;
- Earthquake record selection;
- Analysis methods; and
- Results and data post-processing.

The following sections are intended to identify the primary considerations for each part of the methodology.

3.1 Structure Type Classification

The first step towards estimating the earthquake failure probability of structures within any geographical region is to prepare and design an appropriate database for the relevant structural systems in the study area (Liao et al., 2006). Identifying the structural typology based on this database results in an initial understanding of the characteristics and morphology of the relevant materials and the weaknesses and vulnerabilities of the system. These details and other essential information, such as the existing and intended use of the structure, the structure's location, and its external nature, provide a clear portrayal of the subsequent stages of the overall assessment.

Exposure information for a selected region requires a standard systematic inventory system that classifies the structures according to their type, occupancy, and function. Such a categorisation is necessary to derive realistic estimates of seismic risk and loss (FEMA, 1999). Structural components, such as an inventory data collection and classification systems, that are similar to the system reported in the ATC-13 (ATC, 1985) and HAZUS (FEMA, 1999) publications (see Figure 3.1) can be specifically designed for the region under study. Figure 3.1 clearly indicates that constructed facilities are categorised into four main systems: buildings and facilities, transportation systems, utility systems, and hazardous material systems. Excluding the last three systems, the buildings and facilities are considered to represent the general building stock.

Various parameters (e.g., structural, non-structural, and occupancy) that affect and characterise the damage are considered in HAZUS (FEMA, 2003; FEMA, 2006) and used to classify the general building stock. Table 3.1 clearly shows that HAZUS incorporates 16 basic building types based on the type of lateral force-resisting system and four main

building construction materials (i.e., wood, masonry, concrete, and steel). Moreover, these 16 types are further subdivided into 36 classes according to height. In a similar manner, there is a systematic attempt to characterise buildings in the study region by considering the general building stock facilities (e.g., residential, commercial, and industrial) and leaving other constructed facilities, such as hospitals, schools, dams, power plants, highways, bridges, electric power system, and portable water systems, for future studies.

It is also essential to use the correct data on construction material characteristics for the purposes of structural analysis and to gather data on the geometry and identified structural systems for developing a mathematical representation of the structure. Physical (e.g., porosity, density, and colour), chemical (e.g., percentage of salts, types of salts and their likely origins, silica and other compounds, and percentage of carbonation within concrete), and mechanical (e.g., yield stress, compressive strength, shear modulus, and elastic modulus) properties must be considered to accurately identify the material characteristics used in the construction of the selected structure type. However, the most important characteristics are the mechanical properties, which are usually used in the mathematical representation of the materials for structural analysis tools. These mechanical properties can be a source of aleatoric (inherently random) uncertainty in fragility analysis.

Rapid urban growth has led to the construction of various building types, and 75% of casualties caused by earthquakes are associated with the collapse of these structures (Coburn & Spence, 2002). URM buildings have the highest vulnerability to seismic activity (Grünthal, 1998). Aleatoric uncertainty in the form of variability in the mechanical properties of URM structures is more prevalent and applicable than for other structures, e.g., reinforced concrete and steel frame constructions (Gehl et al., 2013b). Several recent studies have considered the effect of this randomness in material parameters in relation to the seismic response of reinforced concrete buildings. They have found that this effect can be avoided in contrast to the effect of strong motion variability (e.g., Kwon & Elnashai, 2007; Fragiadakis & Vamvatsikos, 2010). However, only a few studies regarding URM buildings have been published; moreover, these works have provided no firm conclusions. For example, Cattari et al. (2010) and Pagnini et al. (2011) found that the variability in the structural capacity is the main source of uncertainty when developing fragility curves for masonry buildings in comparison with the uncertainty in the seismic demand and limit state definition. Moreover, Rota et al. (2010) showed that the variability in material properties is not essential in the presence of variability in ground motions; hence, this variability can be ignored when analysing masonry buildings. However, this issue requires careful clarification and additional

attention because few thematic studies pertaining to URM buildings and relevant firm conclusions have been published.

Given the context, variability in material properties is determined in this study by conducting a number of time history analyses of a single-storey building using a fixed accelerogram scaled to three levels of PGA intensities (0.05, 0.2, and 0.8 g). Three levels of PGA intensities (including 0.1, 0.2, and 0.3 g) were previously considered in the study by Rota et al. (2010). Therefore, in the present study, an extreme level of PGA is selected in addition to low and moderate levels of PGA for the purposes of comparison. It is the author's intention to expand on the available boundaries and assess extreme cases and structural behaviour while knowing that such circumstances would not arise in reality. For each level of intensity, a set of variants has been assigned with respect to the different material properties, including compressive strength, shear modulus, and elastic modulus. A second set of analyses has been completed by actualising the variability in terms of ground motions and ignoring the variability induced by the material properties. Therefore, variability in ground motions has been computed by taking and fixing the mean values of material properties for all selected records. Two standard deviations of the results (in terms of top displacement) for the ground motion and material parameter variabilities have also been calculated. Then, the total standard deviation has been calculated as the square root of the sum of the squares of the two aforementioned standard deviations for each considered PGA level (0.05, 0.2, and 0.8g). A distinct comparison between two sets of analyses should provide the real degree of importance of each set accordingly. A similar approach has been followed in several other studies for combining the uncertainty attributed to capacity and demand when determining the total damage state variability (e.g., FEMA, 1999; RISK-UE, 2004; Rota et al., 2010). PGA is used here as a scaling parameter because several studies (e.g., Vamvatsikos & Cornell, 2002; Dhakal et al., 2007) have indicated that the PGA can be comprehensively utilised for short-period structures without producing a significant dispersion in response. Specifically, Dhakal et al. (2007) insisted that PGA-based scaling should not be used for estimating longer period structures.

Characteristics of the building stock in Kurdistan, the selected procedure for classifying buildings as well as the materials used in the construction of buildings are presented in Chapter 4.

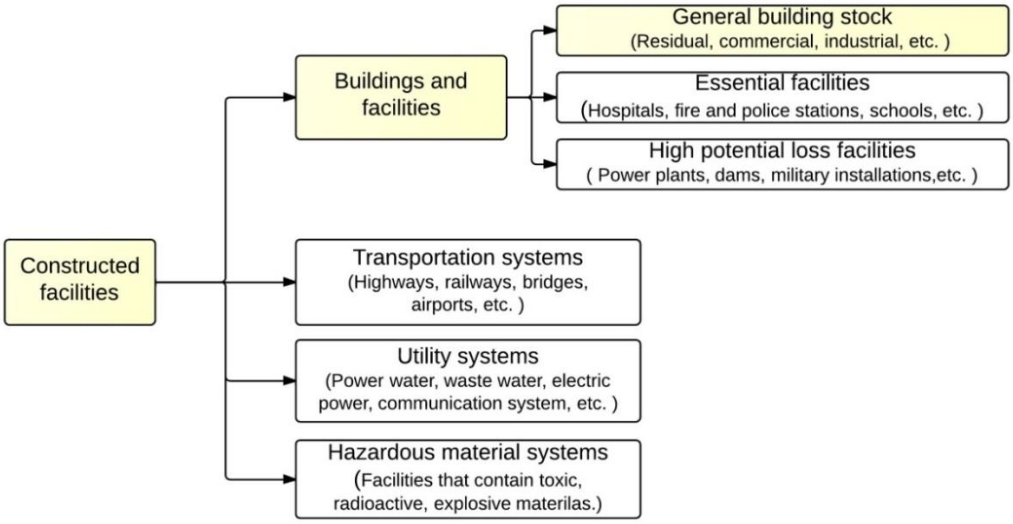


Figure 3.1: A structural inventory classification system (ATC, 1985; FEMA, 1999).

Table 3.1: HAZUS-MH earthquake model building types (FEMA, 2003).

No.	Label	Description	Height			
			Range		Typical	
			Name	Stories	Stories	Feet
1	W1	**Wood, Light Frame (≤ 5,000 sq. ft.)**		1 - 2	1	14
2	W2	**Wood, Commercial and Industrial (>5,000 sq. ft.)**		All	2	24
3	S1L	**Steel Moment Frame**	Low-Rise	1 - 3	2	24
4	S1M		Mid-Rise	4 - 7	5	60
5	S1H		High-Rise	8+	13	156
6	S2L	**Steel Braced Frame**	Low-Rise	1 - 3	2	24
7	S2M		Mid-Rise	4 - 7	5	60
8	S2H		High-Rise	8+	13	156
9	S3	**Steel Light Frame**		All	1	15
10	S4L	**Steel Frame with Cast-in-Place Concrete Shear Walls**	Low-Rise	1 - 3	2	24
11	S4M		Mid-Rise	4 - 7	5	60
12	S4H		High-Rise	8+	13	156
13	S5L	**Steel Frame with Unreinforced Masonry Infill Walls**	Low-Rise	1 - 3	2	24
14	S5M		Mid-Rise	4 - 7	5	60 156
15	S5H		High-Rise	8+	13	
16	C1L	**Concrete Moment Frame**	Low-Rise	1 - 3	2	20
17	C1M		Mid-Rise	4 - 7	5	50
18	C1H		High-Rise	8+	12	120
19	C2L	**Concrete Shear Walls**	Low-Rise	1 - 3	2	20
20	C2M		Mid-Rise	4 - 7	5	50
21	C2H		High-Rise	8+	12	120
22	C3L	**Concrete Frame with Unreinforced Masonry Infill Walls**	Low-Rise	1 - 3	2	20
23	C3M		Mid-Rise	4 - 7	5	50
24	C3H		High-Rise	8+	12	120
25	PC1	**Precast Concrete Tilt-Up Walls**		All	1	15
26	PC2L	**Precast Concrete Frames with Concrete Shear Walls**	Low-Rise	1 - 3	2	20
27	PC2M		Mid-Rise	4 - 7	5	50
28	PC2H		High-Rise	8+	12	120
29	RM1L	**Reinforced Masonry Bearing Walls with Wood or Metal Deck Diaphragms**	Low-Rise	1-3	2	20
30	RM2M		Mid-Rise	4+	5	50
31	RM2L	**Reinforced Masonry Bearing Walls with Precast Concrete Diaphragms**	Low-Rise	1 - 3	2	20
32	RM2M RM2H		Mid-Rise	4 - 7	5	50
33			High-Rise	8+	12	120
34	URML	**Unreinforced Masonry Bearing Walls**	Low-Rise	1 – 2	1	15
35	URMM		Mid-Rise	3+	3	35
36	MH	**Mobile Homes**		All	1	10

3.2 Earthquake Record Selection

The proper selection of seismic inputs corresponding to the seismic and site conditions of the area where the structures are located is also very important for performing a successful fragility analysis. Seismic inputs for ground motions are usually described through time histories (accelerograms), which can be artificial, synthetic or real. Artificial accelerograms are usually derived based on correlations between these time history records and a generated "power spectral density function" that is based on the target mean response spectrum (e.g., code design spectrum). However, generated accelerograms may incorporate unnecessary motion cycles, resulting in a high energy content, which can be unrealistic or non-objective (Iervolino et al., 2008). The use of these records is not recommended in nonlinear analyses based on the warning put forth by Hancock et al. (2006).

The second type of record, i.e., synthetic records, can be simulated from seismological source prototypes by considering the path and site effects (Dumova-Jovanoska, 2000; Singhal & Kiremidjian, 1996). The need for different seismological and geological information, which is rarely entirely accessible, is the primary complexity related to these types of accelerograms (Boore, 2003).

The last type of record represents real or natural ground-motion records obtained directly from real events. These accelerograms incorporate all ground-motion characteristics and the geological and site-related conditions. Moreover, due to the increased accessibility of recorded time histories in online databases [e.g., the Pacific Earthquake Engineering Research Next Generation Attenuation (PEER NGA) flat file database (Chiou et al., 2008)], the use of natural earthquake records has become a prevalent and highly applicable research technique over the last several years (Iervolino & Manfredi, 2008). However, the availability of numerous records representing a specific scenario is viewed as a main challenge for this type of time history (Iervolino & Manfredi, 2008). To overcome this problem, many functional methods and productive procedures have been proposed by various specialists. For example, Haselton et al. (2009) assessed 14 methods selected from a wide variety of available techniques (more than 40 methods) to identify a reliable ground-motion selection and modification (GMSM) method, especially in terms of nonlinear dynamic structure analyses. Four reinforced concrete structural models were used for that purpose by emphasising the maximum inter-storey drift ratio (MIDR) as a structural response parameter. In total, five main groups were classified

according to the objectives of these methods (see Table 3.2). For example, the methods that attempt to estimate the average response or evaluate a structure's full response probability distribution under a specific earthquake scenario were grouped separately. Moreover, the techniques used in each method for defining the earthquake scenarios were also considered in the work of Haselton et al. (2009), for example, by considering $S_a(T1)$, magnitude, and distance. Haselton et al. (2009) insisted that accounting for the record properties and the use of a proper target can lead to an accurate building response estimate. The conditional mean spectrum (CMS) (Baker, 2011a) selection methods (group III in Table 3.2) have been shown to provide the best predicted median response, followed by group IV using some proxy for the CMS. Groups V, II, and I (in Table 3.2) provide the next best levels of accuracy (for additional details, refer to Haselton et al., 2009). However, to estimate the effect of seismic record variance regarding the estimated structural response, which is usually ignored by the CMS, Jayaram et al. (2011) proposed an algorithm based on Monte Carlo simulations and a greedy optimisation technique to select ground motions with response spectra that exhibit the desired mean and variance.

Haselton et al. (2009) and Baker (2011a) showed that the commonly used uniform hazard spectrum (UHS) (e.g., code standard spectrum) is an inappropriate criteria for selecting ground motions to be matched. The inappropriateness of the UHS along with the selection of records based on building codes is associated with the fact that "they conservatively imply that large amplitude spectral values will occur at all periods within a single ground-motion" Baker (2011a). The UHS is obtained from probabilistic seismic hazard analysis by determining the exceeded spectral accelerations with a selected rate over all periods.

Furthermore, generating fragility curves usually involves the consideration of only one ground-motion parameter, such as the peak ground acceleration (PGA), to represent the seismic action in an area of interest, which results in neglecting other important characteristics such as the frequency or energy content (Gehl et al., 2013b). The result is a source of epistemic uncertainty that can interfere with a seismic risk analysis during the construction of fragility curves. The choice of relevant ground-motion parameters always requires additional effort in such studies. Moreover, several studies (e.g., Riddell, 2006; Nanos, 2011, Buratti, 2012; Elenas, 2013) have indicated that different IMs might provide different advantages in predicting structural response, especially when they are used as a potential damage descriptor. Numerous studies (e.g., Krawinkler et al. ,2003; Seyedi et al., 2010; Gehl et al. 2011; Xu & Wen, 2012; Gehl et al., 2013b) have noted that a single

parameter is unable to represent the seismic action of a region and formally describe its complex nature. As a result, substantial uncertainty is present in vulnerability assessments. Krawinkler et al. (2003) noted that no single parameter is ideal for that purpose, which remains an issue among other instrumental challenges that are related to performance-based earthquake engineering. Furthermore, it has been recently demonstrated that increasing from one to two ground-motion parameters could reduce this uncertainty and produce improved vulnerability results for a given building type (Seyedi et al., 2010; Gehl et al., 2011; Gehl et al., 2013b). Consequently, the fragility curves can be functionally converted to a fragility surface by expressing the damage probability as a function of two parameters. Nevertheless, it is essential to consider variability in the selected sets of earthquake records using different ground-motion selection methods before determining the effects of the ground-motion parameters on the fragility curves.

Table 3.2: GMSM methods used by Haselton et al. (2009).

Method name	Description	Group number
S_a(T1) scaling with Bin selection	Select ground motions by magnitude and distance and scaling to S_a(T1)	I
ATC-58 (2011) method		
Building code selection and scaling	Select ground motions with "spectral shapes" that are similar to the spectral shape of the target UHS	II
CMS selection with scaling	Similar to the Group II methods except that a conditional mean spectrum is used as the target spectral shape for selecting ground motions instead of the UHS	III
Genetic algorithm selection to match CMS		
Semi-automated selection & scaling to match CMS		
Design ground-motion library (2008) tool		
Target spectrum based on epsilon (ε) correlations	Similar to the Group III methods except that they use ε as a proxy for the spectral shape of the CMS (i.e select ground-motions whose ε at the fundamental period of the building, denoted ε(T1), is as close as possible to the ε(T1) of the ground-motion scenario)	IV
ε selection with S_d(T1) scaling		
ATC-63(2008) method applied to maximum inter-story drift ratio — Far-field set		
ATC-63(2008) method applied to maximum inter-story drift ratio —Near-field set		
Inelastic spectral displacement scaling	Select ground motions with properties that are related to inelastic spectral displacement and that are as close as possible to corresponding properties estimated for the ground-motion scenario	V
Vector of record properties identified by proxy		
Inelastic response surface scaling — 1st Mode		
Inelastic response surface scaling — 1st-2nd Modes		
Intensity measure $IM_{1II\&2E}$ (Luco and Cornell 2007) scaling		

The aforementioned recommendations from Haselton et al. (2009) were based on predicting the MIDR for reinforced concrete structures; hence, the recommendations may differ for other structural responses (e.g., peak floor accelerations or element force demands) or be technically inefficient for other structural systems and material types. Although numerous methods have been proposed by researchers, no universal method for selecting and modifying an appropriate ground motion are available (Haselton et al., 2009). Furthermore, the selection of seismic inputs that are compatible with the seismic and site conditions of the study area is critically important to nonlinear dynamic analysis methods, specifically if incremental dynamic analysis (IDA) is applied. IDA is a progressive approach that has recently gained substantial attention from contemporary researchers (Vamvatsikos & Cornell, 2002). Therefore, differences in decision making regarding the selection of a suitable method for selecting and scaling the ground motions, especially in nonlinear dynamic analysis, can lead to serious problems in terms of safety and cost management (such as the overdesign or underdesign of structures).

With respect to the number of ground motions, Iervolino and Manfredi (2008) reviewed the current strategies for selecting ground motions to ultimately estimate the physical response of structures in terms of nonlinear dynamic analysis. Based on seismic risk assessment aim, they indicated that the number of records required and the parameters needed for in-depth consideration may vary. The number of records and parameters may be extremely large for a probabilistic structural assessment, considering that the purpose of ground-motion selection is to achieve small computational costs while correctly estimating the response. However, in the code-based design, records are selected to match a previously designed scenario; therefore, the number of records is technically limited. For example, EC8 (CEN, 2003) specifies a set of three accelerograms as the minimum requirement, considering the maximum influence of the records on a structure. If a structure's mean effect is of interest, at least seven records should be used for analysing non-spatial structures and 14 records (representing the two horizontal components of the seismic action with seven records each) for spatial structures. The seismic action's vertical component, which is only utilised in special cases, such as long-span elements, is ignored. In addition, the records should adequately represent a site's seismic and soil conditions. More recently, Gehl et al. (2013a) showed that "a relatively small error is introduced into the final results by the limited number of analyses usually used" in developing fragility curves using non-linear dynamic analysis methods. Their implications were based on the use of different statistical measures aimed to estimating the quality of fragility functions

obtained for a SDOF reinforced concrete model by actualising various numbers of ground motions. In a more focused study, Shome et al. (1998) and Shome (1999) revealed that for a mid-rise building, 10 to 20 records should be sufficient to predict its seismic demand with great confidence. However, upon reducing the number of records, as mentioned by Iervolino and Manfredi (2008), a ground-motion intensity measure that is oriented towards a structure's sensitive response within the established range can be used in addition to considering other vital features such as the earthquake magnitude, the distance to the source or the fault mechanism. This method would also require specific and occasionally extended information about site hazards, which is similar to the procedures associated with the UHS.

Based on the aforementioned facts and considering the variability in GMSM methods during the fragility analysis of URM buildings, the use of the CMS method, the Jayarame et al. (2011) algorithm, and code-based selection method (here, EC8) to identify different sets of ground motions is the focus of this study. Additionally, a target response spectrum that is designed and proposed for the study region (no specific seismic codes and response spectra are yet available for the area, especially for URM buildings) is also utilised in compliance with the three aforementioned methods. As previously indicated, artificial, synthetic and natural time histories can be used in analytical fragility analysis. However, for the purposes of nonlinear analysis, natural time histories are preferable based on the findings of Idris (1991) and Hancock et al. (2006). Thus, real recordings from the PEER NGA database (available at http://peer.berkeley.edu/assets/NGA_Flatfile.xls) are used in this study because no other recordings specific to the given area are available.

Nonetheless, the presence of numerous records representing a specific scenario remains a major challenge with real time histories. Thus, it is essential to rely on a specialised technique that assists in eliminating bias for a certain event, especially when a specific number of records (minimum of seven records based on the EC8 or ASCE (2007) recommendations for predicting the average structural response) are desirable. Furthermore, to consider variability in the number of ground motions, a suite of seven records must be retrieved for comparison with other influential suites of records (more details in section 5.3). For that purpose, a substantial number of available response spectra, which were derived from the PEER NGA flat file database (Chiou et al., 2008), are scaled to match the target spectra defined for the four aforementioned GMSM methods. Based on the recommendations of Shome et al. (1998) and Shome (1999), fifteen records are

selected for each method. Subsequently, the following procedures are used to reduce the number of selected ground motions to a set of seven records:

1- Select the records with the highest number of repetitions;

2- Grant priority to records with the highest number of repetitions;

3- Consider only one relevant record per event; and

4- If the number of repeated records is insufficient or more than one record is correlated with a specific event, select the event record that exhibits the greatest contribution or has properties that are similar to the characteristics of the previously selected records. The final goal is to reduce the gap in the range of desired seismic properties, including the earthquake magnitude, the epicentral distance, and intensity measures such as the peak ground acceleration (PGA).

The majority of the aforementioned GMSM methods require specific information, professional competency in seismology and geology, and engineering expertise. Therefore, a detailed examination of the study area's seismic and geology conditions is required to select proper suites of ground motions, which is needed when using an analytical approach to fragility assessment. Chapter 5 describes such information about the area of study.

3.3 Analytical Methods

Structural analysis relies on data from the previously discussed stages to develop a mathematical representation of a structure. Structural analysis is mainly concerned with ascertaining a structure's behaviour when subjected to a force. Different analysis methods have been used by researchers for that purpose. FEMA-356 (ASCE, 2000) discussed two main approaches, namely, linear and nonlinear methods, as the two most commonly used analytical methods for the static or dynamic evaluation of building responses due to the effects of seismic loads. Depending on the sufficiency of an applied action in terms of the acceleration relative to a structure's natural frequency, it is possible to distinguish between static and dynamic analyses. The analysis can be functionally simplified to a static analysis by ignoring inertial forces if a load is slowly applied. Dynamic analysis, on the other hand, is used to examine the dynamic behaviour of structures when actions or loads operate under high-acceleration conditions (Sharma & Maru, 2014). The following provides a brief

description of the available analytical methods and an indication of their strengths and limitations because it is crucial to select the most suitable method for this study.

3.3.1 Linear and nonlinear static/dynamic procedures

Linear static and linear dynamic methods are usually used in force-based design for developing structures or their components whereby their resisting capacity overcomes the lateral load effects. These methods can be utilised for simple regular structures, i.e. low- to medium-rise buildings (Aldemir, 2010). However, the need to determine a structure's displacement capacity requires other types of analyses such as nonlinear static and nonlinear dynamic analyses. Although linear analysis methods are computationally faster and less demanding than nonlinear methods, they provide poor and insufficient accuracy in terms of capturing a structure's nonlinear behaviour compared with nonlinear analysis methods. These methods are also unable to provide sufficient information on velocity, acceleration, and dissipated energy (ASCE, 2000; Erbay, 2007).

Nonlinear static analysis, which is also known as pushover analysis, is often applied to determine the force-displacement characteristics of structures. The tool technically neglects inertial effects and damping while a structure or building is laterally pushed until one or more of its components yield and enter the plastic region. Once this stage is realised, a structure's capacity or pushover curve is drawn by plotting the base shear versus the structure's roof displacement (Aldemir, 2010). This method is suitable for structures whose behaviour is dominated by the first mode of vibration (TEC, 2007). However, as Sharma and Maru (2014) indicated, the dynamic nature of a load must be considered during the entire design process to mitigate the effects of seismic loads. Similar to a static procedure, nonlinear static analysis neglects this action and ignores higher modes. Furthermore, the limited consideration of ground-motion parameters (e.g., displacement-based demand parameters) and the lack of any information regarding velocity and acceleration are other pitfalls that were mentioned by Erbay (2007) regarding this method. Nevertheless, this method is still computationally faster and less demanding than nonlinear dynamic analysis.

Based on a comprehensive review of the current available methods for assessing a building's seismic collapse capacity, Villaverde (2007) deduced that these methods are not completely satisfactory. The author noticed that the methods based on single-degree-of-freedom models are functionally unreliable because of their focus on an assumed shape for the failure mechanism, which technically complicates early prediction even in terms of pushover analysis. Additionally, Villaverde (2007) claimed that nonlinear static methods

cannot be viewed as reliable because they neglect important effects and are based on incorrect assumptions. Alternatively, Erbay (2007) asserted that nonlinear dynamic analysis or nonlinear time history analysis is the best simulation procedure for determining a structure's dynamic response. Thus it can be used as a powerful tool to overcome all of the aforementioned issues within the other frameworks of analytical methods. The only disadvantage of this method is that it is the most computationally demanding and time-consuming approach. However, progress in computer processing capacities has made it possible to overcome this issue and perform complex analytical work with greater accuracy (Vamvatsikos & Cornell, 2002). Regarding the nonlinear behaviour of materials, giving engineers the opportunity to analyse buildings based on given earthquake time histories, define their capacity curve points, determine the capacity and displacement demands of every member for short time intervals, and consequently obtain the expected damage of individual structural components is a very important advantage of nonlinear time history analysis (Erbay, 2007).

3.3.2 Incremental dynamic analysis

An evaluation technique known as incremental dynamic analysis (IDA) (Vamvatsikos & Cornell, 2002) has gained popularity because it provides realistic opportunities for developing and displaying a structure's complete capacity curve instead of providing capacity curve points for several different earthquakes. Upon using this technique, the time history is scaled to produce multiple levels of intensity. The history is then applied to the structure by performing multiple nonlinear dynamic analyses of a structural model until the complete capacity curve is formed, which is used to illustrate the entire range of structural behaviour, i.e., from elastic to inelastic, ultimately extending to global dynamic instability and collapse. Because the available recorded natural time histories are insufficient for covering all possible intensity levels, scaling by multiplying or dividing the accelerations by a common factor is often used. Several studies, such as Vamvatsikos and Cornell (2002) and Dhakal et al. (2007), have verified that the bias found in the outcomes may differ based on the ground-motion IM used for scaling the time history. These studies showed that various IMs, such as the PGA, can be applied to short-period structures without producing a significant dispersion in the response. Dhakal et al. (2007) insisted that PGA-based scaling requires only a few records to predict the median response for a structure with a period of 0.3 s, whereas S_a (T1, 5%) becomes the most effective IM for a structure with a period of 0.5 s. Furthermore, Vamvatsikos and Cornell (2002) showed that the scaling itself does not

produce bias in the final results if sufficiency of the IM is achieved by identifying its dependence on the magnitude and distance scenarios.

The ability to estimate a structure's global dynamic capacity, an improved structural response behaviour measurement with increasing ground-motion intensity, and a proper understanding of the structural response under more severe and intense ground motions are some of the advantages noted by Vamvatsikos and Cornell (2002) regarding IDA. However, properly selecting the parameters for measuring ground-motion intensities and using numerous ground-motion records are certain problematic areas that should be considered when utilising IDA as a method for an in-depth analysis (Vilaverde, 2007). IDA can be linear or nonlinear based on the elastic or inelastic nature of the materials. The flowchart presented in Figure 3.2 shows an outline of IDA.

Considering the uncertainty in damage states and because there are many variabilities in fragility assessments regarding building geometry, materials, element and material relationships, ground motions, number of ground motions and their parameters needed to represent the seismic action in a region, the required precision for performing a non-linear time history analysis must be identified in this study. In doing so, it is possible to reduce the uncertainty related to the simulation and analysis methods and consequently decrease the bias produced via the summation of all of the involved uncertainties. Given the aforementioned points regarding analytical methods, for viewing the nonlinear behaviour of URM buildings using a necessary and thorough estimation of their performance under seismic loads, incremental dynamic analysis appears to be truly valuable and a vital choice for this study. IDA is used as an alternative to the nonlinear time history because it is functionally and practically impossible to obtain the required natural time histories for all of the intensity levels that are required to develop the fragility curves or surfaces by evaluating their formal compatibility with the study region's seismic signature. PGA is used here as the scaling parameter for the time histories.

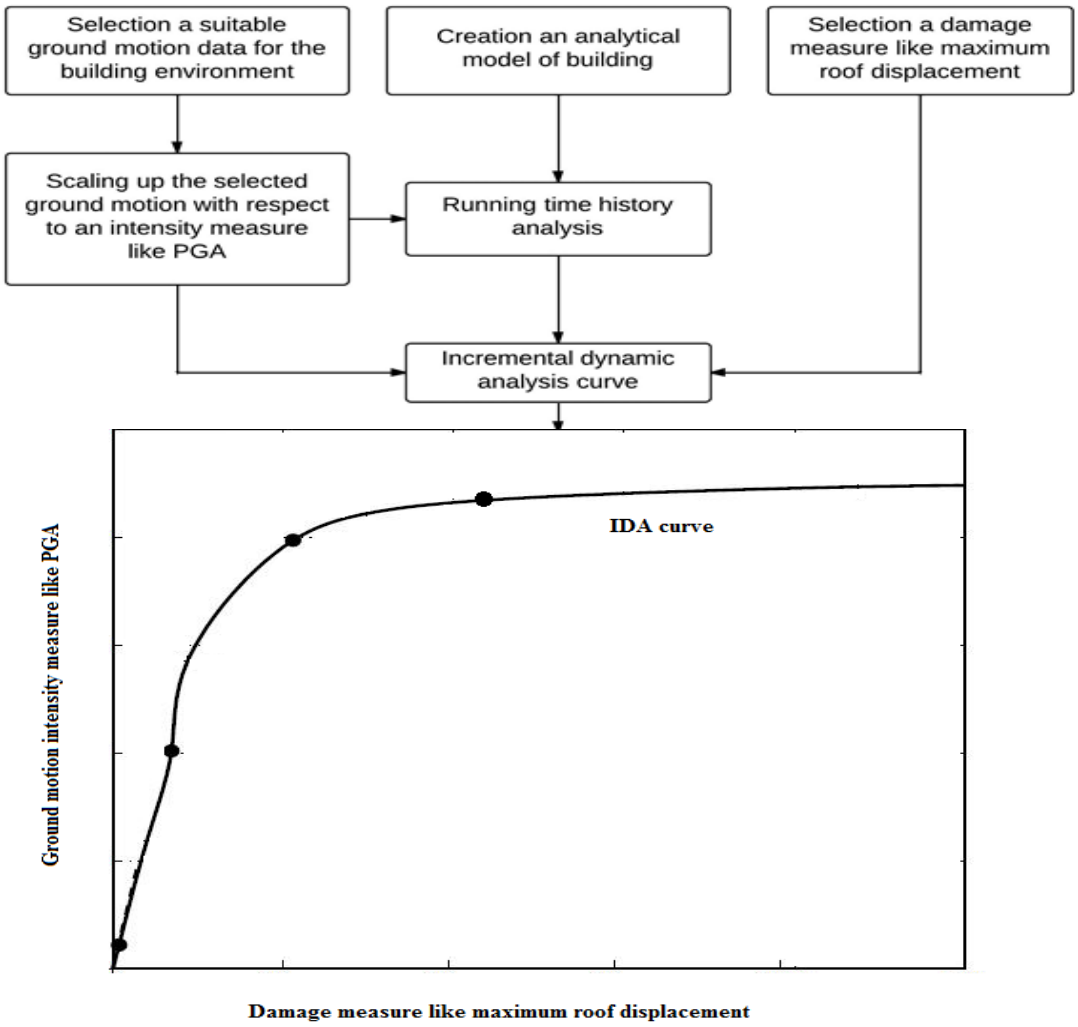


Figure 3.2: Flowchart showing an outline of IDA.

3.4 Data Post-processing

After the IDA of selected buildings, it can be possible to perform some post-processing on the results data. This post-processing allows the analysis of the previous result sets in a formal way. Determining the optimal ground-motion intensity measure and conducting fragility analysis using a suitable regression function to fit data are two essential tools which are needed by this stage of the study. The following provides a detailed explanation of the tools used in this study for that puspose.

3.4.1 Selection of significant seismic parameters

One of the most important issues in designing fragility curves or surfaces is the vast number of IMs that have been proposed by researchers and specialists (as mentioned in the literature review). To determine the best and most relevant seismic parameters, the impact of several uncertainties involved in the process of fragility analysis must be considered in the first step; otherwise, incorrect decisions pertaining to the selection of the optimal IMs may be made. Once the data required for performing a fragility analysis are collected and the structures have been tested via model simulations, the effect of the variability in the mechanical properties of the masonry can be estimated against the effect of the variability in the ground motions. By neglecting less effective variability, the remainder of this study is conducted using only the most effective one, although other variabilities are also examined. Various scenarios are proposed in this context and used to better understand the effect of these variabilities on the selection of the optimal IMs and the consequential effects on the fragility curves:

1- Different ground-motion selection and modification methods (i.e., selection using the EC8-based target response spectrum and the proposed response spectrum for the study region, the CMS method, and the algorithm proposed by Jayaram et al.);

2- Different numbers of ground motions (i.e., 7, 12, 15, 25, 35/38, and 60 records);

3- Different numbers of storeys (one and two) located in different seismic hazardous zones (defined for the specific region); and

4- Scaled records versus unscaled records.

By relying on a suitable regression function (section 3.4.2) and utilising a graphical and statistical representation of the outcomes for all available IMs in this study, the optimal ground-motion parameters are chosen and used to derive the fragility curves and surfaces.

However, considering the large number of existing IMs, their classification in terms of efficiency and sufficiency requires considerable computational efforts and it might also be sensitive to the selected regression functionand. Moreover, it is important to remember that only one ground-motion parameter is used to derive fragility curves, and all other IMs are ignored. This generates certain difficulties in selecting the optimal parameter. Within the field of machine learning, this issue can be addressed by applying a procedure known as feature selection (which is also called attribute selection or variable selection) to identify and remove all irrelevant and redundant information to the greatest extent possible (Hall, 1999). Therefore, the next step is to use beneficial machine learning tools to facilitate the

selection of an optimal IM or IMs and compare the available results with those obtained from statistical regression analysis used in the first step of the operation.

3.4.2 Wrapper method as a machine learning tool for the optimal selection of a ground-motion IM

Feature selection is an effective machine learning method for selecting a good subset of features because it provides options for reducing dimensionality (Yu & Liu, 2003), removing irrelevant data records, increasing learning accuracy, and improving the interpretability of results (Blum & Langley, 1997; Dash & Liu, 1997; Kohavi & John, 1997). Furthermore, the method can also be beneficial in domains that contain numerous features and comparatively few instances or appropriate samples. In addition, as a component of procedural data analysis, feature selection is capable of determining relationships between features and identifying key features in the prediction of results (Padmapriya & Maragatham, 2013).

Many feature subset selection methods have been proposed and examined. In general, these methods can be divided into three main categories: wrapper methods, filter methods, and embedded methods (Kojadinovic & Wottka, 2000). Wrapper method (Figure 3.3) defines several subsets with relevant scores based on a predefined predictive algorithm to train the model. By counting a model's error rate for each subset, it is possible to obtain scores and eventually identify the optimal subset for proper selection. Wrapper method covers feature dependencies that typically provide the highest performing feature set for a specific model. However, this method is subject to various issues, including over-fitting and computational complexity (Saeys et al., 2007), that must be considered. Filter method utilises a proxy instead of an error rate to score the feature subsets. Filter techniques are general pre-processing algorithms that avoid feature dependencies and interactions with classifiers. Filters are typically faster and less computationally intensive than wrappers, although they provide no guarantee of accuracy in terms of classification performance (Saeys et al., 2007). Furthermore, embedded methods incorporate feature selection as a component of the model construction and training process. Similar to wrapper techniques, these methods are specific to a given classifier function (Padmapriya & Maragatham, 2013; Saeys et al., 2007). In terms of computational complexity, embedded approaches are typically less complicated than wrapper methods, although they are more complex than filter methods (Saeys et al., 2007).

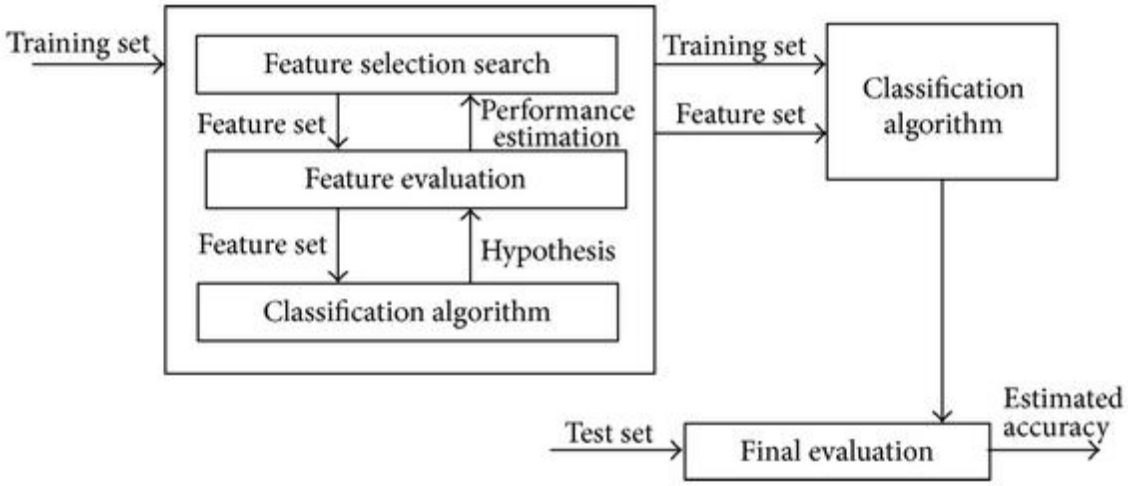


Figure 3.3: Flowchart of a wrapper-based approach for feature subset selection (Kohavi & John, 1997).

Based on a comparative study of automatic feature selection methods, Kohavi and John (1997) recommended the use of the wrapper approach, therein arguing that this approach fits the principles that the optimal features depend on both the specific learning algorithm and the training set at hand. Hence, the wrapper method is applied in this study because the fragility curves are typically obtained for a specific class of structures located in a specified region using particular dataset of earthquake records.

As mentioned above, the wrapper feature selection method is applied in terms of a predefined learning algorithm to sort different available feature subsets. In this method, as shown in Figure 3.3, the training process is used as a black box (i.e., selection is conducted using only an interface without having any knowledge of the algorithm). It is repetitively run on the dataset using a number of feature subsets. To assess its performance on each subset, some technique is used and finally the feature subset with the maximum evaluation is selected as the desired set on which to run the induction algorithm. An independent test set that was not used during the search is then used to evaluate the resulting classifier. Maximisation of classification accuracy on an unseen test set is a typical goal of such supervised learning algorithms (Kohavi & John, 1997). Several of these automated classifier algorithms or learning models are available in the Weka toolbox (Witten & Frank, 1999). A wrapper subset evaluator is included in the Weka toolbox. This tool considers the list of features for the set that yield incorrectly classified instances for the given classifier. The classifiers used in this study, including the Naive Bayes (NB), nearest

neighbour, logistic, multilayer perceptron neural network (MLP), decision table, and decision tree algorithms, are briefly described below.

Naive Bayes (NB) classifier

The NB algorithm (Bayes, 1763) assumes that features are independent with respect to a given class variable. For example, ground motion may be considered as a destructive earthquake activity at a region if it exhibits a magnitude of 8, is located close to the site, and has a long duration. A NB classifier concedes that each of these parameters independently contributes to the probability of this earthquake being very destructive, regardless of the presence or absence of the other parameters. Within the NB algorithm, the most likely class is obtained for an unknown example after representing each class in a probabilistic summary. This algorithm is able to classify data even after only a small amount of training; however, if features are correlated to one another, the NB algorithm may perform poorly (Ratanamahatana & Gunopulos, 2002). A 2D binary classification with the Naive Bayes approach is shown in Figure 3.4 for data with two classes (plus and circle) in two dimensions (x1 and x2). A Gaussian for each class and dimension is estimated using the Naive Bayes classifier so that the estimated distribution can be visualised by enclosing the densest region. Given the input, the decision boundary (coloured in red) represents the location where the probabilities of each class are equal.

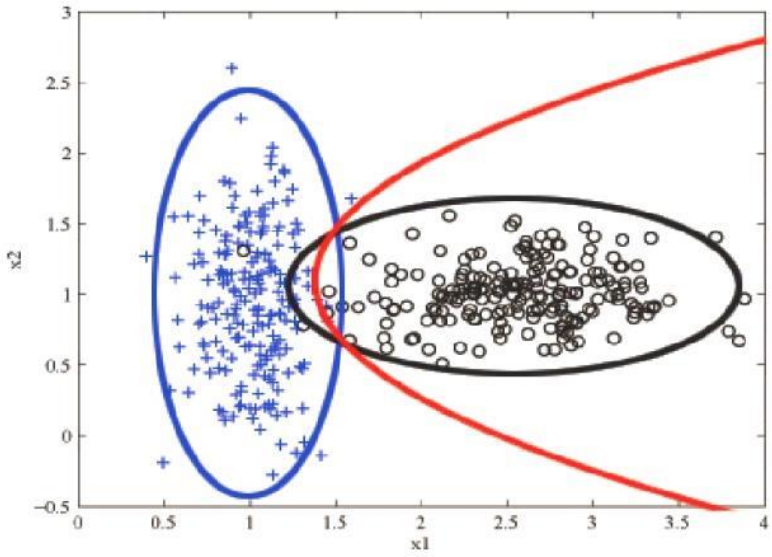


Figure 3.4: 2D binary classification with the Naive Bayes classifier. A density contour is drawn for the Gaussian model of each class; the decision boundary is shown in red.

Nearest neighbour classifier

The nearest neighbour algorithm (Fix & Hodges, 1951) searches the nearest example in the training set and assigns an unknown pattern to the same class of that example. Despite the formal simplicity of the algorithm, it provides optimistic results and is an important benchmark method (Kubat & Cooperson, 2001). In practice, the most common class in the K nearest neighbours is used, where K is a parameter set by the user. The value considers how many neighbours (where a neighbour is defined based on the distance metric that is applied) impact the classification. K is usually an odd number, especially if the number of classes is a multiple of 2. If K=1, then the algorithm is simply titled the nearest neighbour algorithm (Devi et al., 2014). The basic philosophy behind KNN is simple; the method implies that the nearest examples have similar classes. Thus, K should be carefully selected because a large value requires additional computational expense, whereas a very small value can result in noisy results. A simple approach for selecting K is to use the square root of the number of points in the training data set (as a rule of thumb) (Jirina & Jirina, 2011). Still, Hassanat et al. (2014) claimed that the simplified approach is not a proper processing option for the KNN classifier. Furthermore, they indicated that the use of many neighbours does not always assist in improving the KNN classifier's accuracy and objectivity. The results of Hassanat et al. (2014)'s study also indicated that there is no optimal K because each data set tends to prioritise a specific number of neighbours. Hence, after examining the accuracy of the KNN algorithm using different values for K, two KNN classifiers, i.e., K=1 and K=3, are selected for this study.

Relying on the defined value for K, K samples with the shortest distances are selected, and the class with more samples inbound becomes the desired outcome. For example, in Figure 3.5, the test sample (green circle) should be categorised either to the first class of blue squares or to the second class of red triangles. If K= 3 (solid line circle), it is assigned to the second class because only 2 triangles and 1 square are within the inner circle. However, when K is 5 (dashed line circle), the test sample's class is the first class because there are three blue squares inside the circle but only two red triangles. In addition to the fact that K must be a positive integer, the KNN classifier requires a metric distance. Euclidean distance (d_E) metric as a simple and easy-to-implement technique can be used for that purpose other than for further calculation of distances in a multidimensional input space. This method, as noted by Duda et al. (2001), can produce results that are comparable to those generated by the most sophisticated machine learning methods. In

Cartesian coordinates, if x_i and y_i are two points in the Euclidean n-space, then the distance from x to y is given by

$$d_E = \sqrt{\sum_{i=1}^{n}(x_i - y_i)^2}. \qquad \text{Equation 3.1}$$

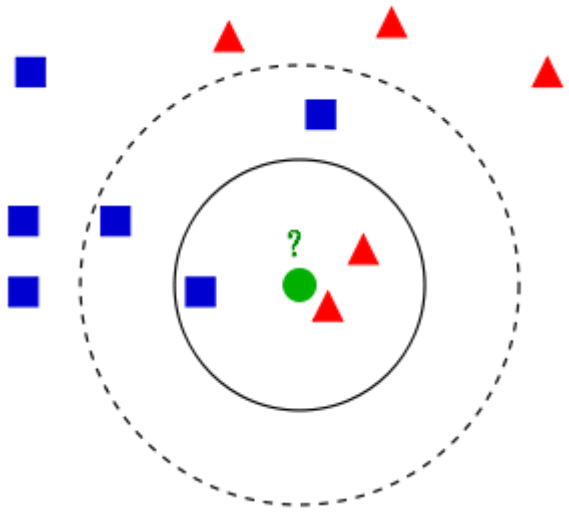


Figure 3.5: An example of KNN classification: solid line circle (when K=3) and dashed line circle (K=5).

Logistic regression classifier

Logistic regression predicts the probability of occurrence for an event, which is why it is typically used to model binary data. The method attempts to measure the relationship between a categorical-dependent variable and one or more independent variables, which are usually (but not necessarily) continuous. For that purpose, logistic regression relies on using probability scores as the predicted values of the dependent variable. The scores mainly take on values between zero and one, which is depicted in Figure 3.6. Weka incorporates the Le Cessie and van Houwelingen (1992) algorithm with selected modifications to build a multinomial logistic regression model based on a ridge estimator. Indeed, ridge regression can essentially contribute to the attainment of more stable estimates in relation to the logistic regression coefficients. Ridge regression is a technique for analysing multiple regression data affected by multicollinearity. In turn, multicollinearity, or collinearity, is the existence of near-linear relationships among independent variables. In the presence of a multicollinearity problem, least squares evaluations are unbiased, while their variances are large, what makes them far from the

correct value. If a degree of bias is added to the regression estimates, the standard errors can then be minimised via ridge regression. As a result, more reliable estimates can be achieved in the process (NCSS, 2007).

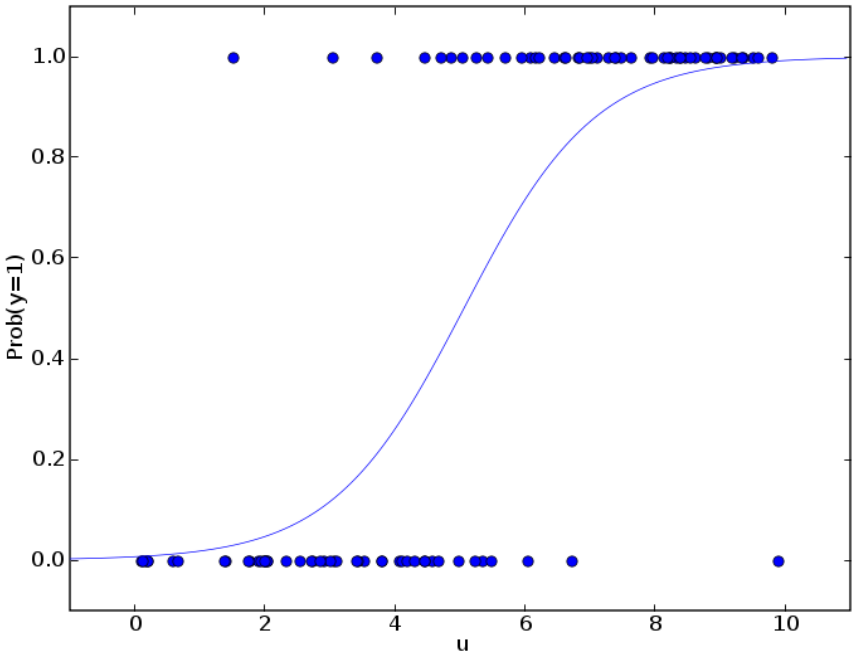


Figure 3.6: An example of logistic regression showing the relationship between a categorical dependent variable (y) and one or more independent variables (u)

Multilayer perceptron neural network (MLP) classifier

The MLP algorithm is a feed-forward artificial neural network (ANN) and receives information from data by mapping input datasets onto a set of desired outputs. ANNs consist of three or more layers: an input layer, an output layer, and one or more hidden layers (Figure 3.7). The error is typically recognised as the summed square error based on the difference between the predicted and desired outputs. A set of changes is subsequently applied to the weighted factors (calculated for the input, hidden, and output layers) via a back-propagation process. The process repeats until the desired performance is achieved (Sivanandam & Deepa, 2012). There are no special guidelines for selecting a network's best configuration, although, allegedly, a trial-and-error approach should be employed when considering the best network performance and average error for the given data.

The neural network can rapidly map a given input to the desired output quantities. Furthermore, this method can also extract information from noisy or poor-quality data and

provide results with relatively high accuracy (Amayreh & Saka, 2005). ANNs have the ability to detect complex nonlinear relationships in an implicit manner by focusing on dependent and independent variables (Tu, 1996); in addition, ANNs learn without a critical need to be reprogrammed (Amayreh & Saka, 2005). However, ANNs are black boxes, which means that it is not possible to interpret the relationships between inputs and outputs; ANNs cannot address uncertainties either. Nevertheless, ANNs are the tools of choice when results are more important than determining how they are obtained and due to their ability to detect and recognise more important inputs (Gopalakrishnan et al., 2009).

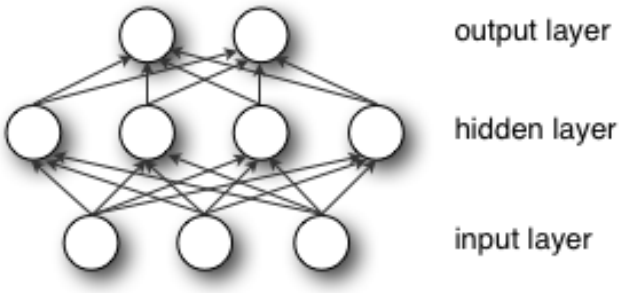


Figure 3.7: A typical artificial neural network.

Decision table classifier

The decision table algorithm attempts to collect a minimum set of features to forecast the class with increased precision. To achieve this objective, the decision table model divides the dataset into cells to ultimately retain identical examples in each cell. As a consequence, the most frequent class in the cell is assigned to the example that has an unknown class (Kohavi, 1995). Thus, a decision table can be recognised as "a table that represents the exhaustive set of mutually exclusive conditional statements within a pre-specified problem area" (Verhelst, 1980). A decision table may suggest several possible actions to be systematically followed by a decision maker based on the outcomes of relevant conditions. The general structure of a decision table is presented in Table 3.3. Each decision table consists of four quadrants, which are shown in Table 3.3 (Witlox, 1998; Witlox et al., 2004). These quadrants are as follows:

1 - A condition set that comprises all relevant conditions or attributes (e.g., inputs, premises, or causes) that may affect the decision-making process.

2 - A condition space, in which all possible combinations of the condition states of the condition are specified.

3 - An action set that contains all possible actions (e.g., outputs, conclusions or consequences) that a decision maker is capable of taking.

4 - An action space of an action that includes all of the possible action states. A decision rule is produced by any element's vertical linking from the condition space with an element from the action space.

If a schematic view of a decision-making process's inference is provided, a decision table can also be called an inference or logical tree. In this tree-like representation, the condition (premises) and actions (conclusions) are depicted as nodes, and the connections between the premises and the conclusions become branches of the tree. To reflect the structure of the if-then rules, the logical operators "AND" and "OR" are utilised. Given the above information, it seems that there is not much difference between a decision table and a decision tree. However, this concept is not conceptually accurate because there are certain important distinctions between the two approaches. The most important strength of using a decision table is the simple process for checking for completeness, correctness and consistency in the input data. Furthermore, such a table contributes to a better understanding of the choice problem because it is capable of providing more compact and exhaustive visual representations. Thus, considering the fact that a decision table implies certain logical constraints, it may be regarded as a special case of a decision tree (Witlox, 1998).

Table 3.3: The general structure of a decision table (Witlox et al., 2004).

Problem area	
CONDITION SET	CONDITION SPACE
ACTION SET	ACTION SPACE

Decision tree classifier

The decision tree CART algorithm implements cost complexity pruning for classification trees (Breiman et al., 1984). The CART model is used within the Weka toolset to build multivariate decision (binary) trees and remove redundant attributes (i.e., pruning). Within this algorithm, a decision tree is formed by splitting each variable into two groups, and the binary partitioning process can be repeatedly applied. The information gained by each split of a decision tree is measured accordingly, and the set with the highest information gain is selected. The process is repeated until the desired threshold is achieved

(Breiman et al., 1984). Figure 3.8 provides a simple example illustrating a classification tree for a problem with a binary response (green star and red circle) and two continuous features (A and B). It is possible to imagine that there is a need to classify the square in the training set shown in Figure 3.8. A possible induced decision tree might be proposed, as shown in Figure 3.8. Evidently, the square record is classified by the decision tree as a red circle given that the record falls on a leaf labelled with circles.

Training-produced decision trees clearly identify the most important features for prediction systems, perform efficiently with redundant variables and mixed data types, and rapidly classify unknown patterns. Furthermore, the pre-selection of variables is not needed because a robust stepwise selection method is used in decision trees. This is particularly useful when the predictors are potentially associated in a non-linear manner. However, correlated and irrelevant attributes may impact the construction of a decision tree and reduce its performance (Perner, 2001; Quinlan, 1996). Michie et al. (1994) argued that among the decision trees (AC^2, NewID, Cal5, C4.5, and IndCART) that are similar in structural framework to the original ID3 algorithm (with partitions being made by splitting an attribute and with an entropy measure for the splits), the IndCART, CART and Cal5 methods are superior because they incorporate costs into their decisions. Both CART and IndCART can address categorical variables, and CART has an important additional feature in that it utilises a systematic method for addressing missing values. Given the arguments above, the CART decision tree classifier is used in this study.

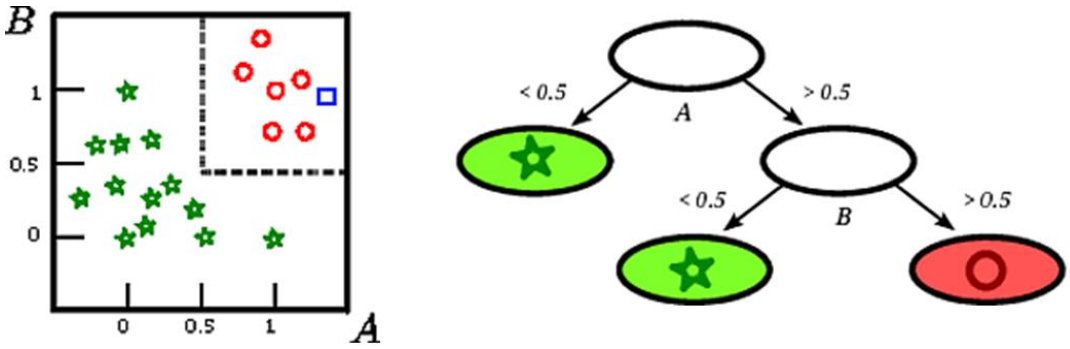


Figure 3.8: An example showing how a decision tree can be used to classify a new instance (blue square): (left) the boundaries (grey dashed lines) on the graphical representation of the training set are used by a tree to classify new instances; (right) decision tree.

To evaluate the performance of subsets as well as to avoid over-fitting errors and to stop the algorithm at the cut-off point when ranking or scoring subsets, a cross-validation

technique is typically applied. In the cross-validation procedure, sample data are shared in two subsets for training and testing/validation purposes. A central analysis is performed on the training set, and the algorithm's performance is tested or validated using the other set (test/validation set). Several iterations are performed by referring to different sets of training and validation samples to reduce variability. Technically, the average value of all validation results is subsequently used to determine the method's merit after several iterations (Efron, 1983). In this study, to limit over-fitting issues and reduce variability, 10 cross-validation iterations are performed in the wrapper method for each classifier; meanwhile, the validation results are averaged over all iterations.

Furthermore, numerous irrelevant features may contribute many biases to the results. Therefore, using fewer attributes can be helpful in minimising these biases and the necessary computing resources while constructing the classification model (Kamal, 2009). Kohavi and John (1997) noted that the optimal selection of a feature set can be influenced and altered if different training set sizes are implemented; thus, they suggested reducing the number of features to a small number of instances and increasing it for a larger number of instances to ultimately reduce the algorithm's variance. Obviously, reducing the number of ground-motion parameters and classifying them can also be useful if only the parameters representing the selected IMs' distribution are desired to be used. Variable clustering is a beneficial technique that can be used for this purpose. Moreover, this technique can be used to quickly summarise the main dimensions in a large dataset and can be used as a variability reduction technique.

In this study, the VARCLUS (variable clustering) procedure, which is implemented by Tanagra software (Rakotomalala, 2005), is used to classify and merge the 36 IMs (specifically selected for this study; additional details are presented in section 7.1.2) within two common clusters. The parameters assigned to each cluster are individually defined for each group of selected records based on a succession of principal component analyses involved in the VARCLUS procedure. The cluster with the closest correlation to the structural response is used as input in the statistical regression method and wrapper subset evaluation procedure.

The VARCLUS procedure is a top-down approach that attempts to divide a set of numeric variables into disjoint or hierarchical clusters. A cluster's linear combination of variables, i.e., the first principal component, is associated with each cluster. The procedure is designed to maximise the variance that is produced by the cluster components, which is summed over all of the clusters. As noted in the Tanagra software documentation, the

group is determined by the first factor of the principal components analysis, which is a weighted average of the variables representing the most variance. In principal components analysis, the correlation coefficient or the squared correlation coefficient is used to compute the similarities between examples, and all variables are treated as equally prioritised. In top-down methods, when a variable is assigned to a group or cluster during the splitting process, it is not verified if the variable is more correlated to other groups; hence, the formulated tree diagram does not correspond to a dendrogram, exhibiting only the succession of the splitting operations. In comparison to other clustering approaches (such as VARHCA and VARKMEANS - variable clustering using hierarchical cluster analysis or K-means), the VARCLUS method is much faster when addressing numerous variables. The process ceases when no relevant subdivisions of the groups are available. The Tanagra software implementation is based on Vigneau and Qannari (2003).

3.4.2 Fragility curve and surface derivation

The final stage of any fragility assessment process is the development of fragility curves or surfaces. A lognormal cumulative distribution function (CDF) is often used to fit the structural analysis data and to provide a graphical representation of the failure's probability as a function of the selected IM (see Figure 2.6). Fitting a lognormal distribution to the observations is performed by determining the lognormal distribution parameters $\boldsymbol{\theta}$ and β to specifically forecast the probabilities obtained from observed functions of ground motions, resulting in a desired failure at each selected IM level (Baker, 2011b):

$$P(D \geq d_i | IM = x) = \Phi\left(\frac{Ln\, x - \theta}{\beta}\right), \qquad \text{Equation 3.2}$$

where $P(D \geq d_i | IM = x)$ is the probability of damage exceeding a specified threshold, d_i, as a function of the ground motion for $IM = x$; Φ( ) is the normal cumulative distribution function; and $\boldsymbol{\theta}$ and β are the mean and standard deviation of ln*IM*, respectively.

Another general function that is relevant for predicting a categorical-dependent variable's outcome (e.g., a class labelled as 'Collapse' or 'Yield') using one or more predictor variables has been occasionally applied to fragility analysis – logistic regression (e.g., Basoz & Kiremidjian, 1998; Bojórquez et al., 2010; Koutsourelakis, 2010).

Bojórquez et al. (2010) relied on a logistic function to design fragility curves and surfaces for steel frames. Furthermore, Koutsourelakis (2010) applied a logistic regression model to obtain vector-valued fragility functions from limited available data. The use of logistic functions for deriving fragility surfaces has been suggested by Koutsourelakis (2010) to produce more accurate predictions.

Logistic regression gauges the relationships between a categorical dependent variable and independent variables, in which probability scores represent the dependent variable's predicted values. Maximum likelihood (ML) is used in logistic regression to identify the best-fit line. Rice (1995), Shinozuka et al. (2000), Baker and Cornell (2005), and Straub and Der Kiureghian (2008) insisted that a more appropriate fitting technique to compute the non-constant variance of a failure's observed fractions (e.g., collapse) relies on the maximum likelihood approach. Different "iterations" are computationally executed (statistical software packages) in the ML method to generate different solutions until the smallest possible deviation, or best fit, is detected. The final value of the deviation is usually referred to as the "negative two log-likelihood" (-2 log-likelihood or -2LL) based on Cohen et al. (2003). Similarly to the chi-square test, which is a measure of goodness of fit for observed and expected values, the -2LL is also used as a measure of goodness of model fit in a given case. Larger differences (or deviations) in the observed values from the expected values correspond to poorer model fits. In this sense, the goal is to minimise the deviation.

However, logistic functions suffer from various drawbacks, especially in addressing parameters equal to zero and estimating nonzero probabilities for them. To overcome this problem, it is useful to use log-logistic distributions by applying logistic regression to the logarithm of the variables, which was discussed by Gehl et al. (2013b). Gehl et al., (2013b) indicated that in comparison with the general shapes of kernel distribution functions, log-logistic and lognormal distribution functions provide similar performance. Therefore, the probability of failure in this study (P_F) is obtained by implementing a log-logistic regression function as follows:

$$P_F = \frac{1}{1+e^{[-\beta_1-\beta_2.Ln(IM)]}} \text{ for fragility curves and} \qquad \text{Equation 3.3}$$

$$P_F = \frac{1}{1+e^{[-\beta_1-\beta_2.Ln(IM_1)-\beta_3.Ln(IM_2)]}} \text{ for fragility surfaces.} \qquad \text{Equation 3.4}$$

Here, β_1, β_2 and β_3 are obtained from a regression analysis of the results that includes the variation in the IMs. For a binary structural response in terms of collapse and

yield, the probability that 'P_F is collapse' is defined by Equations 3.3 and 3.4, whereas the probability that 'P_F is yield' can be determined based on the obtained collapse probability:

$$P_{Yield} = 1 - P_{Collapse} \ . \qquad \text{Equation 3.5}$$

3.5 Summary

To achieve the objectives of this study, a methodology used to perform a seismic analysis on existing URM buildings in the Kurdistan Region (KR) in Iraq, which is the most hazardous area in Iraq, is designed and integrated. Figure 3.9 shows a flow chart of the methodology implemented in the study. An analytical approach is used to perform a fragility analysis of existing buildings in the region because there are no relevant statistics focused on past earthquake records in the region. For that purpose, buildings in the region are classified in Chapter 4, and the damage levels of the two URM (one and two storey) buildings (representing the most common building types in the region) are then evaluated using incremental dynamic analysis. To select proper suites of time histories required as input to the IDA, response spectra for different seismic hazard zones in the Kurdistan region are developed and proposed via a comprehensive review of available information (details in Chapter 5). The reliability of the response spectra is then examined via a comparison between the relevant results and the outcomes from applying the time histories selected using the conditional mean spectrum, the Jayaram et al. (2011) algorithm, and the EC8-code-based selection method based on the damage state prediction of the structural response. Variability in the material properties of the masonry is evaluated by conducting several hundreds of nonlinear time history analyses using different mechanical parameter values for the masonry units. Through a graphical representation of fragility curves, the efficiencies of the IMs are compared. Furthermore, issues concerning the number of ground-motion records that should be used, the method for selecting these records, how they affect the selection of the IMs and, consequently, the derivation of the fragility curves are directly addressed in the study. The derived fragility curves based on scaled (i.e., IDA) and unscaled records are also compared. Additionally, a framework for selecting the best IM or subsets of IMs using the wrapper method is proposed, and its feasibility is thoroughly investigated by comparing results obtained from a statistical regression method. Finally, considering the results obtained herein, the sufficiency of the two best IMs is then inspected and examined through a statistical analysis, while fragility surfaces are derived and compared with the fragility curves.

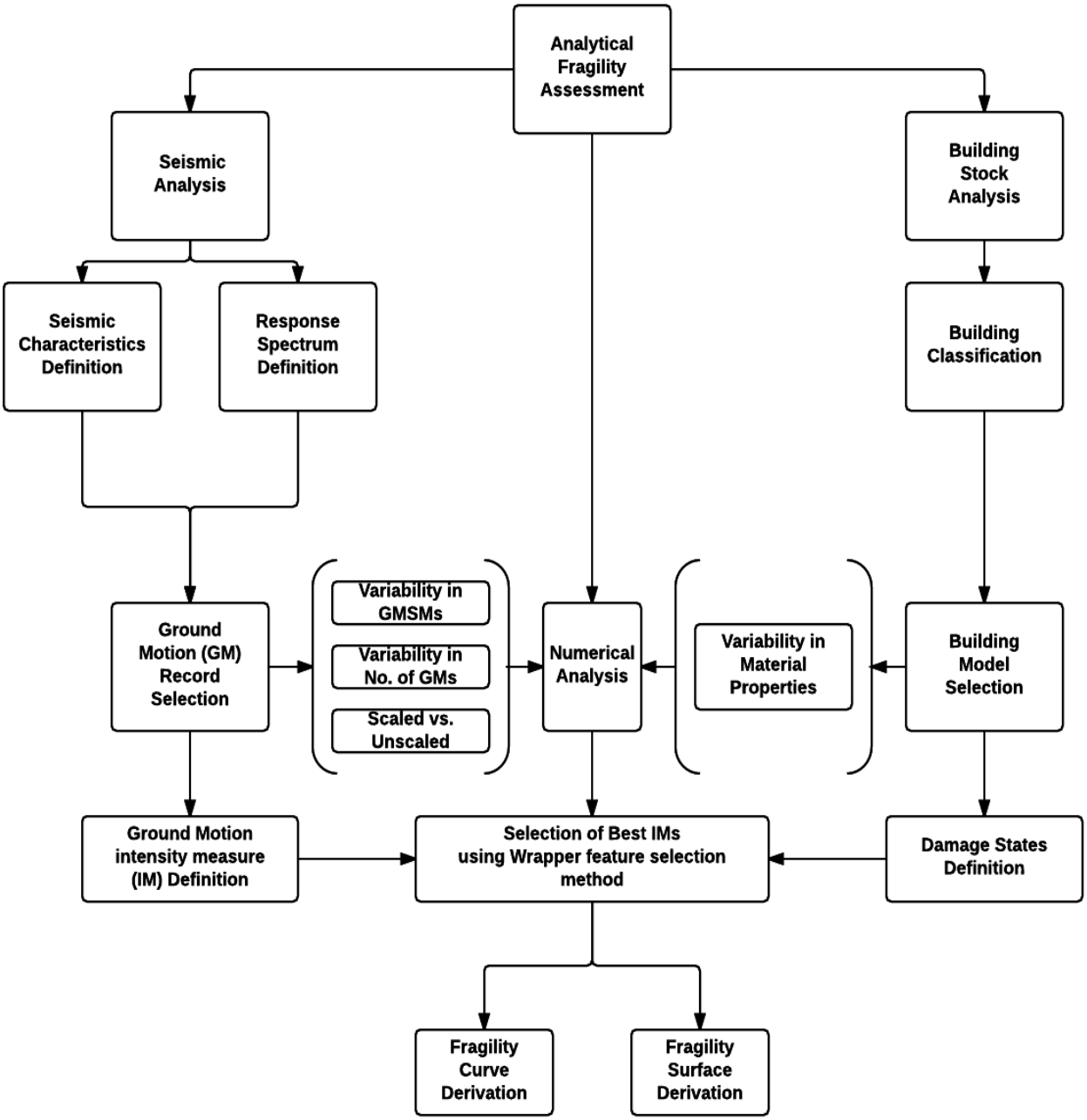


Figure 3.9: Flowchart showing the general methodology used in this study.

CHAPTER 4. BUILDING CLASSIFICATION STAGE

To complete the first stage of any seismic fragility analysis, it is crucial to identify and typify the exposure factors in the study region. The next section provides the characteristics of the building stock in the Kurdistan region as main exposure to seismic hazards. Typical buildings for the test are then selected as representations of the building stock in the region.

4.1 Characteristics of the Building Stock in Kurdistan

Kurdistan is an autonomous region of Iraq with a population of approximately 4.5 million citizens. It borders Iran to the East, Turkey to the North, Syria to the West, and the remainder of Iraq to the South. Due to political issues, the Kurdistan-Iraq border is not clearly recognised yet. Thus, the selected region, i.e., Kurdistan, is represented by three zones (Figure 4.1): (1) an officially recognised region, (2) a disputed region that is mainly under Kurdish control, and (3) the ideal maximum Kurdistan zone. However, this study focuses on the region that is officially governed by the Kurdistan Regional Government (KRG) and consists of the first zone and a small portion of the second zone.

Kurdistan covers 40,643 square kilometres and is located in northern and north-eastern Iraq. Administratively, Kurdistan consists of three governorates (Figure 4.2): the southern governorate (Sulaimaniya), which includes 41% of the total population; the central governorate and capital (Erbil), with 35% of the total population; and the northern governorate (Duhok), representing 24% of the total population (see Table 4.1). The governorates are divided into 32 districts and 128 subdistricts, which are shown in Table 4.2 (Central Statistical Organization [CSO], 2011). A district may consist of one or more subdistricts. Each subdistrict is divided into an urban and rural environment. According to the recent census of buildings, dwelling and establishments conducted by the Central Statistical Organization (CSO) in 2011, the Duhok governorate has more residents (approximately 27%) living within a rural environment than do the other two governorates (Erbil and Sulaimaniya) (see Table 4.1). However, in general, urban areas are the most densely populated parts of Kurdistan, representing 81% of the total population; rural areas contain 19% of the total population (Figure 4.2). Furthermore, the majority of people (more than 60% of the total population) live in subdistricts that are known as district centres; the most populated centres in the list are Sulaimaniya, Erbil, Duhok, and Zakho.



Figure 4.1: Iraqi Kurdistan region limits.

Table 4.1: Iraqi Kurdistan population from CSO (2011).

Governorates	No. of Persons		
	Urban	Rural	Total
Duhok	786,599	285,725	1,072,324
Erbil	1,274,090	257,991	1,532,081
Sulaimaniya	1,515,775	269,078	1,784,853
Total	3,576,464	812,794	4,389,258

Table 4.2: Distribution of districts and subdistricts for the Kurdistan Regional Governorates.

Kurdistan Regional Governorates	Districts	Subdistricts
Duhok	7	26
Erbil	9	41
Sulaimaniya	16	61
Total	32	128

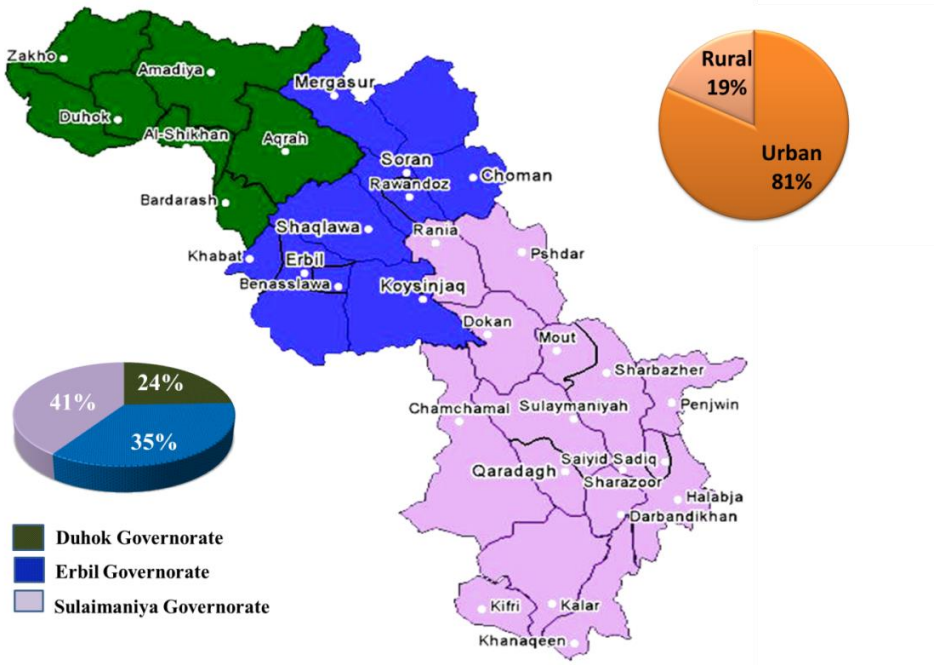


Figure 4.2: Administrative division of Kurdistan showing the population distribution in percentages by governorates (bottom left) and by rural and urban areas (top right)

According to CSO (2011), Kurdistan's building stock consists of approximately 994,585 buildings, 77% of which are widespread in urban areas, and the remaining 23% are scattered in rural zones. Figure 4.3 shows the prevalence of buildings in the region. The Sulaimaniya governorate, with 43% of the total number of buildings, is the largest structured environment in the region; the other two governorates, Erbil and Duhok, have 38 and 19% of the total number of buildings, respectively (see Table 4.3). Because it is not possible to separately analyse each building considering the large number of buildings, it is necessary to group the buildings based on similar characteristics but in different terms. For each building, the following characteristics were considered in CSO (2011): type, occupancy, age, construction material, number of storeys, and other variables. The following sections display the data obtained from CSO (2011) and used to identify the building stock in Kurdistan.

Table 4.3: Number of buildings in Kurdistan (CSO, 2011).

Governorate	Number of Buildings		
	Urban	Rural	Total
Sulaimaniya	344,094	86,739	430,833
Erbil	295,627	82,379	378,006
Duhok	128,784	56,962	185,746
Total	768,505	226,080	994,585

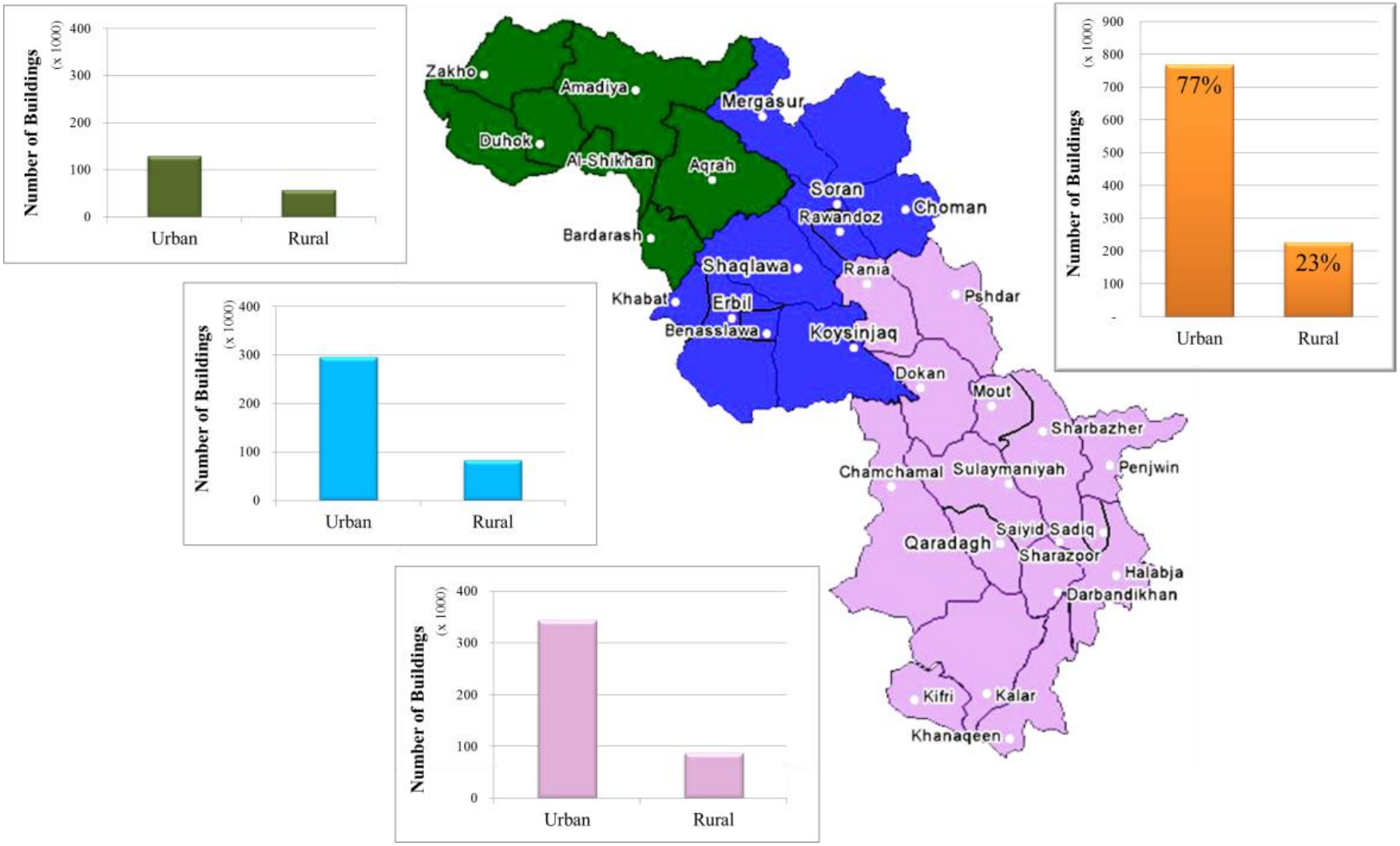


Figure 4.3: Building distribution in Kurdistan by governorate (left) and by region (top right).

4.1.1 Building types

Buildings have been classified in CSO (2011) according to type as (1) conventional buildings, which represent attached buildings and houses used only for residential purposes or for both residence and work (attached buildings are structures that have two or more storeys); (2) mobile or marginal buildings, which represent temporary shelters or independent buildings used for living or both living and work, although they can be removed or shifted; and (3) 'other buildings', which can be used for work (e.g., government or non-government institutions) or economic affairs (e.g., multi-storey commercial/shopping buildings) in addition to all other types of buildings that have not been mentioned among the above categories (for more details, see Table 4.4).

Figure 4.4 shows the following percentages for the different building types in the region based on the data in Table 4.5: 78% are conventional buildings (79% in urban and 21% in rural areas); mobile or marginal types constitute 1% of all buildings (48% in urban and 52% in rural areas); and other buildings constitute 21% of the total number of buildings in the region (71% in urban and 29% in rural areas). Figures A1-A7 display the region's different types of structures based on street surveys in the most populated cities such as Sulaimaniya, Erbil, and Duhok.

Table 4.4: Types of buildings in Kurdistan according to CSO (2011).

Conventional buildings	**Attached Building**: A building composed of two or more storeys, including internal or external staircases used for all storeys. The building may be used for residential purposes only, for commercial purposes only, or for both residence and work.
	House: A single building consisting of one housing unit; this structure is used for residential purposes for only or both residence and work. A house may be attached to another building and may consist of one and or more rooms.
Mobile or Marginal buildings	**Hut or shack**: This is a temporary shelter or an independent building constructed from available crude materials such as clay, reed, palm branches, tin sheets, corrugated plates or wood boards. A hut or shack may be used for living or both living and work.
	Tent: This is a shelter made of camel and/or goat hair and can be moved or shifted easily.
	Caravan: A caravan is a prefab structure that is temporarily fixed and is made of wood, iron or corrugated sheets; it is used for living or work.
Other buildings	**Establishment or Institution**: This is a building used for work and is occupied by government or non-government officials. A building may contain a special part for guard use. An institution may be a collective home designed originally to accommodate many individuals with similar or special conditions such as hospitals or prisons, worker's quarters, and student's hostel. Hotels and hotel apartments are also considered as establishments.
	Mall or shopping centre: This is a multi-storey building designed as an integrated commercial centre; all storeys and components are used for operating economic practices.
	Under construction: A building whose foundation, four walls, ceiling and at least one storey have been built. It may currently be under furnishing. Thus, a building in this group is incomplete and intact, meaning that it is not suitable for use at the time of census.
	Other: This category covers all buildings that are not included in the other types such as a fenced plot of land.

Table 4.5: Number of different types of buildings in Kurdistan (CSO, 2011).

Types of Buildings	Erbil Governorate		Sulaimaniya Governorate		Duhok Governorate		Kurdistan		
	Urban	Rural	Urban	Rural	Urban	Rural	Urban	Rural	Total
Conventional	233966	57905	278694	62600	101017	39837	613677	160342	774019
Mobile or Marginal	1119	1614	1778	566	1104	2169	4001	4349	8350
Other	60542	22860	63622	23573	26663	14956	150827	61389	212216

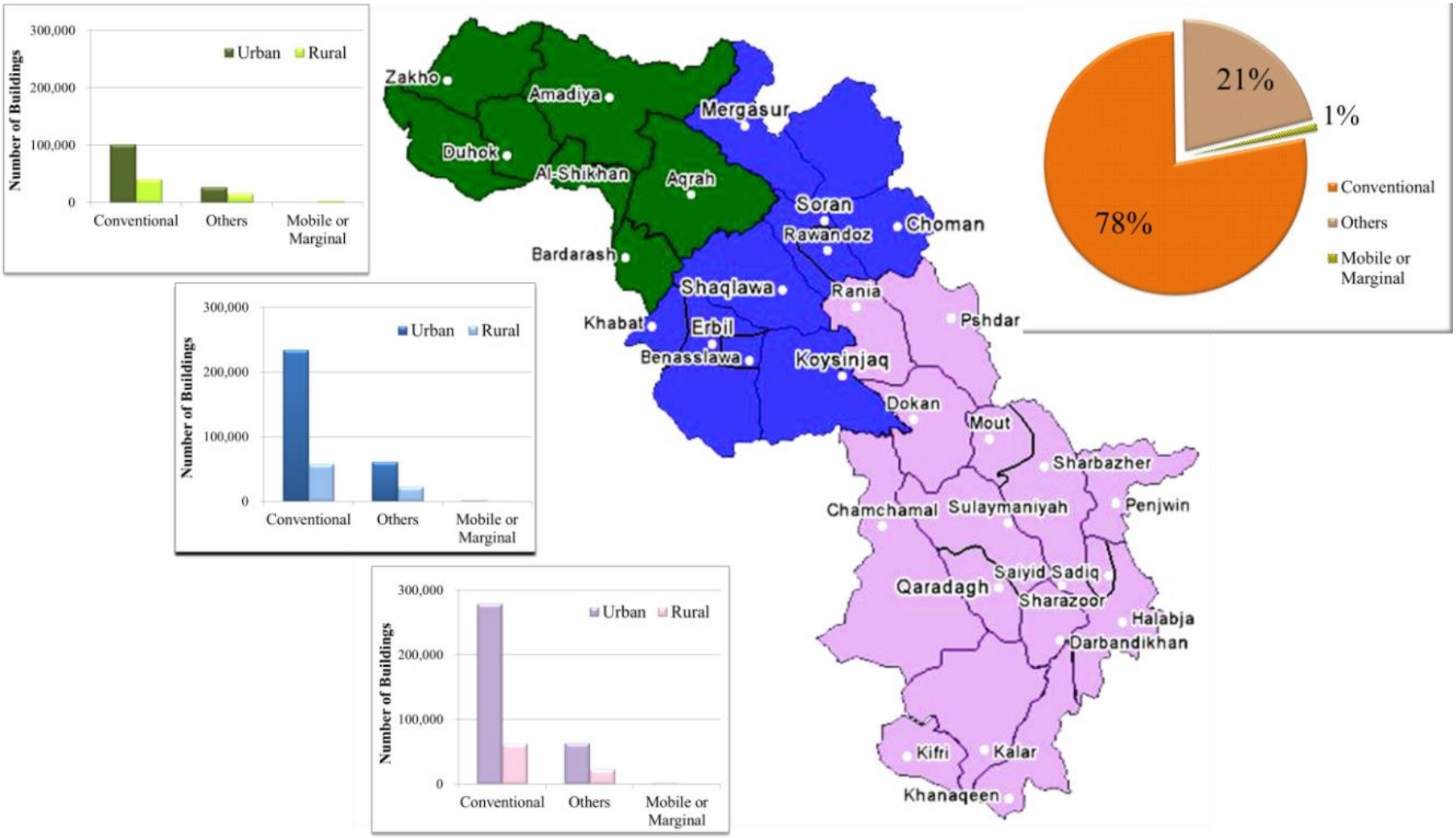


Figure 4.4: Types of buildings by region (right) and by governorates (left).

4.1.2 Building occupancy classifications

Each building type is classified based on occupancy (CSO, 2011) as (1) dwelling units, i.e., buildings that are originally designed for one or more households, or (2) establishments or other units used for performing group activities and assets that are managed by one or more owners or one administration.

An analysis of the inventoried buildings provides the following percentages: the dwelling and establishment/other units represent 89 and 11% of conventional buildings, 23 and 77% of mobile or marginal buildings, and 3 and 97% of other buildings (Figure 4.5).

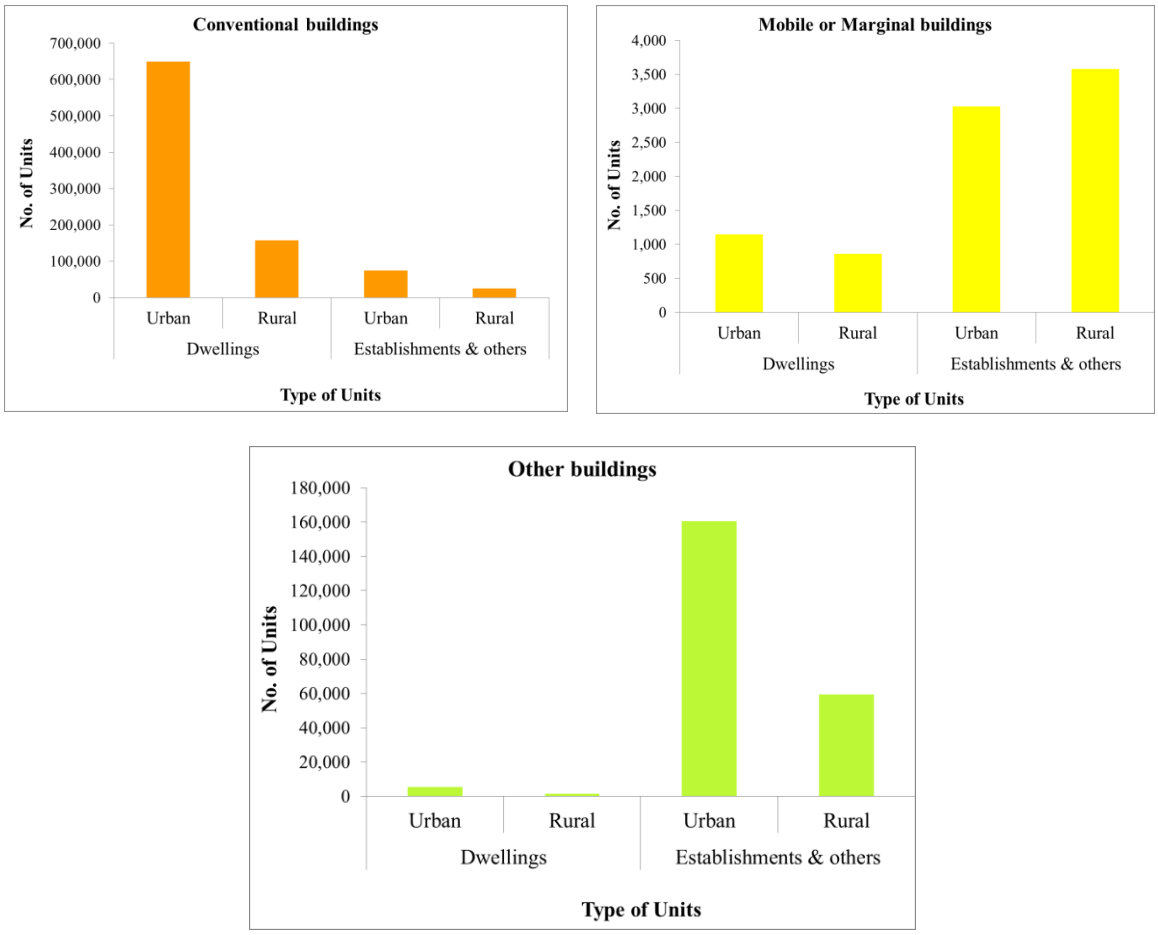


Figure 4.5: Number of units in Kurdistan by occupancy.

4.1. 3 Building classification according to the number of storeys

The data (CSO, 2011) show that 78.85% of the conventional buildings in the research area have one storey; units with 2 and 3 storeys represent 18.55 and 2.46% of all buildings, respectively (Figure 4.6). However, the census data are insufficient to identify the number of storeys for buildings other than conventional buildings. Given the circumstances, a street survey was conducted in the study area. While the findings of this survey indicated that mobile or marginal buildings typically include 1 storey; the majority of 'other buildings' are multi-storey buildings. Thus, it is suggested that establishment buildings, which constitute approximately 70% of the 'other buildings' type (CSO, 2011), are low-rise (1 to 3 storeys) buildings. The other 30% represent commercial centres/malls under construction and other buildings that are either mid-rise or high-rise buildings (i.e., greater than 4 storeys).

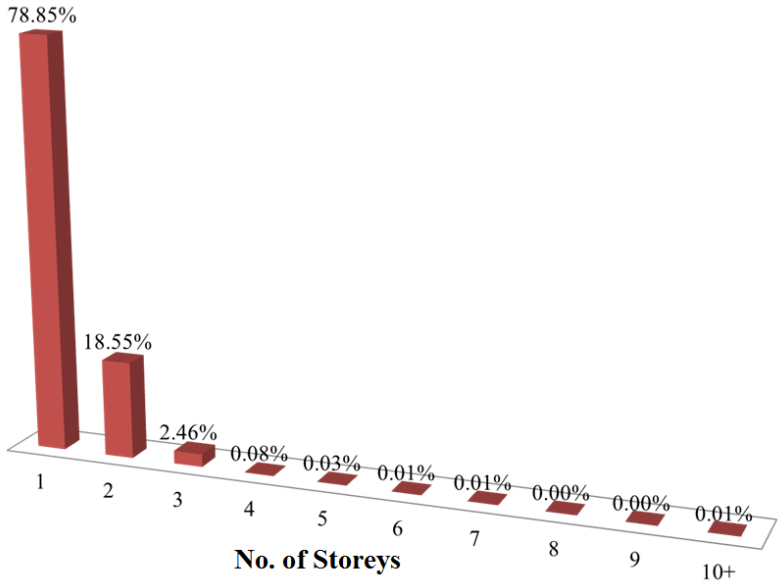


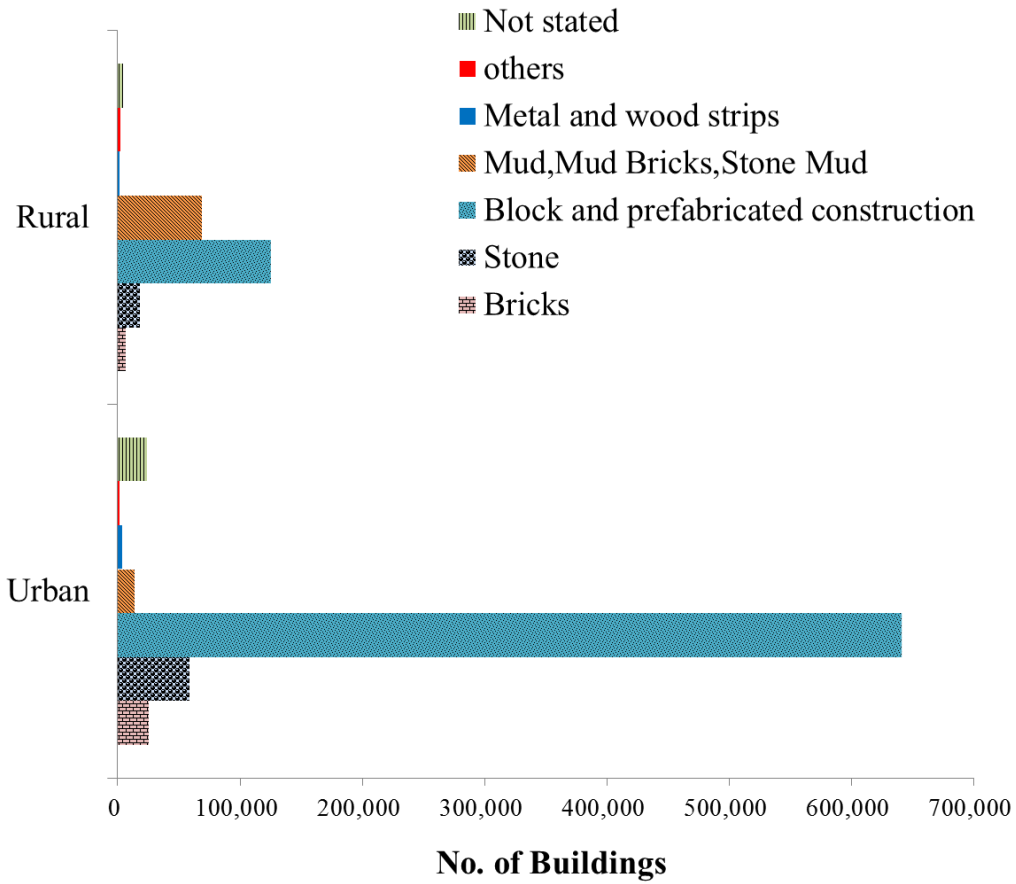
Figure 4.6: Percentage of conventional buildings in Kurdistan by number of storeys.

4.1.4 Building classification according to the material type (external walls)

In CSO (2011), buildings are classified according to the type of materials used when constructing external walls. Different types of materials, such as bricks, stone, concrete block and prefabricated construction, mud (mud bricks and stone mud), metal and wood strips, are considered. Buildings with external block walls constitute the highest proportion of buildings in Kurdistan, i.e., 77.14%, followed by mud, stone, and brick walls

at 8.33, 7.69, and 3.16%, respectively (Figure 4.7). Metal and wood strip walls, including other non-recorded material types, account for approximately 4% of the buildings. Although buildings with clay external walls in rural areas constitute a higher percentage (83%) of buildings compared with those having the same external walls in urban areas (17%), concrete blocks are still regarded as the principal materials for constructing external walls in both urban and rural areas (see Figure 4.7).

Figure 4.8 clearly shows that concrete blocks are the main material used in constructing external walls of conventional buildings. Moreover, the street survey showed that reinforced concrete slabs were and remain the most common type of slab used for roofs, except for in a small percentage of mobile/marginal homes and buildings in rural areas, where mud or other types of roofs are utilised.



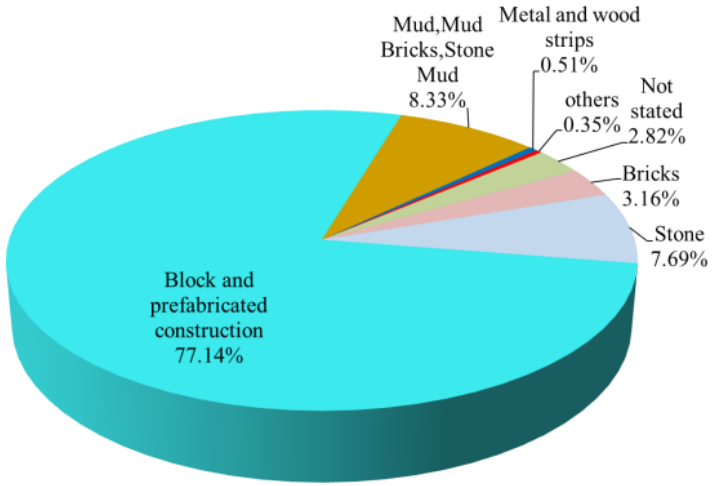


Figure 4.7: Distribution of buildings in Kurdistan with respect to the construction material of external walls.

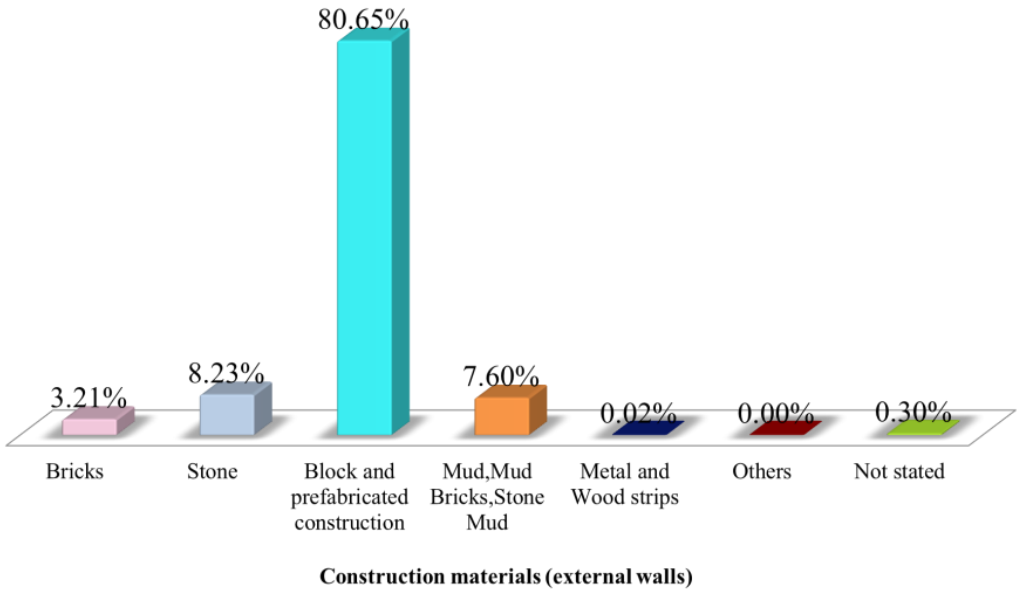


Figure 4.8: Percentage of conventional buildings in Kurdistan according to the building material used for constructing external walls.

4.1.5 Age of buildings

For the first time in administrative practice, a draft of the "Iraqi Building Code requirements for reinforced concrete" was prepared by the Building Research Centre (BRC) in 1978; nine years later, in 1987, the code was finally completed (BRC, 1987). In 1997, the "Iraqi Seismic Code Requirements for Buildings" was arranged and published as a draft to considered seismic-resistant designs for RC structures (BRC, 1997). Since then, this code has remained unchanged, i.e., no amendments, renewals, or updates have been put forth. Regarding the specific area under study (Kurdistan), an independent seismic code is not currently available. No specialised guideline is available concerning time history analyses and seismic code requirements for URM buildings.

To classify buildings with respect to their year of construction, the following periods for published Iraqi building codes are used: (1) before 1978, (2) between 1978 and 1987, (3) between 1987 and 1997, and (4) after 1997. However, the classification is not beneficial for this study in terms of ensuring satisfactory seismic building codes for URM buildings. Nevertheless, it will be helpful for future studies regarding RC structures in the

region. Furthermore, this classification assists in distinguishing between old and recently constructed buildings. Moreover, it can be used to determine the construction rate for each building to focus on those with higher construction rates. According to the available statistical data from the census (CSO, 2011) and as represented in Figure 4.9, approximately 10% of the conventional buildings in the study region were constructed before 1978. For the same building type, the percentage was 15% for construction between 1978 and 1987 and 21% between 1987 and 1997. After 1997, the construction rate of conventional buildings increased to 54% (i.e., between 1997 and 2009). For mobile/marginal homes and 'other buildings', the construction rate during the aforementioned periods is assumed to be the same as the rate of construction of conventional buildings (as mentioned above) because no data regarding the period of their construction are officially provided in the census.

To relate constructed buildings to the number of storeys and the types of construction materials, the following statistics are extracted from CSO (2011). Conventional buildings with 1 storey constitute the highest percentage of buildings whose construction rate increased during the aforementioned periods (Figure 4.10); however, a slight increase in the construction of buildings with 2 storeys is found, ranging from 2% before 1978 to 10% after 1997 (see Figure 4.11). Moreover, buildings with 3 storeys exhibit an increase in the construction rate (approximately 2%) after 1997. Similarly, concrete blocks were used as a main material in the construction of buildings during these periods. However, other materials, such as clay, brick, and stone, were also used in the region, especially before 1997. After this period, the use of these materials decreased gradually, and the use of concrete blocks has ultimately become the only available option for building construction in the region (Figure 4.12).

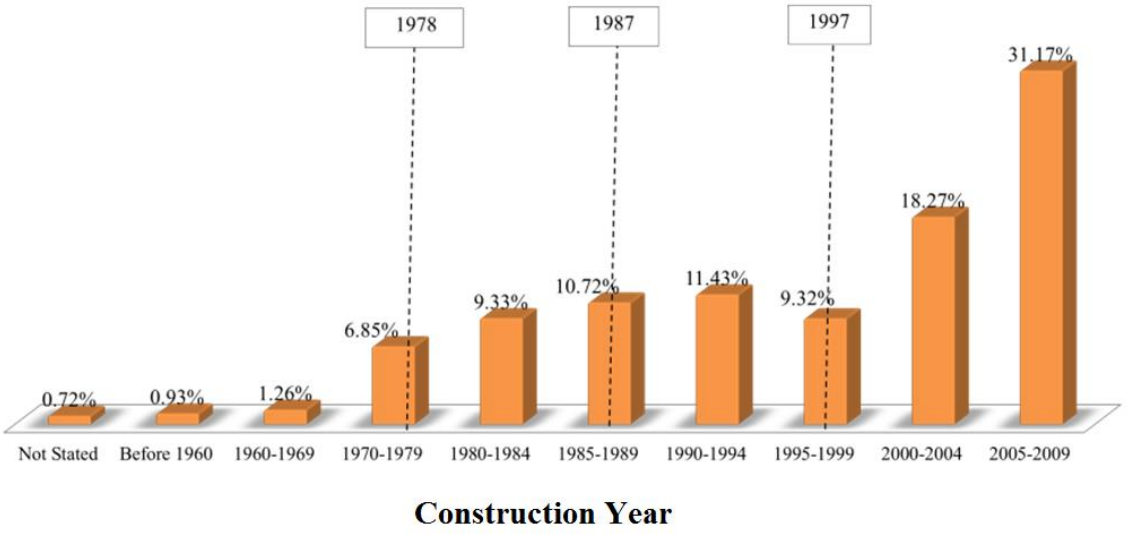


Figure 4.9: Distribution of conventional buildings according to the construction period.

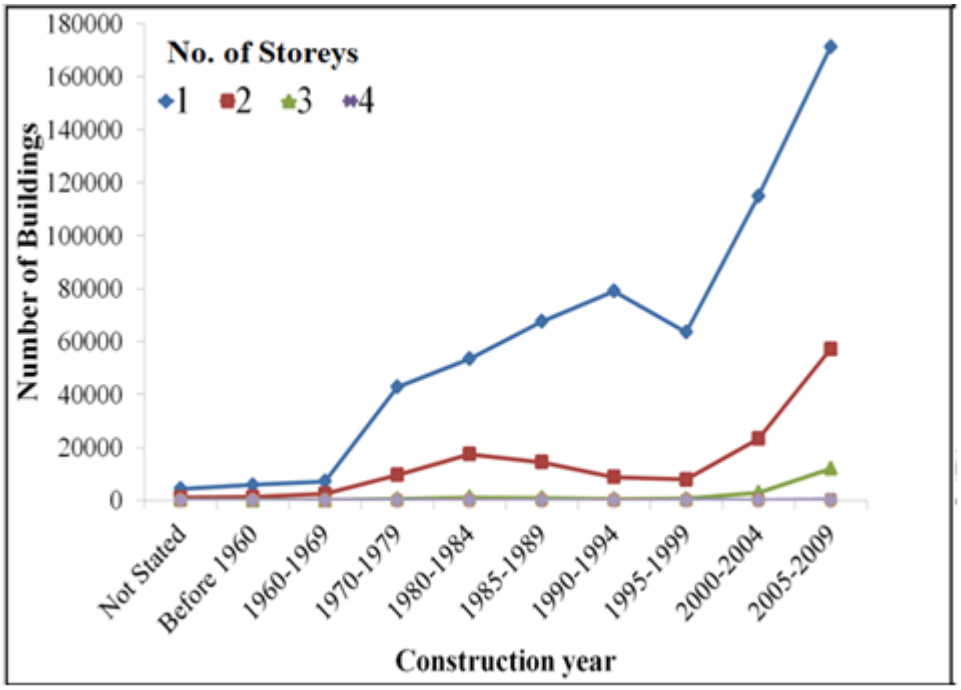


Figure 4.10: Distribution of conventional buildings in Kurdistan according to the construction year and number of storeys.

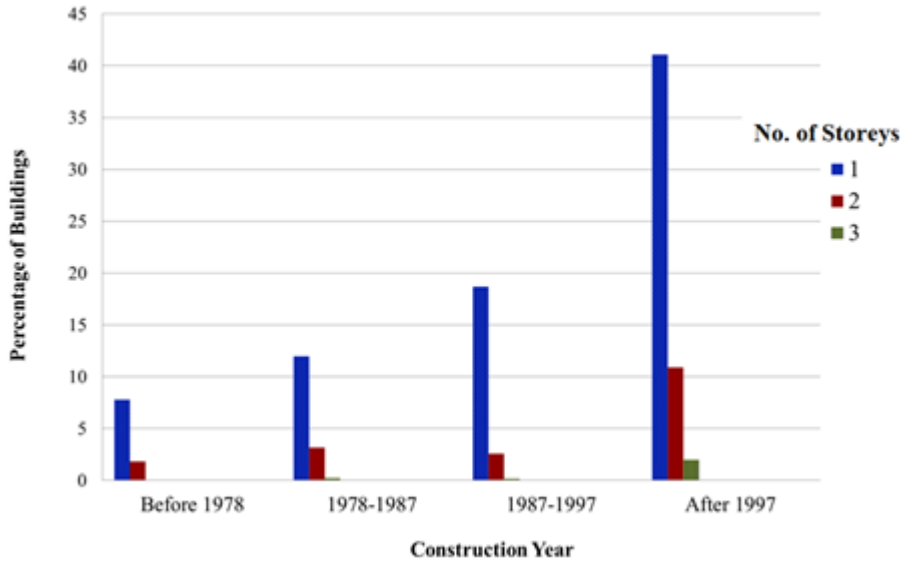


Figure 4.11: Percentage of conventional buildings in Kurdistan according to the number of storeys and construction year.

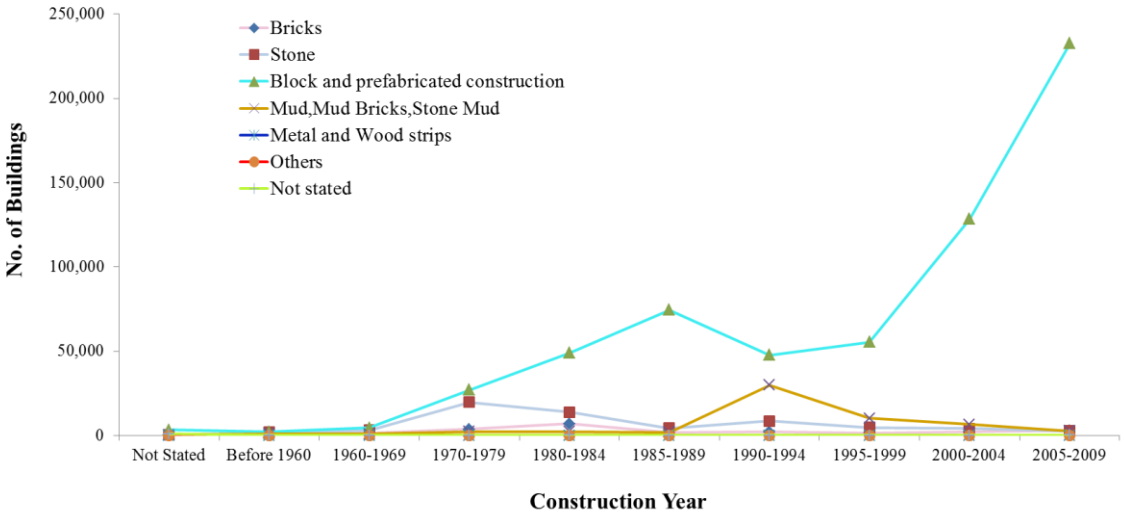


Figure 4.12: Distribution of conventional buildings in Kurdistan according to the construction year and type of materials (external walls).

4.2 Selected Building Model

The building stock data that are relevant to Kurdistan can be properly summarised in accordance with the aforementioned classifications as follows (see Table 4.6): the majority of residents (nearly 80%) in the region live in buildings that are widely scattered

in urban areas. A total of 70% of the buildings are used for dwelling purposes, and 30% are used for other purposes. Moreover, the statistics show that 87% of the buildings in the study area include 1 to 2 storeys, and 13% are greater than 3 storeys and were primarily constructed after 1997. The buildings with less than 3 storeys are mainly URM buildings, which can be classified as low-rise unreinforced masonry (URML) buildings according to the HAZUS-MH building classification model (see Table 3.1). Moreover, buildings with exactly 3 storeys are dual-system buildings with reinforced concrete frames on the base storey and URM storeys at the top levels (based on the street survey). Buildings with 4 or more storeys have reinforced concrete frames with URM infill walls, corresponding to the C3M and C3H types of the HAZUS-MH building classification model. Furthermore, concrete blocks and reinforced concrete are the basic materials used in the construction of building walls and slabs, respectively. Although half of the buildings in the study area were constructed after the implementation of the Iraqi seismic code (1997) specific to RC structures, most of the regional buildings were not originally designed to resist seismic loads.

Two typical URM buildings are selected to represent one- and two-storey residential buildings in urban areas of Kurdistan (Figure 4.13). The building stock for residential purposes ranges from regular to irregular within both horizontal and elevation plans. These buildings are mostly rectangular in terms of their planning design; typical dimensions are 7.5 x 20 m and 10 x 20 m, including a front yard. However, the majority of the URM buildings in the study area are 10 x 20 m (or 10 x 15 m excluding the front yard). Most of the URM buildings have a 15 cm RC slab with a 1:2:4 cement:sand:gravel ratio, which is mixed with an appropriate amount of water. The wall thickness is 200 mm, and the masonry units used in construction are solid concrete blocks with dimensions of 400 x 200 x 150 mm. These units are constructed according to ASTM C140-2006 and IQ.S. No. 1077/1987.

Based on IQ.S. No. 1077/1987, the lower limit of compression strength computed based on the total area is 11 MPa for a one solid concrete block unit and it is 13 MPa for an average of 3 units, where the rate of water absorption does not exceed 10%. Mortar used in construction varies from 1:3 to 1:6 (one part Portland cement and 3 to 6 parts sand). According to IQ.S. No. 5/1984, the compressive strength should not be less than 15 MPa for Portland cement after one day and 23 MPa after 3 days.

The inter-storey height of a single-storey building is typically 3.0 m. Generally, these buildings have lintel beams directly above windows and doors. Although a proper

connection is produced at the intersection of corner walls by block toothing, including meet points between internal walls and the boundary walls, this connection does not ensure a normal strength. Connection at the base of the walls is achieved through strip footing with non-stepped solid block resting on 50 cm to 60 cm x 30 cm to 40 cm 1:2:4 reinforced concrete. Buildings in the non-recently constructed category have ring beams; their footings are below ground. The ring beams are monolithic with the slab; the connection between the walls and slab is achieved using cement and mortar. However, buildings that have been constructed more recently are built directly above the ground without much compaction to the surface soil. The ring beams are ignored for the construction of this building type.

The building selected for the study test, which is shown in Figure 4.13, is a typical URM taken from KR building stock.



Figure 4.13: Unreinforced masonry buildings with one storey (top) and two storeys (bottom).

Table 4.6: Summarised building stock data for the Kurdistan region from CSO (2011).

Type of Buildings		Conventional		Mobile or Marginal		Others		All	
Distribution of Buildings (%)		Urban	Rural	Urban	Rural	Urban	Rural	Urban	Rural
		61.62	16.38	0.48	0.52	14.91	6.09	77.01	22.99
No. of Persons (%)								81	19
Occupancy (%)	Dwellings	69.42		0.23		0.63		70.3	
	Establishments & others	8.58		0.77		20.37		29.7	
No. of Storeys (%)	1	61.50		1.0		4.9		67.4	
	2	14.47		0.0		4.9		19.4	
	3	1.92		0.0		4.9		6.8	
	4+	0.11		0.0		6.3		6.4	
Construction Materials - (External Wall) (%)	Bricks	2.50		0.00		0.66		3.16	
	Stone	6.42		0.00		1.27		7.69	
	Block and prefabricated construction	62.91		0.00		14.23		77.14	
	Mud, Mud Bricks, Stone Mud	5.93		0.33		2.10		8.33	
	Metal and Wood strips	0.01		0.33		0.22		0.56	
	Other	0.00		0.29		0.10		0.39	
	Not stated	0.23		0.04		2.49		2.76	
Period of Construction (%)	Before 1978	7.8		0.1		2.1		10	
	1978-1987	11.1		0.2		4.1		15.4	
	1987-1997	16.4		0.21		4.41		21.02	
	After 1997	43		0.49		10.5		53.99	

CHAPTER 5. GROUND-MOTION SELECTION STAGE

Upon completing the identification and definition of the tested buildings, it is important to select ground motions for the time history analyses. Moreover, to obtain and define the optimal ground-motion intensity measures, correct ground motions that are compatible with the seismic characteristics of the KR must be obtained. For that purpose, the seismology and geology of Kurdistan are reviewed, and sets of ground motions are then sorted.

5.1 Seismic Characteristics of Kurdistan

As noted in section 3.2, the earthquake ground motion in a region and the associated damage to structures directly depend on the geological state of the region and other factors such as earthquake magnitude, fault structure, and distance. Accordingly, a detailed review of the seismic characteristics of the KR is undertaken by focusing on the geology, soil conditions, and region's seismicity. Seismic characteristics and parameters relevant to the regional analysis required for selecting ground motions are further elaborated.

5.1.1 Geology of Kurdistan

The Kurdistan region of Iraq is characterised by a complex tectonic structure and is located in a relatively active seismic zone that stretches along the Zagros-Taurus (Zagros-Bitlis) Belt, where the Arabian plate is subducted under the Anatolian and Eurasian plates in the north and northeast, respectively (Gok et al., 2006; Alsinawi, 2002), which is shown in Figure 5.1. This region is considered to be the most hazardous area in Iraq. According to the data presented in Figure 5.2, Iraq is located at the northern corner of the Arabian plate. This implies that the majority of earthquakes in the coastal part of this plate occur within a 200-km-wide zone. This zone is where the Zagros folded belt is located (Alsinawi, 2006). The Zagros and Taurus fold belts extend for more than 1500 km in the NW-SE and EW directions, passing through Oman, Iran, Iraq, and Turkey (Alsinawi, 2002).

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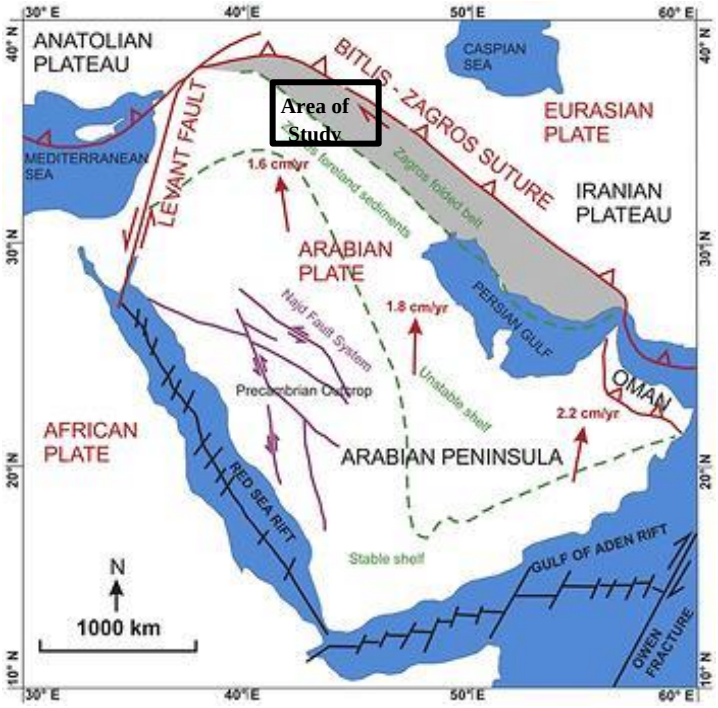


Figure 5.1: Plate tectonic setting of the study area (Frehner et al., 2012).

Geologically, the Middle East, which encompasses the selected Iraqi Kurdistan region, was formed under the processes of subduction, transformation, compression, and extension of four major tectonic plates, including the African, Eurasian, Indian, and Arabian plates, an another smaller plate, i.e., then Anatolian plate. The collision between the Arabian and Eurasian plates was the primary cause of the formation of the fold and thrust belt (Zagros Mountains) in western Iran and north-eastern Iraq (Nilforoushan et al., 2003). Alsinawi (2006) argued that the seismic activity of Iraq is mainly dictated by the Zagros-Taurus system, and this system passes through Kurdistan on its way to Turkey, which was noted earlier (Figures 5.1 and 5.2). Because no specific or existing seismic studies focused on Kurdistan have been performed, Kurdistan's seismicity can be reviewed and examined based on studies for all of Iraq in terms of seismic data and incidents.

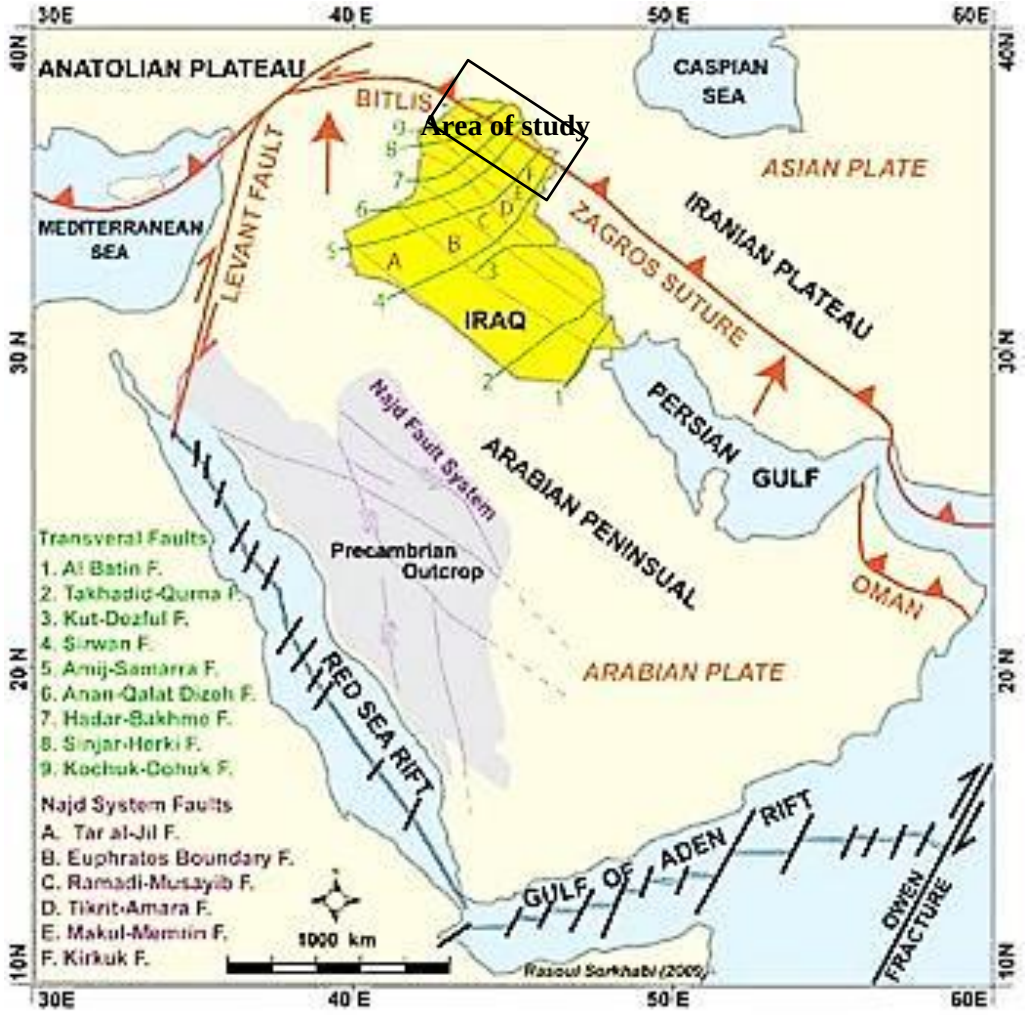


Figure 5.2: Plate tectonic setting of Iraq; the fault systems are shown (Sorkhabi, 2009).

At the tectonic level, Iraq is divided into two main units of the Arabian plate: stable and unstable shelf units. These two units were mainly differentiated during Pliocene Alpine orogeny (Dunnington, 1958). The unstable shelf (in Iraq) is categorised by four zones: the foothill (Low folded) zone, the high folded zone, the imbricated zone and the Zagros and thrust suture zone (Jassim and Buday, 2006). Figure 5.3 shows the tectonic zones and structural elements of the stable and unstable shelf units in Iraq and Kurdistan. Geologically, the unstable shelf is characterised by thick and folded Jurassic-Cretaceous sedimentary coverage that is in parallel with the Zagros-Taurus suture belt, as opposed to the Tertiary unfolded sedimentary coverage of the stable shelf. The different systems of the geologic units spread along the Kurdistan region; their ages can be found in Figure 5.4.

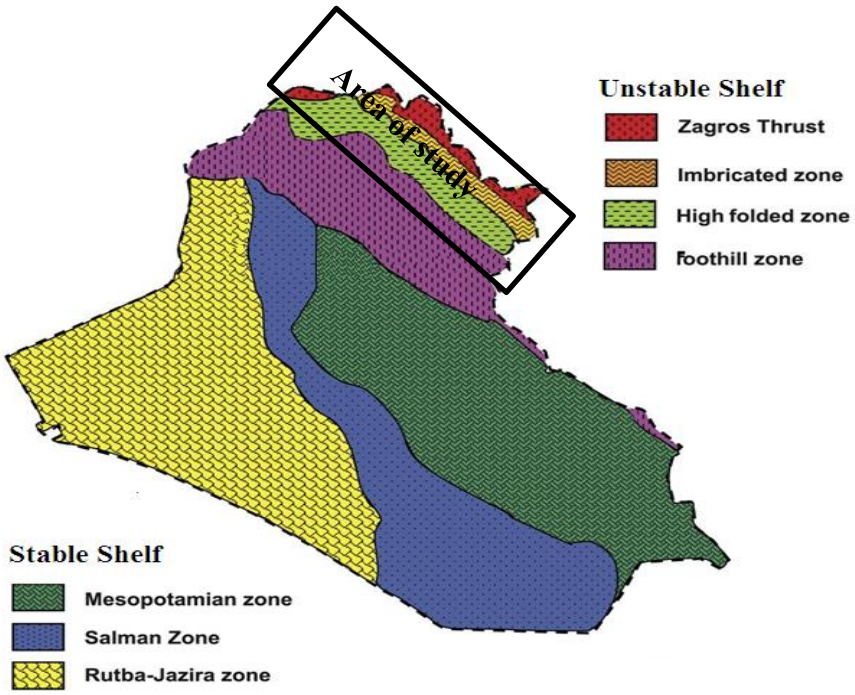


Figure 5.3: Tectonic zones and structural elements of Iraq and the study area.

Based on Figure 5.4, it is clear that the study area (i.e., the Kurdistan region) is mostly covered by the high folded zone with a width of 25 to 50 km; this region consists of harmonic folds with Mesozoic limestone in their cores and Palaeogene and Neogene (Cenozoic) limestone, representing clastic and carbonates, on their flanks. Cretaceous and Tertiary sediments are prevalent sediments and consist of different formations, which are shown in Table 5.1. Tertiary sediments can be characterised by conglomerates, red claystone, siltstone, and sandstone of Bai-Hasan, Muqdadiah, Injana, Fatha, Kolosh, and red bed series formations. Moreover, Cretaceous sediments are characterised by different types of limestone (e.g., Marl, Dolomite, and Neritic), shale, and marl from various formations such as the Shiranish, Tanjero, Aqra-Bekhma, and Balambo formations. Considering other historical periods, such as the Jurassic, Triassic, and Paleozoic, the geological structure of Kurdistan is mainly composed of limestone rocks. Quaternary deposits are covered with clay beds that vary in colour from heavy brown to yellowish brown; in addition, these deposits have variable thicknesses, covering moderately deep inclined rock strata (Jassim & Buday, 2006). For additional details, see Table 5.1 and Figure 5.4. From an engineering and seismological perspective, the Late Cretaceous and

Palaeogene formations can be considered to be stable and good sites for construction purposes due to the high absorption ability of their basic materials, such as conglomerates and red bed materials, to absorb seismic wave energy (Aziz et al., 2001).

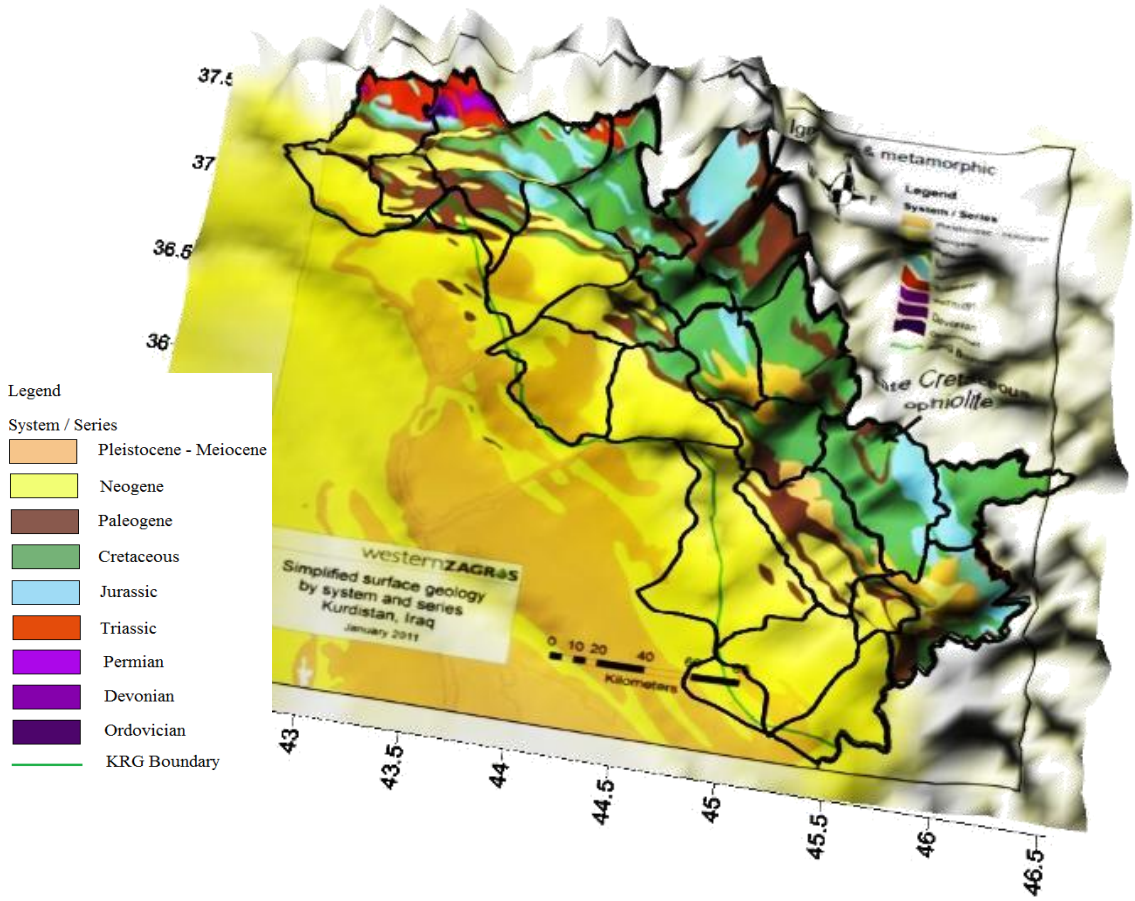


Figure 5.4: Simplified surface geology of the Iraqi Kurdistan region by age and system, modified from Western Zagros (2011) [as cited in Banks, 2011].

Table 5.1: Geological structure of Kurdistan; age, period, formation, and soil type are shown; adapted from Jassim & Goff (2006).

Age			Period	Formation	Soil type description
Cenozoic Era (65 Ma-present)	Quaternary		Holocene	Flood plain deposits, Residual soils	Silt, Clay, Sand
			Pleistocene	Bai Hasan (Upper Bakhtiari)	Conglomerate, Sandstone, Red Claystone
	Tertiary	Neogene	Pliocene	Bai Hasan (Upper Bakhtiari) Muqdadidah (Lower Bakhtiari)	Red Claystone, Sandstone
			Miocene	Muqdadidah (Lower Bakhtiari)	Conglomerate, Sandstone, Red Claystone, Gypsum, Limestone, Siltstone
				Injana (Upper Fars)	
				Fatha (Lower Fars)	
		Paleogene	Eocen Paleocene	Jaddala	Conglomerate, Limestone, Dolomited Limestone, Shale, Marlston, Siltstone, Sandstone, Claystone
				Avanah	
				Pilaspi	
				Gercus	
				Sinjar	
				Khurmala	
				Kolosh	
				Red bed series	
				Aaliji	
Mesozoic Era (251 Ma-65 Ma)	Cretaceous			Shiranish	Limestone, Dolomited Limestone, Marly Limestone, Conglomerate, Sandstone, Black Calcareous Shale
				Tanjero	
				Dokan	
				Gulneri	
				Qamchuqa	
				Bekhma	
				Aqra	
				Kometan	
				Qulqula Radiolarian	Massive Neritic Limestone, Dolomited Limestne, Marly Limestone
				Balambo	
				Sarmord	
	Jurassic			Karimia	Dark Dolomited Limestone, Oolitic Limestone, Black Calcerous Shale, Marlstone, Dark Mudstone
				Makhul	
				Chia Gara	
				Barsarine	
				Naokelekan	
				Najmah	
				Gotnia	
				Sargelu	
				Alan	
				Mus	
				Adaiyah	
				Butmah	
				Sehkanyan	
				Sarki	
	Triassic			Baluti	Dolomitic and Oolitic Limestone, Marly Limestone, Shale
				Kurra chine	
				Gelikhan	
				Beduh	
				Mirga Mir	
Paleozoic Era (542Ma-251Ma)	Permian Carboniferous Devonian Silurian Ordovician Cambrian			Chiazairi	Organic Limestone, Argillacouse Limestone, Marl and Marly Sandstone, Quartzite Sandstone, Micaceous Shale
				Gaara	
				Harur	
				Kaista	
				Ora	
				Pirispiki	
				Akkas F	
				Khabour	

5.1.2 Site classification

The effects of the geological structure in terms of seismic risk assessment are usually recognised as 'site effects' and are primarily based on the dynamic behaviour of soil within a particular site. Amplification and liquefaction are the two most commonly identified impacts of soil conditions or site effects on the peak (acceleration and velocity) parameters of earthquake-related ground motion. An example of 'site effects' in the form of amplification has been reported during the Loma Prieta earthquake in 1989 (M_x=7.1). During the seismic activity, two sites that were located at the same distance from the source of the earthquake but being covered by different types of soils, exhibited different accelerations. The peak ground accelerations at the soft soil site amplified to a value (PGA= 0.16g) much more than its value at the rock site (PGA= 0.06g) (Park, 2004). Analogously, the recent devastating earthquakes in Iran, such as the 1962 M_S 7.3 Buyin-Zara, 1978 M_W 7.3 Tabas, 1981 M_W 7.2 Sirch, 1990 M_W 7.4 Manjil, 1997 M_W 7.2 Ardakul, and 2003 M_W 6.6 Bam earthquakes, have provided examples of site effects in terms of the amplification process. During the seismic activity of the above earthquakes (for example, in 2003, the M_W 6.6 Bam earthquake), adobe houses built on alluvium deposits suffered substantial damage compared to those located on firm soil or rock sites, as noted by Jafari et al. (2005). Furthermore, topographic irregularity can also cause amplification at sites in hilly or flat locations. During the 1985 earthquake in Chile, houses constructed on hilly sites experienced substantial damage, while structures built on flat sites generally exhibited only minor damage (Villaverde, 2009).

Villaverde (2009) indicated that the characteristics of ground motions affected by 'site effects' can be interpreted as (1) a significant influence on the peak ground velocity and response spectrum shape because of subsurface soil conditions; (2) large effects of topographical irregularities on the ground-motion characteristics; (3) and high accelerations at shallow soft soil sites. The other effect of soil conditions, as presented by Villaverde (2009), is a temporary state change of saturated soils in which a shift from solid to liquid occurs during an earthquake. This phenomenon is commonly known as liquefaction. During liquefaction, the soil is deprived of its ability to carry loads for a sufficiently long period of time; consequently, soil failure occurs, which results in widespread damage to the ground and buildings located above. Saturated, loose, and sandy soils are the most common sediments that are vulnerable to liquefaction during seismic processes. Among previous earthquakes, the Niigata, Japan, and the Great Alaska, US,

earthquakes in 1964; the Tangshan, China, earthquake in 1976; the Valparaiso, Chile, earthquake in 1985; and the Chi-Chi, Taiwan, China, earthquake in 1999 represent serious seismic episodes that resulted in substantial damage to the site and structures due to liquefaction effects. Therefore, it can be assumed that site classification is an essential step in obtaining a proper understanding of the 'site effects' in a particular region and for earthquake engineering design purposes.

Laboratory or field tests should be systematically used to define the types of soils in an area of interest by estimating dynamic characteristics (e.g., shear modulus and damping ratio) of the local soil. Among the different parameters that can be utilised in these laboratory or field tests, the shear wave velocity is the most important parameter (Villaverde, 2009). In such tests, the average shear wave velocity measured at a shallow depth (30 m) beneath the soil surface (V_s^{30}) is utilised based on the suggestions of different building codes such as the 1997 UBC (1997) and EC8 codes. Table 5.2 illustrates the site classification proposed by EC8 and UBC (1997) based on V_s^{30}.

Table 5.2: Site classification modified from (a) EC8 and (b) UBC (1997).

(a)			(b)		
Ground Type	Description of Stratigraphic Profile	V_s^{30} (m/s)	Site Class	Soil Profile Name	V_s^{30} (m/s)
A	Rock or other rock-like geological formation, including at most 5 m of weaker material at the surface.	> 800	A	Hard rock	> 1500
			B	Rock	760-1500
B	Deposits of very dense sand, gravel, or very stiff clay, at least several tens of metres in thickness, characterised by a gradual increase of mechanical properties with depth.	360-800	C	Very dense soil and soft rock	360-760
C	Deep deposits of dense or medium-dense sand, gravel or stiff clay with thickness from several tens to many hundreds of metres.	180-360	D	Stiff soil	180-360
D	Deposits of loose-to-medium cohesionless soil (with or without some soft cohesive layers), or of predominantly soft-to-firm cohesive soil.	< 180	E	Soil (soft clay)	< 180
E	A soil profile consisting of a surface alluvium layer with V_s values of type C or D and thickness varying between about 5 m and 20 m, unerlain by stiffer material with V_s > 800 m/s.		F	Soils requiring site specific study (e.g. Liquefiable soils)	
S_1	Deposits consisting, or containing a layer at least 10 m thick, of soft clays/silts with a high plasticity index (PI > 40) and high water content.	< 100 (indicative)			
S_2	Deposits of liquefiable soils, of sensitive clays, or any other soil profile not included in types A-E or S_1				

Because neither a regional seismic site conditions map nor information about the shear wave velocity relevant to the study area have been produced, the topographic slope-based global map derived from Wald and Allen (2007) is used as an acceptable source of classification data for the study region's site conditions. As demonstrated in Table 5.3, Wald and Allen (2007) successfully constructed their global V_s^{30} map (site condition map) by taking a correlation between 30-arc-sec topographic data and the available V_s^{30} measurements in relation to active and stable tectonics areas based on the site classification of the National Earthquake Hazards Reduction Program (NEHRP) (Building Seismic Safety Council, 2004) .

Table 5.3: Summary of the slope ranges for the NEHRP V_s^{30} categories

(Wald & Allen, 2007).

Class	V_s^{30} Range (m/sec)	Slope Range (m/m)	
		Active Tectonic	Stable Continent
E	<180	<0.0001	<0.00002
D	180-240	0.0001-0.0022	0.00002-0.002
	240-300	0.0022-0.0063	0.002-0.004
	300-360	0.0063-0.018	0.004-0.0072
C	360-490	0.018-0.05	0.0072-0.013
	490-620	0.05-0.1	0.013-0.018
	620-760	0.1-0.138	0.018-0.025
B	>760	>0.138	>0.025

Kurdistan, which is predominantly a mountainous region, varies in elevation from 500 – 1000 m above sea level in hilly areas to 1000 – 3600 m above sea level in mountainous areas (Jassim & Goff, 2006). Thus, based on the methodology of Wald and Allen (2007), the site conditions should also change in conjunction with the topographic slope in the region. Therefore, three different types of sites can be identified in the site condition map of the Kurdistan region (Figure 5.5): stiff soil sites (D), with V_s^{30} ranging from 180 to 360 m/s; very dense soil and soft rock sites (C), with V_s^{30} from 360 to 760 m/s; and rock sites (A&B), with V_s^{30} > 760 m/s. The A&B, C, and D types in the 1997 UBC and NEHRP site classifications are equivalent to the A, B, and C types in the EC8 classification, respectively, which is shown in Table 5.2. Similarly, the geology-based classification of sites formally corresponds to the aforementioned classifications and can

be confirmed using the exploratory soil map of Iraq (Figure 5.6) prepared by Buringh (1957). Similar to the topographic slope-based map and as shown in Figure 5.6, the soil types in the geology-based map are deep-phase and medium brown soil (stiff) in the foothill zone and rough broken and stony land (very dense soil and soft rock) in the high folded and imbricated zone; the Zagros Thrust zone is primarily composed of rock.

Considering all of the site classification maps discussed above, the Kurdistan region can be classified as a mountainous land that is primarily covered by different sedimentary rocks. Thus, the amplification and liquefaction effects of potential earthquakes can be ignored in this study. However, there is a formal academic necessity to investigate earthquake-induced landslide effects because of Kurdistan's mountainous environment, which is an area of interest for future studies in the region.

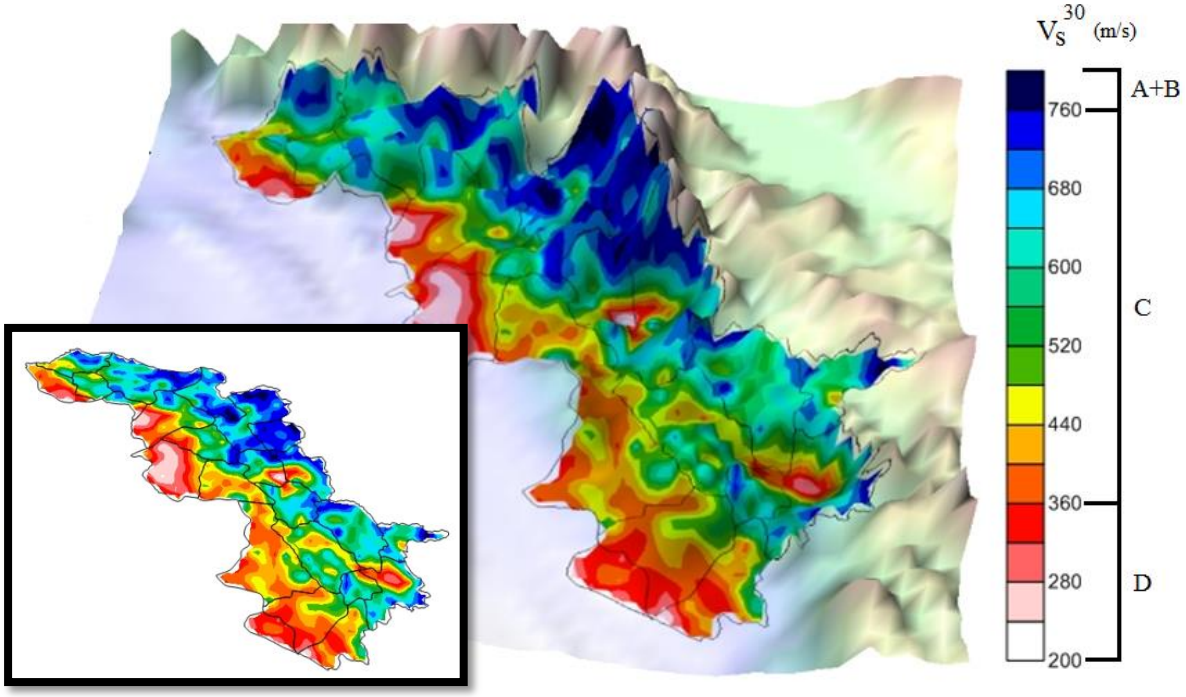


Figure 5.5: Site condition map for Kurdistan based on the topographic slope modified from Wald and Allen (2007).

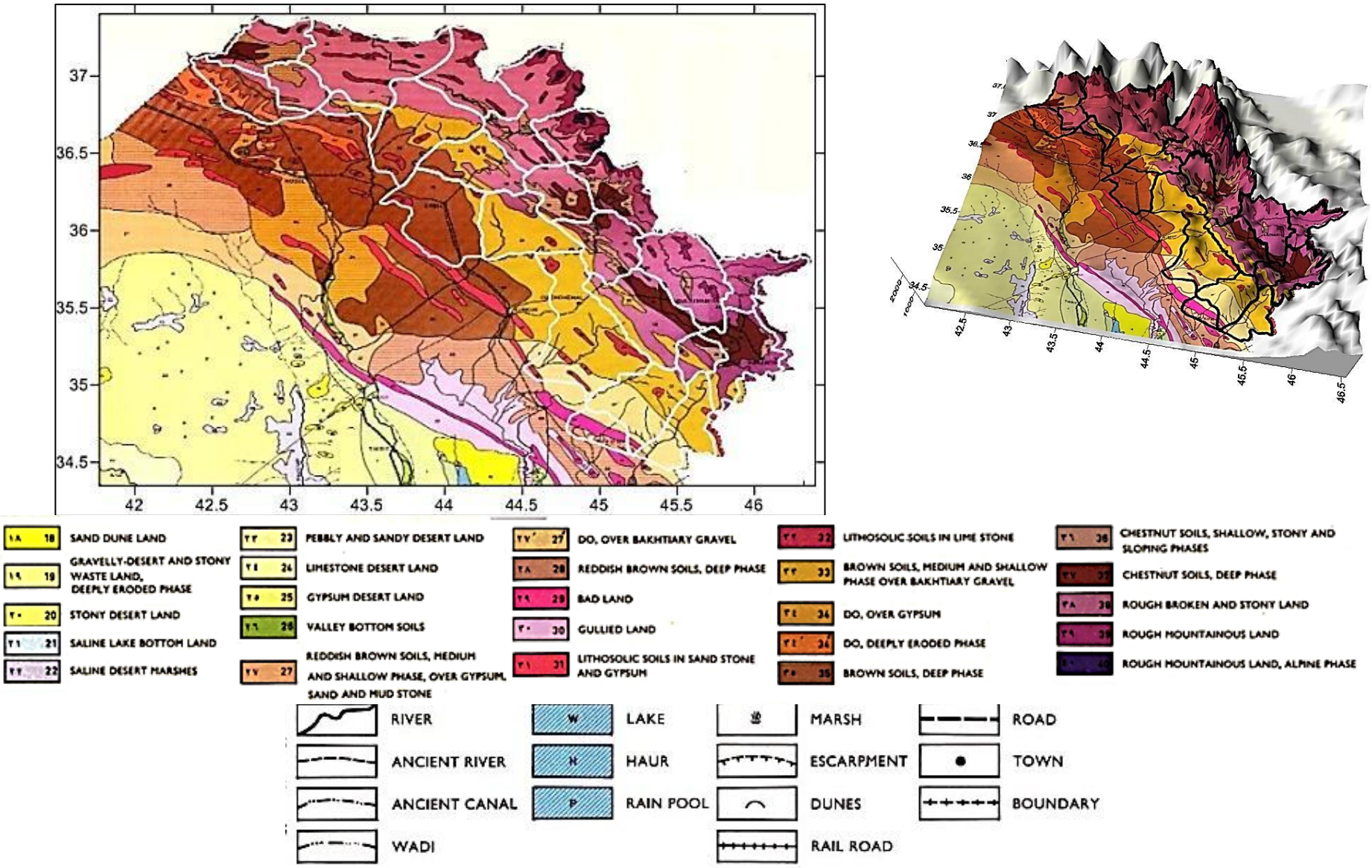


Figure 5.6: Soil map of Kurdistan modified from Buringh (1957).

5.1.3 Seismicity of Kurdistan

The forces driven by movements of the Arabian plate towards the N and NE and its slipping beneath the Eurasian plate at a rate of less than 3 cm/yr (Figure 5.1) are regarded as the main sources of seismic activities in the folded area of Iraq (Alsinawi, 2006). The majority of the moderate and large earthquakes that have been historically recorded in the selected territory occurred along the entire folded area, while all of the seismic intensity maps constructed to date (e.g., Alsinawi & Ghalib, 1975; Alsinawi & Issa, 1986; Alsinawi, 1988; Fahmi & Alabbasi, 1989; Alsinawi, 1997; Alsinawi & Al-Qasrani, 2003) illustrate a good correlation between historical events (before 1900 AD) and recently recorded events (after 1900 AD), which was asserted by Alsinawi and Al-Qasrani (2003) and is shown in Figure 5.7. Figure 5.7 shows the periods both before and after 1900 AD and indicates that Kurdistan is a geological region where the maximum seismic intensity is relatively concentrated compared with other parts of Iraq.

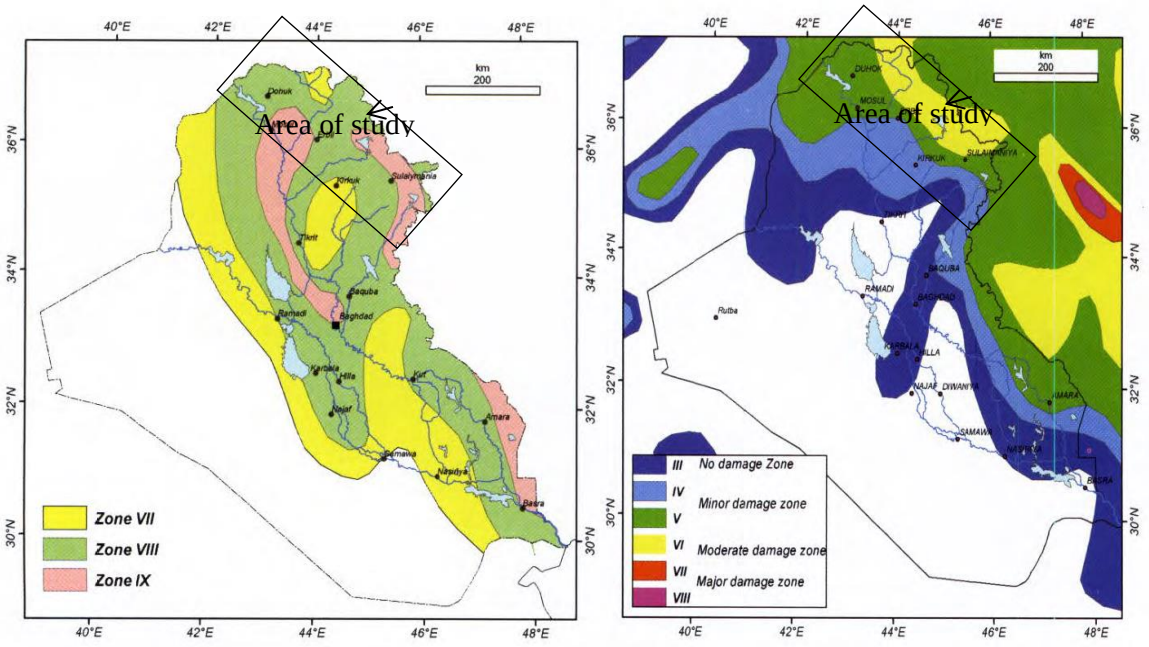


Figure 5.7: Seismic isointensity maps of Iraq (after Alsinawi and Al-Qasrani, 2003): (a) historical (before 1900) and (b) recent (after 1900).

According to the recently compiled Iraqi seismic intensity map (Figure 5.7b), Iraq can be divided into four seismic zones: (1) no damage zone with earthquake intensities equal to or less than III on the Modified Mercalli scale (MM); (2) minor damage zone with MM=IV to V; (3) moderate damage zone with intensities of MM=VI to VII; and 4) major

damage zone of MM ≥VIII (Alsinawi, 2006). Consequently, the Kurdistan region is characterised by intensities of MM=V and VI, representing a clearly moderate damage zone. However, the earthquake intensity map derived from the Office for the Coordination of Humanitarian affairs [OCHA] (2007) indicates a different outcome. The data indicate that Kurdistan is a region with a high probability of occurrence for very strong (MM=VII) earthquakes in the west and destructive (MM=VIII) earthquakes in the east, which is shown in Figure 5.8. The high seismic activity within the Kurdistan region, especially along the folded zone, can also be seen if the seismic epicentres of earthquakes and magnitudes are plotted for Iraq and the surrounding areas for a period of more than 100 years, which is shown in Figure 5.9 (Ali, 2009).

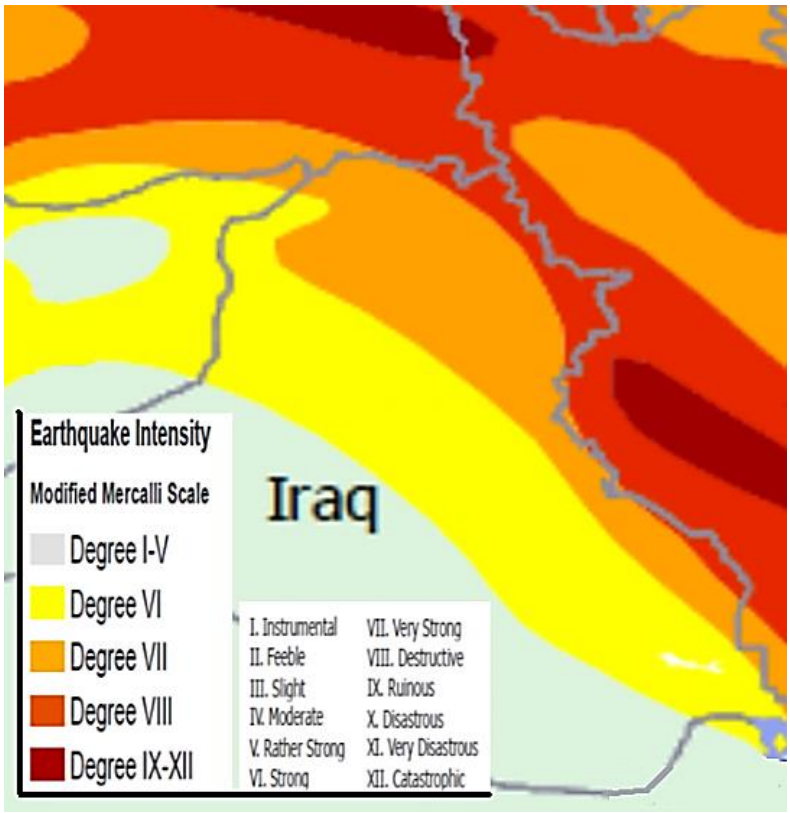


Figure 5.8: Earthquake intensity map for Iraq based on the Modified Mercalli scale.

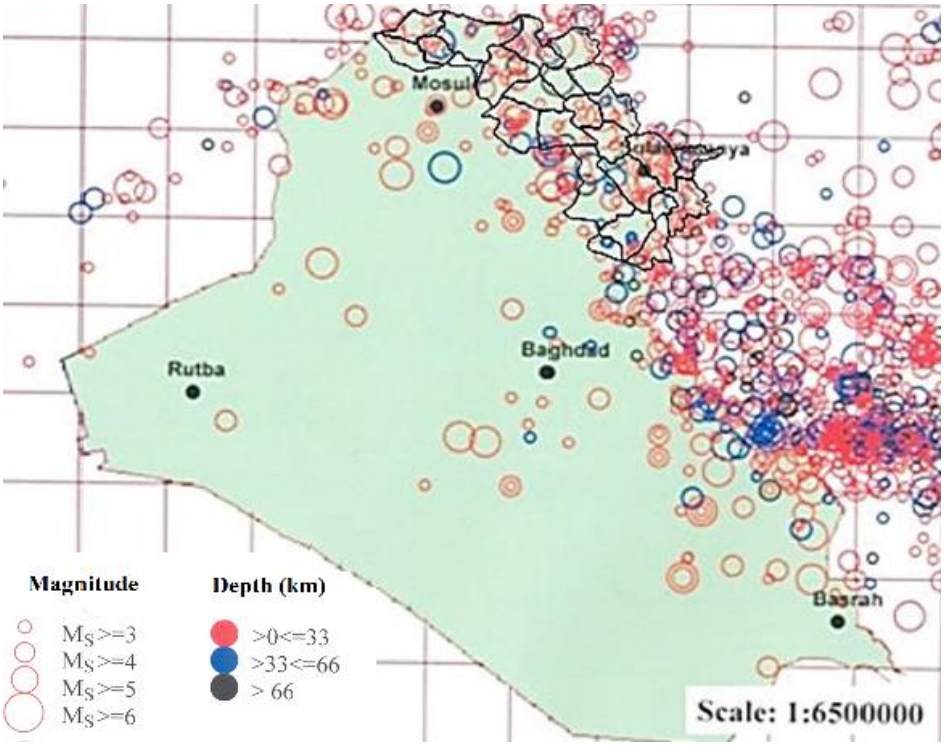


Figure 5.9: Seismic map of Iraq and surrounding area during the period 1900-2008 (Ali, 2009).

Seismic sources

Due to the evident change in the Arabian plate's geometry caused by the collision along its boundaries with the Iranian (Eurasian) and Turkish (Anatolian) plates and its rotational movements (within the Arabian plate itself), Alsinawi (1997) indicated that the distribution of seismic activity in Iraq (particularly along Zagros-Taurus folded belts) is inhomogeneous and mostly scattered (see Figure 5.9). Despite the high seismicity of the Zagros-Taurus belt, which is the main source of seismicity, there are also extensional activities along faults that are still considered active in the selected region. The Sinjar-Herki, Hadhar-Bekhme, Anah-Qalat-Dizaeh and Sirwan transversal faults can be interpreted as distinct examples of these faults (see Figure 5.10).

The fault system in Iraq can be described as being composed of three major fault systems: the N-S Nabitah, the NW-SE Najd, and the NE-SW/E-W Transversal (see Figures 5.2 and 5.10). The faults of these systems emerged primarily during the Late Precambrian Nabitah orogeny as strike-slip faulting systems (Jassim & Buday, 2006). Banks (2011) claimed that the commonly distributed types of faults in the Iraqi Kurdistan region are transversal and thrust faults recognised as normal, strike-slip, and reverse fault

mechanisms. In normal or thrust faults (dip-slip fault), the slip generated at the fault is along both the vertical and horizontal directions (along the dip's direction), whereas in strike-slip faults, which are nearly vertical, the two sides of the fault slip horizontally in the lateral directions. The hanging wall moves down relative to the footwall in normal faults, while in thrust faults, the hanging wall moves up with respect to the footwall. Thrust faults typically have low dip angles, and a high-angle thrust fault is usually called a reverse fault (Thompson & Turk, 1999). Normal and thrust/reverse faults occur mainly in areas where the crust is being extended or shortened, such as divergent and convergent boundaries, respectively (Fossen, 2010).

Considering the geological pattern of faults in the region, Rowshandel et al. (1981) defined five line and four area seismic sources in Iraq, which are shown in Table 5.4 and Figure 5.11. Moreover, Ameer et al. (2005) identified thirteen zonal seismic sources (Figure 5.12) in the Iraq territory and the surrounding area based on seismic features, the geological background, and a tectonic basis of the zones. According to Figures 5.11 and 5.12, the majority of these sources are located in Kurdistan and the surrounding area.

Table 5.4: Seismic sources in Iraq (Rowshandel et al., 1981).

Source type	No. of events	Length/Radius (km)	Avg. focal depth * (km)	Highest magnitude
Line source 1	72	640	36	6.7
Line source 2	66	367	32	7.2
Line source 3	16	220	59	5.2
Line source 4	16	210	42	5.3
Line source 5	8	188	35	6.1
Area source 1	129	80	44	5.6
Area source 2	31	70	33	5.6
Area source 3	31	60	32	6
Area source 4	28	75	38	5.5

* The depth of focus (or hypocentre) from the epicentre is called the focal depth (d_f)

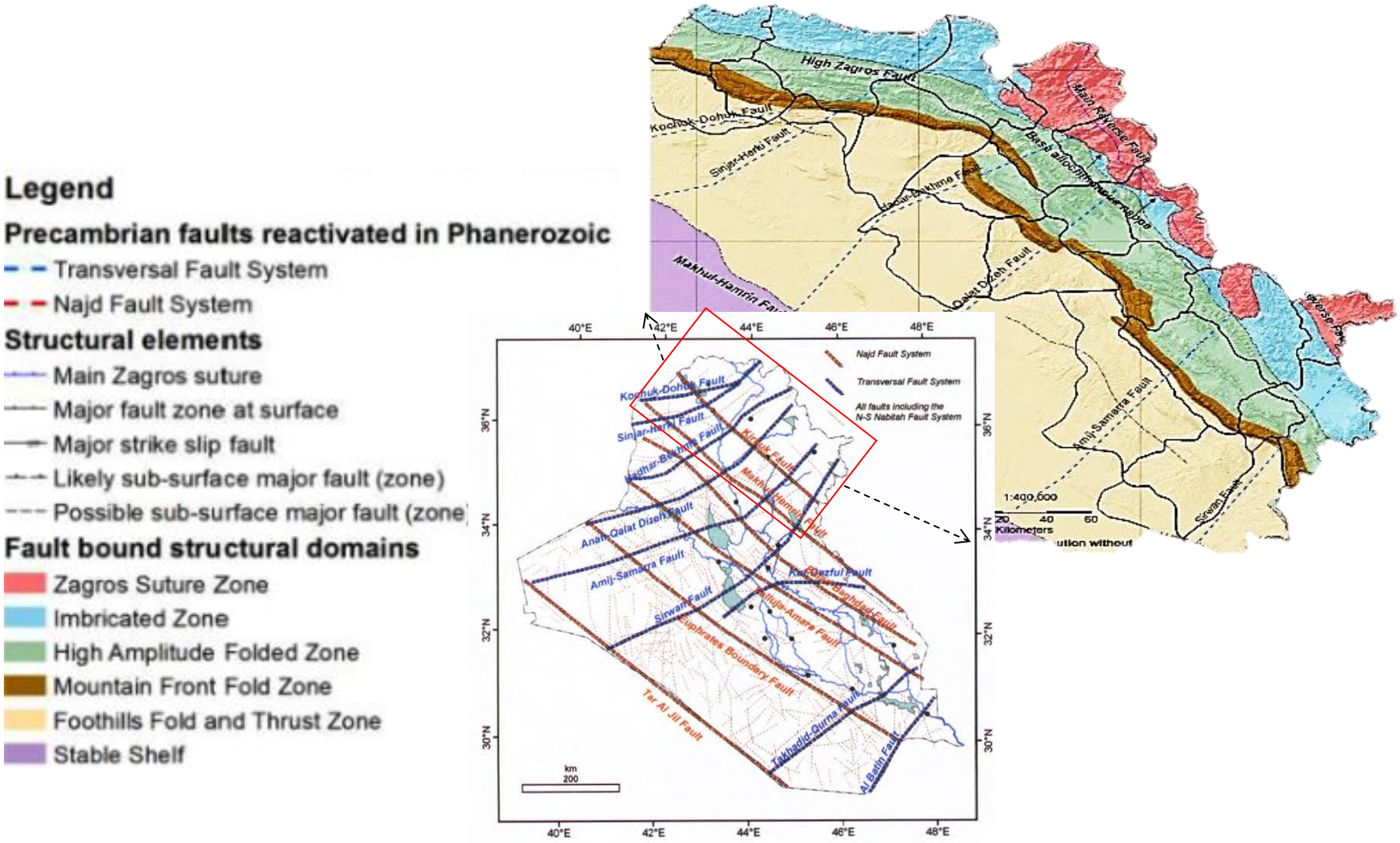


Figure 5.10: Distribution of faults in Iraq (Jassim & Buday, 2006) and the Kurdistan region; adapted from Western Zagros (2011) [as cited in Banks, 2011].

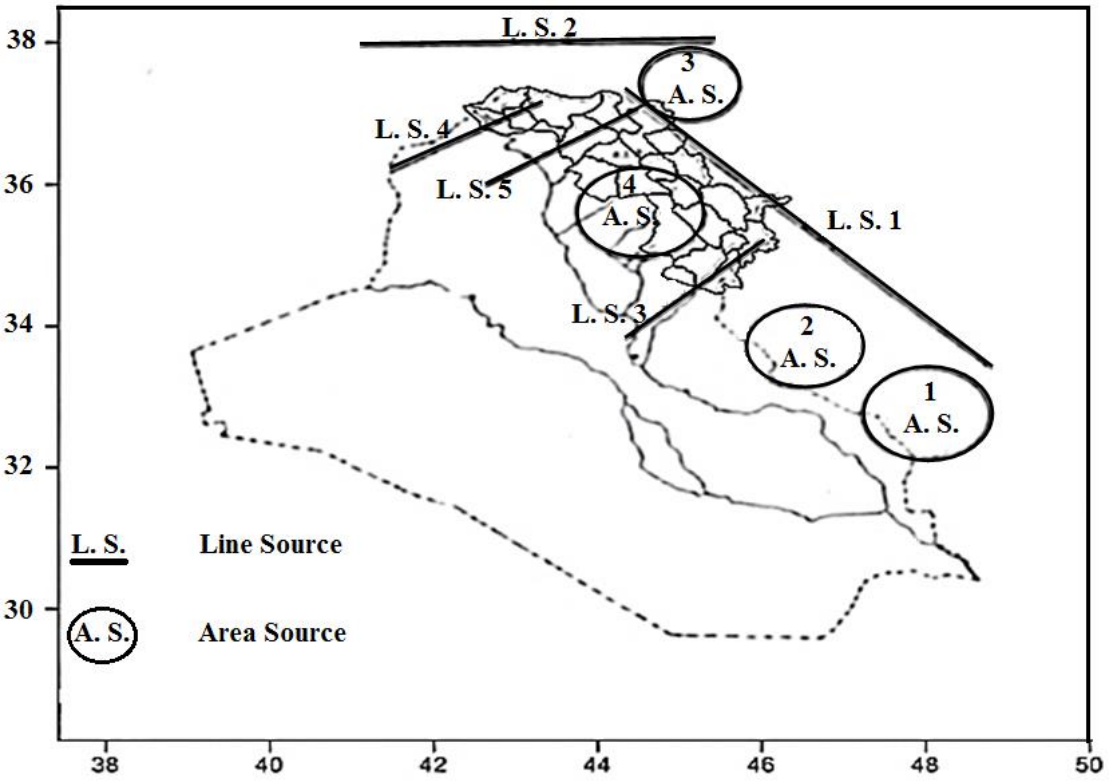


Figure 5.11: Seismic sources in Iraq (modified from Rowshandel et al., 1981).

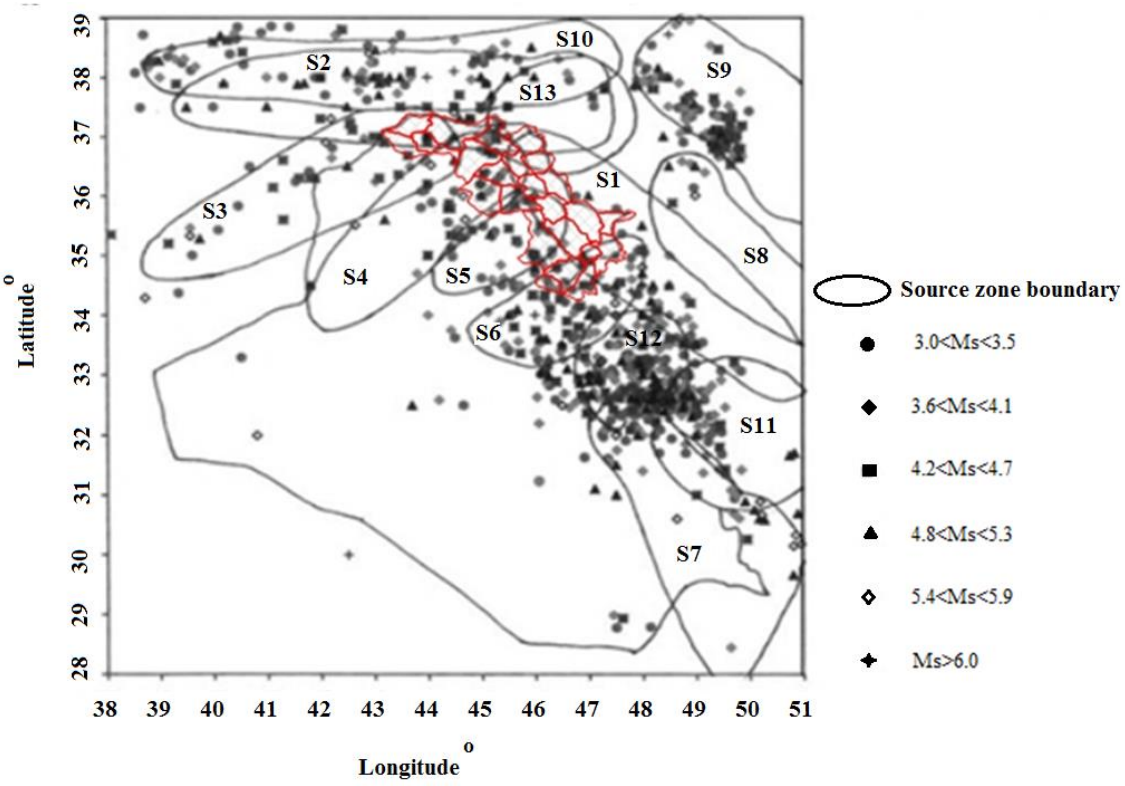


Figure 5.12: Seismic source zones in Iraq and the surrounding areas (Ameer et al., 2005).

Seismic hazard zones

Many seismic maps have been developed and introduced by several studies to illustrate the different seismic hazard zones in Iraq and clarify the most hazardous zones in Iraq. The extra level of seismicity in the area of study (NE Iraq) compared with the other parts of Iraq is shown in Figure 5.13, which is a map of seismic accelerations designed by Alsinawi and Al-Qasrani (2003) for a return period of 100 years. Furthermore, based on a total of 10000 hours of microearthquake recordings in Iraq and the analyses of Alsinawi and Banno (1976), Alsinawi and Al-Heety (1987), and Alsinawi and Al-Ridha (1988) for the period between 1974 and 1990, microseismicity increases towards the NE (the location of the KR), with a daily frequency of occurrence between 1 and 8 events.

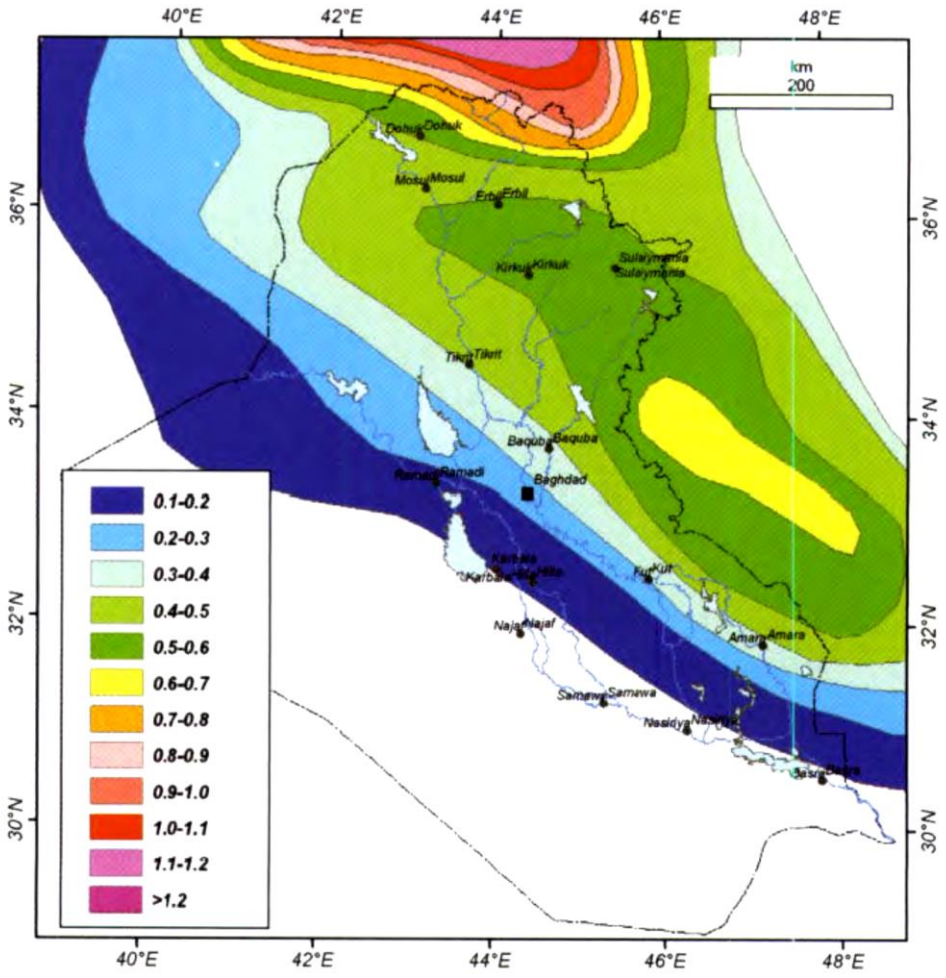


Figure 5.13: Seismic acceleration map (PGA in units of g) for a 100-year return period (Alsinawi & Al-Qasrani, 2003) [as cited in Alsinawi, 2006].

According to the Iraqi seismic building code (BRC, 1997), the Kurdistan region is located within zones II and III (as shown in Figure 5.14), which are the most hazardous zones in Iraq. The seismic zone factors (or Z factors) related to these zones are Z= 0.07 and 0.09, respectively, in contrast to the UBC (1997) seismic zones map of Iraq, which suggest higher seismic zone factors in the region (Z= 0.2 and 0.4 for zones 3 and 4, respectively) (see Figure 5.15). The Z factor is used to represent the peak parameter of ground motion (acceleration or velocity) during numerical calculations of base shear as follows:

$$V = \frac{ZIC}{R_W} W, \qquad \text{Equation 5.1}$$

where V is the shear force, Z is the seismic zone factor, I is the importance factor, R_w is the system performance factor, and W is the total seismic dead load. The factor C is a numerical factor that depends on a site's soil characteristics and a structure's fundamental period.

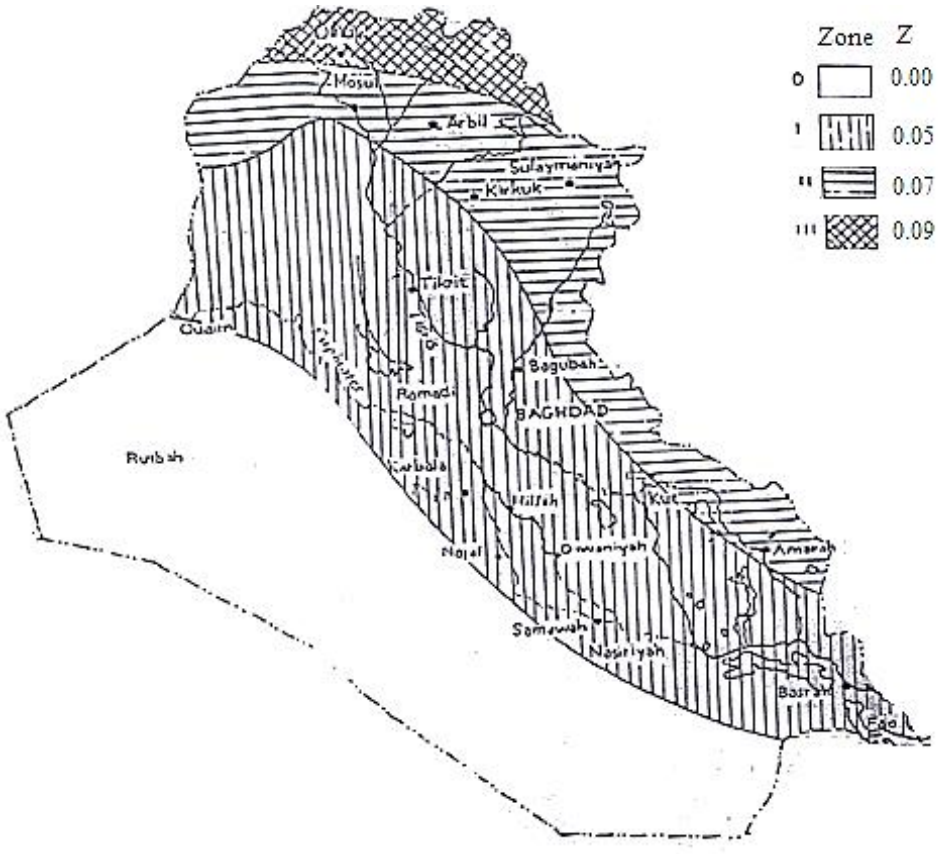


Figure 5.14: Seismic zone map of Iraq showing the seismic zone factors (Z) (BRC, 1997).

The seismic hazard maps created by the Global Seismic Hazard Assessment Program [GSHAP] (Grünthal et al., 1999), World Health Organization [WHO] (2010), and UBC (1997) also illustrate seismic hazards that vary from high to very high levels in the region, which is shown in Figure 5.15. High PGAs ranging from 1.6 to 4.8 m/s^2 (0.163g to 0.49g) are found in Figures 5.15 and 5.16. Figure 5.16 shows the 3D seismicity levels of the study area and demonstrates the high PGAs in the folded zone. To predict ground motions from a large earthquake, which is of great importance in earthquake engineering, many attenuation relations have been proposed based on regression analyses of strong motion recordings. Fahmi and Alabbasi (1989) noted that the equation of ground-motion prediction in terms of the PGA proposed by Ambraseys (1978) is conceptually suitable for seismic hazard mapping in Iraq:

$$\log PGA = 2.24 + 0.3M_S - 1.1 \log EpiD \ . \qquad \text{Equation 5.2}$$

Moreover, considering the focal depth, Bagheri et al. (2011) provided an equation for the peak ground acceleration attenuation relationship for the Zagros zone and for rocky sites (defined by V_s^{30}>375 m/s) in Iran, which is the same Zagros zone that covers most of the Kurdistan region:

$$\log PGA = 2.448 + 0.348M_S - 0.02M_S^2 - 1.329\sqrt{EpiD^2 + d_f^2}, \qquad \text{Equation 5.3}$$

where PGA is the ground acceleration in cm/s^2, EpiD is the epicentral distance in km, and d_f is the focal depth in km.

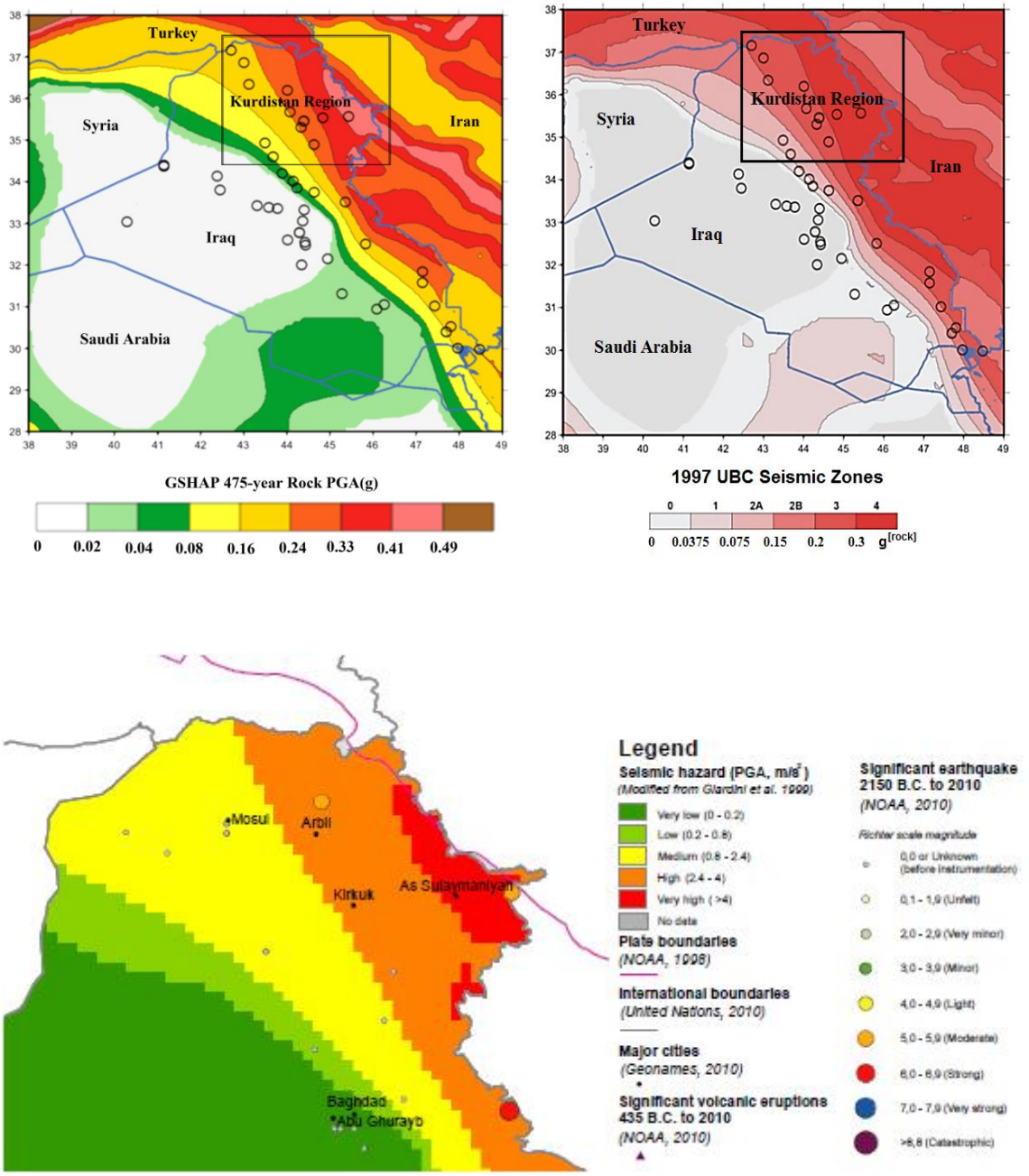


Figure 5.15: Peak ground acceleration (PGA m/s^2) map for rock with a 10% probability of exceedance in 50 years (475-year return period):
(top left) GSHAP (Grünthal et al., 1999); (top right) seismic zones equivalent to UBC (1997); and (bottom) WHO (2010).

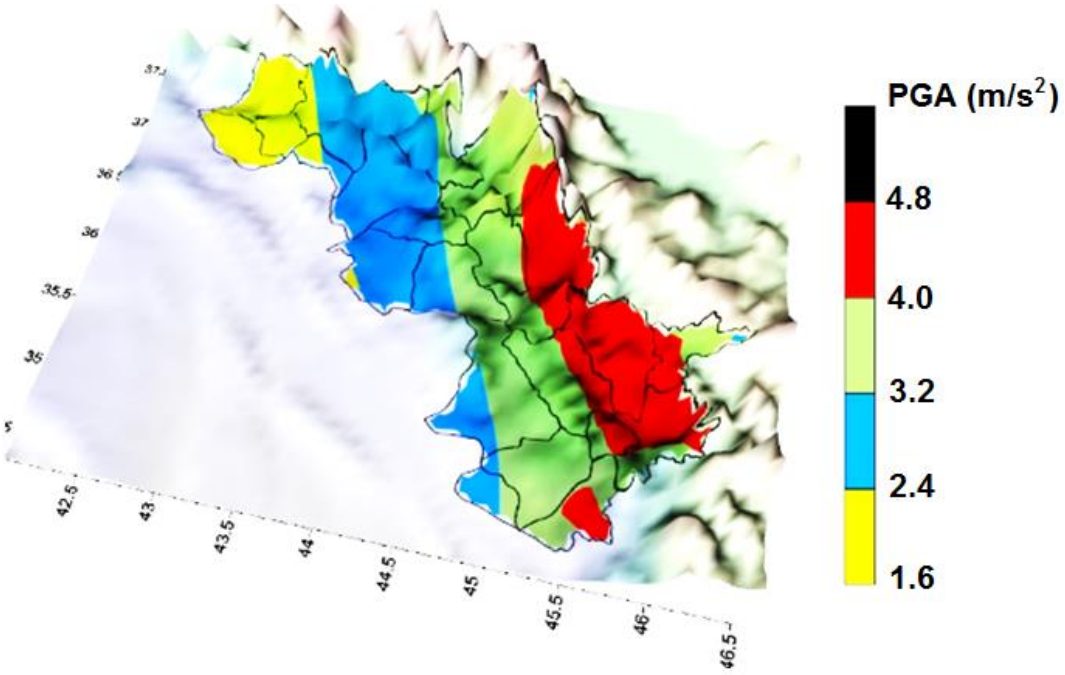


Figure 5.16: 3D map of the Kurdistan region showing the peak ground acceleration (PGA m/s^2) for a 10% probability of exceedance in 50 years (using the GSHAP map created by Grünthal et al., 1999).

Seismic characteristics and parameters

Ameer et al. (2005) and Fahmi et al. (1989) showed that the total number of events recorded in Iraq after 1950 has increased exponentially in comparison to the number of events reported before 1950. Moreover, Alsinawi & Al-Qasrani (2003) and Alsinawi (2006) noted that the majority of earthquakes in Iraq occurred at "intermediate-shallow focus levels" (here, the focus or hypocentre is the point along a fault where slip begins). The events with body wave magnitude (Gutenberg, 1945a,b,c) m_b>5.5 constitute only 8% of the total events, whereas those with magnitudes of 4-5 represent nearly 91%. In total, 76% of the events have focal depths of less than 50 km (see Table 5.4 and Figure 5.9), and 55% of those located in the focal depth range 30-50 km have magnitudes m_b=4 to 5. Furthermore, the seismicity index map (Figure 5.17) constructed by Alsinawi and Al-Qasrani (2003) demonstrates that Kurdistan, which has a high seismic number (greater than 0.2), should expect an earthquake of magnitude $m_b \geq 4.0$ every 2-5 years within an area of 100 km in diameter, whereas the frequency of earthquake occurrence in other parts of Iraq is either five to ten years or more than ten years.

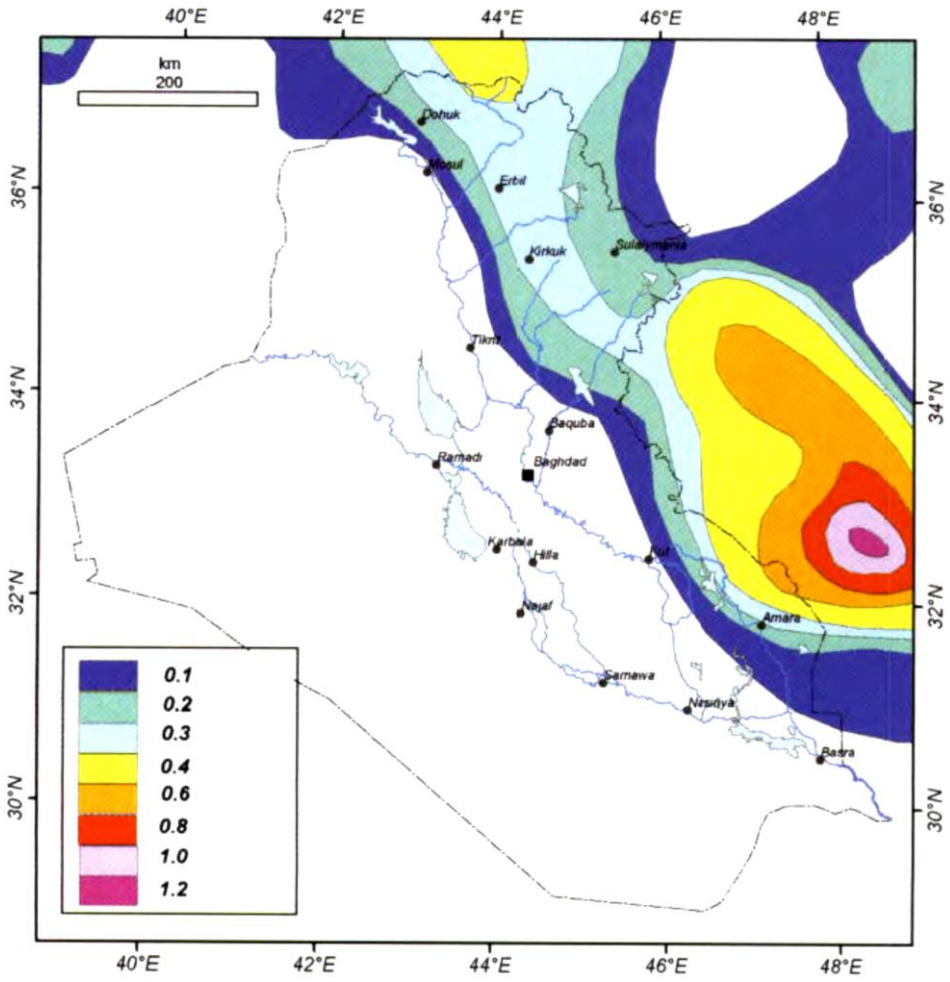


Figure 5.17: Seismicity index map (Alsinawi & Al-Qasrani, 2003)

[as cited in Alsinawi, 2006].

Because of the force activities along the entire Arabian plate and the deformation caused by these factors, destructive earthquakes in NE Iraq (particularly in Kurdistan) are widely expected to occur in the future based on the forecasts put forth by Ameer et al. (2005). The maximum earthquake magnitude M_{max} expected in Iraq is 7.87±0.86, whereas the maximum observed magnitude is M_S=7.2. With regard to the seismic source zones related to the Kurdistan region (1, 2, 3, 4, 5, 6, and 13 in Figure 5.12), the minimum and maximum expectations for M_{max} are 5.85 ± 0.35 (zone 4) and 7.68 ± 0.52 (zone 1), respectively (see Table 5.5). For the same zones, the minimum and maximum magnitudes of observed events are 5.7 and 7.3, respectively (see Table 5.6) (Ameer et al., 2005; Alsinawi, 2006).

Table 5.5: Seismic hazard parameters in Iraq (Ameer at al., 2005).

Seismic zone	λ	b	M_{Smax}	R (yr) M_S=6.0	Po $(T_x= 100)^{Ms=6.0}$
1	8.62±0.72	0.72±0.05	7.68±0.52	34.0	0.95
2	4.65±0.51	0.66±0.06	7.53±0.42	25.3	0.98
3	4.30±0.94	1.09±0.14	6.53±0.91	705.3	0.13
4	1.04±0.35	0.49±0.19	5.85±0.35	----	----
5	1.81±0.41	0.74±0.13	6.45±0.77	232.8	0.35
6	5.16±0.88	0.92±0.10	6.62±0.64	339.5	0.25
7	5.75±0.76	0.95±0.09	6.52±1.10	257.0	0.32
8	2.39±0.37	0.78±0.10	7.92±1.55	157.4	0.47
9	5.18±0.60	0.86±0.08	6.80±0.51	125.7	0.55
10	4.13±0.41	0.59±0.05	7.51±0.39	26.5	0.98
11	14.22±1.16	0.68±0.05	5.66±0.33	----	----
12	12.71±1.60	0.98±0.07	6.99±1.54	114.3	0.58
13	3.44±0.61	0.77±0.10	6.53±0.67	127.9	0.54
Iraq	34.03±1.65	0.82±0.03	7.87±0.86	9.9	1.0

R: the interval of time between events of a given magnitude (return period or recurrence interval), which is defined as R=1/P, where P is the annual exceedance probability of an event; Po =1- $(1-P)^{Tx}$; b: a constant representing the slope of the magnitude-frequency law that indicates the relative number of large and small earthquakes; λ: an average rate of occurrence or mean seismic activity rate; and M_S: surface wave magnitude of earthquakes.

Table 5.6: Input data used in the PSHA undertaken by Ameer et al. (2005) for evaluating the seismic hazard parameters for all of Iraq and its thirteen seismic source zones (see Figure 5.12).

Seismic zone	M_c	Number of events	Standard deviation	Max. observed magnitude M_S
1	3.7	237	0.56	7.3
2	4	137	0.57	7.3
3	4	58	0.44	6
4	4.2	31	0.42	5.7
5	3.9	43	0.49	5.8
6	3.7	89	0.45	6.2
7	3.9	110	0.51	5.8
8	3.8	63	0.58	6.8
9	3.9	122	0.55	6.5
10	3.7	143	0.63	7.3
11	3.5	364	0.49	5.6
12	3.9	189	0.42	6.1
13	3.9	76	0.47	6
Iraq	4	783	0.53	7.2

* M_c: minimum magnitude reported for M_S

The probability of occurrence and the mean return period of earthquakes of several magnitudes for all seismic source zones in Iraq were determined by Ameer at al. (2005) and are depicted in Figures 5.18-5.20. The high probability of occurrence of large magnitude earthquakes in the Kurdistan region, which is represented by seismic source zones 1, 2, 3, 4, 5, 6, and 13 (especially in the high folded zone), is clearly shown in Figures 5.19 and 5.20. Similarly, with respect to the return periods of earthquakes, the seismic activity is highest in the aforementioned zones in Kurdistan, which is shown in Table 5.5 and Figure 5.18. Considering earthquakes with a magnitude M_S=6.0 as an exemplary pattern, the time interval (return period) between events of such magnitude in Kurdistan is the lowest for seismic zone 2 (25 years) and the highest in seismic zone 3 (705 years). Similarly, the probability of occurrence of an event with a magnitude M_S=6.0 during a 100-year return period is approximately 98% in seismic zone 2, whereas a probability of 13% is found for zone 3.

To this end, the Kurdistan region can be characterised by a relatively high frequency of shallow seismic events with surface wave magnitudes as high as 7.7. Therefore, it is essential to thoroughly consider seismic vulnerability in these particular zones, i.e., the focus region for this study – the Kurdistan region. This can be achieved by properly defining the seismic parameters that are of critical importance for the analysis and design of structures.

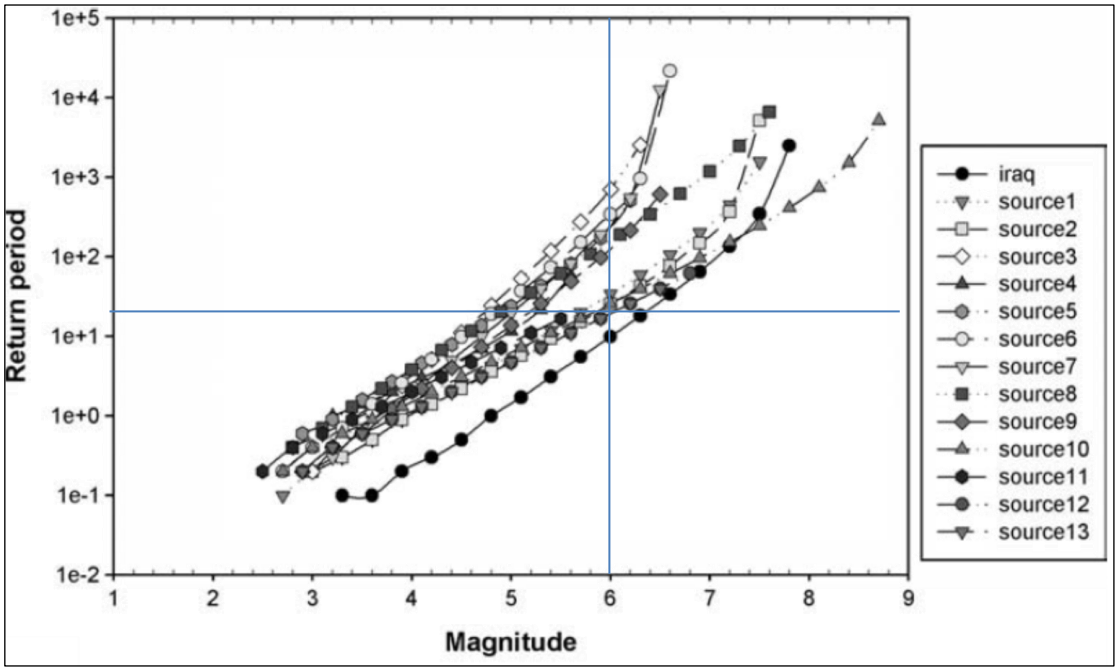


Figure 5.18: Return periods for different earthquakes magnitudes and for all of the seismic source zones in Iraq (Ameer et al., 2005).

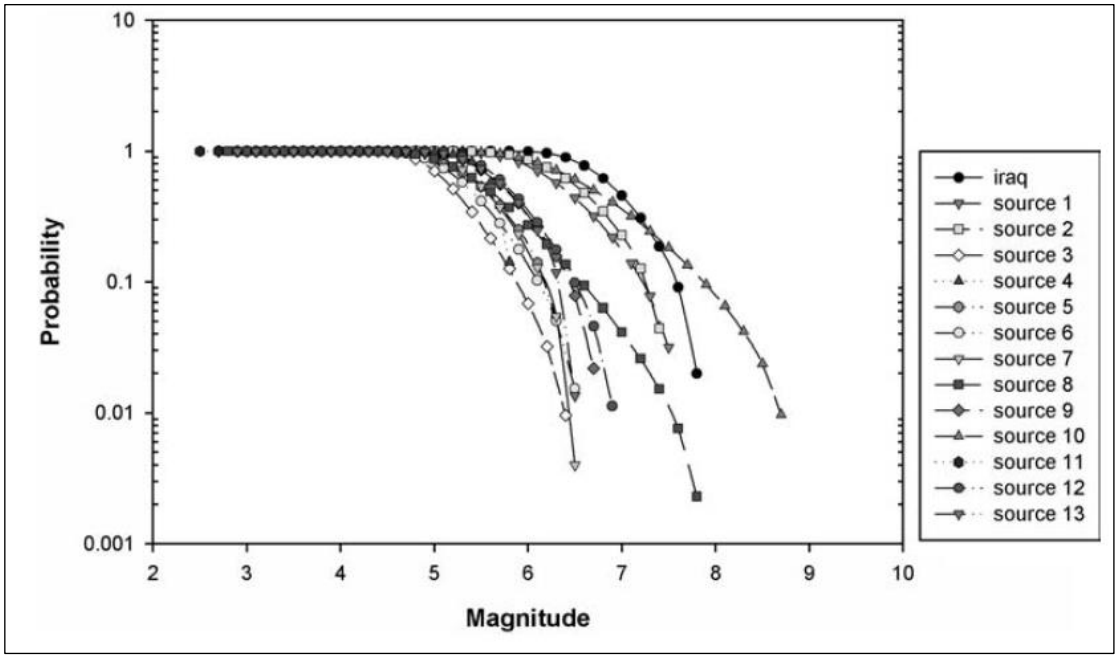


Figure 5.19: 50-year return period probability of occurrence for different earthquake magnitudes and for all of the seismic source zones in Iraq (Ameer et al., 2005).

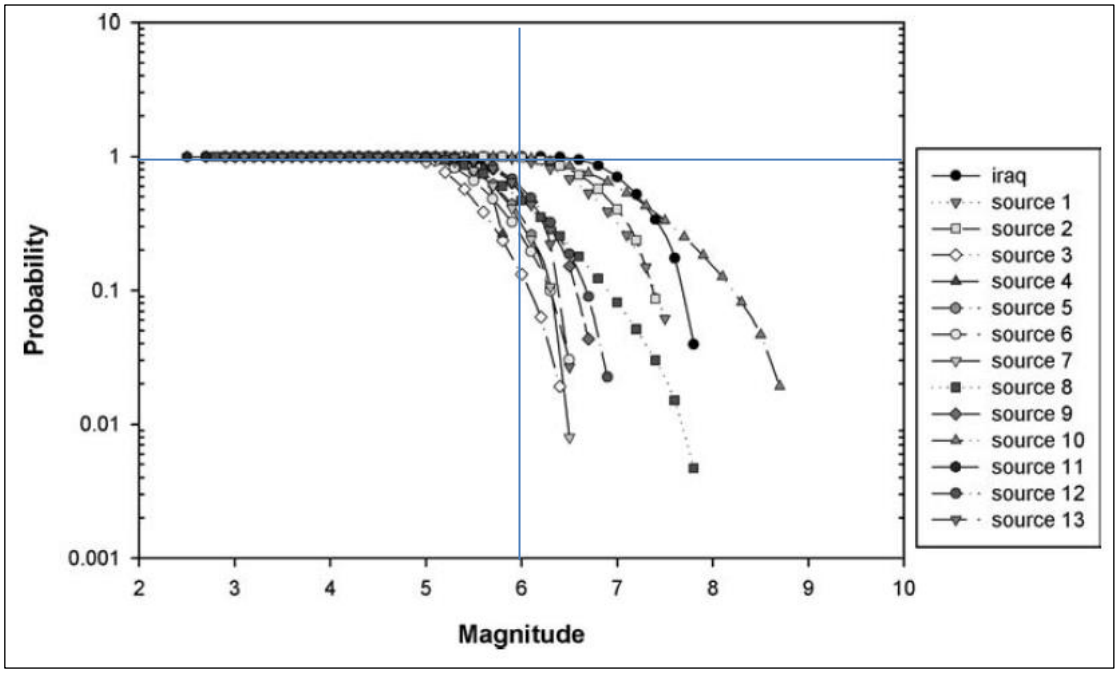


Figure 5.20: 100-year return period probability of occurrence for different earthquake magnitudes and for all of the seismic source zones in Iraq (Ameer et al., 2005).

Probabilistic Seismic Hazard Analysis (PSHA)

An estimation of earthquake parameters remains a practical challenge for engineers because of numerous uncertainties and problematic ground-motion parameters. These uncertainties can be classified as aleatory variability (the inherent randomness in a process, such as variability in the earthquake occurrence of a known fault) and epistemic uncertainty (the scientific ignorance in a process, such as the identification of seismotectonic zones in a study region) (Wen et al., 2003). Notably, probabilistic seismic hazard analysis is viewed as a method that can functionally limit these uncertainties and provide a reasonable degree of accuracy. The PSHA method accounts for the impacts of all earthquakes in a region by estimating the aleatory variability in earthquake parameters such as the magnitude, amplitude, return period time, and epicentre location. However, the other commonly used technique, which is known as the deterministic approach, can also be implemented for estimating the hazard parameters based on a single scenario earthquake. However, a scenario earthquake is not a proper representation of a region's seismicity because there are always different temporal and spatial conditions at each site and for different earthquakes (Cornell, 1968; Gupta, 2007).

PSHA was pragmatically performed in Iraq by Ameer et al. (2005). A newly updated earthquake catalogue that consists of more than 1000 reported events during the period between 1905 and 2000 was used in their study to determine relevant seismic hazard parameters such as the earthquake activity rate λ, the b-value in the Gutenberg-Richter equation (Equation 5.4), the maximum regional magnitude M_{max}, the return period R, and the probability of occurrence of events Po. The input data used in the work of Ameer et al. (2005) and the outcome of their PSHA-based study were discussed in previous section (see Tables 5.5 and 5.6). Moreover, the constants of the Gutenberg-Richter equation (recurrence relationships) for the five different periods of record in Iraq were also determined by Alsinawi and Al-Qasrani (2003) and are provided in Table 5.7.

Table 5.7: The recurrence relationship parameters for Iraq (Alsinawi & Al-Qasrani, 2003).

Recording period	Constant a	Constant b
1900-1930	4.66	0.6
1931-1960	5.55	0.74
1961-1970	6.36	0.96
1971-1980	7.47	1.16
1981-1988	8.01	1.3
1900-1988	6.85	0.89

The variance in the type of data, the geographical areas under study and the time of recording were recognised by Alsinawi and Al-Qasrani (2003) as the three main reasons for discrepancies among the expected b-values for the different historical periods (see Table 5.7). The Gutenberg–Richter formula is a relationship between the size and frequency of earthquakes in a particular region:

$$\log_{10} N_m = a - bM_x \,, \qquad \text{Equation 5.4}$$

where M_x is the size or magnitude of events, N_m is the cumulative number of events with a magnitude $\geq M_x$, and 'a' and 'b' are constants. The constant 'a' reflects the total seismicity rate in the region, and the constant b represents the slope of the relationship, which is typically equal to 1.0, although it can vary between 0.5 and 1.5 (Gutenberg & Richter, 1942). For Iraq, the average recurrence relations suggested by Fahmi et al. (1989), Alsinawi and Al-Qasrani (2003), and Ameer et al. (2005) are given by the following relationships, respectively:

$$\log N_m = 6.33 - 0.87M_S, \qquad \text{Equation 5.5}$$

$$\log N_m = 6.85 - 0.89m_b, \qquad \text{Equation 5.6}$$

and

$$\log N_m = (6.49 \pm 0.12) - (0.89 \pm 0.02)M_S. \qquad \text{Equation 5.7}$$

As shown by Ameer et al. (2005) and Alsinawi (2006), the highly complex seismicity in regional tectonic behaviour can be demonstrated by noting the comparatively high estimated values of 'b' and the large variance between this b-value and the risk level from zone to zone (see Table 5.5).

Historical events

The most significant earthquakes in Iraq over the last several centuries were reviewed by Utsu (2002) and are shown in Table 5.8. According to Table 5.8, earthquakes of moderate and high impact occurred in the Kurdistan region in -600, 1179, 1226, 1227, 1573, 1666, 1946, and 1991. The most devastating earthquakes in this region were those that occurred in 1179, 1226, 1227, 1573, and 1666, which were characterised by high seismic intensity (i.e., IX on the MM or similar scales). However, specific information regarding the number of people killed/injured or houses destroyed during those earthquakes is unavailable.

Despite the fact that dozens of small and moderate earthquakes usually occur in Kurdistan each year, the only recently recorded moderate but devastating earthquake in the region was reported on July 24, 1991, which occurred in the province of Erbil at a distance of 37.2 km from Erbil city, the capital of the Kurdistan region. An event with a moderate magnitude of M_S=5.1 (moment magnitude M_W=5.5) and a focal depth of 26 km caused a number of deaths and substantial damage. Approximately 20 people were killed and many houses were destroyed (Utsu, 2002). Figure 5.21 shows the epicentres of the aforementioned historical earthquakes in Kurdistan and various other devastating earthquakes in neighbouring areas. As depicted in Figure 5.21, almost all of Kurdistan's governorates (Duhok in the north, Erbil in the centre, and Sulaiymaniya in the south) have experienced one or more large earthquakes in recent history.

Table 5.8: The most significant historical earthquakes in Iraq; those occurring in the Kurdistan region are indicated by coloured cells (adapted from Utsu, 2002).

Year	Month	Day	Latitude	Longitude	Depth	M	Dead	Injured	Damage	Source	Remark
-600	-	-	35	45	12	6.5	-	-	some	GEM	Iraq:Sinkarah,Temple of Taras (around -600)
847	-	-	35.5	43	16	7.3	50000	-		GM	Iraq:Mosul(Al-Mawsil)(possibly the same as Apr or Nov/sc/s earthquake?)
872	6	21	34	44	-	6.8	20000	-	extr	GM	Iraq:Samarra Valley (longitude possibly wrong: damage possibly occurred on the following main shock)
958	4	0	34.5	45.9	-	6.4	many	-	cons	AM	Iraq/Iran:Qum,Hulvan(Khalwan),Jibal(Jahbel)
986	11	0	36.3	43.3	14	6.6	many	-		M	Iraq:Mosul(I=9)
1052	6	2	36.2	57.8	s	7	many	-	seve	AM	Iraq:Baihaq
1058	11	0	35.8	43.6	25	7.2	many	-		M	Iraq(I=9) (possibly the same quake as on 02/12)
1058	12	8	34.3	44.7	-	6.4	many	-	seve	AM	Iraq:Hamrin
1130	2	27	33.6	45.7	-	6.8	-	-	seve	AM	Iraq:Jibal,Mosul,Jazirh
1135	7	25	36.1	45.9	30	6.3	many	-	cons	AM	Iran/Iraq:Kurdistan(I=7-8) (also occurred on 13/8)
1136	-	-	36	43.5	-	7	many	-		M	Iraq:(I=9)
1137	9	19	37	38	18	7.4	230000	-	seve	GM	Iraq/Syria:Jazirah,Mosul(I=10) (New Year/sc/s day of 552H)
1150	4	1	34.5	45.9	-	5.9	many	-	cons	AM	Iran/Iraq:Halvan(Hulvan/Khalwan)(I=10)
1151	-	-	32.6	36.7	-	6.2	-	-	some	EMG	Syria?/Iraq?:Nakrura,Hauran,Busra
1179	4	29	36.5	44.2	16	7.1	great	-	seve	MA	Iraq:Arbil(Irbil)(I=9-10)
1194	3	0	32	44.3	-	5	-	-	some	A	Iraq:Najaf
1226	11	18	35.3	46	9	6.3	-	-	cons	AM	Iraq:Shahrizur(Hulwan)(I=9)
1227	-	-	35.8	44.4	35	7.6	many	-		M	Iraq:(I=9)
1430	-	-	32.2	46.4	-	5.3	-	-	some	A	Iraq:Wasit(833H)
1450	-	-	44.3	34.3	20	6.5	-	-		M	Iraq:(I=8-9)
1503	-	-	37.4	43.8	22	6.7	-	-	cons	AM	Turkey:Hakkari/Iraq:Mosul(I=9) (908H)
1573	-	-	35.4	45	20	6.9	-	-	cons	AM	Iraq:Shahrizur(I=9)
1666	11	0	37	43	15	6.8	-	-	seve	GREM	Iraq:Mosul(I=9)
1693	-	-	36.5	41.9	15	6.7	-	-		M	Iraq:(I=9)
1781	-	-	36.5	43	14	6.7	80	-	mode	GM	Iraq:Mosul(1196H)(I=9-10)
1864	12	7	33.3	45.9	-	6.4	-	-	cons	A	Iraq-Iran border
1917	7	15	33.5	45.8	-	5.6	-	-	mode	A	Iran-Iraq border
1927	11	12	32.53	47.38	-	5.6	-	-	mode	A	Iran-Iraq border
1938	1	26	33.1	45.9	-	5.7	-	-	mode	A	Iraq-Iran border
1946	7	27	35.6	45.8	-	5.4	-	-	limi	GAV	Iraq-Iran border
1991	7	24	36.52	44.07	26	5.1	20	many	mode	SG	Iraq:Arbil-Dibs area 5.5W



Figure 5.21: Damaging historical earthquakes in Kurdistan and neighbouring areas (adapted from Utsu, 2002).

The seismic activity increase has been more considerable in the study region over the last decade. For example, during the three-month period from March 1999 to June 1999, a group of 73 moderate magnitude earthquakes ranging from 2 to 4.7 degrees struck Sulaimaniya city in the south. Notably, the energy released by this swarm of activity was equivalent to an earthquake with a magnitude of 5.1. It was believed that the process of the renewal of the Sulaimani-Sitak transversal fault beneath Sulaimaniya city, which is located at the intersection of another fault, i.e., the Chaq-Chaq strike-slip fault (Figure 5.22), was the source of this catastrophic phenomenon. The focal depth and epicentre distance of the earthquakes were estimated to be 2-11 km and 5-11 km, respectively (Aziz et al., 2001). Besides the faults related to the Bitlis-Zagros belt that were previously discussed, there are also many faults that remain active and are being adjacent to the northern cities in the Kurdistan region. As an example, Figure 5.23 shows a fault on the northern limb of the Duhok Anticline, which is located to the south of Duhok city. Such weak zones (faults) are considered to be naturally hazardous zones from geological and engineering perspectives, which means that construction specialists should avoid these zones.

Each year, hundreds of low and moderate magnitude earthquakes are also recorded along the Iraq-Turkey border, which follows the Bitlis/Taurus belt. Many of these events are felt in Duhok, the northern governorate in the Kurdistan region. Some of these events, such as the Oct 23, 2011, Van (Turkey) earthquake, which had a magnitude of M_W=7.2, increased public concern and resulted in trepidation for engineers and the governorate of the region. The 2011 Van earthquake occurred at a distance of approximately 16 km to the north-northeast of Van and at a focal depth of 5 to 10 km (16 km according to USGS, 2011). At least 600 people died in addition to the collapse of more than 1000 buildings. The peak acceleration, intensity, and total duration of the rupture were estimated to be approximately 0.5g, VIII, and 50 seconds, respectively (Kandilli Observatory and Earthquake Research Institute, 2011). An oblique-thrust mechanism that is consistent with the Bitlis-Zagros faulting system and created by the collision of the Arabian and Eurasian Plates was the source of the Van cataclysm (USGS, 2011). As a consequence of earthquake's shallow focal depth and substantial intensity, large ground motions over a wide area were produced by the seismic activity (see Figure 5.24). Fine-grained loose soil deposits (Quaternary alluvium) with V_s^{30} less than 360 m/s played an important role in the amplification (site effect) of the shaking; this resulted in more substantial damage in the cities of Van and Ercis compared with the villages located on stiffer soil and positioned at a short distance from the earthquake's epicentre (Zaré et al., 2011). Among the cities that

responded to the 2011 Van shaking, the cities of Duhok and Zakho in the north and Erbil in the central region of Kurdistan experienced an intensity of nearly IV (MM scale) despite their long distance (more than 150 km) from the earthquake's epicentre (Figure 5.24).

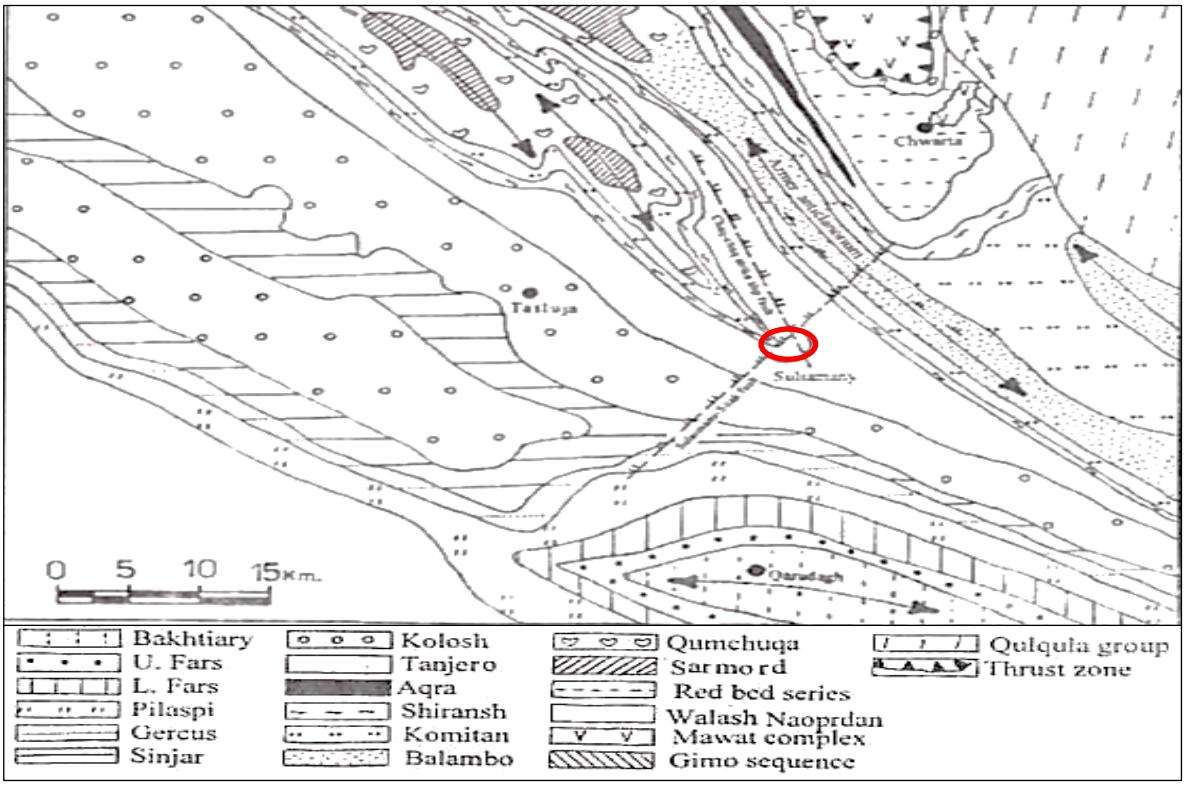


Figure 5.22: Geological map of the Sulaimaniya governorate and the location of faults beneath the city of Sulaimaniya (Lawa and Aziz, 2000) [as cited in Aziz et al., 2001].

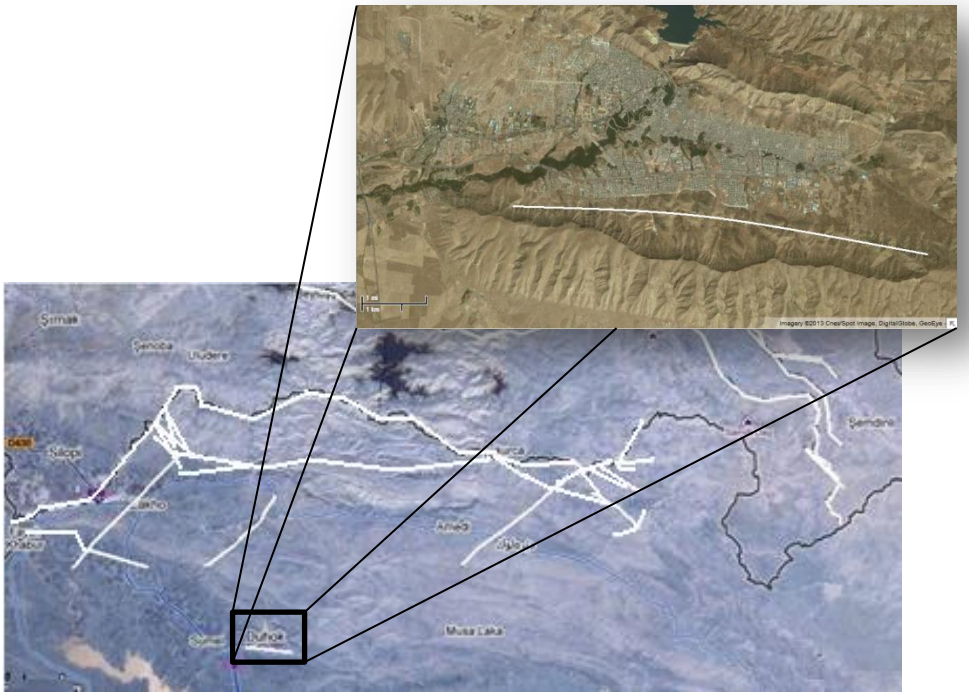


Figure 5.23: Active fault existing on the northern limb of the Duhok Anticline south of Duhok city; the white lines correspond to the faults (adapted from Turkish Republic Disaster and Emergency Management Presidency, Earthquake Department-AFAD, 2012).

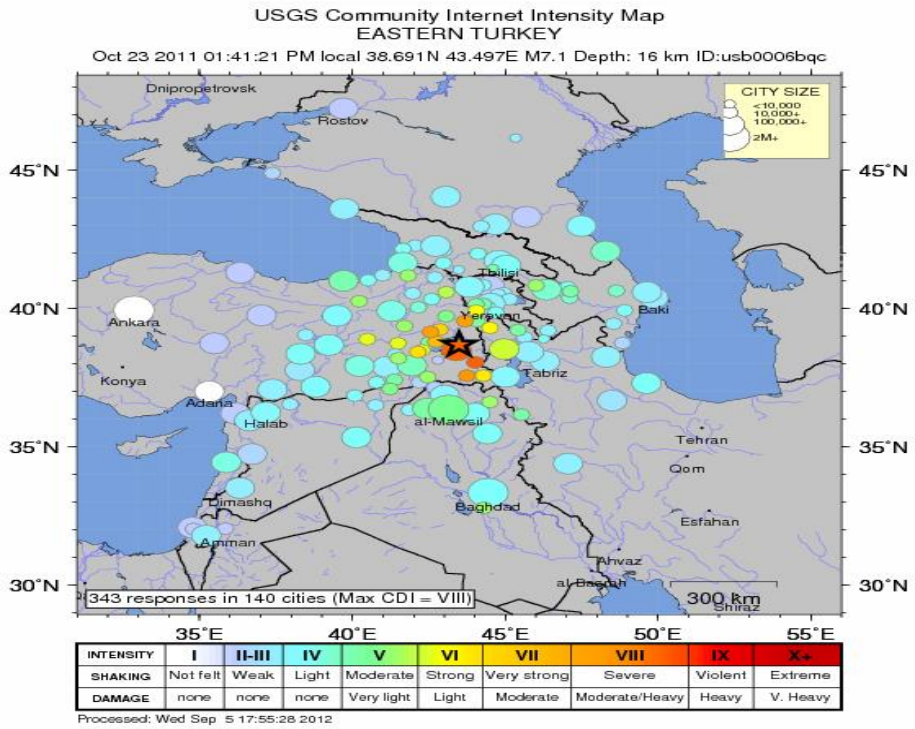


Figure 5.24: Intensity map for the Oct 23, 2011, Van, Turkey, earthquake and the surrounding area (USGS, 2011).

5.1.4 Summary

The key step in performing a comprehensive seismic analysis of any region of interest is to classify the sites in the selected area. It has already been shown that site conditions (e.g., amplification) can have a large effect on the characteristics of ground motions. A review of the available studies related to Iraq have illustrated that the study region (Kurdistan) is a mountainous area that is mostly covered by different Cretaceous sedimentary rocks, despite the presence of stiff to very stiff soil sites (Quaternary and Tertiary geologic periods) along a low folded zone in the western part of the region. These types of sites are formally represented by categories B and C in the EC8 site classification system. The next significant step to be considered in terms of a seismic study is to identify the seismic sources in a particular region.

From a seismic perspective, the Kurdistan is the region where the maximum seismic intensity is concentrated in Iraq. The seismicity of Iraq is mainly affected by the Zagros-Taurus system that was developed during the collision process between the Arabian and Eurasian plates. The Kurdistan region in N and NE Iraq is located in the northern corner of the Arabian plate and along the Zagros-Taurus system. Thus, this area is considered to be the most hazardous part of Iraq. The large-scale (MM=VII) earthquakes in the west and the destructive (MM=VIII) earthquakes in the east (along the folded zone) of the Kurdistan region are highly predictable in nature. In addition to the high seismicity of the Zagros-Taurus belt, there are additional activities along various other active faults that are prevalent in the study area. Transversal and thrust faults are the two common fault types in the region that involve normal, strike-slip and reverse focal mechanisms. Such weak zones are regarded as naturally hazardous zones from engineering and seismic perspectives, which implies that construction should be prohibited in these zones to ensure the safety of the public.

Kurdistan has been subject to many moderate and large magnitude earthquake events over the last several centuries, including the events of -600, 1226, 1227, 1573, 1666, 1946, and 1991. The highest seismic intensity and magnitude were IX (MM scale) and 7.6 respectively, whereas the most recently recorded moderate but devastating earthquake in the region was the M_S5.1 (M_W5.5) July 24, 1991, event, which took place at a focal depth of 26 km and a distance of 37.2 km from Erbil city, the capital of the Kurdistan region. In general, all three governorates in the region have experienced at least one or more devastating large earthquakes in recent history. Earthquake records in Iraq for the period

1974-1990 illustrate that microseismicity has increased in the Kurdistan region, with a daily frequency of occurrence between 1 and 8 events. The peak ground acceleration (PGA) has been estimated to range between 0.16 and 0.49g for a 475-year return period (10% probability in 50 years). The minimum and maximum PGAs are found in the foothill zone in western Kurdistan and the thrust zone in eastern Kurdistan, respectively. Moreover, the 1997 seismic building code of Iraq indicates that the Kurdistan region has the two strongest seismic zones (i.e., II and III) in Iraq.

In light of the aforementioned facts, the occurrence of destructive earthquakes in the study region is highly anticipated in the future, especially due to the active forces along the Arabian plate and the related deformation caused by these forces. Considering the existing details of the region's seismicity, a seismic hazard zone map can be created from the available data from different studies and used to summarise the overall implications. Thus, it was important to prepare a table containing various seismic information linked to the six seismic source zones in the study area (taken from Table 5.5), which is shown in Table 5.9. The coordinates of the districts within each seismic source zone were collected and are presented in the table. Then, the available data were correlated to each zone and district, thus providing individual scores. The mean scores from all of the data were computed. The minimum and maximum mean scores were obtained, and the range between them was categorised into two main zones, 'A' and 'B', as shown in Figure 5.25. These zones reflect areas with a high (zone B) to very high (zone A) level of seismic hazard. Characteristics of each zone are summarised in Table 5.10.

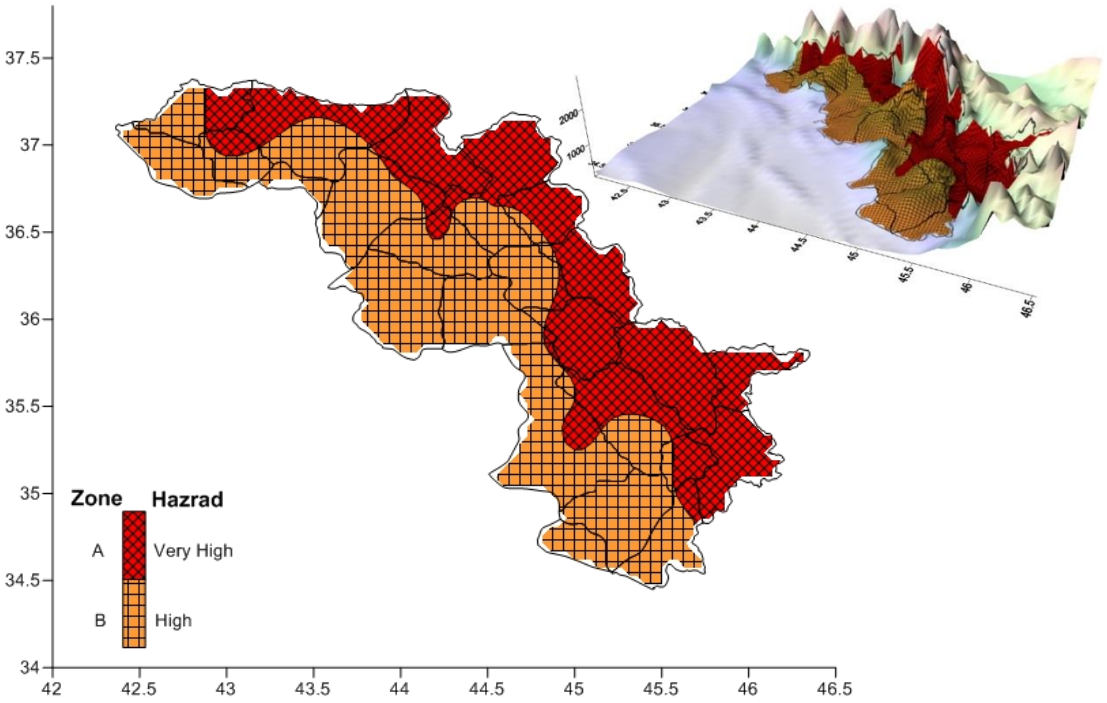


Figure 5.25: Seismic hazard zonation map proposed in this study for the Kurdistan region.

Table 5.9: Seismic hazard zones proposed for Kurdistan based on the mean score computed for each seismic source zone using information from various studies in the literature.

Seismic Sources	Iraqi Kurdistan Region	Coordinates		Max. Earthquake Magnitude			Max. PGA(g)	Intensity (MM)		Vs^{30} (m/s)	Hazard Zone			Damage	Focal Depth (km)		Epicenteral Distance (km)	Score based on Return Period related to each Seismic Source Zones (Table 5.5 and Figure 5.18)	Result	Seismic Hazard Zone
Zone (Table 5.5)	Governorates	Longitude	Latitude	Historical (Table 5.8)	Observed (Table 5.4) For equivalented source zones of Table 5.5	Expected (Table 5.5)	GSHAP (Grünthal et al., 1999) [Figure 5.15]	OCHA(2007) [Figure 5.8]	Alsinawi & Al-Qasrani (2003) [Figure 5.7b]	EC8 Classification (Figure 5.5)	WHO(2010) [Figure 5.15]	UBC (ICBO,1997) [Figure 5.15]	Iraqi Seismic Code (BRC,1997) [Figure 5.14]	Level (Table 5.8)	Historcial events (Table 5.8)	Average depth taken from Table 5.4 and assigned to the equivalented source zones of Table 5.5	Equation 5.2		Mean Score	This Study
1	Sulaimaniya	45.4333	35.5500		7.30	7.68	0.49	VIII	VI	B	Very High	4	II			36	39	25	25	A
		45.6905	35.3170				0.49	VIII	V	B	Very High	4	II				39		24	A
		45.8724	35.3464				0.49	VIII	V	B	Very High	4	II				39		24	A
		45.9833	35.1833				0.49	VIII	V	B	Very High	4	II				39		24	A
		45.9491	35.6239				0.49	VIII	VI	B	Very High	4	II				39		25	A
		45.6037	35.7590				0.49	VIII	VI	B	Very High	4	II				39		25	A
		45.3400	35.8592				0.49	VIII	VI	B	Very High	4	II				39		25	A
		45.1528	36.3028				0.49	VIII	V	B	Very High	4	II				39		24	A
		44.9758	35.9279				0.49	VIII	VI	B	Very High	4	II				39		25	A
		45.6858	35.1155				0.49	VIII	V	B	Very High	4	II				39		24	A
	Erbil	44.8893	36.6374				0.49	VIII	V	B	Very High	4	II				39		24	A
	Sulaimaniya	45.8000	35.6000	5.4			0.49	VIII	VI	B	Very High	4	II	Limit			39		22	A
		46.0000	35.3000	6.3			0.49	VIII	V	B	Very High	4	II	Cons.	9		39		24	A
2	Duhok	43.0000	36.8667		7.30	7.53	0.24	VII	V	B	Medium	3	II			32	74	30	18	B
		42.6983	37.1381				0.24	VII	V	B	Medium	3	II				74		18	B
		43.4872	37.0925				0.33	VII	VI	B	High	3	II				56		21	A
		43.8933	36.7414				0.33	VII	VI	B	High	3	II				56		21	A
	Erbil	44.5378	36.6547				0.41	VIII	VI	B	High	3	II				46		23	A
		44.1459	36.9996				0.41	VIII	VI	B	High	3	II				46		23	A
	Duhok	43.0000	37.0000	6.8			0.24	VII	V	B	Medium	3	III	Severe	15		74		20	B
		42.8501	36.8583				0.24	VII	V	C	Medium	3	II				74		17	B
3	Duhok	43.3691	36.7493		6.00	6.53	0.33	VII	V	B	High	3	II			42	56	10	18	B
4	Erbil	44.2000	36.5000	7.1	5.70	5.85	0.41	VIII	V	B	High	3	III	Severe	16	35	46	5	20	B
	Duhok	43.5848	36.5018				0.33	VII	V	C	High	3	II				56		15	B
	Erbil	44.0092	36.1911				0.33	VII	V	C	High	3	II				56		15	B
		44.1884	36.1333				0.41	VII	V	B	High	3	II				46		17	B
		44.3404	36.3964				0.41	VII	VI	B	High	4	II				46		18	B
		43.6733	36.2724				0.41	VII	V	C	High	3	II				46		16	B
		44.5244	36.6125				0.41	VII	VI	B	High	3	II				46		18	B
		44.0700	36.5200	5.1			0.33	VII	VI	C	High	3	II	Mod.	26		56		16	B
5	Sulaimaniya	45.3904	35.3104		5.80	6.45	0.41	VII	V	B	High	4	II			38	46	20	20	B
		44.6941	36.3389				0.41	VII	VI	B	High	4	II				46		20	B
		44.8333	35.5333				0.41	VII	V	B	High	4	II				46		20	B
	Erbil	44.6333	36.0833				0.41	VII	V	B	High	4	II				46		20	B
		44.4000	35.8000	7.6			0.41	VII	V	B	High	4	II	----	35		46		19	B
	Sulaimaniya	45.0000	35.4000	6.9			0.41	VII	V	B	High	4	II	Cons.	20		46		20	B
6	Sulaimaniya	45.3276	34.6308		6.20	6.62	0.41	VII	V	B	High	4	II			59	46	15	20	B
		44.9667	34.6833				0.41	VII	V	C	High	4	II				46		19	B
		45.4220	34.4746				0.49	VII	V	B	High	4	II				39		21	A
		45.0000	35.0000	6.5			0.41	VII	V	C	High	4	II	Some	12		46		19	B

Table 5.10: Seismic characteristics of the seismic hazard zones A and B for the Kurdistan region that are proposed in this study.

Zone	Mag.	PGA (g)	Focal depth (km)	Vs^{30} (m/s)	EC8 Site class	Fault Mechanism
A	5.4-7.7	> 0.4	< 35	> 400	B	Thr./Rev., S.S., Nor.
B	5.1-7.5	0.25-0.4	< 60	> 200	B & C	Thr./Rev., S.S., Nor.

Thr.: thrust; Rev.: reverse; S.S.: strike slip; and Nor.: normal.

The majority of regional earthquakes occur at a depth of less than 60 km; hence, they can be categorised as shallow-focus earthquakes. Furthermore, the region is generally impacted by both crustal and subduction earthquakes, which was mentioned earlier in this chapter. It is now possible to argue that suites of strong motion records that exhibit these characteristics can be gathered based on the properties of the different seismic sources in the study region. These properties include the earthquake magnitude ranges that may be produced, the fault mechanisms and systems and the depths. Moreover, the site classification and a well-known earthquake intensity measure (such as the PGA) associated with each seismic hazard zone within the region must be highlighted. Most of these parameters were identified in this section and are summarised in Table 5.10.

5.2 Ground Motion Selection

As previously discussed in Chapter 3, natural earthquake records are considered to be extremely useful in this study. Given the objectives of this study, the seismic characteristics of the study region were defined through a detailed review of available information regarding the selection of records. However, selecting and modifying an appropriate ground motion is a challenging and complicated process that requires careful attention. Furthermore, to estimate the performance of a structure, it is often necessary to scale the different earthquake records to a given target intensity as prescribed by incremental dynamic analysis. Scaling a time series to a target spectrum and modifying it to match the entire response spectrum are two commonly used approaches in this area. Another approach is to sort the natural records to meet specific ground-motion properties and parameters that are related to a predefined earthquake scenario with at least three

magnitude, distance, and site condition parameters. Nevertheless, similar to record selection, there are many criteria without systematic standards, and no consistent principles for record scaling are currently available (Haselton et al., 2009).

The use of scaled real records is recommended in a majority of the studies in the literature. For example, Galin (2012) investigated the effects of the selection and scaling of ground motions on the response of three 4-storey, 10-storey and 16-storey RC buildings, i.e., reinforced concrete frame buildings, based on linear and nonlinear time history analyses. The author asserted that the scaled real records seem to be preferable in such an analysis and that a simulated accelerogram should be used only when real records are unavailable. Moreover, Iervolino and Cornell (2005) indicated that the amplitude (S_a-based) scaling of records may further reduce the response's variability without biasing the estimation of this response. However, for cases in which a record's S_a deviates by ε (epsilon) from the forecasted S_a (based on the ground-motion prediction equation), the scaling will induce a bias in the structure's calculated response (Baker & Cornell, 2006a). Although the deviation (i.e., ε) is not an essential ground-motion feature for high IM levels, it is important to select records based on the correct epsilon because the IMs that contribute directly to rare maximum inter-storey drift ratios are linked to the large epsilon values, which was mentioned by Iervolino and Manfredi (2008). To overcome this issue, Baker and Cornell (2006a) proposed a method to account for the influence of magnitude, distance and epsilon, although, according to the authors, the impact of such earthquake record characteristics (e.g., magnitude and distance) on the response conditioned to $S_a(T1)$ or another IM (independent of magnitude and distance) has not yet been proven.

Iervolino et al. (2008) also noted that the scaling of spectra can reduce the variability in spectral ordinates within a set of selected records. Scale factors limited to and/or approaching unity are generally recommended. According to Julian and Acevedo (2004), a typical range from 2 to 4 is preferred for scaling ground-motion records. Moreover, Watson-Lamprey and Abrahamson (2006) assumed that the scaling should be limited to time histories based solely on the magnitude, distance, and site condition; otherwise, if other properties (e.g., the PGA, PGV, and IA) are used in addition to the magnitude, distance, and site condition, large scale factors can also be implemented without biasing the estimated response, meaning they can still lead to an average response. EC8 underlines that the average spectrum of the scaled records should not be less than 90% of the target spectrum in the periods between 0.2T1 and 2.0T1 without providing an

upper limit. Additionally, the value of the average spectrum at zero period should be larger than the value of the target spectrum at the same period. According to ASCE (2007), the average spectrum of the selected records should be scaled by 1.3 times the target spectrum for periods between 0.2T1 and 1.5T1. However, Baker and Cornell (2008) indicated that periods longer than 1.5T1 are effective for nonlinear buildings. Moreover, Baker (2011a) suggested that periods between 0.2T1 and 2T1 are effective for mid-rise buildings. The lower limit (0.2T1) represents the effect of higher vibrational modes, whereas the upper limit (1.5-2.0T1) accounts for stiffness degradation and softening effects due to the inelastic response of structures (Chen, 2011). Moreover, Stewart et al., (2002) and Fahjan (2008) indicated that appropriately selected earthquake magnitudes, fault-site distances, and site conditions are critical for selecting ground motions because they significantly impact the central characteristics of ground motions, including the frequency content, duration of shaking, and both near-fault and amplification effects.

To this end and as mentioned above, real time histories can be scaled without biasing the results if seismic characteristics, such as earthquake magnitudes, fault-site distances, site conditions, and ground-motion intensity measures, are accurately defined. The following sections explain the four GMSM methods used in this study to select real time histories based on seismic characteristics of the KR.

5.2.1 Proposed response spectra for the Kurdistan region

Based on the aforementioned facts, scaling a time series to a target spectrum and modifying it to match the entire response spectrum is a proper technique for the accurate selection of earthquake records that are representative of the seismic signature of a particular region. A target spectrum, which is an acceleration response curve, is used to graphically represent the seismic action of a site, with an option of depicting the measured or estimated peak horizontal acceleration of a SDOF over time. It is important to learn about the local geological conditions, previous earthquakes and available seismological data of a region to eventually develop such site-specific curves.

For sites with insufficient records regarding significant events, such as the KR, curve development is based on events recorded in regions with similar seismic and tectonic characteristics. Because the Iraqi seismic building code (BRC, 1997) has not been updated for approximately fifteen years and due to the lack of KR-specific response curves, a

seismic hazard zone map was constructed by referring to a comprehensive study of available region-specific information (given in section 5.1). Then, the sets of records could be selected for each seismic hazard zone (A and B) proposed in this study by considering general record properties (e.g., the magnitude, PGA, distance, depth, site soil, and fault mechanism) of each zone. The PEER NGA flat file database (Chiou et al., 2008), which includes more than 3,500 ground motions recorded for more than 100 shallow crustal earthquakes in active global tectonic regimes, is implemented in the study. Approximately 150 records representative of the seismic characteristics of zones A and B were compiled. Figures 5.26 and 5.27 indicate the number of records with respect to the earthquake magnitude and the PGA, respectively. In addition, the relationship between the epicentral distance and the selected earthquake magnitudes is also represented in Figure 5.28.

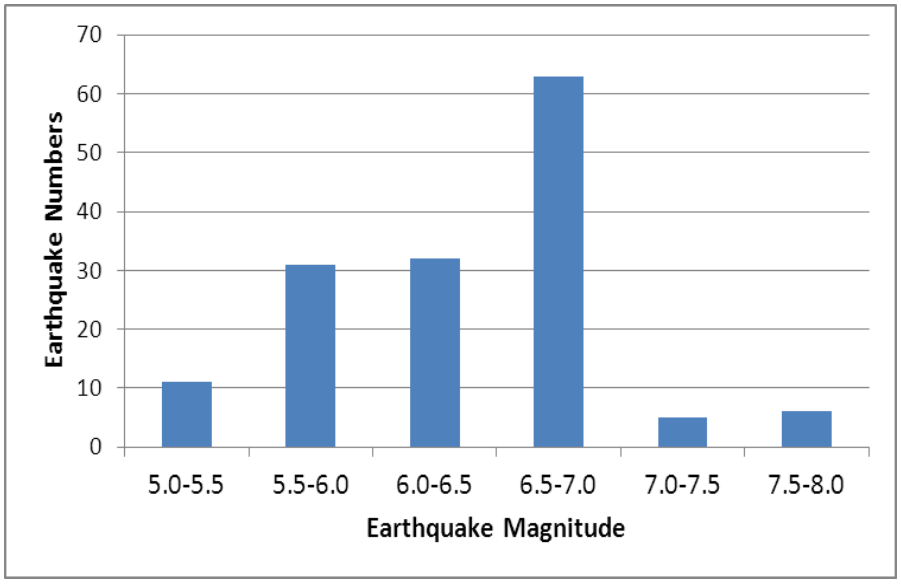


Figure 5.26: Number of selected records with respect to the earthquake magnitudes.

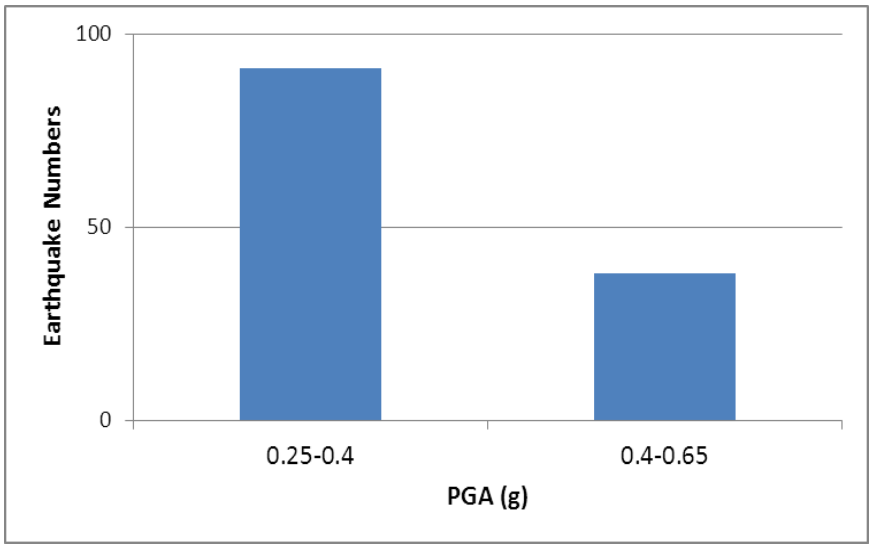


Figure 5.27: Number of selected records with respect to the PGAs.

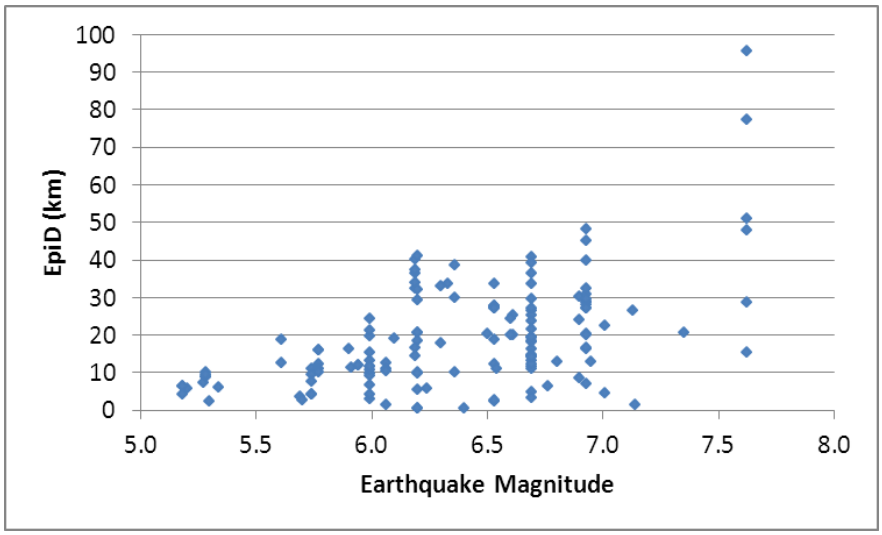


Figure 5.28: Relationship between the epicentral distance and the selected earthquake magnitudes.

With regard to Figure 5.29, the shape of a measured or estimated response spectrum is introduced as a smoothed response spectrum that can be defined by three main parameters: the PGA, the maximum spectral acceleration, and the periods corresponding to constant acceleration (T_B), velocity (T_C), and displacement (T_D) regions. To define these parameters, 95th-percentile confidence intervals and median response spectra for the records corresponding to each zone were computed (Figure 5.30); subsequently, the response spectra for the KR were constructed.

Three smoothed response spectra for seismic hazard zones A and B were identified based on the soil sites in each zone (Figure 5.31). However, considering the maximum spectral acceleration and the general shape of the response spectra, two types of acceleration response spectra developed from the parameters listed in Table 5.11 were recommended for the seismic analysis of KR buildings.

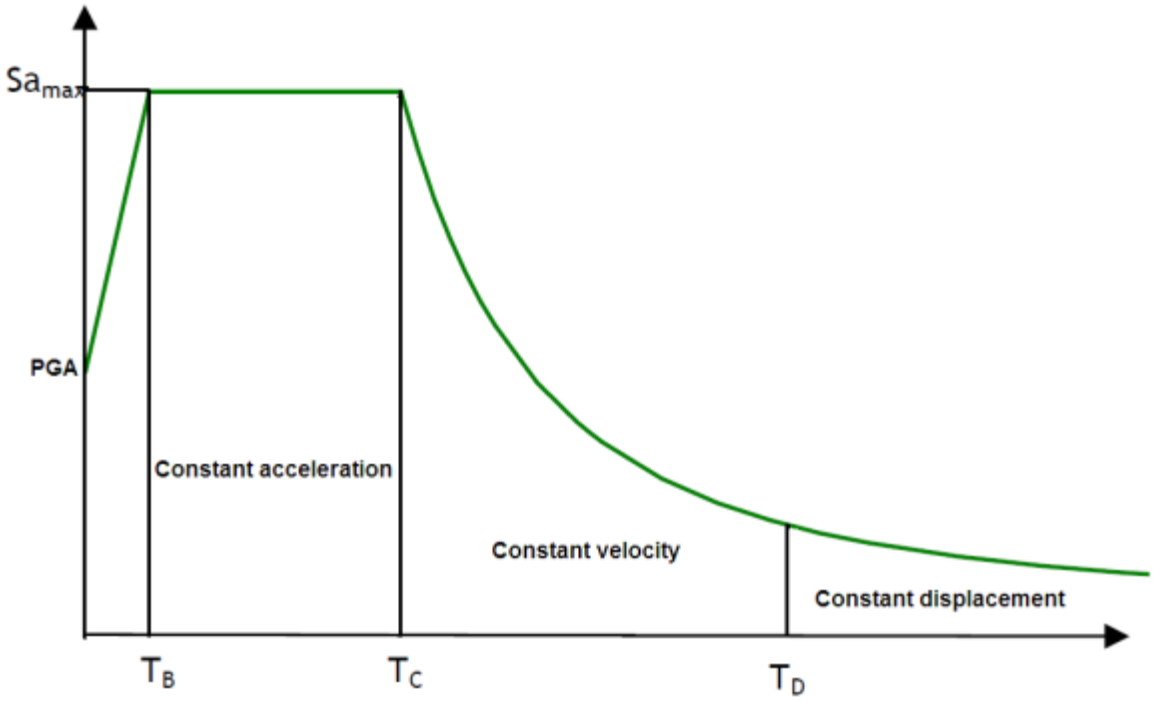


Figure 5.29: Typical smoothed response spectrum.

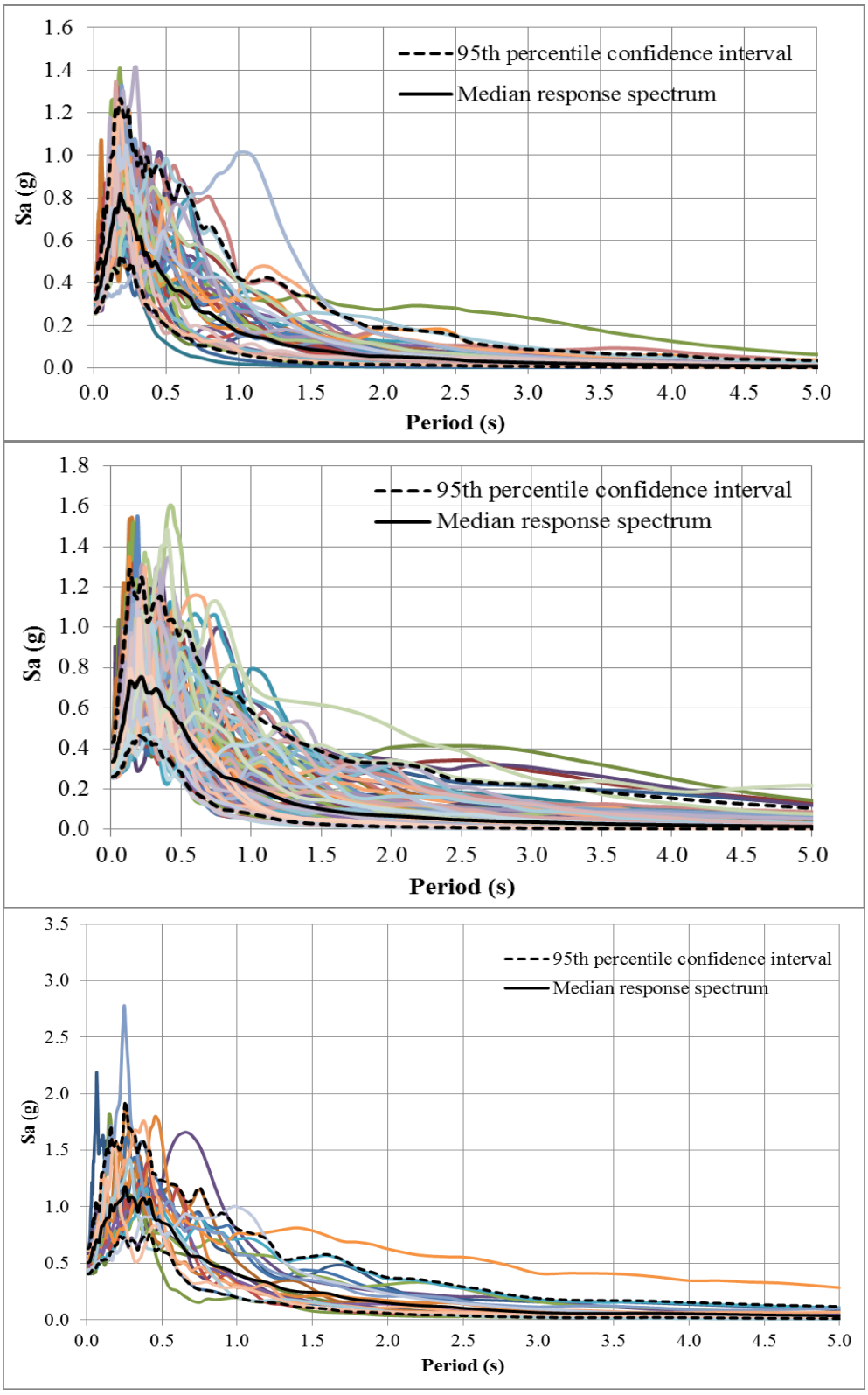


Figure 5.30: Response spectra of selected records for a rock site in zone B (top), a soil site in zone B (middle), and a rock site in zone A (bottom).

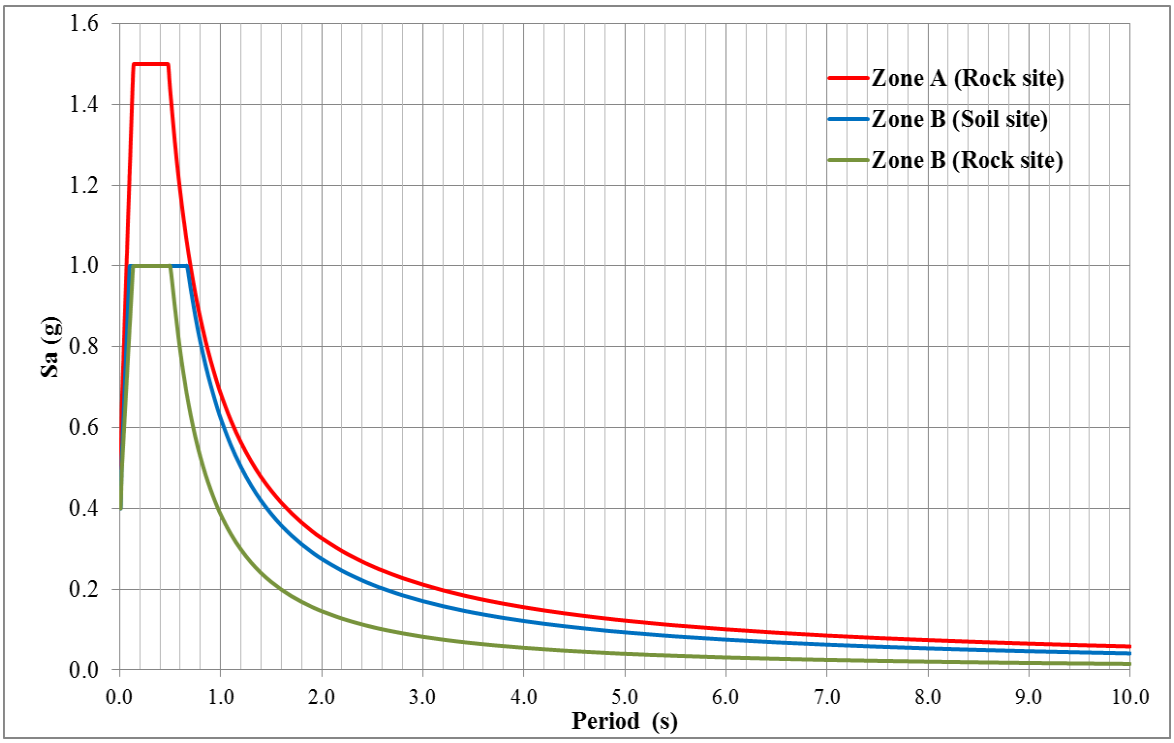


Figure 5.31: Proposed response spectra for the KR.

Table 5.11: Parameters for the proposed response spectra relevant to the KR based on this study.

Seismic Hazard Zone	PGA (g)	Sa_{max} (g)	T_B (s)	T_C (s)	S_a ($T_x > T_C$) (g)
A	0.5	1.5	0.1	0.5	$1.046\ T_x^{-1.183}$
B	0.4	1.0	0.1	0.7	$0.62\ T_x^{-1.176}$

5.2.2 Conditional mean spectrum (CMS)

Baker (2011a) recently proposed the conditional mean spectrum as an alternative to the UHS as a target spectrum in ground-motion record selection. The UHS is approximately replicated by design spectra in building codes. In terms of the CMS method, "the response spectrum is derived by conditioning on spectral acceleration at a period of interest" (Baker, 2011a). Moreover, the author provided four steps to obtain a CMS: (1) finding the target spectra (S_a) at the period of interest (T') and the magnitude of the related ground motion, the source-to-site distance, and the number of standard deviations by which a given $\ln S_a$ value differs from the mean predicted $\ln S_a$ value for a given magnitude and

distance [termed as ε(T')]; (2) calculating the mean and standard deviation of log spectral acceleration values covering all periods for the target magnitude and distance [e.g., $\theta_{lnS_a}(Magnitude, Distance, T_i)$ and $\beta_{lnS_a}(T_i)$]; (3) computing the conditional mean epsilon covering other periods ε (T_i); and (4) generating the CMS by determining the exponential of the values obtained in the three aforementioned steps using the following equation:

$$\theta_{lnS_a(T_i)|lnS_a(T')} = \theta_{lnS_a}(Magnitude, Distance, T_i) + \rho(T_i, T')\varepsilon(T')\beta_{lnS_a}(T_i) ,$$

Equation 5.8

Here, $\theta_{lnS_a(T_i)|lnS_a(T')}$ is the predicted mean value of the expected spectral acceleration's natural logarithm at period T_i , assuming that the natural logarithm of the target S_a(T') is given; θ_{lnS_a} and β_{lnS_a} are the predicted mean and standard deviation, respectively, of lnS_a at period T_i; and $\rho(T_i,T')$ is the correlation coefficient between the values of ε at the two periods, which can be derived as follows:

$$\rho(T_{min}, T_{max}) = 1 - cos\left(\frac{\pi}{2} - \left(0.359 + 0.163 I_{(T_{min}<0.189)} ln\frac{T_{min}}{0.189}\right) ln\frac{T_{max}}{T_{min}}\right) ,$$

Equation 5.9

Here, $I_{(T_{min}<0.189)}$ is an indicator function that is equal to 1 if $T_{min} < 0.189$ s and 0.0 otherwise, and T_{min} and T_{max} denote the smaller and larger of the two periods of interest, respectively (Baker & Cornell, 2006b). This prediction is valid for periods between 0.05 and 5 seconds. However, a more refined (although more complicated) correlation model that is valid over a wider range of periods (i.e., 0.01 to 10 seconds) is also available (Baker & Jayaram, 2008). Nevertheless, Equation 5.9 is nearly equivalent if only periods between 0.05 and 5 seconds are interest. An example CMS is illustrated in Figure 5.32 from Baker (2011a).

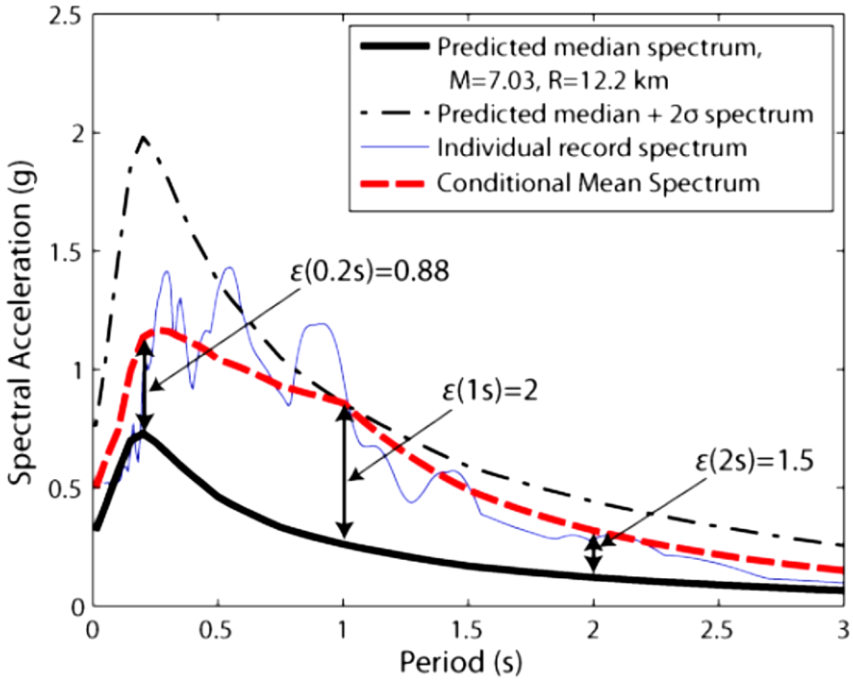


Figure 5.32: Conditional mean values of spectral acceleration covering all periods, assuming that Sa(1 s) is given and the example Castaic Old Ridge Route ground motion (from Baker, 2011a).

Several previous studies (e.g., Haselton et al., 2009; FEMA, 2009; Goulet et al., 2008; Zareian & Krawinkler, 2006) have successfully shown that accounting for the CMS target or the target ε (T') is very important, especially when estimating the safety of structures against physical collapse. However, the CMS approach is very sensitive to the given T'; previous information about this period must be known because S_a may vary if the given T' changes. This issue becomes a practical challenge in scientific applications, especially before completing a structure's analysis and design calculations. Within such contexts, the UHS approach may be used because of its invariance to the given periods (Baker, 2011a). Nevertheless, the UHS represents a nearly impossible earthquake scenario, particularly for higher hazard stages, because the rate of detecting a high positive ε at all periods is much lower than the rate of observing a high ε at any single period (Baker & Cornell, 2006b). The formal conservatism of the UHS has also been addressed by other researchers such as Reiter (1990) and Naeim and Lew (1995). Furthermore, Baker and Cornell (2006c) noted that the CMS is inconsistent during periods that are longer or shorter than the period of interest T' for conditioning the response spectrum. Thus, there is an internal need for modification if a proper variance in building response is required.

Similarly, Jayaram et al. (2011) asserted that the selection of ground motions based only on a target response spectrum mean (e.g., the CMS) and neglecting the inherent variance within the response spectrum may also lead to technical variations in the response of buildings. Therefore, Jayaram et al. (2011) recently proposed an approach for selecting real ground motions that match the target response spectrum mean and variance (see section 5.2.3).

To generate the conditional mean spectrum in this study, the geometric mean of the spectral acceleration values predicted by the Abrahamson and Silva (2008) and Boore and Atkinson (2008) equations, which were developed for the Next Generation Attenuation (NGA) project (http://peer.Berkeley.edu/products/nga_project.html), were used (see Figure 5.33). The purpose of the NGA project was to develop empirical equations to predict the main shock ground motions in shallow crustal earthquakes within active tectonic regions. Shoja–Taheri (2010) indicated that the NGA models are generally applicable to the currently available Iranian dataset. Because the Zagros zone is more active than the other main seismic zones in Iran, such as the Alborz and Eastern zones (Shoja-Taheri & Niazi, 1981), and because the seismicity of the KR is also largely affected by this zone (Zagros mountains), it can be concluded that the aforementioned equations (i.e., Abrahamson and Silva (2008) and Boore and Atkinson (2008) equations) are sufficient for the KR's strong motion dataset when used to measure the horizontal component of ground motions as a function of earthquake magnitude, source-to-site distance, local average shear-wave velocity in soil, and fault mechanism (i.e., normal, strike-slip, and reverse).

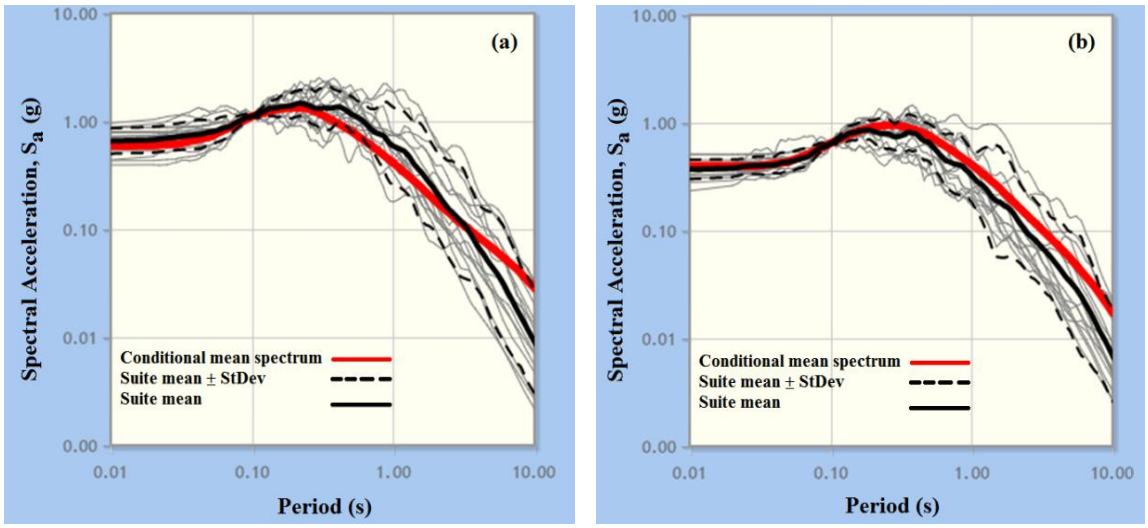


Figure 5.33: Conditional mean spectra used in the study for selecting the ground motions that are relevant to zone A (a) and zone B (b).

5.2.3 Ground-motion selection algorithm for matching the target response spectrum mean and variance

Selecting ground motions based on a target response spectrum mean (e.g., a CMS) and ignoring the inherent variance within the response spectrum may result in variations in a building's response (Jayaram et al., 2011). Thus, to estimate the effect of seismic record variance on the expected structural response, Jayaram et al. (2011) proposed an algorithm based on a Monte Carlo simulation and a Greedy Optimization technique. In this algorithm, numerous response spectra are probabilistically generated using a Monte Carlo simulation based on the target means and variances (Figure 5.34a). A ground motion compatible with each of the already generated response spectra is subsequently selected (Figure 5.34b). Furthermore, to improve the matching between the target and the sample means and variances, a greedy optimisation technique is used by substituting each selected ground motion with another one that better reflects the target (Figure 5.34c). Ultimately, the ground motions with response spectra having the desired mean and variance are selected (Figures 5.34d and 5.34e). Jayaram et al. (2011) utilised a set of selected ground motions based on this method to assess the response of sample single-degree-of-freedom (SDOF) and multiple-degree-of-freedom (MDOF) structures. They found that this algorithm can be used when the dispersion of the response and, consequently, the damage state and loss estimation are in the area of interest, especially in terms of numerous performance-based engineering evaluations. The proposed algorithm can be used for any target mean and covariance obtained from various existing methods of ground-motion selection and modification (such as the CMS method), which was discussed by Jayaram et al. (2011). A MATLAB application for the proposed algorithm (available at http://www.stanford.edu/~bakerjw/gm_selection.html) is used to select the ground motions that are representative of the seismic characteristics of the Kurdistan region for each seismic hazard zone (i.e., A and B).

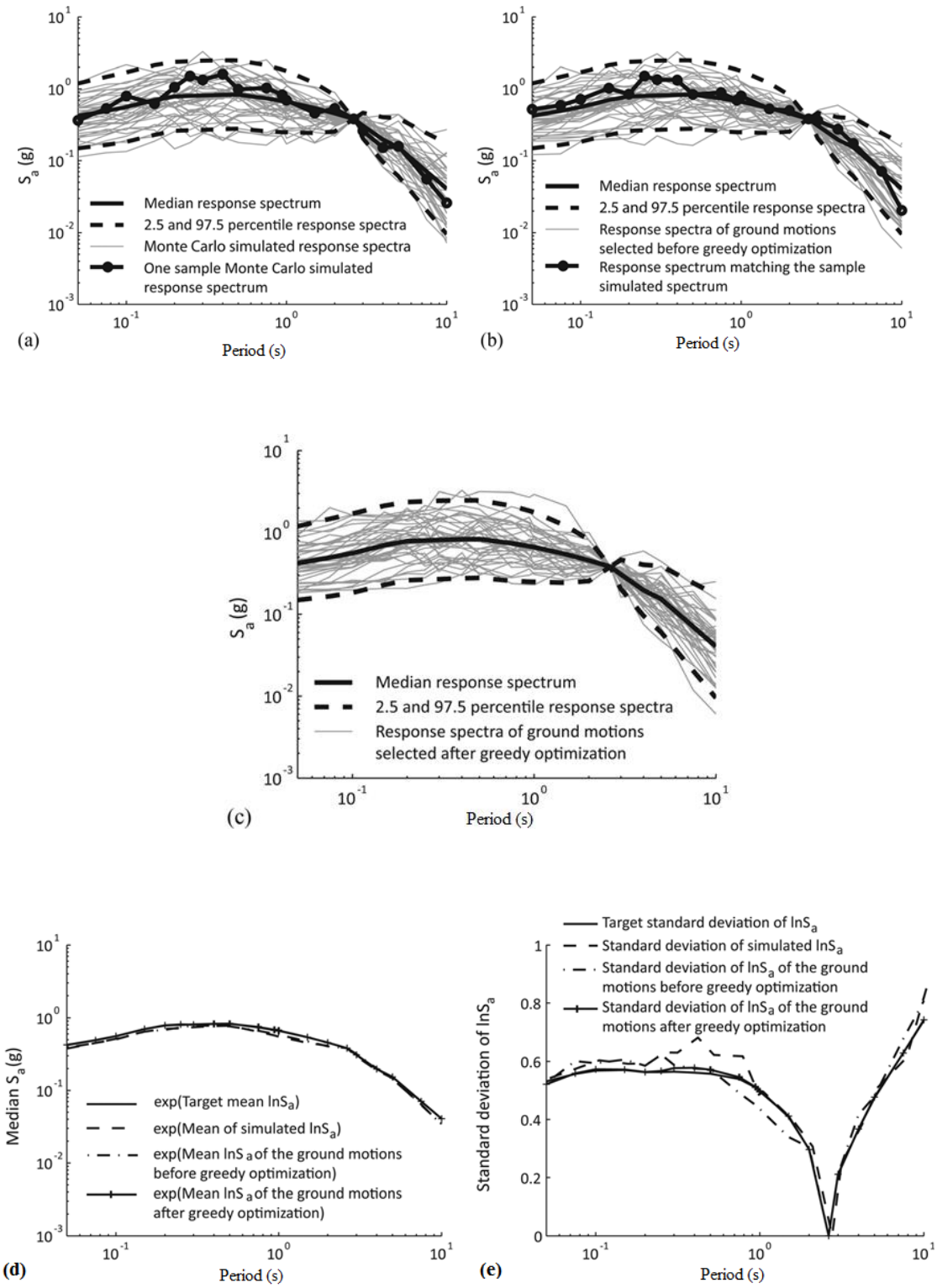


Figure 5.34: Response spectra for 40 selected ground motions used in Jayaram et al. (2011): (a) simulated, (b) before greedy optimisation, (c) after greedy optimisation, (d) response spectrum mean and (e) response spectrum standard deviation.

5.2.4 EC8 building code selection and scaling method

Considering that there is no specific seismic code relevant to the study region and that the draft version of the Iraqi seismic building code has not been updated for more than 15 years, the design spectra proposed by the EC8 building code are used in this study for matching and selecting earthquake records. Furthermore, for the purpose of comparison, this code is used to compare the results obtained using the proposed response spectra of the KR.

Two standard design response spectra are proposed in the EC8 building code. Type 1 spectra are suggested for high-seismicity regions with a surface-wave magnitude (M_S) greater than 5.5. Moreover, type-2 spectra are proposed for low- and moderate-seismicity areas with M_S less than or equal to 5.5. Because the estimated maximum earthquake magnitude in the Kurdistan region exceeds 5.5, type-1 spectra are used in the present study based on the parameters in Table 5.12 and according to the EC8 code. The spectral shape for the type-1 spectrum in the EC8 code is as follows:

$$\begin{cases} 0.0 \le T_x \le T_B : S_e(T_x) = a_g S_s \left[1 + \frac{T_x}{T_B}(\eta 2.5 - 1)\right] \\ T_B \le T_x \le T_C : S_e(T_x) = a_g S_s \eta 2.5 \\ T_C \le T_x \le T_D : S_e(T_x) = a_g S_s \eta 2.5 \left[\frac{T_C}{T_x}\right] \\ T_D \le T_x \le 4sec. : S_e(T_x) = a_g S_s \eta 2.5 \left[\frac{T_C T_D}{{T_x}^2}\right] \end{cases} , \qquad \text{Equation 5.10}$$

where a_g is the design ground acceleration on type-A ground; S_s is the soil factor; T_B, T_C, and T_D are the lowest periods of the constant acceleration, velocity and displacement spectral portion, respectively; η is the damping correction factor, which is equal to 1 for 5% viscous damping; $S_e(T_x)$ is the elastic response spectrum; and T_x is the SDOF vibration period. The effects of local site conditions are taken into account by considering five ground types (see Table 5.2a).

Table 5.12: Parameters selected for constructing elastic design spectra relevant to the seismic hazards in zones A and B according to the EC8 building code.

Seismic Hazard Zone	a_g (g)	Soil Type	S_s	T_B (s)	T_C (s)	T_D (s)
A	0.5	B	1.2	0.15	0.5	2
B	0.4	C	1.15	0.1	0.6	2

5.3 Selected Ground-motion Data Set

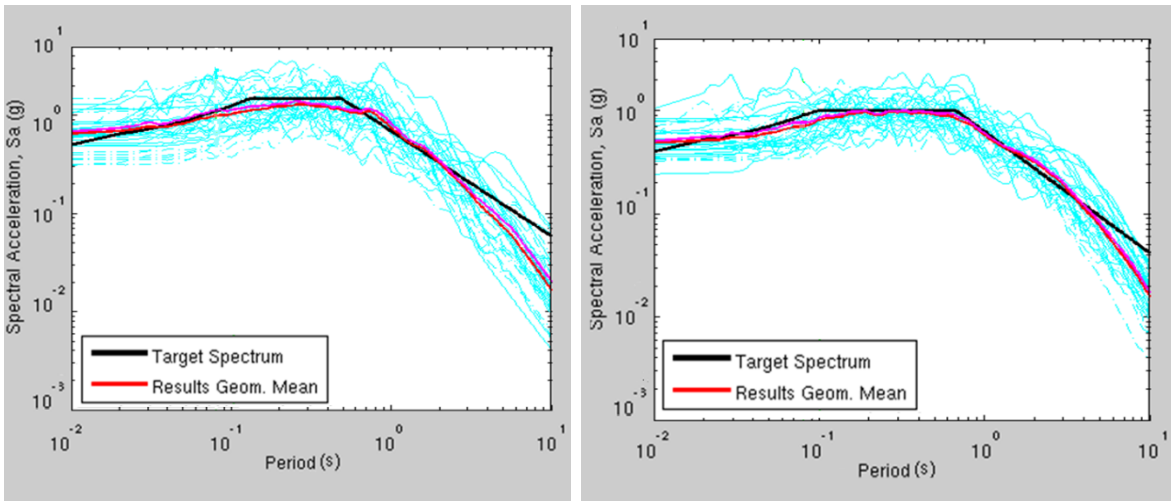
During this stage, selecting suitable ground motions for the analysis of the selected buildings in the Kurdistan region using the aforementioned methods becomes a priority. To this end, the web-based PEER ground-motion database (http://peer.berkeley.edu/peer_ground_motion_database/) is used. This database allows engineers to search, select, and scale records to match a target response spectrum based on specific requirements and search criteria. Therefore, target response spectra of real earthquakes were selected to comply with the EC8 building code, the CMS, and the algorithm proposed by Jayaram et al. (2011) and to match the proposed response spectra for seismic hazard zones A and B. As a result, a database of 60 real earthquake records compatible with the seismic characteristics of the KR was selected. This database was organised by combining records obtained from seismic hazard zones A and B after selecting four sets of fifteen records, where each record was matched with the defined target spectra and seismic characteristics relevant to the corresponding seismic zone (see Tables 5.13 and 5.14). Figures 5.35 and 5.36 show the matched response spectra for the selected ground motions and their geometric mean to the target response spectrum proposed in this study and to other spectra defined according to the EC8 code for the proposed seismic hazard zones A and B, respectively.

Table 5.13: Selected sets of fifteen records for each GMSM method used in this study for zone A.

Proposed Response Spectrum					EC8 Response Spectrum				
NGA Record No.	Earthquake Name	Magnitude	EpiD(km)	PGA(g)	NGA Record No.	Earthquake Name	Magnitude	EpiD(km)	PGA(g)
126	Gazli, USSR	6.80	12.82	0.60	126	Gazli, USSR	6.80	12.82	0.60
143	Tabas, Iran	7.35	55.24	0.84	143	Tabas, Iran	7.35	55.24	0.84
568	San Salvador	5.80	7.93	0.88	568	San Salvador	5.80	7.93	0.88
779	Loma Prieta	6.93	18.46	0.97	779	Loma Prieta	6.93	18.46	0.97
825	Cape Mendocino	7.01	10.36	1.50	825	Cape Mendocino	7.01	10.36	1.50
828	Cape Mendocino	7.01	4.51	0.59	828	Cape Mendocino	7.01	4.51	0.59
879	Landers	7.28	44.02	0.72	963	Northridge-01	6.69	40.68	0.57
983	Northridge-01	6.69	13.00	0.57	983	Northridge-01	6.69	13.00	0.57
1086	Northridge-01	6.69	16.77	0.60	1004	Northridge-01	6.69	8.48	0.75
1111	Kobe, Japan	6.90	8.70	0.51	1085	Northridge-01	6.69	13.60	0.83
1197	Chi-Chi, Taiwan	7.62	32.67	0.65	1086	Northridge-01	6.69	16.77	0.60
1504	Chi-Chi, Taiwan	7.62	28.70	0.50	1231	Chi-Chi, Taiwan	7.62	31.65	0.97
1508	Chi-Chi, Taiwan	7.62	21.42	0.49	1507	Chi-Chi, Taiwan	7.62	15.42	0.57
1509	Chi-Chi, Taiwan	7.62	19.08	0.60	1508	Chi-Chi, Taiwan	7.62	21.42	0.49
1517	Chi-Chi, Taiwan	7.62	8.90	1.16	1517	Chi-Chi, Taiwan	7.62	8.90	1.16
Conditional Mean Spectrum					Jayaram et al. Algorithm				
NGA Record No.	Earthquake Name	Magnitude	EpiD(km)	PGA(g)	NGA Record No.	Earthquake Name	Magnitude	EpiD(km)	PGA(g)
126	Gazli, USSR	6.80	12.82	0.60	139	Tabas, Iran	7.35	20.63	0.33
802	Loma Prieta	6.93	27.23	0.51	143	Tabas, Iran	7.35	55.24	0.84
825	Cape Mendocino	7.01	10.36	1.50	178	Imperial Valley-06	6.53	28.65	0.27
828	Cape Mendocino	7.01	4.51	0.59	180	Imperial Valley-06	6.53	27.80	0.52
963	Northridge-01	6.69	40.68	0.57	183	Imperial Valley-06	6.53	28.09	0.60
983	Northridge-01	6.69	13.00	0.57	779	Loma Prieta	6.93	18.46	0.97
1004	Northridge-01	6.69	8.48	0.75	879	Landers	7.28	44.02	0.72
1085	Northridge-01	6.69	13.60	0.83	1231	Chi-Chi, Taiwan	7.62	31.65	0.97
1086	Northridge-01	6.69	16.77	0.60	1402	Chi-Chi, Taiwan	7.62	88.84	0.31
1197	Chi-Chi, Taiwan	7.62	32.67	0.65	1503	Chi-Chi, Taiwan	7.62	26.70	0.81
1507	Chi-Chi, Taiwan	7.62	15.42	0.57	1505	Chi-Chi, Taiwan	7.62	47.86	0.57
1508	Chi-Chi, Taiwan	7.62	21.42	0.49	1524	Chi-Chi, Taiwan	7.62	95.70	0.38
1511	Chi-Chi, Taiwan	7.62	16.00	0.30	1549	Chi-Chi, Taiwan	7.62	14.16	1.00
1512	Chi-Chi, Taiwan	7.62	5.00	0.44	1596	Chi-Chi, Taiwan	7.62	14.16	0.96
1513	Chi-Chi, Taiwan	7.62	7.60	0.74	1633	Manjil, Iran	7.37	40.43	0.51

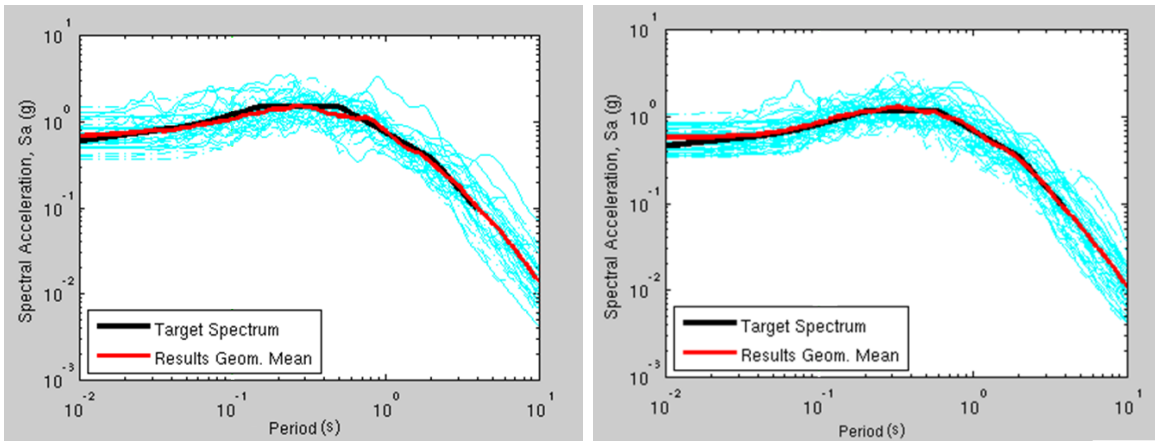
Table 5.14: Selected sets of fifteen records for each GMSM method used in this study for zone B.

Proposed Response spectrum					EC8 Response Spectrum				
NGA Record No.	Earthquake Name	Magnitude	EpiD(km)	PGA(g)	NGA Record No.	Earthquake Name	Magnitude	EpiD(km)	PGA(g)
126	Gazli, USSR	6.80	12.82	0.60	126	Gazli, USSR	6.80	12.82	0.60
169	Imperial Valley-06	6.53	33.73	0.24	184	Imperial Valley-06	6.53	27.23	0.35
179	Imperial Valley-06	6.53	27.13	0.36	568	San Salvador	5.80	7.93	0.88
180	Imperial Valley-06	6.53	27.80	0.52	828	Cape Mendocino	7.01	4.51	0.59
182	Imperial Valley-06	6.53	27.64	0.46	953	Northridge-01	6.69	13.39	0.42
183	Imperial Valley-06	6.53	28.09	0.60	963	Northridge-01	6.69	40.68	0.57
723	Superstition Hills-02	6.54	15.99	0.46	983	Northridge-01	6.69	13.00	0.57
779	Loma Prieta	6.93	18.46	0.97	1013	Northridge-01	6.69	11.79	0.51
802	Loma Prieta	6.93	27.23	0.51	1044	Northridge-01	6.69	20.27	0.58
821	Erzican, Turkey	6.69	8.97	0.50	1048	Northridge-01	6.69	3.42	0.37
828	Cape Mendocino	7.01	4.51	0.59	1085	Northridge-01	6.69	13.60	0.83
959	Northridge-01	6.69	4.85	0.36	1086	Northridge-01	6.69	16.77	0.60
1048	Northridge-01	6.69	3.42	0.37	1119	Kobe, Japan	6.90	38.60	0.69
1086	Northridge-01	6.69	16.77	0.60	1602	Duzce, Turkey	7.14	41.27	0.73
1602	Duzce, Turkey	7.14	41.27	0.73	1605	Duzce, Turkey	7.14	1.61	0.35
Conditional Mean Spectrum					Jayaram et al. Algorithm				
NGA Record No.	Earthquake Name	Magnitude	EpiD(km)	PGA(g)	NGA Record No.	Earthquake Name	Magnitude	EpiD(km)	PGA(g)
126	Gazli, USSR	6.80	12.82	0.60	174	Imperial Valley-06	6.53	29.44	0.36
169	Imperial Valley-06	6.53	33.73	0.24	179	Imperial Valley-06	6.53	27.13	0.36
179	Imperial Valley-06	6.53	27.13	0.36	729	Superstition Hills-02	6.54	29.41	0.18
180	Imperial Valley-06	6.53	27.80	0.52	802	Loma Prieta	6.93	27.23	0.51
181	Imperial Valley-06	6.53	27.47	0.41	821	Erzican, Turkey	6.69	8.97	0.50
182	Imperial Valley-06	6.53	27.64	0.46	1086	Northridge-01	6.69	16.77	0.60
183	Imperial Valley-06	6.53	28.09	0.60	1148	Kocaeli, Turkey	7.51	53.68	0.22
184	Imperial Valley-06	6.53	27.23	0.35	1244	Chi-Chi, Taiwan	7.62	31.96	0.35
802	Loma Prieta	6.93	27.23	0.51	1485	Chi-Chi, Taiwan	7.62	77.50	0.47
803	Loma Prieta	6.93	27.05	0.25	1489	Chi-Chi, Taiwan	7.62	38.91	0.29
821	Erzican, Turkey	6.69	8.97	0.50	1492	Chi-Chi, Taiwan	7.62	39.58	0.35
879	Landers	7.28	44.02	0.72	1505	Chi-Chi, Taiwan	7.62	47.86	0.57
983	Northridge-01	6.69	13.00	0.57	1508	Chi-Chi, Taiwan	7.62	21.42	0.49
1085	Northridge-01	6.69	13.60	0.83	1510	Chi-Chi, Taiwan	7.62	20.67	0.33
1602	Duzce, Turkey	7.14	41.27	0.73	1605	Duzce, Turkey	7.14	1.61	0.35



(a) (b)

Figure 5.35: Response spectra for the selected ground motions matching the target spectrum proposed for the Kurdistan region for seismic hazard zones A (a) and B (b).



(a) (b)

Figure 5.36: Response spectra for the selected ground motions matching the target spectrum defined according to the EC8 code for seismic hazard zones A (a) and B (b).

For the purpose of comparison and achieving this study's objectives, other sets of records with different numbers of records in each set must be utilised. A combination of records from four sets of fifteen records results in a set of 35 records for zone A and another set of 38 records for zone B. Additionally, two sets of seven records can also be obtained if the procedure discussed in section 3.2 is applied to each zone (A and B). For that purpose, the most frequent records among the four sets of fifteen records

corresponding to each zone should be selected; subsequently, a set of seven earthquake records for each seismic hazard zone can be successfully obtained.

Based on the aforementioned procedure (in section 3.2), the records that are repeated between two and four times were identified (see Table 5.15) to select a set of seven earthquake records for zone B. Because no records that are repeated four times were identified, the selection should start from records that are repeated three times. While six records from six different events are available, only one more record that is repeated twice is then required. Among the ten available records that satisfy this condition, the record that differs from the six events selected in the first step is taken. Thus, seven records from seven different earthquake events that are representative of the seismic properties of zone B were collected, as shown in Table 5.16.

Table 5.15: Number of repeated records among all GMSM methods used in this study for zone B.

Repeated three times		Repeated two times	
NGA Record No.	Earthquake Name	NGA Record No.	Earthquake Name
126	Gazli, USSR	169	Imperial Valley-06
179	Imperial Valley-06	180	Imperial Valley-06
802	Loma Prieta	182	Imperial Valley-06
821	Erzican, Turkey	183	Imperial Valley-06
1086	Northridge-01	184	Imperial Valley-06
1602	Duzce, Turkey	828	Cape Mendocino
		983	Northridge-01
		1048	Northridge-01
		1085	Northridge-02
		1605	Duzce, Turkey

Table 5.16: Selected set of seven records in zone B.

No.	NGA Record No.	Earthquake Name	PGA (g)	Magnitude	EpiD (km)
1	126	Gazli, USSR	0.60	6.8	12.8
2	179	Imperial Valley-06	0.36	6.5	27.1
3	802	Loma Prieta	0.51	6.9	27.2
4	821	Erzican, Turkey	0.49	6.7	9.0
5	828	Cape Mendocino	0.59	7.0	4.5
6	1086	Northridge-01	0.60	6.7	16.8
7	1602	Duzce, Turkey	0.72	7.1	41.3

An equivalent procedure is applied to zone A. Table 5.17 provides detailed information on six events with eight records for a group of records that are repeated three times; four records are associated with four different events, and the remaining four records are correlated with two other events. During this step, one may also select six records: four records corresponding to the four non-repeating events and two records corresponding to the two remaining repeated events. Based on the PGA or EpiD values of the first four selected records, two other records can be selected. At the end of this step, six records are found to exhibit commonality; thus, an extra record is required to complete the set. The remaining record is selected from a group of records that are repeated two times. One of the events that was not previously selected can be obtained similarly to what was undertaken for zone B; alternatively, both records can be selected to produce two sets, each with seven records (Table 5.18).

At this point, several sets of earthquake records are identified, including sets of 35/38, 15(per each GMSM method used in the study), and 7 records for each zone and sets of 60, 25 (per each GMSM methods used in the study) and 12 records derived from a combination of records from both zones.

Table 5.17: Number of repeated records among all GMSM methods used in this study for zone A.

Repeated three times		Repeated two times	
NGA Record No.	Earthquake Name	NGA Record No.	Earthquake Name
126	Gazli, USSR	568	San Salvador
143	Tabas, Iran	879	Landers
779	Loma Prieta	963	Northridge-01
825	Cape Mendocino	1004	Northridge-01
828	Cape Mendocino	1085	Northridge-01
983	Northridge-01	1197	Chi-Chi, Taiwan
1086	Northridge-01	1231	Chi-Chi, Taiwan
1508	Chi-Chi, Taiwan	1507	Chi-Chi, Taiwan
		1517	Chi-Chi, Taiwan

Table 5.18: Selected sets of seven records for zone A: (a) option 1 and (b) option 2.

(a) Option 1

No.	NGA Record No.	Earthquake	PGA (g)	Mag.	EpiD (km)
1	126	Gazli, USSR	0.6	6.8	12.8
2	143	Tabas, Iran	0.8	7.4	55.2
3	568	San Salvador	0.9	5.8	7.9
4	779	Loma Prieta	1.0	6.9	18.5
5	828	Cape Mendocino	0.6	7.0	4.5
6	983	Northridge-01	0.6	6.7	13.0
7	1508	Chi-Chi, Taiwan	0.5	7.6	21.4

(b) Option 2

No.	NGA record no.	Earthquake	PGA (g)	Mag.	EpiD (km)
1	126	Gazli, USSR	0.6	6.8	12.8
2	143	Tabas, Iran	0.8	7.4	55.2
3	779	Loma Prieta	1.0	6.9	18.5
4	828	Cape Mendocino	0.6	7.0	4.5
5	879	Landers	0.7	7.3	44.0
6	983	Northridge-01	0.6	6.7	13.0
7	1508	Chi-Chi, Taiwan	0.5	7.6	21.4

CHAPTER 6. TREMURI CODE FOR ANALYSIS OF UNREINFORCED MASONRY BUILDINGS

URM is considered to be a brittle material; thus, when the stress state within a wall (which is caused by horizontal loads rather than gravity loads) exceeds the masonry strength, brittle failure usually occurs. Furthermore, masonry is known to poorly resist shear and bending forces. As a result, URM walls are vulnerable to earthquakes and are usually subject to collapse during seismic actions (Virdi & Rashkoff, 2014). Therefore, this type of structure requires a careful design before construction proceeds and an accurate assessment of existing structures to increase their strength.

Modelling of existing masonry structures under seismic loads is not an easy task because of their geometrical non-linear behaviours when subjected to strong ground motions. However, as noted in section 3.3.2, non-linear dynamic analysis and, more specifically, IDA is used in this study to consider the nonlinear behaviour of URM buildings and to provide a more thorough estimation of their performance under seismic loads for reducing the uncertainty related to the simulation and analysis methods. Nonlinear dynamic analysis requires the use of many time history simulations when considering the nonlinear behaviour of masonry as opposed to a pushover curve approach, which is a nonlinear static analysis technique. Thus, the TREMURI software (Lagomarsino et al., 2008), which enables faster dynamic computations compared to other more detailed modelling software products, is useful for studies attempting to reduce computational costs.

Over the last two decades, analytical methods (more specifically, finite element analyses) used to model the seismic response of masonry structures have made significant progress at the procedural level (e.g., Calderini & Lagomarsino 2008; Brasile et al., 2007; Lourenço, 2002; Milani et al., 2006). However, a proper understanding of the range of validity within each model and the requirements of expert judgment when using such tools are still viewed as the main issues facing these analytical approaches. Additionally, in many cases, these methods can be applied only to problematic areas that are limited in size. Therefore, several methods have been developed that require a small to moderate computational burden and provide accurate outcomes. Among the possible modelling strategies proposed in the literature, the equivalent frame model (Magenes, 2000; Roca et al., 2005) has been implemented in the TREMURI computer program using the macro-

element method. This model is more attractive, and its pragmatic framework has been highlighted in various seismic codes (e.g., NTC, 2008, and EC8).

The equivalent frame approach provides a comprehensive analysis of complete 3D buildings, mainly referring to their in-plane strength and stiffness contributions, with a reasonable computational effort that requires a limited number of degrees of freedom. Moreover, by introducing other structural elements, such as reinforced concrete beams or columns, combined with masonry elements, a structure can be readily idealised as an equivalent frame, especially for modelling mixed structures. Thus, this approach appears to be more suitable and beneficial for practical engineering purposes. According to the equivalent frame approach, resistant masonry walls are subdivided into deformable masonry panels that are subjected to deformation and nonlinear responses and to rigid areas that connect these deformable panels. The reliable prediction of the overall behaviour of a masonry wall after idealising it as a set of structural elements is consequently achieved via the appropriate representation of the properties of each structural member (Lagomarsino et al., 2013).

The TREMURI software, which is a nonlinear macro-element model initially proposed by Gambarotta and Lagomarsino (1996) and later improved by Penna (2002) and Galasco et al. (2006), is used to discretise a masonry wall into three main components, including vertical elements, which are called ”piers”; horizontal ones, which are called ”spandrels”; and rigid undamageable zones (see Figure 6.1). Rigid nodes/elements link piers and spandrels together, while spandrels join a wall’s piers in one plane. Piers tend to resist both dead and horizontal forces, and spandrels couple piers in the case of seismic loads (Galasco et al., 2004; Roca et al., 2005). This masonry wall discretisation results from post-earthquake observations of URM structures and the damaged caused by earthquake activities.

According to Holmes et al. (1990), Lizundia et al. (1993), and Bruneau (1994), the three main damage patterns observed in URM buildings can be classified as follows: in-plane, out-of-plane, and system-level damage patterns. Walls that are parallel and perpendicular to the direction of shaking are typically vulnerable to in-plane and out-of-plane damage, respectively, whereas system-level damage is caused by a combination of the other two damage patterns (i.e., in-plane or out-of-plane damage patterns).

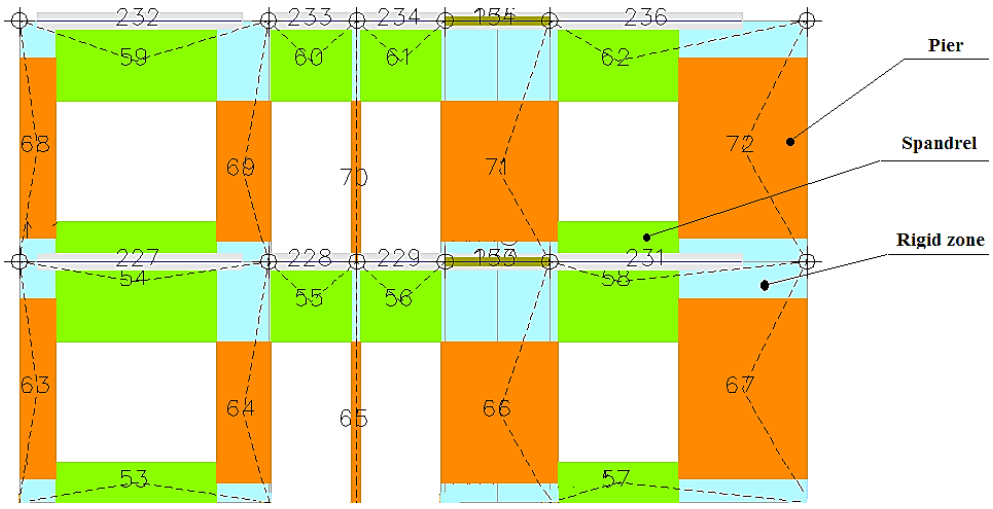


Figure 6.1: Macro-element modelling of a masonry panel used in the TREMURI software.

Masonry walls resisting in-plane loads tend to exhibit the following three failure modes (Figure 6.2): 1) sliding failure: experienced by a wall with an aspect ratio (height/width) of 1:1 (usually) or less (1:1.5), low axial load, weak mortar, and poor shear strength that is subjected to reversed seismic actions; 2) diagonal tension/shear failure (most common): experienced by a wall with an aspect ratio of 1:1 (usually), high axial load and strong mortar; the tensile stresses (as a result of vertical and horizontal loads) exceed the tensile strength of the masonry; 3) bending failure or rocking (more ductile and more preferable mechanism): experienced by a wall with improved shear resistance or a wall with a larger aspect ratio (i.e., 2:1), including small vertical loads and high shear resistance (Erbay, 2007; Virdi & Rashkoff, 2014; Magenes & Calvi, 1997; Turnsek & Cacovic, 1971).

The out-of-plane mechanism is primary caused by out-of-plane actions such as displacements and accelerations. This type of damage is commonly observed at the floor or roof levels or at the mid-height of walls that are perpendicular to the direction of shaking (Erbay, 2007), which is depicted in Figure 6.3. Erbay (2007) noted that the different combinations of the aforementioned damage patterns (in-plane or out-of-plane failure modes) can be a source of another type of damage: system-level damage. Soft storey, corner damage, non-structural failures, and roof/floor collapse are typical forms of this damage pattern. For example, piers of a storey losing their stiffness due to in-plane deformations or rocking result in a weakness in the vertical direction, resulting in building collapse, as shown in Figure 6.4.

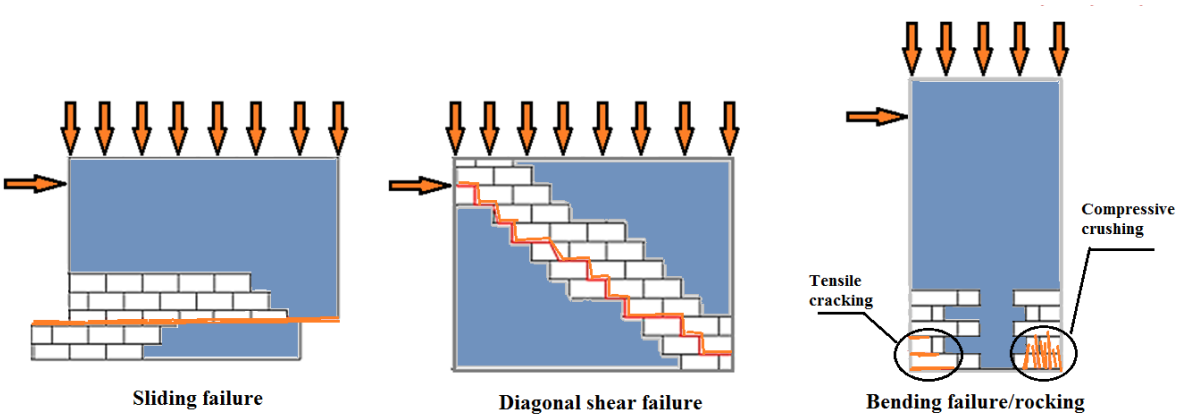


Figure 6.2: In-plane damage patterns.

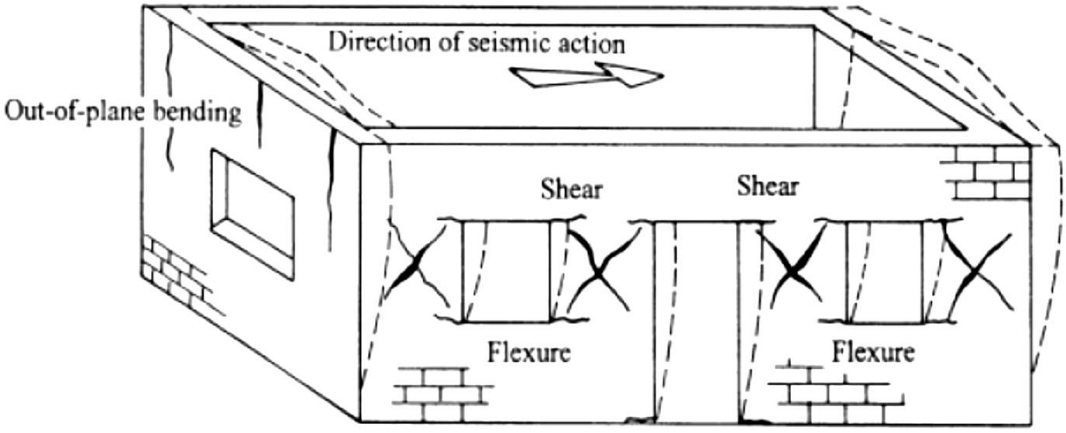


Figure 6.3: Behaviour of a "box" masonry structure showing out-of-plane and in-plane failure patterns (from Tomazevic, 1999).

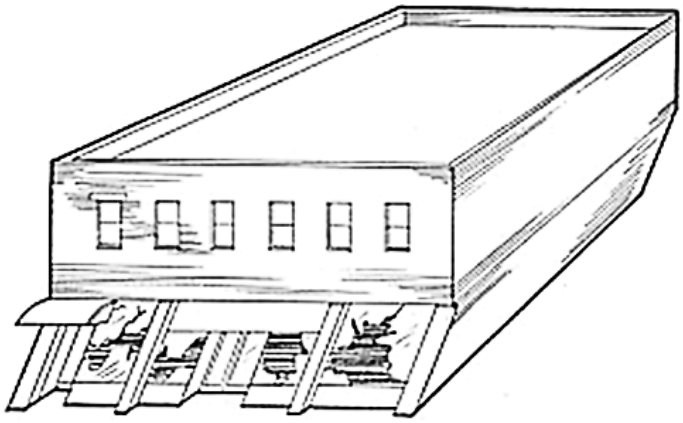


Figure 6.4: Soft storey failure (from Holmes et al., 1990).

In the TREMURI code, only the global building response in terms of a wall's in-plane behaviour is considered. The assumption is that the structure exhibits the behaviour of a typical "box" structure (Figure 6.3) when the slab is rigid. As a result, the lateral forces (seismic loads) are distributed between the walls based on their stiffness properties. Thus, a global seismic analysis can be conducted based on the response of the entire structure, which is governed by the lateral strength and deformation capacities of the corresponding macro-elements. In addition, an equivalent homogeneous material is assumed as a part of these macro-elements, while the flexural strength of these macro-elements is determined by assuming that no tensile resistance occurs for very low tensile strengths (Lagomarsino et al., 2008).

The TREMURI program's technical ability to describe the seismic response of masonry buildings is verified through a comparison between real (experimental) and numerically simulated damage patterns (simulated using a finite element model and the TREMURI program), which was undertaken by Lagomarsino et al. (2013), who focused on several URM buildings. Likewise, in the study by Ademović (2011), a detailed assessment of typical five-storey masonry buildings constructed between the 1950s and the 1970s across the entire ex-Yugoslavia region was performed using numerical modelling with nonlinear analyses (pushover and time history analyses) in two software programs, i.e., DIANA, which is based on the finite element method, and 3MURI, which is based on macro-elements. The use of the 3MURI program has been recommended for the future analysis of these structures because of its ability to obtain representative and reliable results with less computation time.

6.1 Analytical Modelling

In this section, the theory of modelling masonry structures using a macro-element model is discussed; the theory has largely been summarised from Penna (2002), Galasco et al. (2004) and Lagomarsino et al. (2008).

As noted above, it is practically possible to represent the two main in-plane damage mechanisms of a masonry panel, i.e., shear (sliding or diagonal) and bending failures, by implementing a non-linear macro-element model to represent a complete masonry model. The average properties of a masonry panel are given in terms of a direct correlation between the mechanical properties of the masonry elements and the parameters defined for

a continuous model in a macro scale. A macro-element model is an example of a continuous model that is composed of three substructures (shown in Figure 6.5): a central part 2, where shear deformations are observed, and two layers (inferior 1 and superior 3), where bending and axial effects are concentrated. Considering three degrees of freedom for each node 'i' and 'j' (i.e., axial displacement *w*, horizontal displacement *u* and rotation φ), and two degrees of freedom for the central node (axial displacement δ and rotation ϕ), the complete 2D kinematic model for a macro-element can be characterised by 8 degrees of freedom, as shown in Figure 6.5.

By assuming zero thickness (Δ=0) for the inferior and superior layers, as shown in Figure 6.5, the bending flexibility is assigned to the ends; the shear flexibility is only considered for the central fragment; and the axial flexibility is considered in all parts of the macro-element (Penna, 2002). For the adjacent piers, a compatibility condition at the edges is assumed, in which the same vertical displacement can be achieved by connecting the macro-element nodes directly to the floor elements (thus representing a connection between the floors and walls).

The main assumptions used in developing the non-linear constitutive equations for the macro-elements are based on the concept that extreme parts lack tensile strength, and sliding shear resulting in damage occurs in the central part. Therefore, the axial and flexural responses are uncoupled from the shear response. Using a mono-lateral elastic contact between interfaces 1 and 3, the overturning mechanism (because the materials do not have a tensile strength) can be formally modelled.

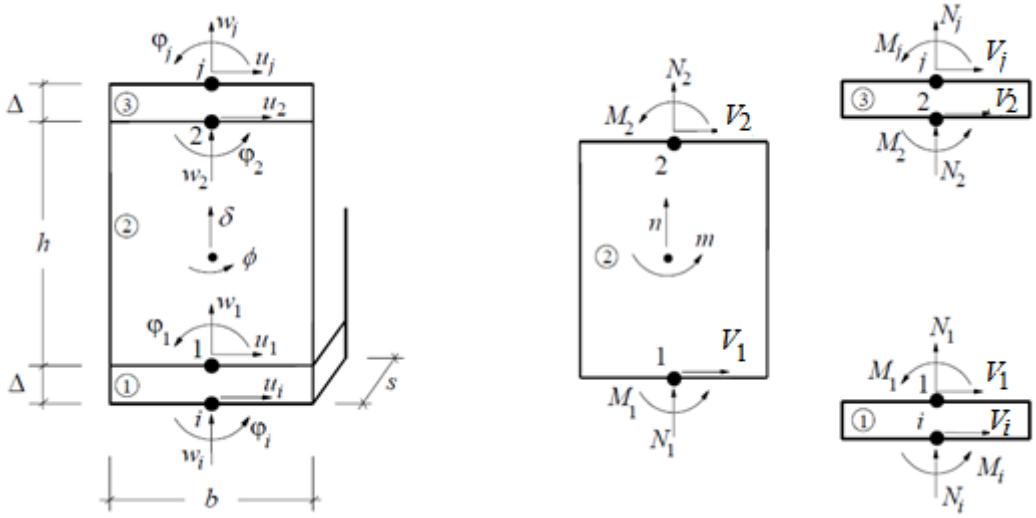


Figure 6.5: Kinematic and static parameters for the macro-element (TREMURI).

If a panel's cross section is completely compressed, then the constitutive equations used for establishing a relationship between the kinematic and static parameters in extreme parts of the macro-element are uncoupled as a result of the application of the axial force to the central core of inertia. In other words, the axial force and the bending moment can be expressed in terms of the following equations. When subject to an axial force (N), the rotation increases linearly with the bending moment (M) until the static and kinematic conditions for this stage ($\left|\frac{M}{N}\right| \leq \frac{b}{6}$) are attained:

$$N = \frac{2\,E}{h}\, b\, s\, w\,, \qquad \text{Equation 6.1}$$

$$M = \frac{E}{6\,h}\, b^3\, s\, \varphi, \qquad \text{Equation 6.2}$$

and

$$\left|\frac{M}{N}\right| = \frac{\left|\frac{E}{6\,h} b^3\, s\, \varphi\right|}{\left|\frac{2\,E}{h} b\, s\, w\right|} = \frac{\frac{E}{6\,h} b^3\, s\, |\varphi|}{\frac{2\,E}{h} b\, s\, |w|} = \frac{b^2\, |\varphi|}{12\, |w|} \leq \frac{b}{6}. \qquad \text{Equation 6.3}$$

Considering the kinematics and no tensile resistance for masonry in bending (i.e., $w < 0$), we find

$$|\varphi| \leq -\frac{2\,|w|}{b}\,. \qquad \text{Equation 6.4}$$

The cross-section will crack if these conditions are not met. Due to the assumption of small displacements, the effect of eccentric compression as inelastic contributions of the internal forces (i.e., M* and N*) can be calculated using the following equations:

$$N = \frac{2\,E}{h}\, b\, s\, w + N^* \qquad \text{Equation 6.5}$$

and

$$M = \frac{E}{6\,h}\, b^3\, s\, \varphi + M^*\,, \qquad \text{Equation 6.6}$$

where

$$N^* = \frac{2\,E\,s}{h}\left[b\, w - \frac{(|\varphi|\, b + 2\, w)^2}{8\,|\varphi|}\right] \qquad \text{Equation 6.7}$$

and

$$M^* = \frac{E\,s}{6\,h}\left[b^3|\varphi| + \frac{(-|\varphi|\, b + w)}{2\,\varphi\,|\varphi|}(|\varphi|\, b + 2\, w)^2\right] \qquad \text{Equation 6.8}$$

To model a panel's shear response, a uniform shear deformation distribution is considered in the central part (2), while a relationship between the kinematic quantities *u* and ϕ, including the shear strength *V*, is imposed. When the Coulomb limit friction

condition is achieved, cracking damage usually begins along a diagonal direction. This diagonal is where displacement occurs along the joints, and an inelastic deformation component is used for its visual representation:

$$V = \frac{G\ b\ s}{h}\ \ (u + \emptyset\ h) + V^{*} \qquad \text{Equation 6.9}$$

and

$$V^{*} = \frac{-\ G\ b\ s\ c\ \alpha}{(1+c\ \alpha)\ h}\left(u + \emptyset\ h + \frac{h\ f}{G\ bs}\right), \qquad \text{Equation 6.10}$$

where V^* is the inelastic component, G is the shear modulus, f is the global friction coefficient, c is the non-dimensional coefficient that controls the inelastic deformation, and α is a damage parameter to describe the damage effects upon panel's mechanical characteristics, that grows according to a pre-defined failure criteria:

$$\Phi_d = Y(S) - F(\alpha) \leq 0.0\ . \qquad \text{Equation 6.11}$$

Here, Y is the damage energy release rate, F is the resistance function, and $S = \{t\ n\ m\}^T$ is the internal stress vector. Assuming F is an increasing function of α to a critical value α_C=1 and decreases for higher values, the model can provide the stiffness degradation, the strength degradation and the pinching effect.

If the macro-element's shear force reaches the ultimate shear strength calculated by the Mohr-Coulomb criterion (Equation 6.12), then the shear sliding mechanism is achieved. However, if the shear force reaches the ultimate shear force defined by the Turnsek-Cacovic criterion (Equation 6.13), then the diagonal cracking is activated. The following equations represent the Mohr-Coulomb and the Turnsek-Cacovic criteria, respectively:

$$V_u = l't\ f_{vo} + \ \mu\ N, \qquad \text{Equation 6.12}$$

where t is the panel's thickness, l' is the actual width subjected to compression, f_{vo} is the shear resistance of the masonry without compression, μ is the Mohr-Coulomb friction coefficient, and N is the axial compression force on the panel.

$$V_u = l\ t\ \frac{1.5\ \tau_o}{b'}\sqrt{1 + \frac{N}{1.5\ \tau_o\ l\ t}}\ , \qquad \text{Equation 6.13}$$

where l is the panel's width, b' is a shape factor (height-to-width ratio), and τ_o is the shear strength at the first diagonal crack (tensile strength is reached), then diagonal cracking occurs.

To represent flexural failure, the following equation is used to calculate a masonry panel's ultimate bending moment:

$$M_u = \frac{l^2\, t\, \sigma_o}{2}\left(1 - \frac{\sigma_o}{0.85\, f'_m}\right) = \frac{N\, l}{2}\left(1 - \frac{N}{N_u}\right), \qquad \text{Equation 6.14}$$

where σ_o is a normal compressive stress over the entire area and f'_m is the average resistance of the masonry to compression.

The toe-crushing effect in TREMURI is modelled using a phenomenological non-linear constitutive law with actual stiffness deterioration in compression. The impact of this modelling on the cyclic vertical displacement-rotation interaction is represented in Figure 6.6 (Lagomarsino et al., 2008).

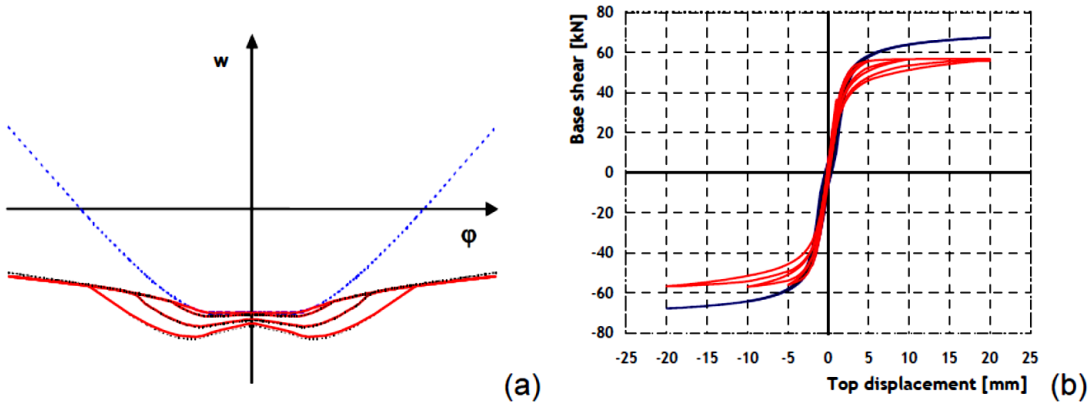


Figure 6.6: (a) Cyclic vertical displacement-rotation interaction with (red line) and without (blue dots) toe-crushing effects from Penna (2002); (b) rocking panel with (red line) and without (blue line) crushing effects (Lagomarsino et al., 2008).

To this end, the in-plane behaviour of a masonry panel can be modelled using macro-elements and forming a "frame" structure, which is shown in Figure 6.5. The structure is discretised into macro-elements by subdividing the walls into piers and spandrels that are linked via rigid nodes. Generally, piers and spandrels experience inelastic and damage mechanisms associated with both the opening of cracks and shear dissipative sliding. Notably, rigid portions experience only slight (if any) damage. A 3D model can be obtained by joining these in-plane masonry walls together. In that case, the

walls are viewed as bearing elements, while the floors distribute the horizontal load to the walls (a rigid floor is assumed by considering only the in-plane behaviour of the walls). Different walls are connected at the corners and intersections by nodes with five degrees of freedom (u_x, u_y, u_z, φ_x, and φ_y). The rotational degree of freedom around the vertical Z-axis, φ_z, can be disregarded because the membrane behaviour is adjusted between the walls and the floors (Lagomarsino et al., 2008).

6.2 Application of Incremental Dynamic Analysis

To perform IDA, scaled time histories of the ground motions selected in Chapter 5 must be applied to the selected buildings discussed in Chapter 4. The numerical model is developed using the TREMURI code, which was reviewed in the preceding section.

The exact geometry of a typical URM building is designed to represent typical one- and two-storey buildings in the study region. The mechanical properties, such as the compressive strength of masonry units and mortar, were either evaluated using physical tests performed in regional construction laboratories or selected from common features of masonry units (i.e., solid concrete blocks). In addition, minimum compression strength limits, according to IQ.S. No. 1077/1987 and IQ.S. No. 5/1984, were also considered for the solid concrete blocks and cement mortar (see section 4.2). Due to the lack of experimental data, the modulus of elasticity of the masonry material was assumed to be $E = 750f_k$, which is between the two recommended values provided by the EC6 (CEN 2005) and the FEMA 274 (1997) building codes, i.e., $E = 1000f_k$ and $E = 550f_k$, respectively. This value was also proposed by Paulay and Priestley (1992) as well as it ocures in the range between $200f_k$ and $2000f_k$ that was suggested by Tomazevic (1999). Here, f_k represents the characteristic compressive strength (in MPa) of the masonry and can be computed using the equation proposed by the EC6 building code:

$$f_k = k f_b^{0.7} f_m^{0.3}, \qquad \text{Equation 6.15}$$

where k is a constant value that ranges from 0.4 to 0.6 depending on the masonry and mortar types and f_b and f_m are the compressive strengths of the masonry units and mortar, respectively. The shear modulus G is 40% of the masonry modulus E.

The studied one-storey building is 3.0 m high with plan dimensions of 15 × 10 m, whereas the studied two-storey building is twice as high assuming a repeated plan

structure. The parapet and penthouse loads are transferred to the model with equivalent vertical loads on the roof level. After studying the effect of variability in the material properties on the structural response (details in section 7.1.1), the mean values of these material properties (Young's modulus E=4350 N/mm^2, shear modulus G=1740 N/mm^2, and specific weight 21k N/m^3) are used for the masonry units. A 3D view of the tested building is shown in Figure 6.7. The out-of-plane damage mechanism for the masonry walls is not interpreted because the structure's global behaviour is assumed to be governed by the in-plane failure mechanism of masonry panels.

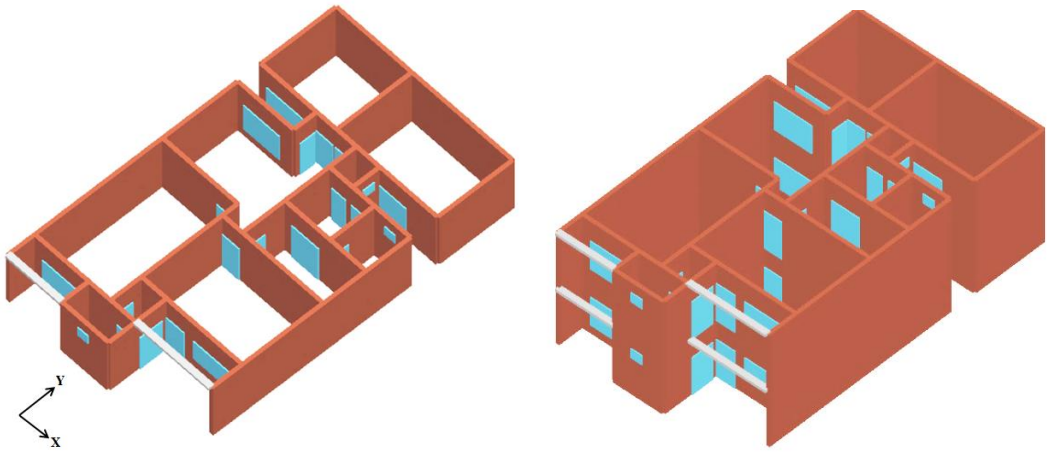


Figure 6.7: 3D view of the one- and two-storey buildings modelled in the TREMURI software.

To understand the dynamic behaviour of buildings under earthquake conditions, modal analysis is used to determine a building's vibrational characteristics (fundamental period and mode shapes) before performing IDA. A total of 90% of the effective mass required by CEN (2003) was eventually obtained by the first 3 modes of vibration, which is shown in Table 6.1. Evidently, the first mode is the prevailing mode of the response, where the largest effective mass contribution occurs in the x direction for both the one- and two-storey buildings. This is the "typical" box behaviour of a masonry structure with stiff connections between its walls and slab. First mode shape's periods (i.e., fundamental natural periods) for the one- and two-storey buildings were found to be 0.08 and 0.13 s, respectively.

Table 6.1: Cumulative percentage of the effective mass in the x and y directions.

Mode	Cumulative Percentage of the Effective Mass			
	One-storey building		Two-storey building	
	Mx [%]	My [%]	Mx [%]	My [%]
1	87.6	0.18	78.49	0.23
2	3.68	54.8	1.59	65.22
3	2.15	42.82	8.92	4.21

The next step is to perform a pushover analysis to determine a building's global strength and ultimate top displacement. A progressive lateral force is applied along the x direction according to the first mode shape. The ultimate displacement, as noted in previous studies (e.g., Lu et al., 2001, and Kaltakci et al., 2007), is obtained when the shear force decreases to 80% of its maximum value. For defining the yield point, which is required by the Milutinovic and Trendafiloski (2003) criterion for identifying damage states, an idealised bilinear curve of the force-deformation capacity curve is generated using the equal-energy criterion. Subsequently, the elastic limit (yield displacement) is reached, as shown in Figures 6.8 and 6.9 for the one- and two-storey buildings. Finally, using the criterion of Milutinovic and Trendafiloski (2003), the damage states can be computed (see Table 6.2).

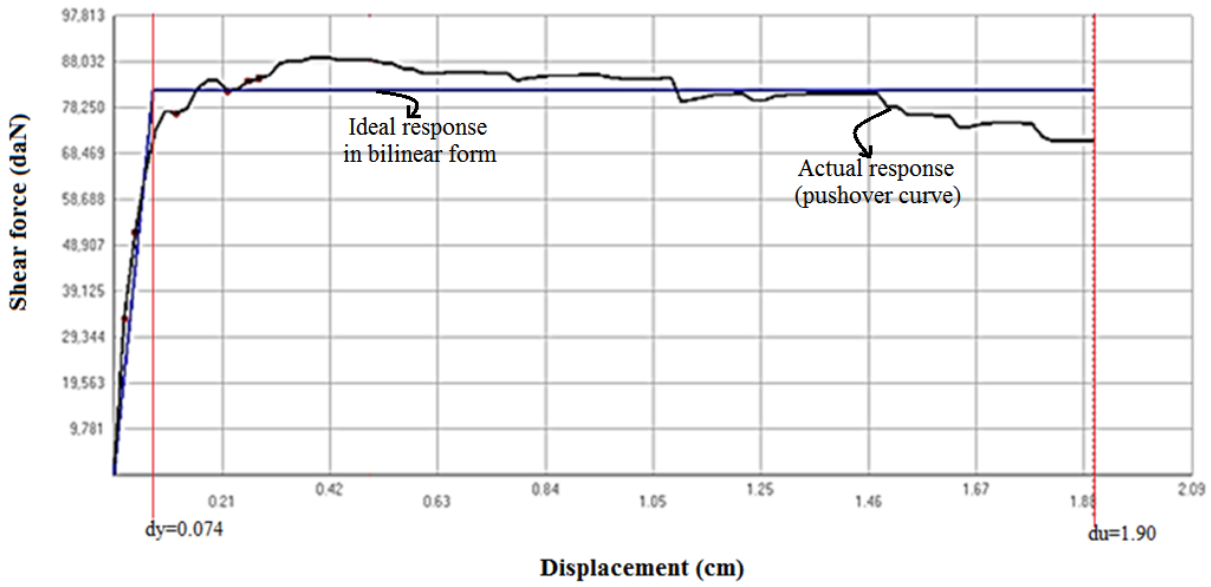


Figure 6.8: Pushover curve for the one-storey building.

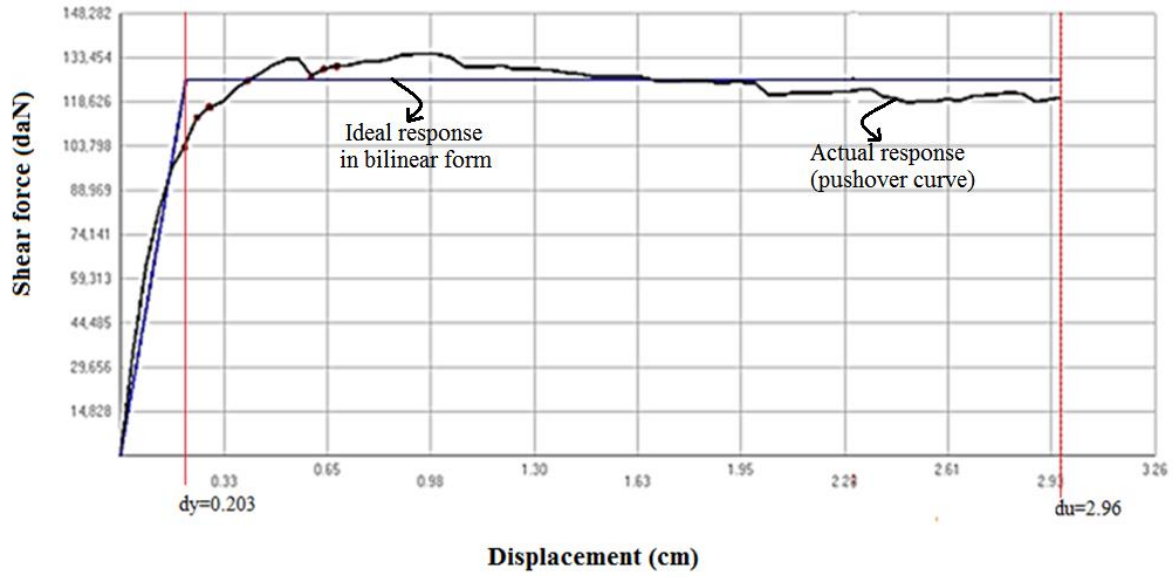


Figure 6.9: Pushover curve for the two-storey building.

Table 6.2: Performance levels for URM buildings proposed by Milutinovic and Trendafiloski (2003).

Damage state	Displacement limit equation	Limit for d (mm) One-storey building	Limit for d (mm) Two-storey building
Slight	$d = 0.7d_y$	0.51	1.42
Moderate	$d = 0.7d_y + 0.05(0.9d_u - 0.7d_y)$	1.34	2.55
Extensive	$d = 0.7d_y + 0.2(0.9d_u - 0.7d_y)$	3.83	5.93
Very heavy	$d = 0.7d_y + 0.5(0.9d_u - 0.7d_y)$	8.79	12.7
Collapse	$d = 0.9d_u$	17.1	26.64

To perform an accurate IDA, the structural model is subjected to several ground-motion time histories by increasing and scaling each time history to several intensity levels. To complete this task, the time history records selected for seismic hazard zones A and B are scaled to eight levels, i.e., 0.025g, 0.05g, 0.1g, 0.2g, 0.4g, 0.6g, 0.8g, and 1.0g of the PGA. In total, approximately 570 time histories are developed for both zones and subsequently applied to the base of the one- and two-storey buildings in the x direction. The maximum top displacement is obtained for each time history; thereafter, the corresponding damage state is identified. As a result, more than 1100 time history analyses are performed for both buildings. Using the criterion of Milutinovic and Trendafiloski (2003), the performance levels (damage states) for URM buildings can be classified into

five states: slight, moderate, extensive, very heavy, and collapse (see Table 6.2). However, the outcomes indicate that it is not beneficial or rational to draw a clear boundary between the very heavy and collapse levels; similarly, a boundary cannot be drawn between the slight and moderate damage states because these failure patterns are observed less frequently than are the other damage states. Thus, a boundary between the two states may produce palpable scatter in the final representation. As a result, the damage state limits are reduced to two levels instead of five, as proposed by Gehl et al. (2013b), by merging the damage states from 'slight' to 'extensive' into 'yield' and the damage states of 'very heavy' and 'collapse' into 'collapse'. The collapse state is a level for which building repair is not feasible, whereas a structure can still be used, assuming that suitable repairs are undertaken, in the yield state. Tables 6.3 and 6.4 present the results for the set of seven earthquake records in seismic hazard zones A and B, respectively. All other auxiliary results for different combinations of records can be found in the Appendix at the end of this study (see Table A1).

Furthermore, to examine the effect of variability in material properties on the structural response, an additional 1100 time history analyses are applied to the one-storey building in the x direction by referring to a fixed accelerogram scaled to three PGA intensities (0.05g, 0.2g, and 0.8g). Moreover, 60 analyses are also conducted for the unscaled time histories to determine the effects of scaling (i.e., IDA) on the results and to examine the reliability of the proposed response spectra for the KR in terms of structural responses' mean prediction. More details are provided in Chapter 7.

Table 6.3: IDA for the one- and two-storey buildings using a set of seven earthquake time histories selected for seismic hazard zone A.

NGA Record No.	PGA (g)	One-storey building			Two-storey building		
		Max. x displacement (cm)	Damage state	Merged damage states	Max. x displacement (cm)	Damage state	Merged damage states
126	0.025	0.0065	No damage	None	0.0189	No damage	None
	0.05	0.0110	No damage	None	0.0335	No damage	None
	0.1	0.0215	No damage	None	0.0613	No damage	None
	0.2	0.0512	No damage	None	0.1637	Slight	Yield
	0.4	0.2790	Moderate	Yield	1.0343	Extensive	Yield
	0.6	0.9427	Very heavy	Collapse	1.9791	Very heavy	Collapse
	0.8	1.6400	Very heavy	Collapse	3.3052	Collapse	Collapse
	1.0	2.3800	Collapse	Collapse	3.9765	Collapse	Collapse
143	0.025	0.0050	No damage	None	0.0214	No damage	None
	0.05	0.0086	No damage	None	0.0365	No damage	None
	0.1	0.0160	No damage	None	0.0657	No damage	None
	0.2	0.0480	No damage	None	0.2781	Moderate	Yield
	0.4	0.3780	Moderate	Yield	0.9073	Extensive	Yield
	0.6	0.8500	Extensive	Yield	2.4018	Very heavy	Collapse
	0.8	1.8700	Collapse	Collapse	4.1526	Collapse	Collapse
	1.0	2.3400	Collapse	Collapse	5.7074	Collapse	Collapse
779	0.025	0.0034	No damage	None	0.0153	No damage	None
	0.05	0.0068	No damage	None	0.0238	No damage	None
	0.1	0.0110	No damage	None	0.0419	No damage	None
	0.2	0.0214	No damage	None	0.0876	No damage	None
	0.4	0.0837	Slight	Yield	0.6776	Extensive	Yield
	0.6	0.3300	Moderate	Yield	1.4878	Very heavy	Collapse
	0.8	0.8700	Extensive	Yield	4.1858	Collapse	Collapse
	1.0	2.3200	Collapse	Collapse	4.4184	Collapse	Collapse
828	0.025	0.0038	No damage	None	0.0202	No damage	None
	0.05	0.0070	No damage	None	0.0365	No damage	None
	0.1	0.0134	No damage	None	0.0677	No damage	None
	0.2	0.0290	No damage	None	0.1739	Slight	Yield
	0.4	0.2000	Moderate	Yield	0.8536	Extensive	Yield
	0.6	0.7100	Extensive	Yield	2.5449	Very heavy	Collapse
	0.8	1.5500	Very heavy	Collapse	4.5537	Collapse	Collapse
	1.0	2.8600	Collapse	Collapse	4.1536	Collapse	Collapse
983	0.025	0.0044	No damage	None	0.0148	No damage	None
	0.05	0.0060	No damage	None	0.0256	No damage	None

	0.1	0.0110	No damage	None	0.0518	No damage	None
	0.2	0.0280	No damage	None	0.1572	Slight	Yield
	0.4	0.3500	Moderate	Yield	1.3031	Extensive	Yield
	0.6	1.3800	Very heavy	Collapse	2.7626	Collapse	Collapse
	0.8	2.2300	Collapse	Collapse	3.9100	Collapse	Collapse
	1.0	2.4150	Collapse	Collapse	5.2107	Collapse	Collapse
1508	0.025	0.0036	No damage	None	0.0167	No damage	None
	0.05	0.0059	No damage	None	0.0280	No damage	None
	0.1	0.0110	No damage	None	0.0559	No damage	None
	0.2	0.0650	Slight	Yield	0.1750	Slight	Yield
	0.4	0.6110	Extensive	Yield	1.7101	Very heavy	Collapse
	0.6	1.4000	Very heavy	Collapse	3.3744	Collapse	Collapse
	0.8	2.2600	Collapse	Collapse	5.6544	Collapse	Collapse
	1.0	3.2000	Collapse	Collapse	7.3218	Collapse	Collapse
568	0.025	0.0037	No damage	None	0.0197	No damage	None
	0.05	0.0066	No damage	None	0.0354	No damage	None
	0.1	0.0131	No damage	None	0.0675	No damage	None
	0.2	0.0308	No damage	None	0.1420	No damage	None
	0.4	0.0956	Slight	Yield	0.7633	Extensive	Yield
	0.6	0.3489	Moderate	Yield	2.2882	Very heavy	Collapse
	0.8	1.1656	Very heavy	Collapse	3.9766	Collapse	Collapse
	1.0	2.0700	Collapse	Collapse	5.3355	Collapse	Collapse

Table 6.4: IDA for the one- and two-storey buildings using a set of seven earthquake time histories selected for seismic hazard zone B.

NGA Record No.	PGA (g)	One-storey building			Two-storey building		
		Max. x displacement (cm)	Damage state	Merged damage states	Max. x displacement (cm)	Damage state	Merged damage states
126	0.025	0.0065	No damage	None	0.0189	No damage	None
	0.05	0.0110	No damage	None	0.0335	No damage	None
	0.1	0.0215	No damage	None	0.0613	No damage	None
	0.2	0.0512	No damage	None	0.1637	Slight	Yield
	0.4	0.2790	Moderate	Yield	1.0343	Extensive	Yield
	0.6	0.9427	Very heavy	Collapse	1.9791	Very heavy	Collapse
	0.8	1.6400	Very heavy	Collapse	3.3052	Collapse	Collapse
	1.0	2.3800	Collapse	Collapse	3.9765	Collapse	Collapse
828	0.025	0.0038	No damage	None	0.0202	No damage	None
	0.05	0.0070	No damage	None	0.0365	No damage	None
	0.1	0.0134	No damage	None	0.0677	No damage	None
	0.2	0.0290	No damage	None	0.1739	Slight	Yield
	0.4	0.2000	Moderate	Yield	0.8536	Extensive	Yield
	0.6	0.7100	Extensive	Yield	2.5449	Very heavy	Collapse
	0.8	1.5500	Very heavy	Collapse	4.5537	Collapse	Collapse
	1.0	2.8600	Collapse	Collapse	4.1536	Collapse	Collapse
802	0.025	0.0045	No damage	None	0.0210	No damage	None
	0.05	0.0077	No damage	None	0.0377	No damage	None
	0.1	0.0142	No damage	None	0.0701	No damage	None
	0.2	0.0320	No damage	None	0.1641	Slight	Yield
	0.4	0.2160	Moderate	Yield	0.7157	Extensive	Yield
	0.6	0.5700	Extensive	Yield	1.8897	Very heavy	Collapse
	0.8	1.2967	Very heavy	Collapse	2.4584	Very heavy	Collapse
	1.0	1.8000	Collapse	Collapse	4.1135	Collapse	Collapse
1086	0.025	0.0035	No damage	None	0.0093	No damage	None
	0.05	0.0051	No damage	None	0.0241	No damage	None
	0.1	0.0115	No damage	None	0.0544	No damage	None
	0.2	0.0249	No damage	None	0.1274	No damage	None
	0.4	0.1850	Moderate	Yield	1.0621	Extensive	Yield
	0.6	0.8000	Extensive	Yield	2.5748	Very heavy	Collapse
	0.8	1.7300	Collapse	Collapse	3.6862	Collapse	Collapse
	1.0	3.2200	Collapse	Collapse	5.8653	Collapse	Collapse
179	0.025	0.0044	No damage	None	0.0187	No damage	None
	0.05	0.0073	No damage	None	0.0327	No damage	None

	0.1	0.0143	No damage	None	0.0625	No damage	None
	0.2	0.0337	No damage	None	0.1370	No damage	None
	0.4	0.1724	Moderate	Yield	0.9054	Extensive	Yield
	0.6	0.6099	Extensive	Yield	5.6646	Collapse	Collapse
	0.8	2.0022	Collapse	Collapse	9.9294	Collapse	Collapse
	1.0	9.1170	Collapse	Collapse	16.2144	Collapse	Collapse
821	0.025	0.0037	No damage	None	0.0192	No damage	None
	0.05	0.0070	No damage	None	0.0351	No damage	None
	0.1	0.0130	No damage	None	0.0649	No damage	None
	0.2	0.0316	No damage	None	0.1905	Slight	Yield
	0.4	0.2890	Moderate	Yield	1.4518	Very heavy	Collapse
	0.6	1.0170	Very heavy	Collapse	4.0135	Collapse	Collapse
	0.8	2.0330	Collapse	Collapse	4.5051	Collapse	Collapse
	1.0	3.8210	Collapse	Collapse	5.3769	Collapse	Collapse
1602	0.025	0.0037	No damage	None	0.0165	No damage	None
	0.05	0.0059	No damage	None	0.0249	No damage	None
	0.1	0.0127	No damage	None	0.0464	No damage	None
	0.2	0.0281	No damage	None	0.1280	No damage	None
	0.4	0.1650	Moderate	Yield	0.8899	Extensive	Yield
	0.6	0.7611	Extensive	Yield	1.9899	Very heavy	Collapse
	0.8	1.3490	Very heavy	Collapse	3.1633	Collapse	Collapse
	1.0	1.8250	Collapse	Collapse	4.6233	Collapse	Collapse

CHAPTER 7. DATA POST-PROCESSING

This chapter provides a comprehensive summary of data post-processing and statistical analysis activities. It begins with a presentation of data post-processing procedures used in database development. This section incorporates information on the steps taken to identify and determine the impact of the three main uncertainties involved in the fragility assessment process of unreinforced masonry buildings considered in this study. The optimal selection of ground-motion intensity measures for deriving fragility curves is then performed using statistical regression and machine learning tools. The remainder of the chapter is dedicated to summarising the outcomes pertaining to the comparison analysis using two ground-motion IMs instead of one IM. Finally, the probability of failure for the two tested URM buildings in the KR is provided in terms of fragility curves and surfaces.

7.1 Fragility Curves and Selection of the Best IMs

Once the data required for performing a fragility analysis are collected and the test structures are modelled and developed, a fragility assessment of the buildings can be properly conducted. As noted earlier, fragility curves are commonly used to provide a graphical representation of exceeding a damage state with respect to one IM. By defining damage states and selecting a suitable probability distribution function for the data, fragility curves can be more accurately derived, especially if a more accurate IM is used to represent a region's seismic activity. Based on the observed IDA results and as discussed in section 3.4.2, the binary log-logistic regression (Equation 3.3) is selected to fit the data because the response of a building in terms of the damage state was classified according to two possible types: "collapse" and "yield". However, to select an optimal IM, it is important to determine and estimate the role of various uncertainties such as the variability in material properties and ground-motion intensity measures, the variability in ground-motion selection and modification methods, the number of records, the scaling of records and other factors that may be influence the IM selection process. However, before taking these measures, reducing the number of ground-motion parameters in this study (i.e., 36 IMs) and classifying them to assist in minimising the time required to study all IMs can be beneficial; likewise, this can also be useful when finding parameters that are representative of the selected IM distribution. This chapter details the results of this study and considers all of the aforementioned procedural aspects.

7.1.1 Variability in material properties

The variability effect of the material properties of masonry units (i.e., solid concrete blocks and cement mortar as the main materials used in the construction of the tested buildings) on the structural response for various ground motions is analysed by performing approximately 1100 time history analyses applied to a one-storey building in the x direction using a fixed accelerogram scaled to three PGA intensities (0.05g, 0.2g, and 0.8g). For each intensity level, a particular set of variants is assigned with respect to the different material properties shown in Table 7.1. Typically, 20 samples (having minimum and maximum modulus of elasticity values of E=2700 MPa and E=6000 MPa, respectively, for masonry walls) are determined for each ground-motion intensity measure and for various material properties, which are provided in Table 7.1 for each sample.A second set of analyses are conducted to consider the variability due to ground motions by neglecting the variability induced by the material properties. Therefore, variability in ground motions is generated by taking and fixing the mean values of the material properties for all of the selected records. The two standard deviations of the outcomes (in terms of the top displacement) related to the variability of the ground motions and material parameters are subsequently computed. Then, the total standard deviation is calculated for the results based on each of the considered PGA levels (0.05g, 0.2g, and 0.8 g) as the square root of the sum of the squares for both of the aforementioned standard deviations. Based on the results obtained and referring to Figure 7.1, it is clear that the effect of ground-motion variability is more important than the effect of material property variability, especially for moderate to high PGAs. This concept can be explained by the possibility of structural response modification due to the increased nonlinearity behaviour of the building while ground-motion severity increases. Thus, the variability in material parameters is neglected, and the mean values of the material properties are implemented for the remainder of this study.

Furthermore, the influence of the elasticity modulus, shear resistance, specific weight, shear ultimate drift ratio, and rocking ultimate drift ratio on the structural response is presented in Figures 7.2-7.6. Figure 7.2 illustrates the variation in the modulus of elasticity while other parameters are fixed. However, Figures 7.3-7.6 display the effect of change in material properties other than the modulus of elasticity and with respect to each of 20 samples with different values for the modulus of elasticity (as mentioned above). Except for the elasticity modulus, all of the other parameters do not have a pronounced

effect on the results. Figure 7.2 shows that the one-storey building can resist seismic action with greater subsequent integrity if the masonry's modulus of elasticity is increased.

Table 7.1: Different values considered for the mechanical parameters of the masonry units.

Masonry's modulus of elasticity (N/mm^2)	Comp. strength of cement mortar (N/mm^2)	Comp. strength of solid concrete blocks (N/mm^2)	Shear resistance (N/mm^2)	Specific weight (kN/m^3)	Shear ultimate drift ratio (%)	Rocking ultimate drift ratio (%)
2700-6000	8-23	11-25	0.15,0.25,0.35	18,20,22	0.4,0. 6,0.8	0.6-1.0

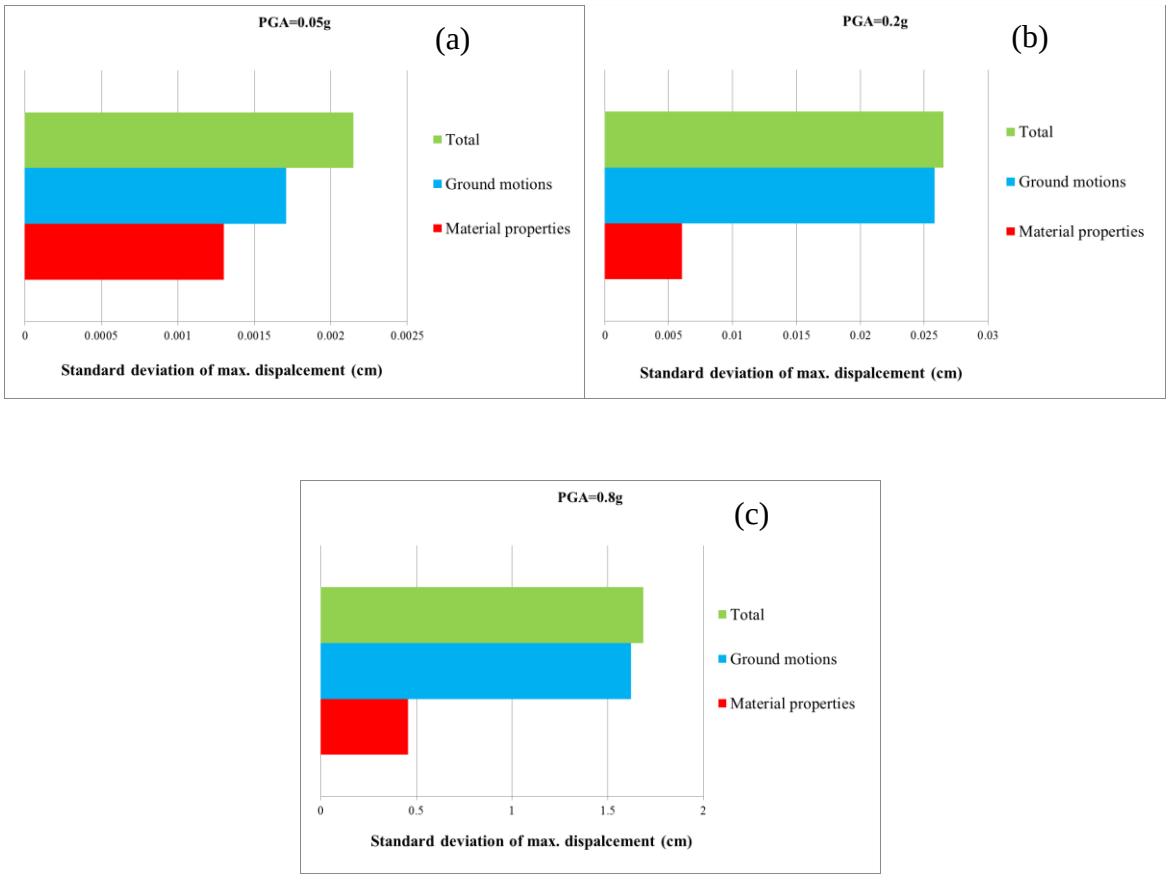


Figure 7.1: Comparison of the standard deviations obtained for various material properties and ground motions at different PGAs: a) 0.05g, b) 0.2g, and c) 0.8g.

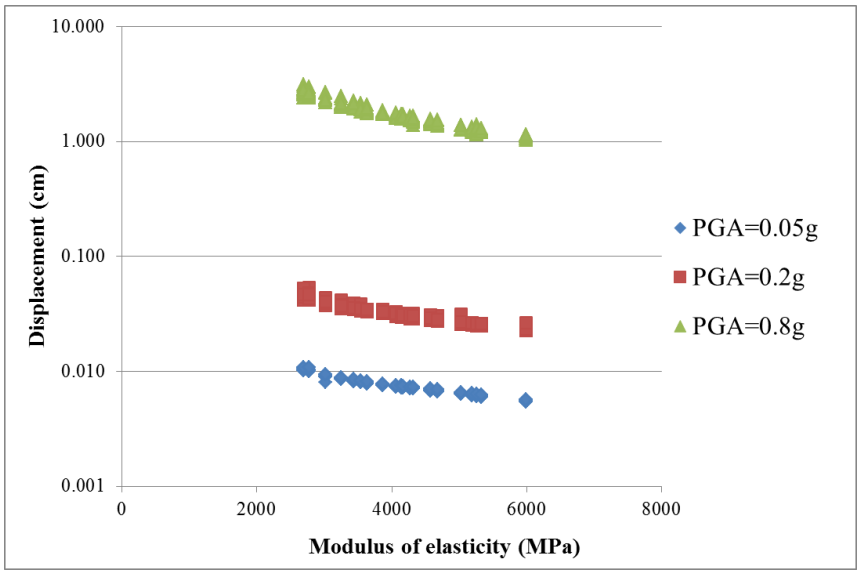


Figure 7.2: Influence of the modulus of elasticity of the masonry units on the response of the studied one-storey building in the x direction.

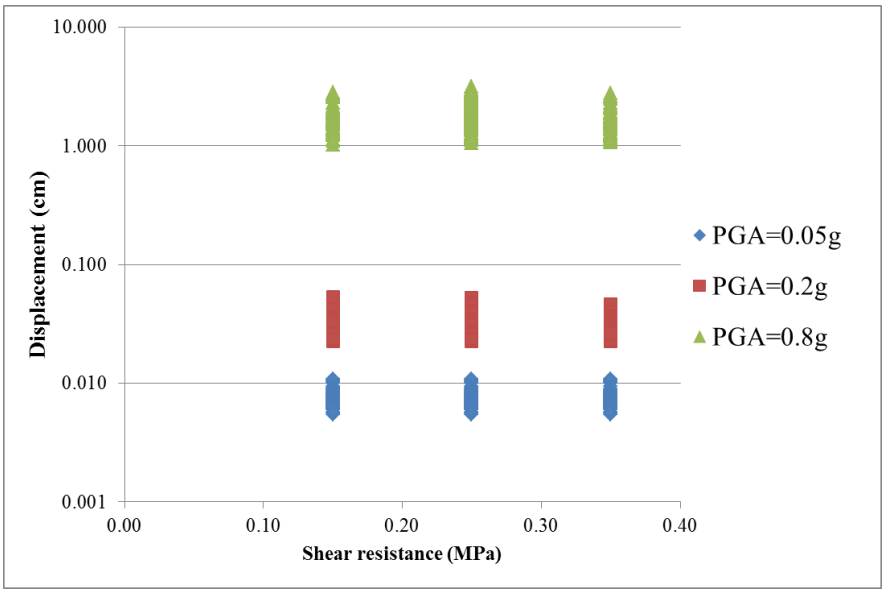


Figure 7.3: Influence of the shear resistance on the response of the studied one-storey building in the x direction.

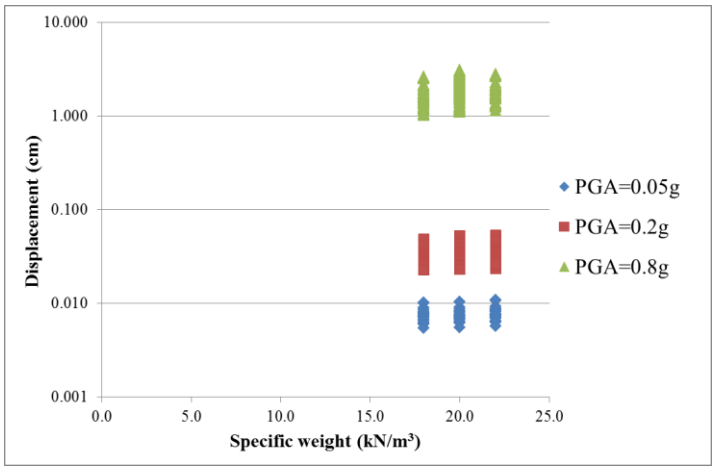


Figure 7.4: Influence of the specific weight on the response of the studied one-storey building in the x direction.

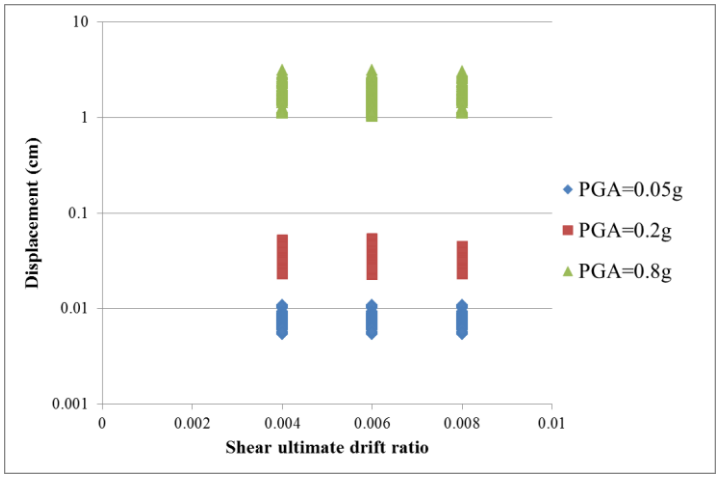


Figure 7.5: Influence of the shear ultimate drift ratio on the response of the studied one-storey building in the x direction.

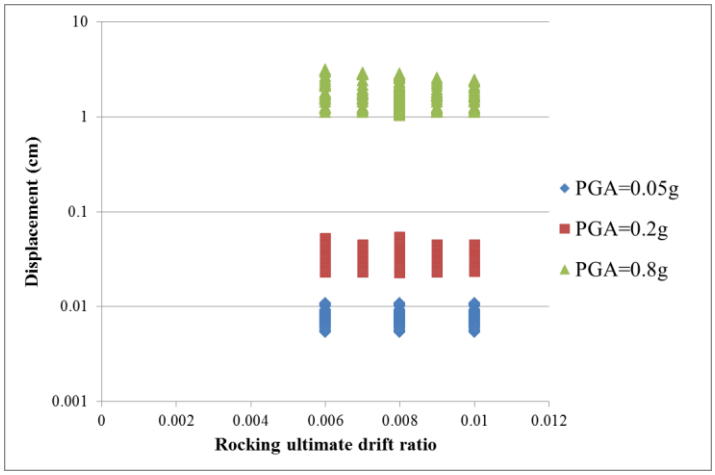


Figure 7.6: Influence of the rocking ultimate drift ratio on the response of the studied one-storey building in the x direction.

7.1.2 Clustering of IMs

As shown in Table 7.2, 36 IMs are estimated and extracted from each of the time histories used in the analysis. Using the Tanagra software, the VARCLUS method is used to classify these IMs and merge them into two main clusters based on different scenarios that are typical for the study area. Figure 7.7 displays the common succession of the splitting operations. The correlation coefficients between the top displacement and the two clusters are then obtained and are presented in Tables 7.3-7.7 for each dataset.

Table 7.2: Ground-motion IMs considered in this study.

IMs	Name
Acceleration based	Peak ground acceleration (PGA), root-mean-square acceleration (RMSA), Arias intensity (IA), characteristic intensity (IC), cumulative absolute velocity (CAV), acceleration spectrum intensity (ASI), sustained maximum acceleration (SMA), effective design acceleration (EDA), A95 parameter, and spectral acceleration at different periods S_a(nT1), where n=1, 2, 3, 4, 8, 16, and 32 and T1 is a structure's fundamental period.
Velocity based	Peak ground velocity (PGV), root-mean-square velocity (RMSV), specific energy density (SED), velocity spectrum intensity (VSI), sustained maximum velocity (SMV), and Housner Intensity (IH).
Displacement based	Peak ground displacement (PGD), root-mean-square displacement (RMSD), spectral displacement at different periods S_d(nT1), where n=1, 2, 3, 4, 8, 16, and 32 and T1 is a structure's fundamental period.
Hybrid	PGV/PGA, PGA/PGV, and PGV^2/PGA.
Duration	Predominant period (Tp) and mean period (Tm).

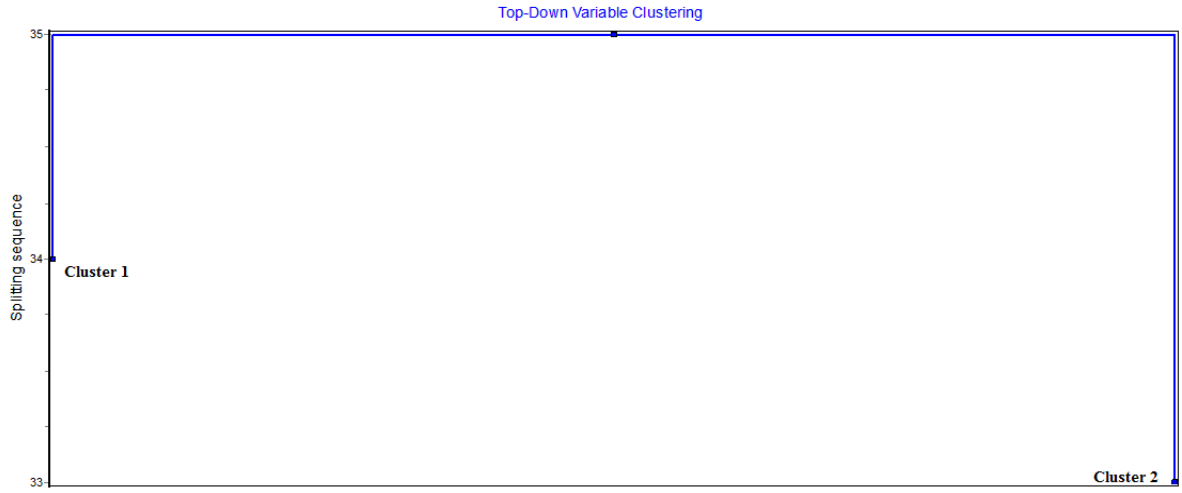


Figure 7.7: Top-down approach showing the splitting process and arrangement of clusters.

Table 7.3: Clusters generated from the accelerogram dataset and the corresponding ground-motion parameters; the correlation coefficients between these clusters and the drift obtained from the simulations are provided (one-storey building in zone A).

No. of earthquake records	Cluster	Ground-motion parameters	Top displacement
7 records	1	PGA, PGV, PGD, PGV^2/PGA, RMSA, RMSV, RMSD, IA, IC, SED, CAV, ASI, VSI, IH, SMA, SMV, EDA, and A95	0.916
	2	S_a(T1), S_a(2T1), S_a(3T1), S_a(4T1), S_a(8T1), S_a(16T1), S_a(32T1), S_d(T1), S_d(2T1), S_d(3T1), S_d(4T1), S_d(8T1), S_d(16T1), S_d(32T1), Tp, Tm, PGV/PGA, and PGA/PGV	0.1396
15 records CMS	1	PGA, PGV, PGD, PGV/PGA, PGA/PGV, PGV^2/PGA, RMSA, RMSV, RMSD, IA, IC, SED, CAV, ASI, VSI, IH, SMA, SMV, EDA, A95, Tp, and Tm	0.9413
	2	S_a(T1), S_a(2T1), S_a(3T1), S_a(4T1), S_a(8T1), S_a(16T1), S_a(32T1), S_d(T1), S_d(2T1), S_d(3T1), S_d(4T1), S_d(8T1), S_d(16T1), S_d(32T1)	0.1228
15 records EC8	1	PGA, PGV, PGD, PGV^2/PGA, RMSA, RMSV, RMSD, IA, IC, SED, CAV, ASI, VSI, IH, SMA, SMV, EDA, and A95	0.9028
	2	S_a(T1), S_a(2T1), S_a(3T1), S_a(4T1), S_a(8T1), S_a(16T1), S_a(32T1), S_d(T1), S_d(2T1), S_d(3T1), S_d(4T1), S_d(8T1), S_d(16T1), S_d(32T1), Tp, Tm, PGV/PGA, and PGA/PGV	0.1187
15 records Jayaram et al.	1	PGA, PGV, PGD, PGV/PGA, PGA/PGV, PGV^2/PGA, RMSV, RMSD, IA, SED, CAV, VSI, IH, SMV, EDA, A95, and Tm	0.8667
	2	S_a(T1), S_a(2T1), S_a(3T1), S_a(4T1), S_a(8T1), S_a(16T1), S_a(32T1), S_d(T1), S_d(2T1), S_d(3T1), S_d(4T1), S_d(8T1), S_d(16T1), S_d(32T1), Tp, RMSA, IC, ASI, and SMA	0.0597
15 records Proposed response spectra	1	PGA, PGV, PGD, PGV^2/PGA, RMSA, RMSV, RMSD, IA, IC, SED, CAV, ASI, VSI, IH, SMA, SMV, EDA, A95	0.8917
	2	S_a(T1), S_a(2T1), S_a(3T1), S_a(4T1), S_a(8T1), S_a(16T1), S_a(32T1), S_d(T1), S_d(2T1), S_d(3T1), S_d(4T1), S_d(8T1), S_d(16T1), S_d(32T1), Tp, Tm, PGV/PGA, and PGA/PGV	-0.0411
35 records	1	PGA, PGV, PGD, PGV/PGA, PGA/PGV, PGV^2/PGA, RMSA, RMSV, RMSD, IA, IC, SED, CAV, ASI, VSI, IH, SMA, SMV, EDA, A95, and Tm	0.8842
	2	S_a(T1), S_a(2T1), S_a(3T1), S_a(4T1), S_a(8T1), S_a(16T1), S_a(32T1), S_d(T1), S_d(2T1), S_d(3T1), S_d(4T1), S_d(8T1), S_d(16T1), S_d(32T1), and Tp	-0.0333

Table 7.4: Clusters generated from the accelerogram dataset and the corresponding ground-motion parameters; the correlation coefficients between these clusters and the drift obtained from the simulations are provided (two-storey building in zone A).

No. of earthquake records	Cluster	Ground-motion parameters	Top displacement
7 records	1	PGA, PGV, PGD, PGV^2/PGA, RMSA, RMSV, RMSD, IA, IC, SED, CAV, ASI, VSI, IH, SMA, SMV, EDA, and A95	0.9148
	2	$S_a(T1)$, $S_a(2T1)$, $S_a(3T1)$, $S_a(4T1)$, $S_a(8T1)$, $S_a(16T1)$, $S_a(32T1)$, $S_d(T1)$, $S_d(2T1)$, $S_d(3T1)$, $S_d(4T1)$, $S_d(8T1)$, $S_d(16T1)$, $S_d(32T1)$, Tp, Tm, PGV/PGA, and PGA/PGV	0.2407
15 records CMS	1	PGA, PGV, PGD, PGV/PGA, PGA/PGV, PGV^2/PGA, RMSA, RMSV, RMSD, IA, IC, SED, CAV, ASI, VSI, IH, SMA, SMV, EDA, and A95	0.9093
	2	$S_a(T1)$, $S_a(2T1)$, $S_a(3T1)$, $S_a(4T1)$, $S_a(8T1)$, $S_a(16T1)$, $S_a(32T1)$, $S_d(T1)$, $S_d(2T1)$, $S_d(3T1)$, $S_d(4T1)$, $S_d(8T1)$, $S_d(16T1)$, $S_d(32T1)$, Tp, and Tm	0.3077
15 records EC8	1	PGA, PGV, PGD, PGV^2/PGA, RMSA, RMSV, RMSD, IA, IC, SED, CAV, ASI, VSI, IH, SMA, SMV, EDA, A95, and Tm	0.8722
	2	$S_a(T1)$, $S_a(2T1)$, $S_a(3T1)$, $S_a(4T1)$, $S_a(8T1)$, $S_a(16T1)$, $S_a(32T1)$, $S_d(T1)$, $S_d(2T1)$, $S_d(3T1)$, $S_d(4T1)$, $S_d(8T1)$, $S_d(16T1)$, $S_d(32T1)$, Tp, PGV/PGA, and PGA/PGV	0.2759
15 records Jayaram et al.	1	PGA, PGV, PGD, PGV/PGA, PGA/PGV, PGV^2/PGA, RMSV, RMSD, IA, IC, SED, CAV, VSI, IH, SMA, SMV, EDA, A95, Tm	0.9228
	2	$S_a(T1)$, $S_a(2T1)$, $S_a(3T1)$, $S_a(4T1)$, $S_a(8T1)$, $S_a(16T1)$, $S_a(32T1)$, $S_d(T1)$, $S_d(2T1)$, $S_d(3T1)$, $S_d(4T1)$, $S_d(8T1)$, $S_d(16T1)$, $S_d(32T1)$, Tp, RMSA, and ASI	0.0032
15 records Proposed response spectra	1	PGA, PGV, PGD, PGV^2/PGA, RMSA, RMSV, RMSD, IA, IC, SED, CAV, ASI, VSI, IH, SMA, SMV, EDA, A95, and Tm	0.9098
	2	$S_a(T1)$, $S_a(2T1)$, $S_a(3T1)$, $S_a(4T1)$, $S_a(8T1)$, $S_a(16T1)$, $S_a(32T1)$, $S_d(T1)$, $S_d(2T1)$, $S_d(3T1)$, $S_d(4T1)$, $S_d(8T1)$, $S_d(16T1)$, $S_d(32T1)$, Tp, PGV/PGA, and PGA/PGV	0.0509
35 records	1	PGA, PGV, PGD, PGV/PGA, PGA/PGV, PGV^2/PGA, RMSA, RMSV, RMSD, IA, IC, SED, CAV, ASI, VSI, IH, SMA, SMV, EDA, A95, and Tm	0.9041
	2	$S_a(T1)$, $S_a(2T1)$, $S_a(3T1)$, $S_a(4T1)$, $S_a(8T1)$, $S_a(16T1)$, $S_a(32T1)$, $S_d(T1)$, $S_d(2T1)$, $S_d(3T1)$, $S_d(4T1)$, $S_d(8T1)$, $S_d(16T1)$, $S_d(32T1)$, and Tp	0.0699

Table 7.5: Clusters generated from the accelerogram dataset and the corresponding ground-motion parameters; the correlation coefficients between these clusters and the drift obtained from the simulations are provided (one-storey building in zone B).

No. of earthquake records	Cluster	Ground-motion parameters	Top displacement
7 records	1	PGA, S_a(8T1), S_a(16T1), S_a(32T1), S_d(8T1), S_d(16T1), S_d(32T1), RMSA, IA, IC, CAV, ASI, VSI, IH, SMA, EDA, and A95	0.649
	2	PGV, PGD, PGV/PGA, PGA/PGV, PGV^2/PGA, RMSV, RMSD, SED, SMV, S_a(T1), S_a(2T1), S_a(3T1), S_a(4T1), S_d(T1), S_d(2T1), S_d(3T1), S_d(4T1), Tp, and Tm	0.6363
15 records CMS	1	PGA, PGV, PGD, PGV/PGA, PGA/PGV, PGV^2/PGA, RMSA, RMSV, RMSD, IA, IC, SED, CAV, ASI, VSI, IH, SMA, SMV, EDA, A95, S_a(32T1), S_d(32T1), Tp, and Tm	0.8208
	2	S_a(T1), S_a(2T1), S_a(3T1), S_a(4T1), S_a(8T1), S_a(16T1), S_d(T1), S_d(2T1), S_d(3T1), S_d(4T1), S_d(8T1), and S_d(16T1)	0.4321
15 records EC8	1	PGA, PGV, PGD, PGV/PGA, PGA/PGV, PGV^2/PGA, RMSA, RMSV, RMSD, IA, IC, SED, CAV, ASI, VSI, IH, SMA, SMV, EDA, and A95	0.9314
	2	S_a(T1), S_a(2T1), S_a(3T1), S_a(4T1), S_a(8T1), S_a(16T1), S_a(32T1), S_d(T1), S_d(2T1), S_d(3T1), S_d(4T1), S_d(8T1), S_d(16T1), S_d(32T1), Tp, and Tm	0.2748
15 records Jayaram et al.	1	PGA, PGV, PGD, PGV/PGA, PGA/PGV, PGV^2/PGA, RMSA, RMSV, RMSD, IA, IC, SED, CAV, ASI, VSI, IH, SMA, SMV, EDA, A95, Tp, and Tm	0.8952
	2	S_a(T1), S_a(2T1), S_a(3T1), S_a(4T1), S_a(8T1), S_a(16T1), S_a(32T1), S_d(T1), S_d(2T1), S_d(3T1), S_d(4T1), S_d(8T1), S_d(16T1), S_d(32T1)	0.1056
15 records Proposed response spectra	1	PGA, PGV, PGD, PGV/PGA, PGA/PGV, PGV^2/PGA, RMSA, RMSV, RMSD, IA, IC, SED, CAV, VSI, IH, SMV, EDA, A95, S_a(T1), S_a(2T1), S_a(3T1), S_d(T1), S_d(2T1), S_d(3T1), and Tm	0.8259
	2	S_a(4T1), S_a(8T1), S_a(16T1), S_a(32T1), S_d(4T1), S_d(8T1), S_d(16T1), S_d(32T1), ASI, SMA, and Tp	0.2754
38 records	1	PGA, PGV, PGD, PGV/PGA, PGA/PGV, PGV^2/PGA, RMSA, RMSV, RMSD, IA, IC, SED, CAV, ASI, VSI, IH, SMA, SMV, EDA, A95, Tp, and Tm	0.8932
	2	S_a(T1), S_a(2T1), S_a(3T1), S_a(4T1), S_a(8T1), S_a(16T1), S_a(32T1), S_d(T1), S_d(2T1), S_d(3T1), S_d(4T1), S_d(8T1), S_d(16T1), S_d(32T1)	0.1452

Table 7.6: Clusters generated from the accelerogram dataset and the corresponding ground-motion parameters; the correlation coefficients between these clusters and the drift obtained from the simulations are provided (two-storey building in zone B).

No. of earthquake records	Cluster	Ground-motion parameters	Top displacement
7 records	1	PGV, PGD, PGV/PGA, PGA/PGV, PGV^2/PGA, RMSV, RMSD, VSI, IH, SMV, SED, S_a(T1), S_a(2T1), S_d(T1), S_d(2T1), and Tm	0.8702
	2	PGA, S_a(3T1), S_a(4T1), S_a(8T1), S_a(16T1), S_a(32T1), S_d(3T1), S_d(4T1), S_d(8T1), S_d(16T1), S_d(32T1), RMSA, IA, IC, CAV, ASI, SMA, EDA, A95, and Tp	0.6471
15 records CMS	1	PGA, PGV, PGD, PGV/PGA, PGA/PGV, PGV^2/PGA, RMSA, RMSV, RMSD, IA, IC, SED, CAV, ASI, VSI, IH, SMA, SMV, EDA, A95, and Tp	0.8578
	2	S_a(T1), S_a(2T1), S_a(3T1), S_a(4T1), S_a(8T1), S_a(16T1), S_a(32T1), S_d(T1), S_d(2T1), S_d(3T1), S_d(4T1), S_d(8T1), S_d(16T1), S_d(32T1), and Tm	0.4946
15 records EC8	1	PGA, PGV, PGD, PGV^2/PGA, RMSA, RMSV, RMSD, IA, IC, SED, CAV, ASI, VSI, IH, SMA, SMV, EDA, A95, S_a(T1), S_a(3T1), S_a(32T1), S_d(T1), S_d(3T1), and S_d(32T1)	0.9308
	2	PGV/PGA, PGA/PGV, S_a(2T1), S_a(4T1), S_a(8T1), S_a(16T1), S_d(2T1), S_d(4T1), S_d(8T1), S_d(16T1), Tp, and Tm	0.2019
15 records Jayaram et al.	1	PGA, PGV, PGD, PGV/PGA, PGA/PGV, PGV^2/PGA, RMSV, RMSD, IA, IC, SED, CAV, VSI, IH, SMV, EDA, A95, Tp, and Tm	0.9393
	2	S_a(T1), S_a(2T1), S_a(3T1), S_a(4T1), S_a(8T1), S_a(16T1), S_a(32T1), S_d(T1), S_d(2T1), S_d(3T1), S_d(4T1), S_d(8T1), S_d(16T1), S_d(32T1), RMSA, ASI, and SMA	0.2534
15 records Proposed response spectra	1	PGV, PGD, PGV/PGA, PGA/PGV, PGV^2/PGA, RMSV, RMSD, SED, VSI, IH, SMV, S_a(T1), S_a(2T1), S_d(T1), S_d(2T1), and Tm	0.8882
	2	PGA, S_a(3T1), S_a(4T1), S_a(8T1), S_a(16T1), S_a(32T1), S_d(3T1), S_d(4T1), S_d(8T1), S_d(16T1), S_d(32T1), RMSA, IA, IC, CAV, ASI, SMA, EDA, A95, and Tp	0.6478
38 records	1	PGA, PGV, PGD, PGV/PGA, PGA/PGV, PGV^2/PGA, RMSV, RMSD, IA, IC, SED, CAV, VSI, IH, SMA, SMV, EDA, A95, Tp, and Tm	0.9284
	2	S_a(T1), S_a(2T1), S_a(3T1), S_a(4T1), S_a(8T1), S_a(16T1), S_a(32T1), S_d(T1), S_d(2T1), S_d(3T1), S_d(4T1), S_d(8T1), S_d(16T1), S_d(32T1), RMSA, and ASI	0.1921

Table 7.7: Clusters generated from the accelerogram dataset and the corresponding ground-motion parameters; the correlation coefficients between these clusters and the drift obtained from the simulations are provided (based on a set 60 earthquake records from both zones A and B).

Cluster	Ground-motion parameters	Top displacement (one-storey building)	Top displacement (two-storey building)
1	PGA, PGV, PGD, PGV/PGA, PGA/PGV, PGV^2/PGA, RMSA, RMSV, RMSD, IA, IC, SED, CAV, ASI, VSI, IH, SMA, SMV, EDA, and A95	0.88	0.91
2	S_a(T1), S_a(2T1), S_a(3T1), S_a(4T1), S_a(8T1), S_a(16T1), S_a(32T1), S_d(T1), S_d(2T1), S_d(3T1), S_d(4T1), S_d(8T1), S_d(16T1), S_d(32T1), Tp, and Tm	0.26	0.27

The obtained results show that cluster 1 has the highest correlation with the structural response in nearly all cases, whereas cluster 2 is primarily indicative of the spectral acceleration, spectral displacement and duration parameters being poorly correlated with this response. This first analysis indicates that the relevant ground parameter should be selected from the first cluster. For brevity and comparison purposes, only 20 parameters are used to cover cluster 1, which are obtained from the dataset of 60 records (shown in Table 7.7); these parameters are common for all cases in cluster 1. As mentioned above, in most cases, the spectral accelerations and displacements at different periods being poorly correlated with the damage measure, although this correlation should be superior to the other IMs (as mentioned in the literature review), especially near a building's natural period. This behaviour can be explained by the possibility of variations and increases in a structure's natural period due to progressive damage and the loss of rigidity. Because of a lack of consideration of spectral shape and hence neglect of spectral ordinates for other periods of shaking, which are important for nonlinear systems, parameters mainly found in cluster 2 (shown in Table 7.7) are avoided in the rest of this study.

Based on the aforementioned procedure, 36 IMs are reduced to 20 IMs. To select a single IM, it is crucial to analyse the correlation between these variables and the structural response. The parameter with the best correlation ratio should be selected. The following sections provide a detailed review of the techniques used to achieve the established goal.

7.1.3 Variability in the GMSM methods and the number of records

Relying on the results obtained from the time history analysis, the fragility curves are constructed based on all of the parameters in cluster 1 listed in Table 7.7. The collapse fragility curves are then derived by applying log-logistic regression to fit the data according to equation 3.3. The collapse fragility curves obtained for the studied one- and two-storey buildings located in zone B are shown in Figures 7.8 and 7.9 with respect to the suites of 7, 15 (for each method used in the selection of ground motions) and 38 records. The yield fragility curves are obtained using equation 3.5. However, only the collapse curves are presneted in the study.

There are several key points to discuss regarding Figures 7.8 and 7.9. These figures are designed and scaled in such a manner as to be able to present a qualitative comparison among the different ground-motion IM-based fragility curves. Because of log-logistic regression is used, the prediction of binary damage states (i.e., yield and collapse) is considered to be more accurate if the slope of the curves becomes steeper. In all of the scenarios shown in Figure 7.8 involving the one-storey building, the ASI indicates less sensitivity to changes in the number of records; a lesser impact is found for the different ground-motion selection methods used in the study. Although the PGA in the CMS, the EDA in the EC8 code and A95 in the 7-record cases produces steeper curves, the ASI produces reduced dispersions for the other cases (the 38-record, Jayaram et al. and proposed response spectrum cases) because the slope of the fragility curves become steeper than the other IM-based fragility curves. With respect to the studied two-storey building, the VSI tends to be the best ground-motion parameter for deriving fragility curves (see Figure 7.9) due to its reduced uncertainty (steeper slope). Generally, it can be concluded that all of the parameters exhibit less sensitivity to the variabilities in the GMSM methods and the number of records, except for the displacement-based, hybrid and peak IMs. Furthermore, acceleration-based IMs provide a better performance than do other IMs for the considered scenarios. Because the shape of a curve cannot be easily exploited, a measure known as the -2 log-likelihood (-2LL), which is used in logistic regression, is applied to the analysis as a statistical measure that is similar to the sum of the squares in regression. The -2LL is designed to gauge the ability of an independent variable (in this case, an IM) to accurately predict the dependent variable (in this case, the damage state); a smaller log-likelihood indicates greater predictive capabilities. Instead of finding the best-fitting line by minimising the squared residuals, as with ordinary (or linear) least-squares

regression, a different approach, i.e., maximum likelihood (ML), with logistic is used, as mentioned previously in section 3.4.2. ML is a way of finding the smallest possible deviance between the observed and the predicted values using calculus. The final value for the deviance is -2LL. Table 7.8 introduces the -2LL measures for all of the tested IMs using different sets of earthquake records for the studied one- and two-storey buildings located in zone B. According to Table 7.8, the ASI and VSI are the optimal parameters for the one- and two-storey buildings, respectively, because they produce smaller -2LL values than the other IMs for most of the available cases.To this end it should be mentioned that, the -2LL values provided here, are used only to investigate and clarify the predictability of different IMs within each suits of records without having an intention to provide a comparison measure among different suits of records used in the study.

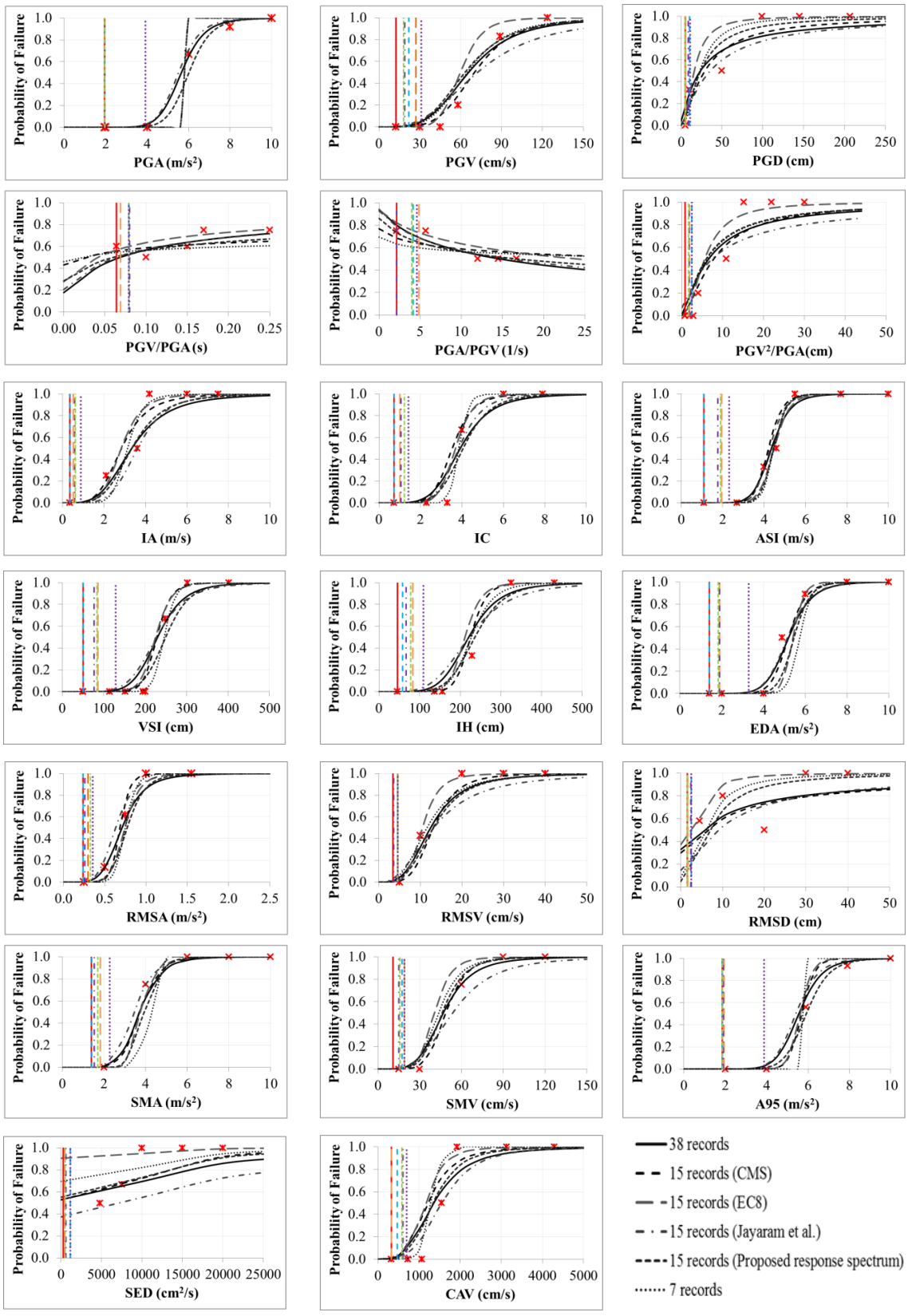


Figure 7.8: Collapse fragility curves for the studied one-storey building in zone B. Vertical dashed lines show the boundary between the damage and no-damage zones; stars show several values of the observed failure probability as examples for each IM.

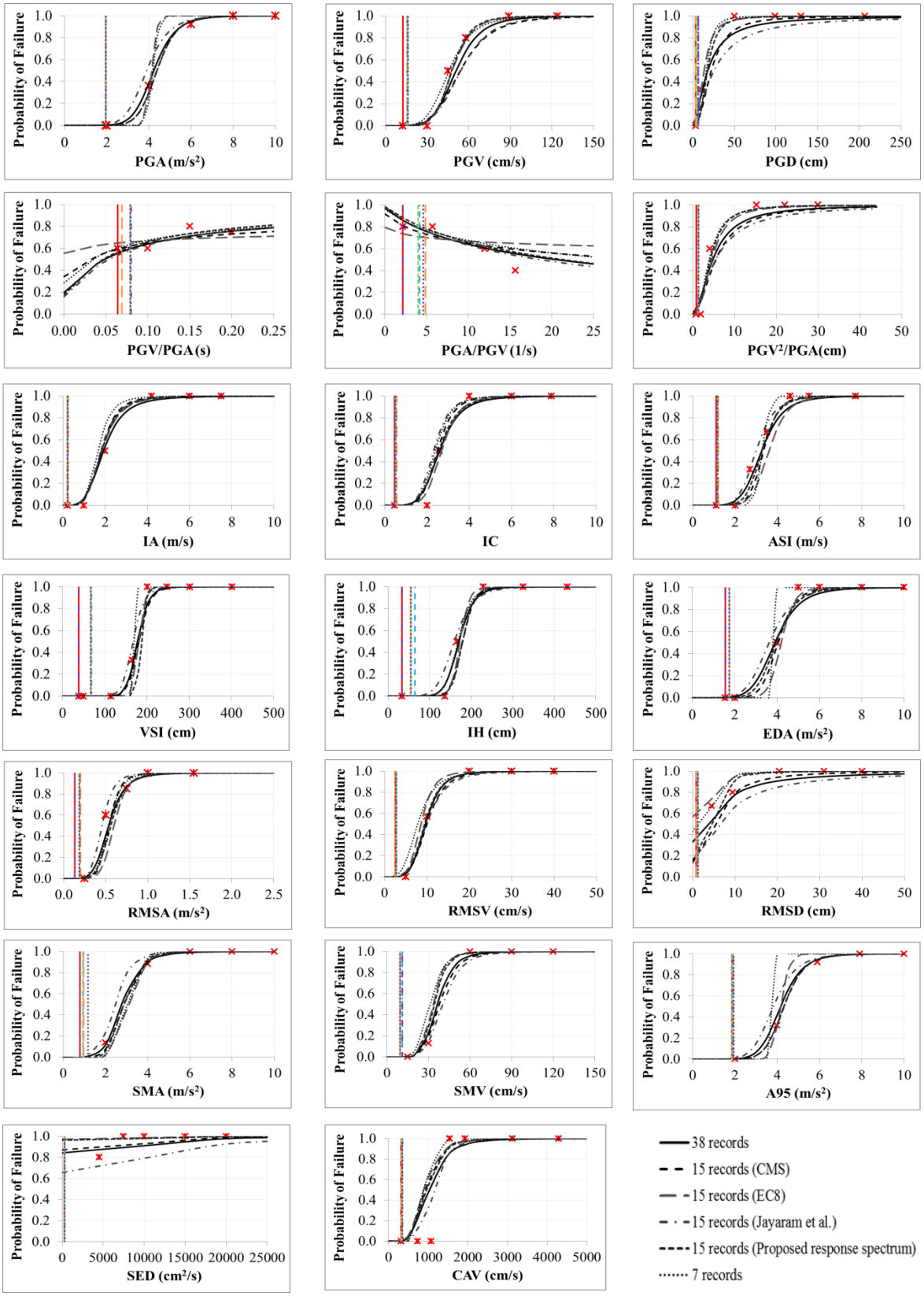


Figure 7.9: Collapse fragility curves for the studied two-storey building in zone B. Vertical dashed lines show the boundary between the damage and no-damage zones; stars show several values of the observed failure probability as examples for each IM.

Table 7.8: -2 Log-likelihood (-2LL) measure used to demonstrate how poorly one IM fits the data for different sets of earthquake records applied to the studied one- and two-storey buildings located in zone B.

IM	Zone B											
	38 records		CMS		EC8		Jayaram et al.		Proposed response spectrum		7 records	
	One storey	Two storey	One storey	Two storey	One storey	Two storey	One storey	Two storey	One storey	Two storey	One storey	Two storey
PGA	76.4	72.9	18.0	24.9	18.4	15.0	26.7	28.6	28.4	24.9	5.0	5.8
PGV	130.8	90.5	46.8	40.1	34.3	32.1	53.5	34.7	53.8	31.1	24.4	17.1
PGD	190.1	172.6	74.2	60.5	62.1	60.2	66.4	64.9	71.0	52.5	31.6	23.0
PGV/PGA	214.6	225.9	85.5	86.6	78.0	89.4	79.1	84.9	83.8	85.6	38.2	39.4
PGA/PGV	214.6	225.9	85.5	86.6	78.0	89.4	79.1	84.9	83.8	85.6	38.2	39.4
PGV2/PGA	177.3	159.2	69.0	62.2	59.5	59.3	68.0	60.8	70.7	55.9	33.0	26.5
IA	91.5	68.8	30.8	23.5	24.1	25.7	27.1	19.6	32.2	24.8	8.2	10.2
IC	90.0	69.9	26.1	23.1	21.2	23.1	28.1	19.7	31.1	26.3	5.9	11.1
ASI	75.1	87.2	22.6	23.9	22.6	27.1	23.1	28.6	20.4	27.3	5.7	6.5
VSI	90.9	44.9	20.4	17.1	23.0	9.4	37.0	16.7	33.2	8.4	8.9	0.0
IH	103.9	57.5	24.2	18.2	26.6	15.5	45.0	25.4	40.8	18.5	16.7	10.6
EDA	77.9	85.6	22.8	24.5	16.2	17.3	29.5	31.8	22.1	28.7	20.2	0.0
CAV	143.4	113.8	55.5	40.0	43.1	42.2	47.4	34.3	54.8	41.6	14.8	12.5
RMSA	118.8	96.2	24.3	27.4	32.1	28.5	42.6	30.0	37.6	33.5	12.1	15.1
RMSV	140.4	97.9	46.8	39.1	42.5	37.2	56.7	40.6	54.6	38.2	26.1	21.5
RMSD	197.1	178.0	79.2	63.0	60.5	60.3	67.3	65.7	67.7	51.5	30.0	24.9
SMA	98.3	89.9	34.5	32.5	22.8	27.2	37.7	30.9	29.7	29.3	6.3	11.0
SMV	123.3	85.6	39.3	34.3	36.6	29.8	49.2	32.3	45.3	29.5	21.2	15.4
A95	76.6	74.2	20.9	25.6	19.7	15.3	27.6	29.2	29.5	25.6	2.8	0.0
SED	148.5	112.7	50.5	36.6	40.6	37.0	57.2	46.0	54.4	31.3	25.9	18.1

With respect to zone A and the one-storey building dataset, the EDA parameter exhibits the smallest sensitivity to the considered variabilities in terms of the GMSM methods and the number of records. Nevertheless, the ASI provides more accurate probability predictions with less uncertainty, which is depicted in Figure 7.10. The smaller -2LL measure for the ASI shown in Table 7.9 also indicates that the ASI is more accurate for predicting the structural response than the other parameters.

In the case of the studied two-storey building in zone A (Figure 7.11), even if an IM is not perfect (i.e., being unaffected by the considered scenarios), the ASI and VSI can be considered as good parameters compared with the other options because they exhibit less sensitivity to the considered scenarios and less uncertainty for predicting the structural response. This conclusion can also be ascertained from the computed -2LL scales in Table 7.9.

Furthermore, to better understand the effect of different number of earthquake records on optimal selection of the ground motion parameters, auxiliary sets of 60, 25 (per each GMSM methods used in the study) and 12 records are used to construct collapse fragility curves for the two selected buildings in the study. As illustrated in Figures 7.12 and 7.13, both the ASI and VSI predict the response of the one- and two-storey buildings with greater accuracy compared with the other potential classifiers. This accuracy is achieved because these fragility curves have a steeper slope (i.e., less uncertainty) than do the other IM fragility curves. The EDA is another candidate for one-storey buildings because it exhibits good performance for different numbers of selected records; however, this IM has a slightly shallower slope compared to the ASI curves, indicating the presence of greater error in the data classification compared to the ASI. Furthermore, Tables 7.10 and 7.11 demonstrate that the ASI and VSI are the optimal classifiers for the studied one- and two-storey buildings, respectively, because they have smaller -2LL measures compared to the other IMs.

Moreover, to make it clear and more comprehensible, an extra set of fragility curves are also plotted for 60 (zones A and B), 35 (zone A), 25 (zones A and B - CMS method), 12 (zones A and B), and 7 (zone A) records for the main IMs, i.e., PGA, PGV, PGD, IA, IC, ASI, VSI, IH, and EDA; the two buildings are separately examined (see Figures 7.14 and 7.15). When a correct IM is selected, it does not matter how many records are used, even if the minimum number of records (7 records) is used.

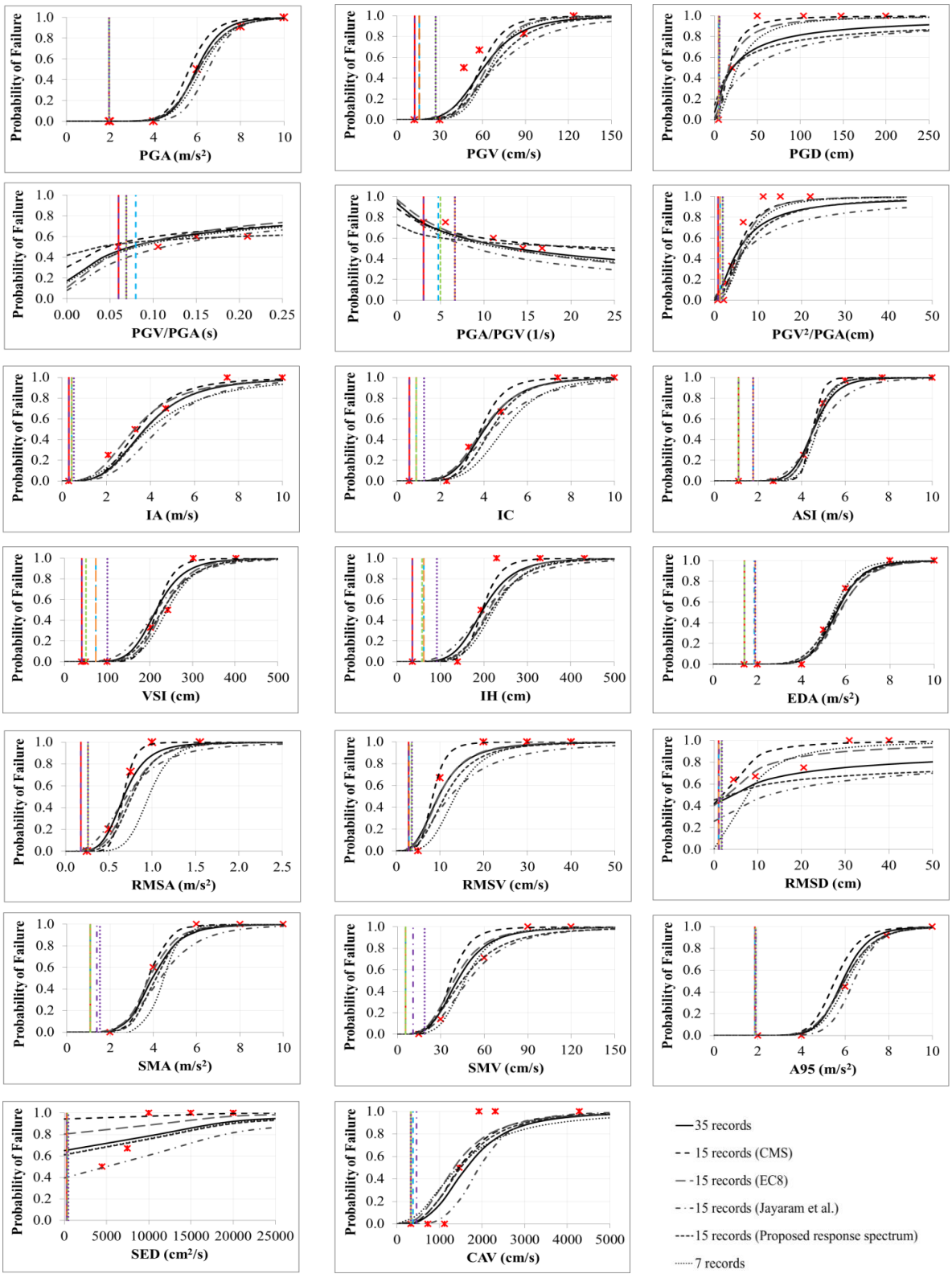


Figure 7.10: Collapse fragility curves for the studied one-storey building in zone A. Vertical dashed lines show the boundary between the damage and no-damage zones; stars show several values of the observed failure probability as examples for each IM.

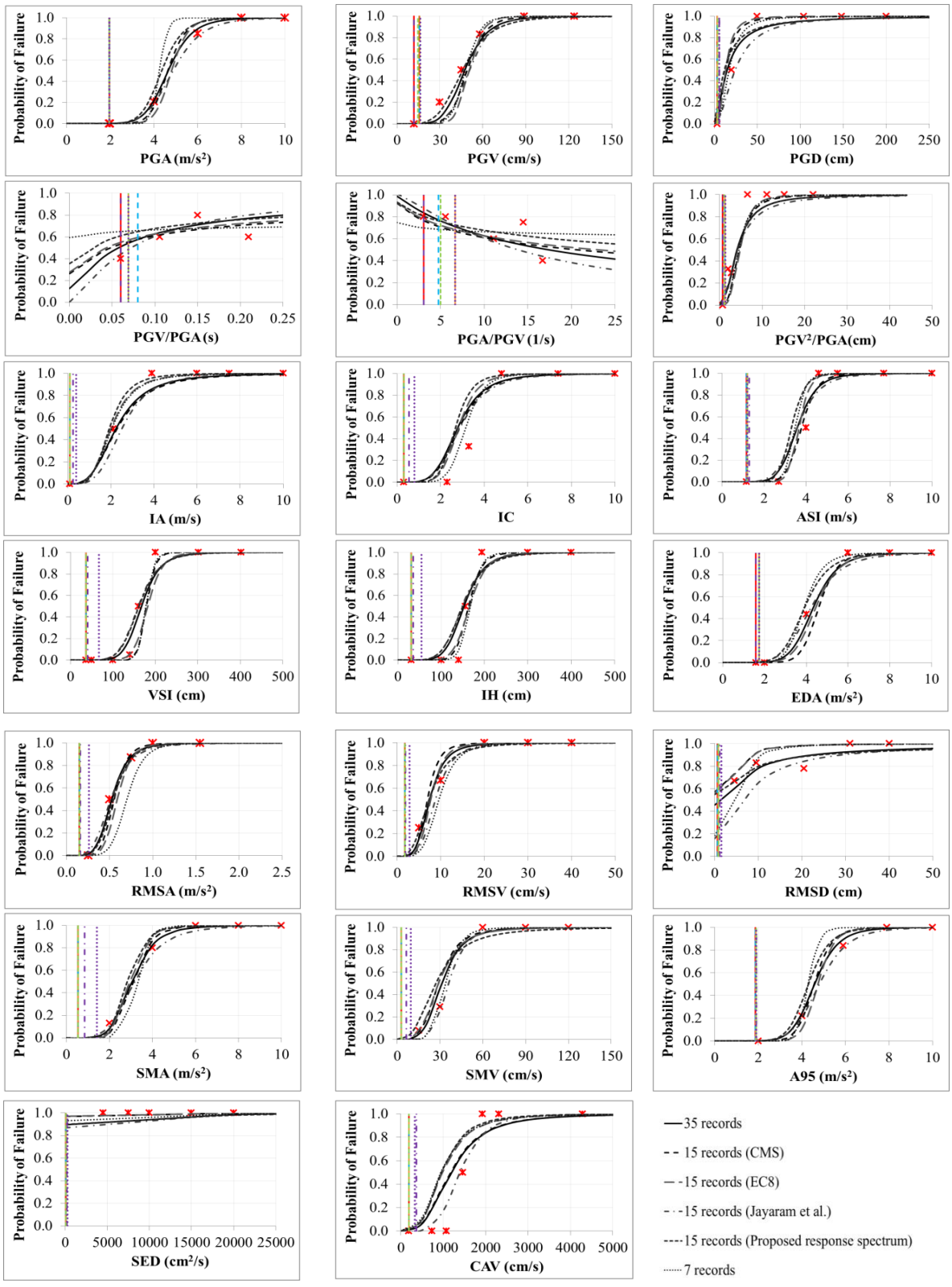


Figure 7.11: Collapse fragility curves for the studied two-storey building in zone A. Vertical dashed lines show the boundary between the damage and no-damage zones; stars show several values of the observed failure probability as examples for each IM.

Table 7.9: -2 Log-likelihood (-2LL) measure used to demonstrate how poorly one IM fits the data for different sets of earthquake records applied to the studied one- and two-storey buildings located in zone A.

IM	Zone A											
	35 records		CMS		EC8		Jayaram et al.		Proposed Response spectrum		7 records	
	One storey	Two storey	One storey	Two storey	One storey	Two storey	One storey	Two storey	One storey	Two storey	One storey	Two storey
PGA	70.4	65.1	27.8	19.2	32.3	14.5	29.7	34.0	32.3	24.2	15.7	6.3
PGV	111.3	82.4	38.6	30.6	39.2	22.3	53.9	32.8	46.5	37.5	19.0	11.4
PGD	175.9	161.8	65.8	64.0	66.9	54.5	77.2	64.5	73.8	64.8	31.9	26.8
PGV/PGA	194.3	207.0	85.3	97.7	78.7	86.2	86.4	87.2	79.2	79.9	39.6	42.0
PGA/PGV	194.3	207.0	85.3	97.7	78.7	86.2	86.4	87.2	79.2	79.9	39.6	42.0
PGV2/PGA	156.4	138.5	61.9	56.8	56.7	46.7	71.7	59.8	64.4	54.7	28.6	24.3
IA	85.2	79.7	32.3	35.5	38.2	27.4	40.7	34.1	32.9	22.9	21.0	13.2
IC	83.5	74.9	25.3	31.6	36.0	22.3	48.5	36.5	29.4	20.9	16.8	9.8
ASI	69.3	69.6	15.4	23.7	24.8	18.0	33.7	37.0	25.1	19.5	11.1	9.0
VSI	87.0	64.2	24.5	13.0	34.8	21.2	48.5	32.2	37.8	28.0	19.0	5.8
IH	96.9	74.5	27.6	21.2	37.8	25.4	53.1	37.2	40.1	31.1	20.5	9.5
EDA	70.8	73.5	30.9	22.1	31.9	28.4	34.1	41.3	32.4	31.4	12.6	13.1
CAV	130.3	126.4	55.4	57.0	55.9	49.9	50.7	45.9	52.4	43.0	31.0	24.8
RMSA	113.3	92.6	27.5	33.0	47.2	30.3	64.7	49.5	44.5	35.3	19.0	14.8
RMSV	135.6	103.7	42.0	42.4	55.4	38.4	66.1	43.3	55.3	47.3	25.9	19.9
RMSD	185.6	174.9	69.0	68.6	71.5	61.2	82.6	71.4	77.1	70.0	32.3	26.7
SMA	95.2	88.6	28.9	30.6	37.0	28.8	53.5	48.7	32.8	27.1	13.2	10.8
SMV	117.6	98.2	39.9	44.6	49.6	43.5	56.8	36.4	52.5	47.4	22.9	15.8
A95	74.6	65.3	29.1	19.6	33.1	15.1	29.3	33.9	32.7	24.6	15.9	6.3
SED	127.5	103.9	35.9	37.9	47.9	35.5	59.2	37.3	53.4	44.4	27.4	20.2

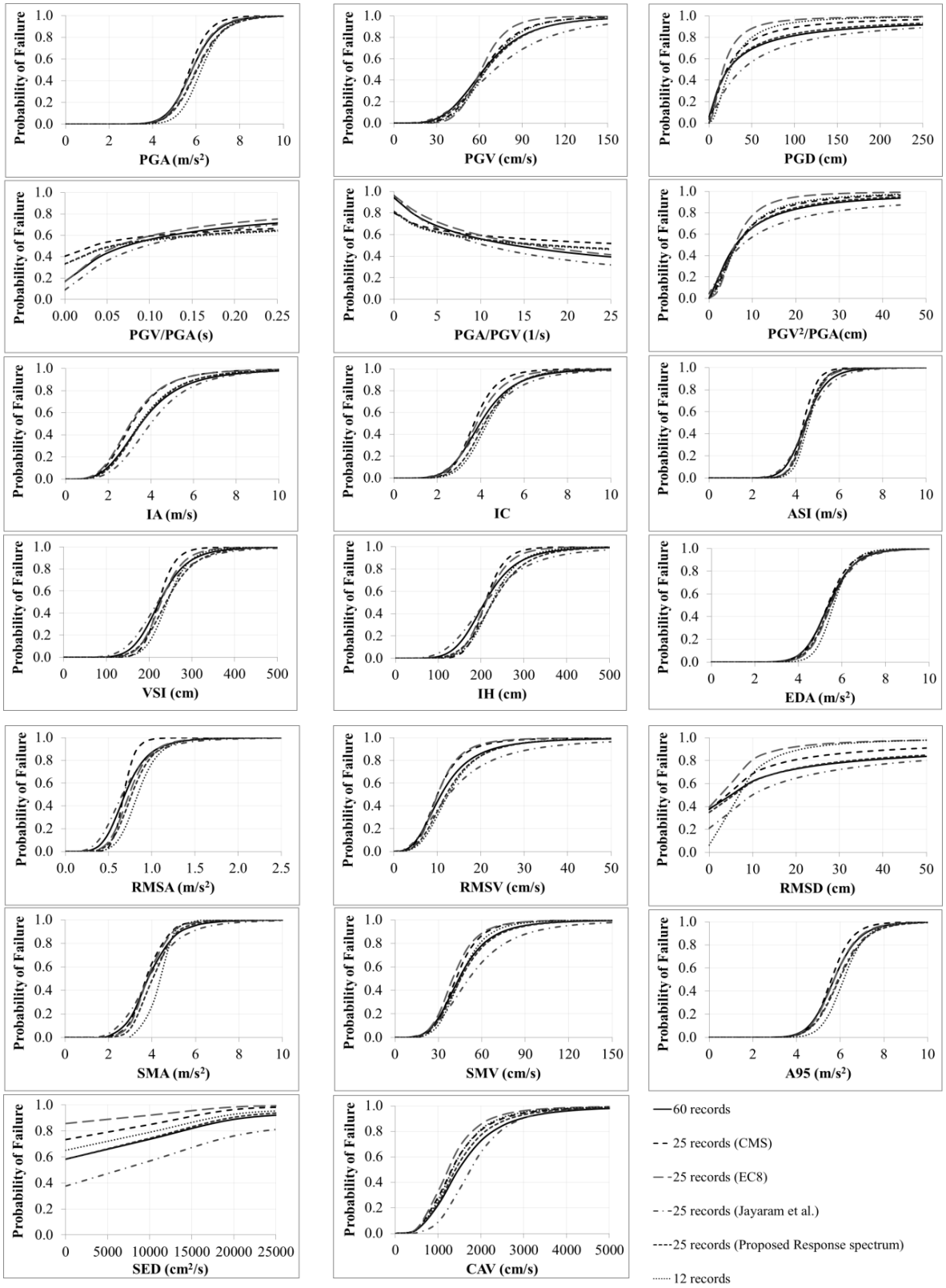


Figure 7.12: Collapse fragility curves for the studied one-storey building using a combination of records from both zones A and B.

Table 7.10: -2 Log-likelihood (-2LL) measure used to demonstrate how poorly one IM fits the data for different sets of earthquake records from both zones A and B applied to the studied one-storey building.

IM	One storey building							
	60 records	25 records-CMS	25 records-EC8	25 records-Jayaram et al.	25 records-Proposed response spectrum	12 records	7 records-zone A	7 records-zone B
PGA	152.1	49.0	55.6	63.2	61.1	25.3	15.7	5.0
PGV	244.2	90.7	74.1	107.9	100.8	44.1	19.0	24.4
PGD	366.3	146.9	129.4	143.9	146.7	63.7	31.9	31.6
PGV/PGA	409.0	170.9	157.4	166.4	163.1	78.0	39.6	38.2
PGA/PGV	409.0	170.9	157.4	166.4	163.1	78.0	39.6	38.2
PGV2/PGA	335.0	135.6	116.8	139.9	135.7	62.9	28.6	33.0
IA	178.1	64.9	65.9	70.8	66.1	34.3	21.0	8.2
IC	173.8	52.2	60.3	80.2	60.9	29.2	16.8	5.9
ASI	141.0	40.1	47.4	71.9	45.9	20.6	11.1	8.9
VSI	179.4	45.7	59.9	86.1	72.5	29.9	19.0	8.7
IH	203.9	55.6	66.1	98.6	81.8	37.6	20.5	16.7
EDA	151.0	55.5	52.7	65.9	56.3	22.6	12.6	20.2
CAV	277.8	113.5	103.5	101.6	108.6	54.0	31.0	14.8
RMSA	232.8	52.1	81.3	111.1	82.7	37.5	19.0	12.1
RMSV	282.9	105.1	99.7	123.0	111.2	52.8	25.9	26.1
RMSD	383.5	154.9	134.7	152.3	151.9	62.9	32.3	30.0
SMA	195.2	63.5	62.5	99.1	63.5	21.5	13.2	6.3
SMV	243.8	87.3	88.5	106.5	100.2	44.4	22.9	21.2
A95	148.9	51.0	56.8	63.8	62.4	25.5	15.9	2.8
SED	277.8	96.5	89.3	117.3	108.4	53.4	27.4	25.9

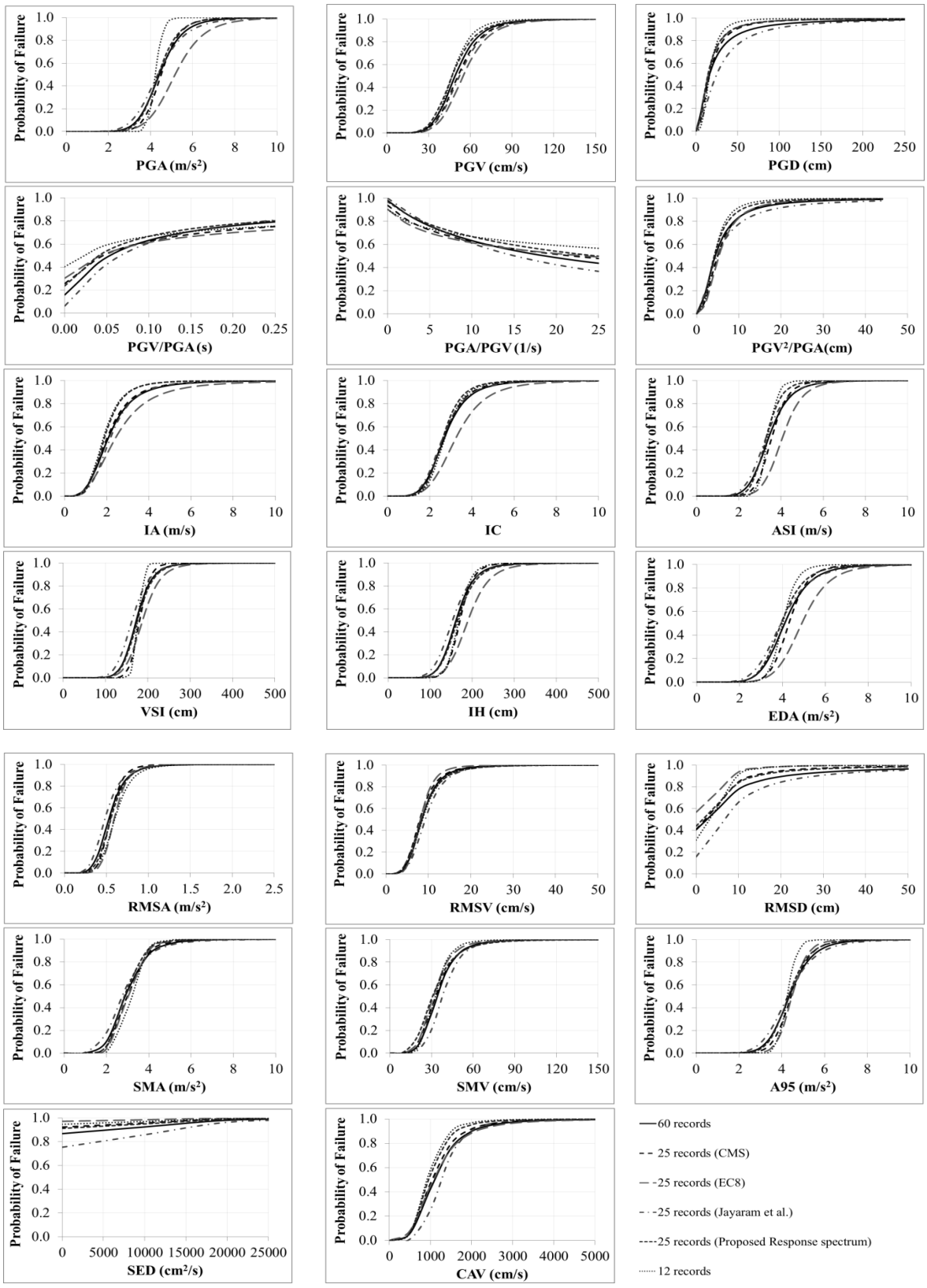


Figure 7.13: Collapse fragility curves for the studied two-storey building using a combination of records from both zones A and B.

Table 7.11: -2 Log-likelihood (-2LL) measure used to demonstrate how poorly one IM fits the data for different sets of earthquake records from both zones A and B applied to the studied two-storey building.

IM	Two storey building							
	60 records	25 records-CMS	25 records-EC8	25 records-Jayaram et al.	25 records-Proposed response spectrum	12 records	7 records-zone A	7 records-zone B
PGA	140.2	45.0	66.9	68.9	49.1	11.6	6.3	5.8
PGV	174.0	73.8	79.8	68.6	69.7	29.5	11.4	17.1
PGD	335.3	130.6	130.6	129.9	122.1	50.7	26.8	23.0
PGV/PGA	433.0	184.4	169.6	172.4	165.5	81.5	42.0	39.4
PGA/PGV	433.0	184.4	169.6	172.4	165.5	81.5	42.0	39.4
PGV2/PGA	300.1	122.9	119.2	122.1	110.6	51.4	24.3	26.5
IA	151.6	62.8	75.9	55.9	47.8	24.5	13.2	10.2
IC	145.9	57.0	70.5	60.3	47.5	23.5	9.8	11.1
ASI	159.3	50.1	61.0	69.2	47.1	16.2	9.0	6.5
VSI	116.4	30.4	60.9	52.7	46.5	8.8	5.8	0.0
IH	136.7	41.1	64.6	63.1	55.7	20.2	9.5	10.6
EDA	162.4	49.6	65.1	75.7	59.9	19.2	13.1	0.0
CAV	246.7	102.8	109.6	84.5	85.7	41.3	24.8	12.5
RMSA	188.9	60.6	58.8	82.7	69.1	33.8	14.8	15.1
RMSV	212.2	91.6	76.6	84.9	89.4	42.5	19.9	21.5
RMSD	354.7	138.4	121.8	137.2	131.3	52.5	26.7	24.9
SMA	180.6	63.5	56.2	87.6	56.5	22.3	10.8	11.0
SMV	188.9	83.8	76.9	69.8	84.1	31.5	15.8	15.4
A95	141.7	46.0	66.5	69.4	50.2	12.0	6.3	0.0
SED	218.4	79.3	72.5	85.9	81.4	38.4	20.2	18.1

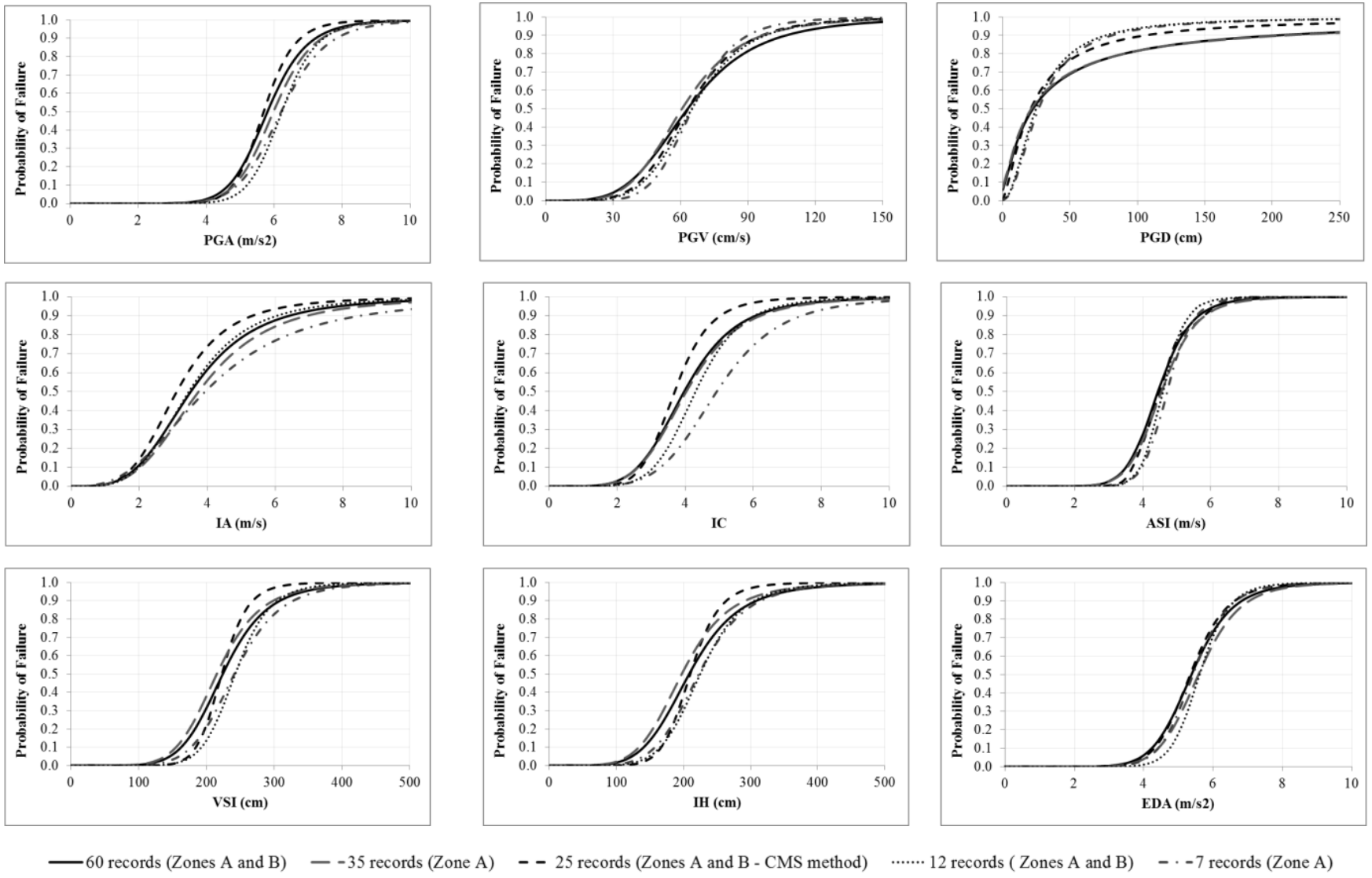


Figure 7.14: Collapse fragility curves for the one-storey building using sets of 60, 35, 25, 12, and 7 records.

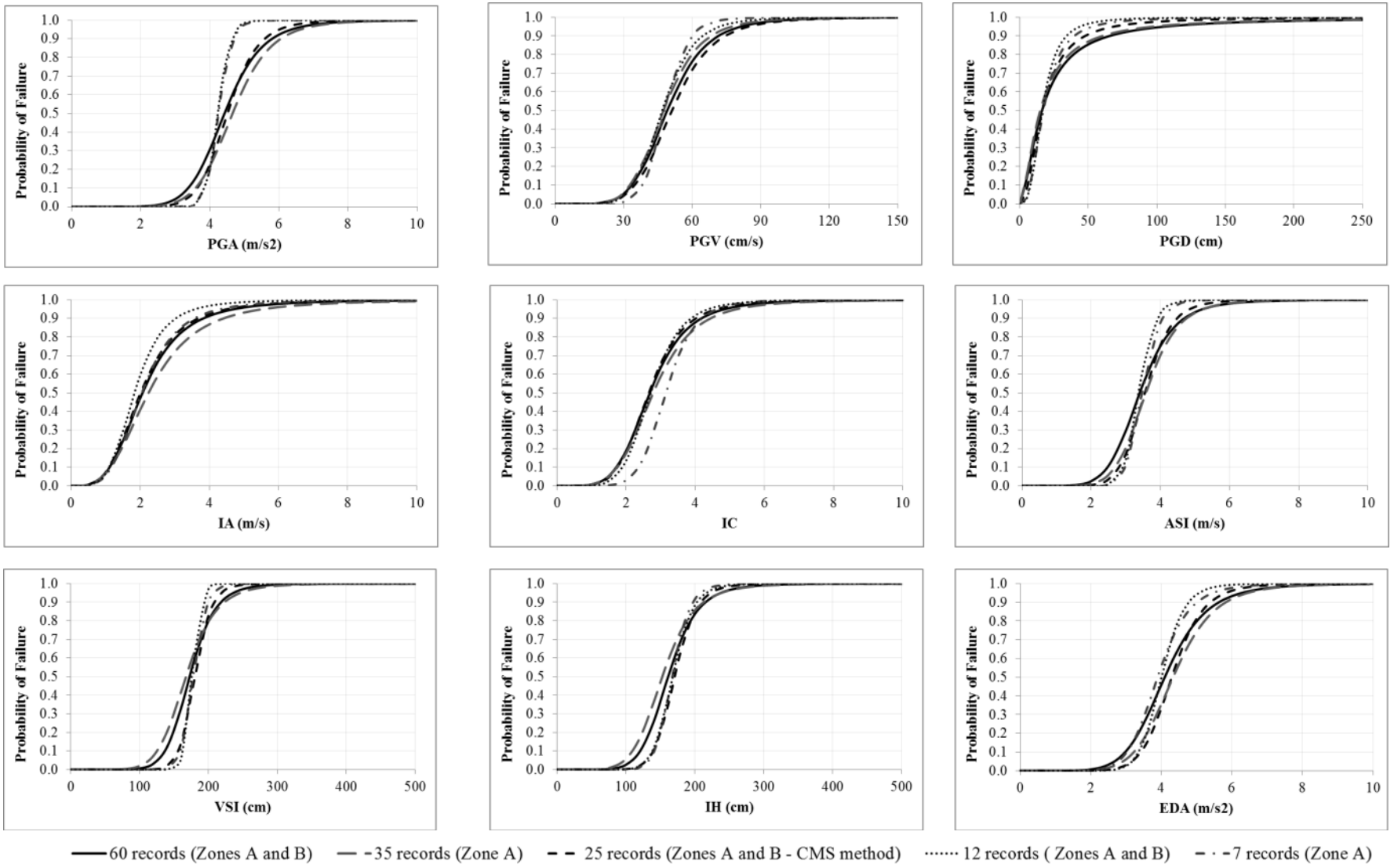


Figure 7.15: Collapse fragility curves for the two-storey building with sets of 60, 35, 25, 12, and 7 records.

When the -2LL values for the 4 GMSM methods used in this study are plotted for the 4 main IMs, including the PGA, EDA, ASI, and VSI, the effect of ground-motion selection method on the optimal IM selection is more pronounced (see Figure 7.16). The aforementioned IMs are selected based on what was discussed in preceding section. Regarding the number of records, variability in the ground-motion selection method may lead to the selection of different IMs for the same building type and the same seismic hazard zone. For example, for zone A and the two-storey building, three different IMs, namely, the VSI, PGA, and ASI, may be recommended according to three different ground-motion selection methods. These methods are the CMS, the EC8 code, and the proposed response spectrum for the KR, especially if 15 records are used for each method. However, when the number of records is increased to 25 records, the effect obtained when selecting a GMSM method disappears; this is especially true for the one-storey building located in zone B. Nevertheless, for the one-storey building in zone A and the two-storey building in zone B, the ground-motion selection method has little impact on the correct selection of an IM.

With respect to the number of storeys, for the same seismic hazard zone and using the same ground-motion selection method, different IMs are selected for each building with a different number of storeys. For example, when the VSI was selected as the best IM for the two-storey building located in zone B, three different IMs (i.e., the PGA, EDA, and ASI) were identified for the same zone except for the one-storey building (based on the selected GMSM method).

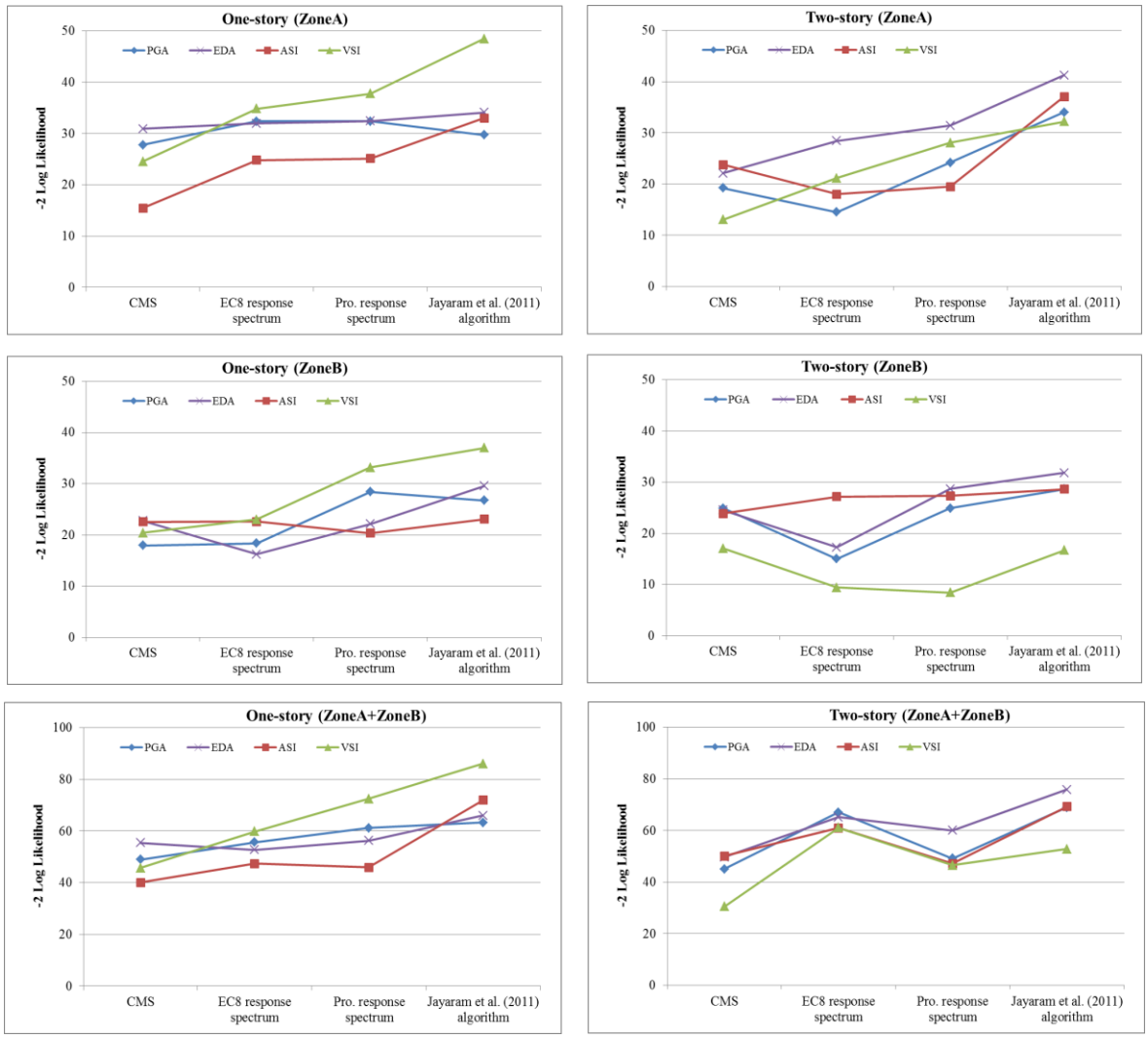


Figure 7.16: Effects of variations in the GMSM methods on the selection of IMs with respect to different numbers of storeys, different seismic hazard zones, and different numbers of records.

7.1.4 Scaled vs. unscaled records

A comparison of collapse fragility curves derived for the studied one- and two-storey buildings based on scaled (IDA) and unscaled records and considering 60 (zones A and B), 35 (zone A) and 38 (zone B) ground-motion records, clearly indicates that: The two are virtually the same in performance once the same IM is used by the both two methods (i.e., scaled and unscaled). Furthermore, Figures 7.17 and 7.18 demonstrate that the scaling of records does not bias the results; however, the correct selection of an IM should have a substantial effect on the obtained outcomes. This is true even if a large (60 records) or small (35 or 38 records) number of records is used.

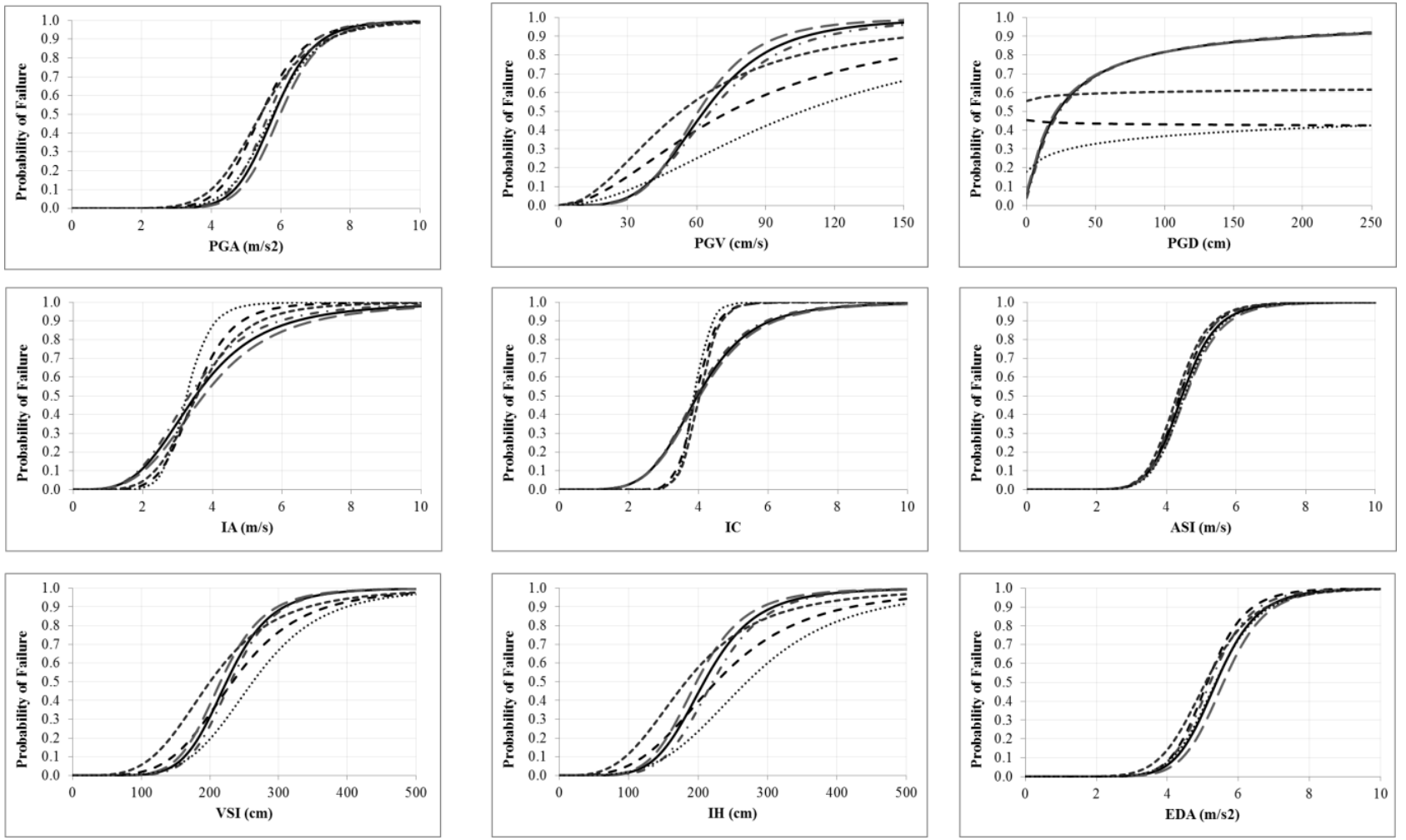


Figure 7.17: Comparison of the fragility curves derived for the studied one-storey building based on scaled (IDA) and unscaled records with 60 (zones A and B), 35 (zone A), and 38 (zone B) ground-motion records.

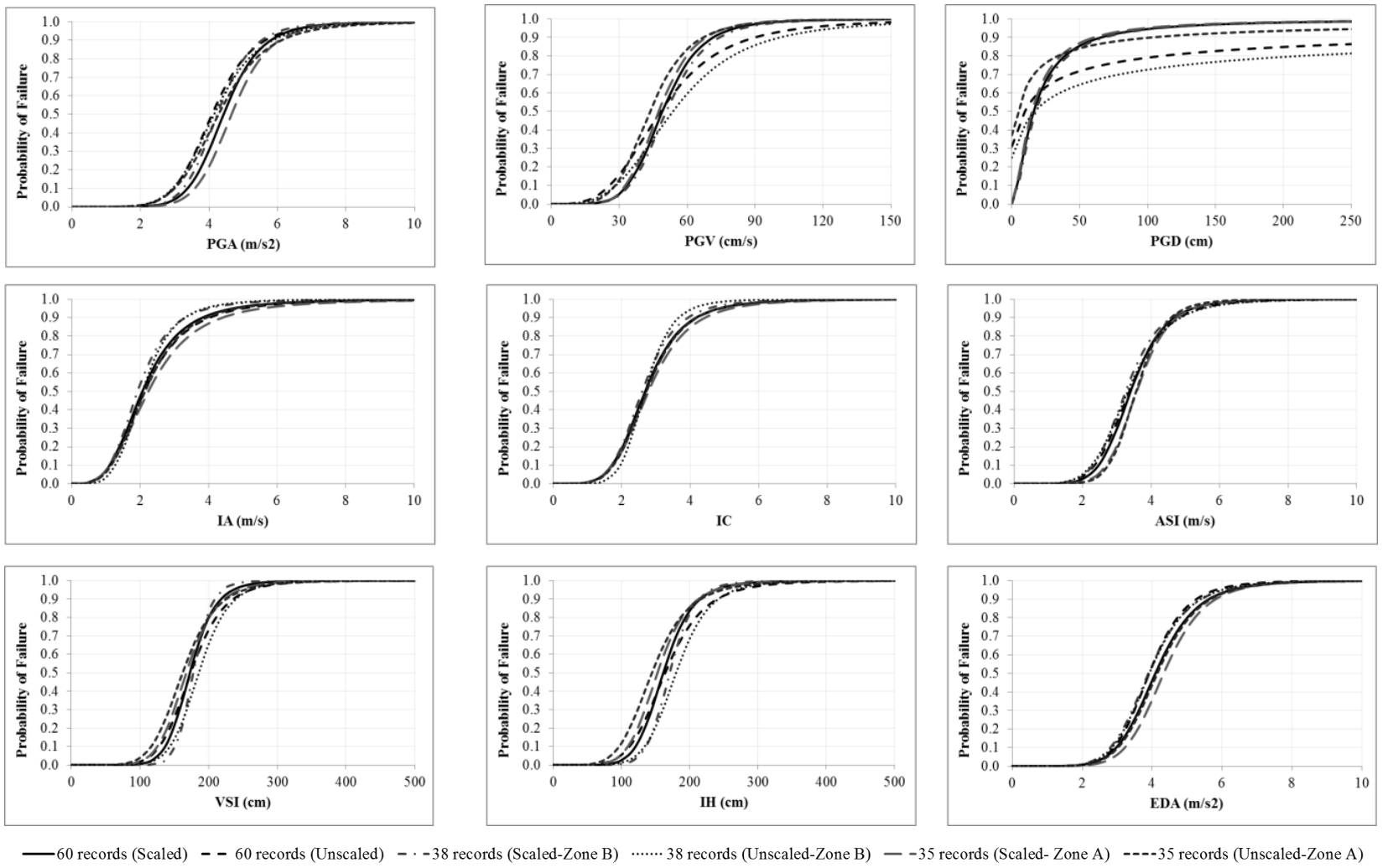


Figure 7.18: Comparison of the fragility curves derived for the studied two-storey building based on scaled (IDA) and unscaled records with 60 (zones A and B), 35 (zone A), and 38 (zone B) ground-motion records.

To this end and after verifying the efficiency of the two best selected ground-motion parameters, i.e., the ASI and VSI, the sufficiency of them is also essential to be determined in this stage. For that purpose, their dependence on the other ground-motion parameters such as the magnitude (M_x) and source distance (e.g., EpiD) is examined by analysing the relationship between them.

For that purpose, the residuals of the responses conditioned on the ASI or VSI are calculated via the arithmetic difference between the observed, and the predicted responses from a regression procedure. Then, the dependency of these residuals on the ground-motion magnitude and its source distance is calculated using the p-value of an F-test as a numerical measure of this independence. The p-value is the estimated probability of rejecting the null hypothesis when the hypothesis is viewed as true. The p-value is usually utilised to evaluate the strength of the evidence against the null hypothesis: a small p-value (typically ≤ 0.05) indicates strong evidence against the null hypothesis, leading to rejection of the null hypothesis; a large p-value (> 0.05) is indicative of weak evidence against the null hypothesis, leading to acceptance of the null hypothesis.

The estimated p-values for the ASI–M_x and ASI-EpiD relationships are 0.485 and 0.195 for the one-storey building and 0.414 and 0.892 for the two-storey building, respectively. For the VSI-M_x and VSI-EpiD relationships, the p-values are 0.61 and 0.403 for the one-storey building and 0.521 and 0.226 for the two-storey building, respectively. Because the p-values are greater than 0.05, there is no statistically significant correlation between the IMs and M_x or EpiD. As a result, the sufficiency of the selected parameters, i.e., the ASI and VSI, is verified.

Tables A2-A9 (given in the Appendix) provide the statistical test results obtained when checking the sufficiency of the ASI- and VSI-based displacement responses with respect to the earthquake moment magnitude (M_x) and source distance (e.g., EpiD) for the 60 unscaled earthquake records.

7.1.5 Wrapper feature selection method

The feasibility of detecting the most accurate ground-motion IM for different sets of earthquake records is determined via a wrapper feature subset selection method. The same 20 IMs from cluster 1 that were selected in section 7.1.2 (shown in Table 7.7) and after reducing the set of 36 IMs defined for each earthquake record using the VARCLUS procedure are used for this purpose. Sets of 7, 15 (related to the CMS and EC8 methods), and 35 or 38 records are derived for the one- and two-storey buildings located in seismic hazard zones A and B; the set of 60 records for each building is also used. The required datasets are obtained by using IDA and applying the aforementioned sets of records to the one- and two-storey URM buildings.

Several subsets of IMs are defined and scored based on a predefined predictive algorithm to train a model based on the wrapper method. By recording the model's error rate for each subset, the subsets are scored, and the optimal subset is selected. Using the Weka 3.6.1 software, the WrapperSubsetEval tool is used as an attribute evaluator to estimate attribute sets via a learning scheme. Cross-validation is used to assess the learning scheme's accuracy for a set of attributes and to avoid over-fitting issues related to the process. The available options for this evaluator are as follows: a classifier (7 different classifiers are used in this study) for estimating the accuracy of subsets, the number of "xval" (cross-validation) folds (5 are used in this study) for estimating subset accuracy, a seed (1 is used in this study) for randomly generating xval splits, and a threshold (0.01 is used in this study) to repeat xval, specifically if the standard deviation of the mean exceeds this value. The seven selected classifiers, namely, the Naive Bayes (NB), nearest neighbour, logistic, multilayer perceptron neural network (MLP), decision table, and decision tree algorithms, were briefly discussed in section 3.4.1.

The optimum feature subsets identified using the wrapper feature selection method are summarised in Tables 7.12-7.19 for the dataset relevant to zone B. For most algorithms, high accuracy is achieved during data classification. Normally, 20 IMs are reduced to subsets containing between 1 and 6 IMs for the different algorithms and sets of records. The highest error rate is approximately 7.0%, and the lowest error rate is 0%. The nearest neighbour method (1-NN) is observed to be the best classifier, with only 0.0 to 4.0% error based on the selected sets of records. Moreover, the NB and 3-NN algorithms also provide good performance in comparison to other classifier types. However, the

higher precision of the 1-NN method indicates that this algorithm provides better results compared to the other classifiers.

Table 7.12: Results of the classifier analysis using the optimum feature subsets chosen for each classifier and the one-storey building dataset (38 records for zone B); the wrapper method is used to obtain these values.

Classifiers	Feature set	TP rate	FP rate	Precision	ROC area	Error rate
1-NN	RMSA, VSI, IH, ASI, and A95	0.957	0.056	0.959	0.95	4.1
Trees (Simple CART)	RMSV, EDA, and A95	0.957	0.062	0.957	0.912	4.3
Logistic	PGA, PGA/PGV, and ASI	0.945	0.064	0.945	0.979	5.5
MLP	PGA, VSI, SMA, and EDA	0.939	0.074	0.939	0.966	6.1
3-NN	RMSV, IA, and EDA	0.951	0.054	0.951	0.943	4.9
NB	PGA/PGV, ASI, VSI, SMA, and A95	0.933	0.06	0.936	0.98	6.7
Decision table	PGA, RMSV, IC, and EDA	0.945	0.082	0.948	0.949	5.5

Table 7.13: Results of the classifier analysis using the optimum feature subsets chosen for each classifier and the one-storey building dataset (15 records from the CMS method for zone B); the wrapper method is used to obtain these values.

Classifiers	Feature set	TP rate	FP rate	Precision	ROC area	Error rate
1-NN	PGA, PGV, PGD, RMSA, and ASI	0.969	0.034	0.969	0.967	3.12
Trees (Simple CART)	RMSA	0.953	0.059	0.953	0.899	4.7
Logistic	PGA and VSI	0.969	0.034	0.969	0.989	3.12
MLP	PGA, RMSA, RMSD, IA, and IC	0.953	0.044	0.954	0.984	4.7
3-NN	PGA, PGD, PGA/PGV, ASI, and IH	0.984	0.024	0.985	0.987	1.6
NB	PGA, ASI, and VSI	0.953	0.059	0.953	0.993	4.7
Decision table	RMSA	0.953	0.059	0.953	0.899	4.7

Table 7.14: Results of the classifier analysis using the optimum feature subsets chosen for each classifier and the one-storey building dataset (15 records from the EC8 method for zone B); the wrapper method is used to obtain these values.

Classifiers	Feature set	TP rate	FP rate	Precision	ROC area	Error rate
1-NN	RMSV and EDA	1.0	0.0	1.0	1.0	0.0
Trees (Simple CART)	EDA	0.967	0.017	0.97	0.955	3.3
Logistic	PGA, PGA/PGV, ASI, and A95	0.951	0.071	0.951	0.969	4.92
MLP	VSI and A95	0.951	0.094	0.954	0.927	4.92
3-NN	RMSA and ASI	0.951	0.048	0.952	0.938	4.92
NB	PGV, ASI, SMA, and EDA	0.967	0.017	0.97	0.995	3.3
Decision table	EDA	0.967	0.017	0.97	0.964	3.3

Table 7.15: Results of the classifier analysis using the optimum feature subsets chosen for each classifier and the one-storey building dataset (7 records for zone B); the wrapper method is used to obtain these values.

Classifiers	Feature set	TP rate	FP rate	Precision	ROC area	Error rate
1-NN	PGA and ASI	0.964	0.048	0.966	0.958	3.57
Trees (Simple CART)	ASI	0.964	0.048	0.966	0.919	3.57
Logistic	PGA, RMSA, ASI, and VSI	1.0	0.0	1.0	1.0	0.0
MLP	RMSA, ASI, VSI, and EDA	1.0	0.0	1.0	1.0	0.0
3-NN	RMSA, RMSV, ASI, and VSI	1.0	0.0	1.0	1.0	0.0
NB	RMSA, ASI, VSI, and EDA	1.0	0.0	1.0	1.0	0.0
Decision table	ASI	0.964	0.048	0.966	0.964	3.57

Table 7.16: Results of the classifier analysis using the optimum feature subsets chosen for each classifier and the two-storey building dataset (38 records for zone B); the wrapper method is used to obtain these values.

Classifiers	Feature set	TP rate	FP rate	Precision	ROC area	Error rate
1-NN	IC, CAV, and VSI	0.973	0.022	0.974	0.976	2.7
Trees (Simple CART)	VSI	0.946	0.062	0.947	0.922	5.4
Logistic	VSI and EDA	0.962	0.045	0.963	0.99	3.8
MLP	PGA and VSI	0.957	0.057	0.0957	0.993	4.3
3-NN	SED, VSI, EDA, and A95	0.957	0.067	0.957	0.969	4.3
NB	PGA/PGV, IA, ASI, VSI, IH, and EDA	0.968	0.024	0.969	0.992	3.2
Decision table	RMSD, IA, VSI, and SMA	0.946	0.081	0.946	0.987	5.4

Table 7.17: Results of the classifier analysis using the optimum feature subsets chosen for each classifier and the two-storey building dataset (15 records from the CMS method for zone B); the wrapper method is used to obtain these values.

Classifiers	Feature set	TP rate	FP rate	Precision	ROC area	Error rate
1-NN	VSI and SMV	1.0	0.0	1.0	1.0	0.0
Trees (Simple CART)	PGA	0.929	0.057	0.934	0.898	7.14
Logistic	PGA, VSI, and A95	0.943	0.075	0.943	0.929	5.7
MLP	IA, IC, VSI, and IH	0.929	0.057	0.934	0.989	7.14
3-NN	VSI, EDA, and A95	0.971	0.038	0.971	0.962	2.85
NB	PGA, IA, CAV, VSI, and SMA	0.957	0.044	0.958	0.986	4.28
Decision table	PGA	0.929	0.057	0.934	0.887	7.14

Table 7.18: Results of the classifier analysis using the optimum feature subsets chosen for each classifier and the two-storey building dataset (15 records from the EC8 method for zone B); the wrapper method is used to obtain these values.

Classifiers	Feature set	TP rate	FP rate	Precision	ROC area	Error rate
1-NN	PGA and VSI	0.986	0.029	0.986	0.978	1.4
Trees (Simple CART)	VSI	0.972	0.036	0.972	0.959	2.81
Logistic	RMSV and VSI	0.958	0.066	0.958	0.98	4.22
MLP	RMSA, VSI, and EDA	0.986	0.007	0.986	0.995	1.4
3-NN	VSI	0.972	0.036	0.972	0.986	2.81
NB	PGA, RMSV, and VSI	0.986	0.007	0.987	0.997	1.4
Decision table	VSI	0.972	0.036	0.972	0.943	2.81

Table 7.19: Results of the classifier analysis using the optimum feature subsets chosen for each classifier and the two-storey building dataset (7 records for zone B); the wrapper method is used to obtain these values.

Classifiers	Feature set	TP rate	FP rate	Precision	ROC area	Error rate
1-NN	VSI	0.969	0.014	0.972	0.977	3.12
Trees (Simple CART)	PGA	0.969	0.014	0.972	0.957	3.12
Logistic	VSI	0.969	0.014	0.972	1.0	3.12
MLP	PGA	0.969	0.014	0.972	0.955	3.12
3-NN	PGA	0.969	0.014	0.972	0.961	3.12
NB	VSI	0.969	0.014	0.972	1.0	3.12
Decision table	PGA	0.969	0.014	0.972	0.966	3.12

Because the goal is to find a single feature (IM) for deriving the fragility curves, the subset with the minimum error rate of each dataset is selected; receiver operating characteristics (ROC) analysis is then used to determine which of the included IMs has the greatest effect on the structural response. ROC analysis provides a tool for selecting potentially optimal models and to discard suboptimal models independently from the class distribution. It is a primary tool for diagnostic test assessment. Based on the ROC curves obtained from the ROC analysis, the satisfactory IMs are those that are as close as possible to the (0,1) corner. Alternatively, the IMs can be analysed by measuring the area under the ROC curve, which is commonly known as the AUC indicator. An AUC that is close to

unity indicates an efficient model. The ROC curves show the true positive (TP) rates (sensitivity) versus the false positive (FP) rates (1-specificity) for a given classification procedure.

For example, the ROC analysis for the one-storey building in zone B with a set of 38 earthquake records is shown in Figure 7.19 and indicates that the ASI is the closest parameter to the (0,1) corner and has an AUC of 0.964 (see Table 7.20). Therefore, the ASI produces better results compared to the other parameters, i.e., the VSI, IH, A95, and RMSA, all of which are selected according to the 1-NN algorithm. For the same scenario but involving the two-storey building, the ROC curve (shown in Figure 7.20) suggests that the VSI is the closest parameter to the (0,1) corner. Furthermore, as shown in Tables 7.20 and 7.21, the ASI and VSI have the highest AUC values for both the one- and two-storey buildings in the greatest number of cases.

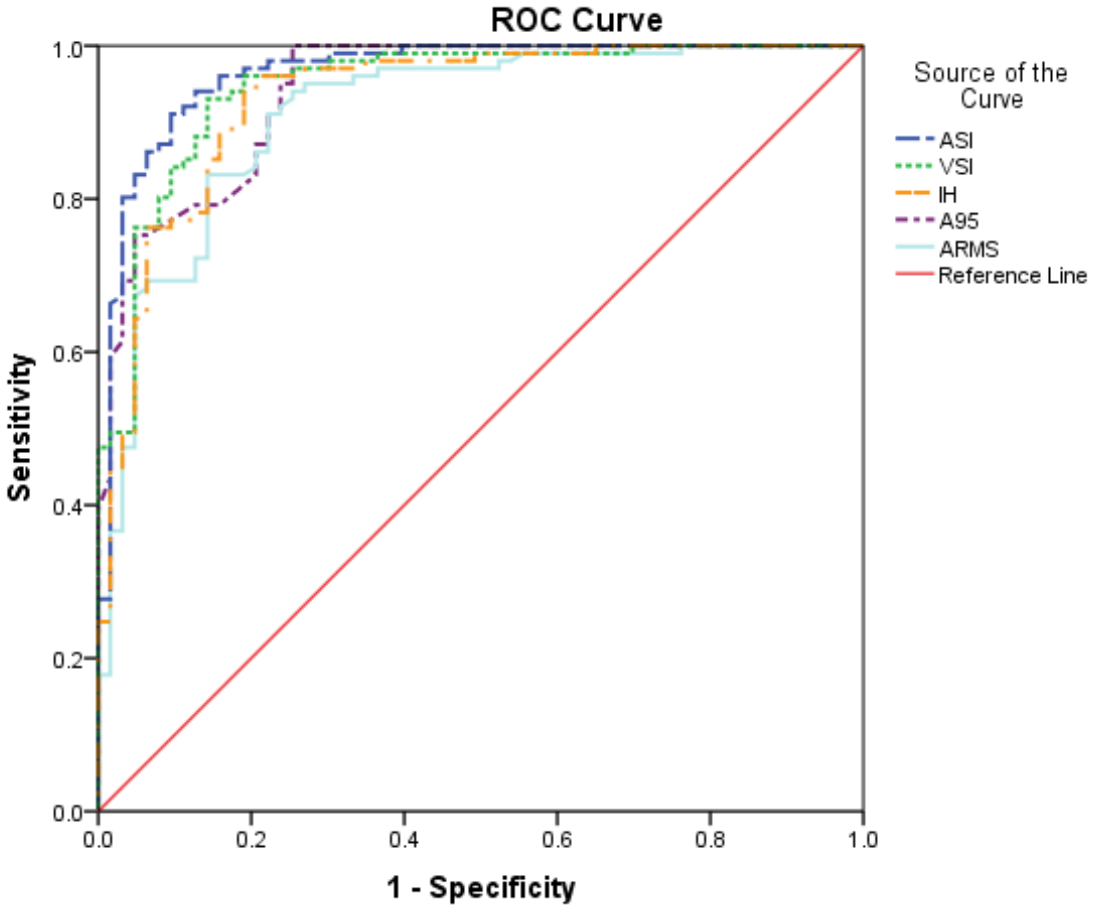


Figure 7.19: ROC analysis of the variables (IMs) selected according to the 1-NN algorithm using the wrapper method for the one-storey building located in zone B and for a set of 38 records.

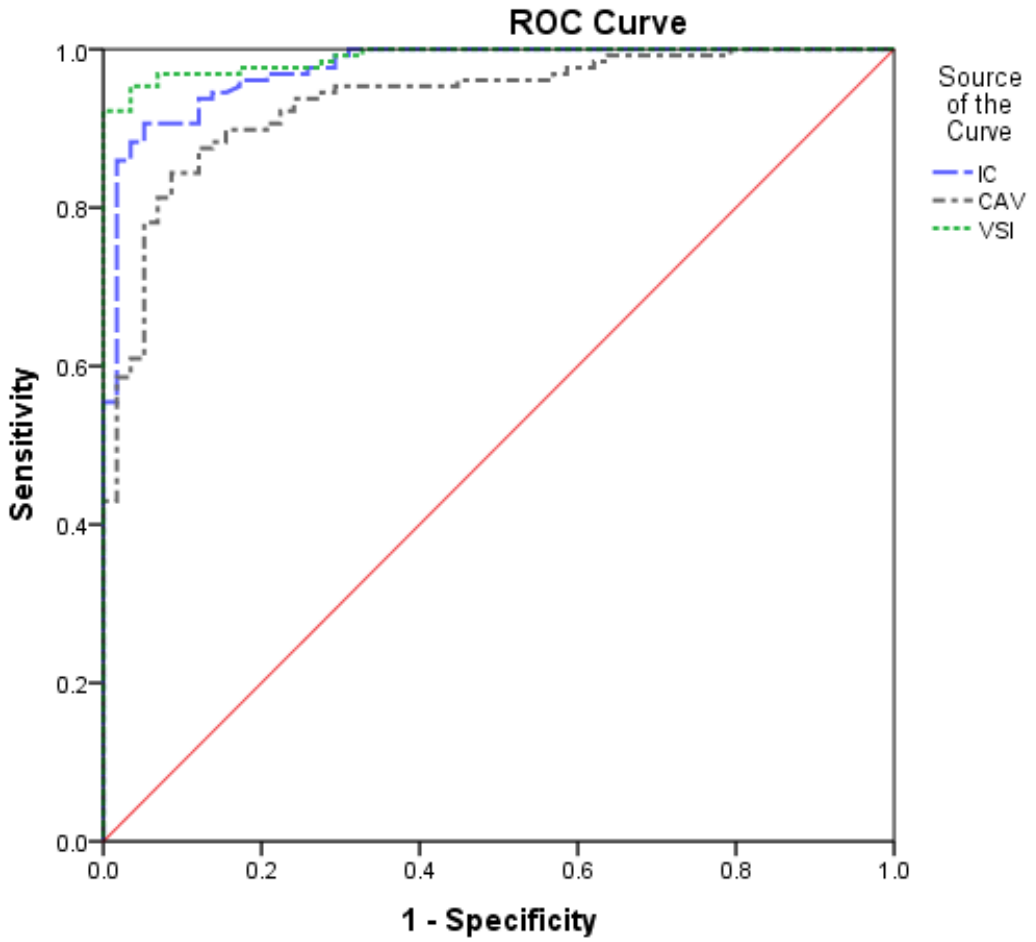


Figure 7.20: ROC analysis of the variables (IMs) selected according to the 1-NN algorithm using the wrapper method for the two-storey building located in zone B and for a set of 38 records.

Table 7.20: Area under the ROC curve [AUC] for the variables selected according to the nearest neighbour algorithm using the wrapper method for the one-storey building located in zone B and various sets of records.

	ASI	VSI	IH	A95	PGA	PGA/PGV	PGD	RMSA	RMSV	EDA
38 records (1-NN)	0.964	0.947	0.931	0.941				0.912		
15 records - CMS (3-NN)	0.982		0.974		0.975	0.474	0.759			
15 records - EC8 (1-NN)									0.913	0.990
7 records (3-NN)	0.99	0.979						0.969	0.854	

Table 7.21: Area under the ROC curve [AUC] for variables selected by the nearest neighbour algorithm using the wrapper method for the two-storey building located in zone B and various sets of records

	VSI	IC	CAV	SMV	PGA
38 records (1-NN)	0.989	0.975	0.934		
15 records - CMS(1-NN)	0.989			0.954	
15 records - EC8 (1-NN)	0.996				0.984
7 records (1-NN)	1.0				

In the second test, the datasets obtained from the time history analysis of the one- and two-storey buildings located in zone A are utilised. With the help of several sets of earthquake records relevant to zone A and for the same 20 IMs selected in the first test, the wrapper subset selection procedure is applied to the aforementioned datasets. The results of the evaluation of the classifiers for determining the optimum feature subsets are listed in Tables 7.22-7.29. The 1-NN, 3-NN and NB classifiers exhibit the best performance among the considered classifiers based on the selected sets of records. The 20 IMs are reduced to a subset with a size ranging between 1 and 7 IMs. Similar to the first test for zone B, the ROC analysis is conducted for the subsets defined by classifiers with the lowest error rate; as a result, the optimal IM is selected. The results of the ROC analysis are presented in Tables 7.30 and 7.31, which show that the ASI and VSI are the optimal IMs for the one- and two-storey buildings, respectively, because they have the highest AUC values in most cases.

Table 7.22: Results of the classifier analysis using the optimum feature subsets chosen for each classifier and the one-storey building dataset (35 records for zone A); the wrapper method is used to obtain these values.

Classifiers	Feature set	TP rate	FP rate	Precision	ROC area	Error rate
1-NN	RMSV and EDA	0.944	0.07	0.947	0.937	5.55
Trees (Simple CART)	EDA	0.896	0.118	0.897	0.879	10.42
Logistic	RMSV, ASI, IH, SMA, SMV, and EDA	0.944	0.058	0.944	0.976	5.55
MLP	IA, SED, ASI, VSI, IH, and SMA	0.938	0.067	0.937	0.966	6.25
3-NN	PGA, IA, IC, and ASI	0.951	0.049	0.952	0.938	4.86
NB	PGA, PGA/PGV, ASI, and VSI	0.938	0.063	0.938	0.977	6.25
Decision table	IC, ASI, and EDA	0.903	0.101	0.903	0.946	9.72

Table 7.23: Results of the classifier analysis using the optimum feature subsets chosen for each classifier and the one-storey building dataset (15 records from the CMS method for zone A); the wrapper method is used to obtain these values.

Classifiers	Feature set	TP rate	FP rate	Precision	ROC area	Error rate
1-NN	RMSA, ASI, and VSI	0.984	0.024	0.985	0.98	1.56
Trees (Simple CART)	ASI	0.969	0.034	0.969	0.938	3.125
Logistic	ASI	0.969	0.034	0.969	0.987	3.125
MLP	ASI and SMA	0.969	0.02	0.971	0.982	3.125
3-NN	PGA, IC, and ASI	0.953	0.044	0.954	0.969	4.68
NB	ASI	0.938	0.054	0.94	0.989	6.25
Decision table	ASI	0.969	0.034	0.969	0.94	3.125

Table 7.24: Results of the classifier analysis using the optimum feature subsets chosen for each classifier and the one-storey building dataset (15 records from the EC8 method for zone A); the wrapper method is used to obtain these values.

Classifiers	Feature set	TP rate	FP rate	Precision	ROC area	Error rate
1-NN	EDA	0.879	0.13	0.879	0.875	12.07
Trees (Simple CART)	PGV, PGD, RMSA, and ASI	0.879	0.13	0.879	0.868	12.07
Logistic	ASI and IH	0.914	0.085	0.915	0.959	8.62
MLP	RMSA, ASI, and SMA	0.897	0.107	0.897	0.924	10.35
3-NN	RMSA, ASI, and VSI	0.931	0.072	0.931	0.94	6.9
NB	ASI and VSI	0.931	0.072	0.931	0.965	6.9
Decision table	ASI	0.879	0.13	0.879	0.865	12.07

Table 7.25: Results of the classifier analysis using the optimum feature subsets chosen for each classifier and the one-storey building dataset (7 records for zone A); the wrapper method is used to obtain these values.

Classifiers	Feature set	TP rate	FP rate	Precision	ROC area	Error rate
1-NN	ASI and EDA	0.966	0.042	0.968	0.962	3.45
Trees (Simple CART)	ASI	0.931	0.085	0.939	0.861	6.89
Logistic	ASI and EDA	0.931	0.07	0.931	0.931	6.89
MLP	SED and ASI	0.931	0.085	0.939	0.913	6.89
3-NN	ASI	0.931	0.085	0.939	0.928	6.89
NB	ASI	0.897	0.098	0.899	0.962	10.35
Decision table	ASI	0.931	0.085	0.939	0.875	6.89

Table 7.26: Results of the classifier analysis using the optimum feature subsets chosen for each classifier and the two-storey building dataset (35 records for zone A); the wrapper method is used to obtain these values.

Classifiers	Feature set	TP rate	FP rate	Precision	ROC area	Error rate
1-NN	PGA, IC, SED, and EDA	0.944	0.061	0.945	0.941	5.55
Trees (Simple CART)	A95	0.926	0.079	0.927	0.883	7.4
Logistic	PGA, IC, IH, and A95	0.951	0.058	0.951	0.984	4.95
MLP	PGA, IH, and A95	0.944	0.061	0.945	0.975	5.55
3-NN	PGA, PGV, PGD, and IH	0.944	0.054	0.946	0.961	5.55
NB	PGA, RMSA, VSI, and IH	0.951	0.05	0.951	0.986	4.93
Decision table	RMSV, SMV, and A95	0.938	0.057	0.94	0.968	6.17

Table 7.27: Results of the classifier analysis using the optimum feature subsets chosen for each classifier and the two-storey building dataset (15 records from the CMS method for zone A); the wrapper method is used to obtain these values.

Classifiers	Feature set	TP rate	FP rate	Precision	ROC area	Error rate
1-NN	ASI and VSI	0.986	0.008	0.987	0.989	1.35
Trees (Simple CART)	VSI	0.973	0.016	0.975	0.97	2.7
Logistic	VSI	0.986	0.008	0.987	0.979	1.35
MLP	RMSA, SED, and VSI	0.986	0.008	0.987	0.993	1.35
3-NN	IC and VSI	0.986	0.008	0.987	0.983	1.35
NB	VSI	0.986	0.008	0.987	0.992	1.35
Decision table	VSI	0.973	0.016	0.975	0.97	2.7

Table 7.28: Results of the classifier analysis using the optimum feature subsets chosen for each classifier and the two-storey building dataset (15 records from the EC8 method for zone A); the wrapper method is used to obtain these values.

Classifiers	Feature set	TP rate	FP rate	Precision	ROC area	Error rate
1-NN	PGA, PGV/PGA, RMSD, SMA, and SMV	1.0	0.0	1.0	1.0	0.0
Trees (Simple CART)	A95	0.97	0.035	0.97	0.938	3.03
Logistic	PGA and SED	0.955	0.044	0.955	0.994	4.5
MLP	PGA, IH, and SMA	0.97	0.035	0.97	0.993	3.03
3-NN	PGA, PGV, PGD, PGV/PGA, RMSA, RMSD, and SMA	1.0	0.0	1.0	1.0	0.0
NB	PGA, RMSV, and VSI	0.985	0.009	0.985	0.994	1.5
Decision table	A95	0.97	0.035	0.97	0.938	3.03

Table 7.29: Results of the classifier analysis using the optimum feature subsets chosen for each classifier and the two-storey building dataset (7 records for zone A); the wrapper method is used to obtain these values.

Classifiers	Feature set	TP rate	FP rate	Precision	ROC area	Error rate
1-NN	IA and A95	0.97	0.015	0.972	0.977	3.03
Trees (Simple CART)	PGA	0.97	0.015	0.972	0.957	3.03
Logistic	PGA	0.97	0.015	0.972	0.959	3.03
MLP	PGA	0.97	0.015	0.972	0.959	3.03
3-NN	PGA	0.97	0.015	0.972	0.963	3.03
NB	PGA, IA, and VSI	1.0	0.0	1.0	1.0	0.0
Decision table	PGA	0.97	0.015	0.972	0.961	3.03

Table 7.30: Area under the ROC curve [AUC] for the variables selected according to the nearest neighbour algorithm using the wrapper method for the one-storey building located in zone A and various sets of records.

	ASI	VSI	IA	IC	PGA	RMSA	EDA
35 records (3-NN)	0.958		0.944	0.946	0.952		
15 records - CMS (1-NN)	0.991	0.977				0.967	
15 records - EC8 (3-NN)	0.97	0.941				0.882	
7 records (1-NN)	0.971						0.962

Table 7.31: Area under the ROC curve [AUC] for the variables selected according to the nearest neighbour algorithm and Naive Bayes using the wrapper method for the two-storey building located in zone A and various sets of records.

	VSI	IH	ASI	RMSA	PGA	RMSD	SMA	SMV	IA	PGV/PGA
35 records (NB)	0.972	0.965		0.941	0.966					
15 records - CMS (1-NN)	0.992		0.983							
15 records - EC8 (1-NN)					0.987	0.834	0.969	0.943		0.544
7 records (NB)	0.992				0.979				0.975	

Based on the wrapper feature selection, the identified optimum feature subsets are summarised in Tables 7.32 and 7.33 for the third dataset (using the set of 60 records). Again, for most of the algorithms, high accuracy is achieved in the data classification, and the nearest neighbour method (1-NN) is found to be the best classifier, with only 2% error for the one-storey building and 3% error for the two-storey building. The average rate of incorrectly assigned records for all classifiers is approximately 5 and 4.0% for the one- and two-storey buildings, respectively. Between two and seven features are selected by the different classifiers. Thus, to obtain a single feature (IM) for deriving the fragility curves, although it is possible to plot the ROC curve for the subset selected by the classifier with the minimum error rate (namely, the 1-NN subset), as performed for the first and second tests above, the frequency at which each feature is selected in all of the wrapper evaluation

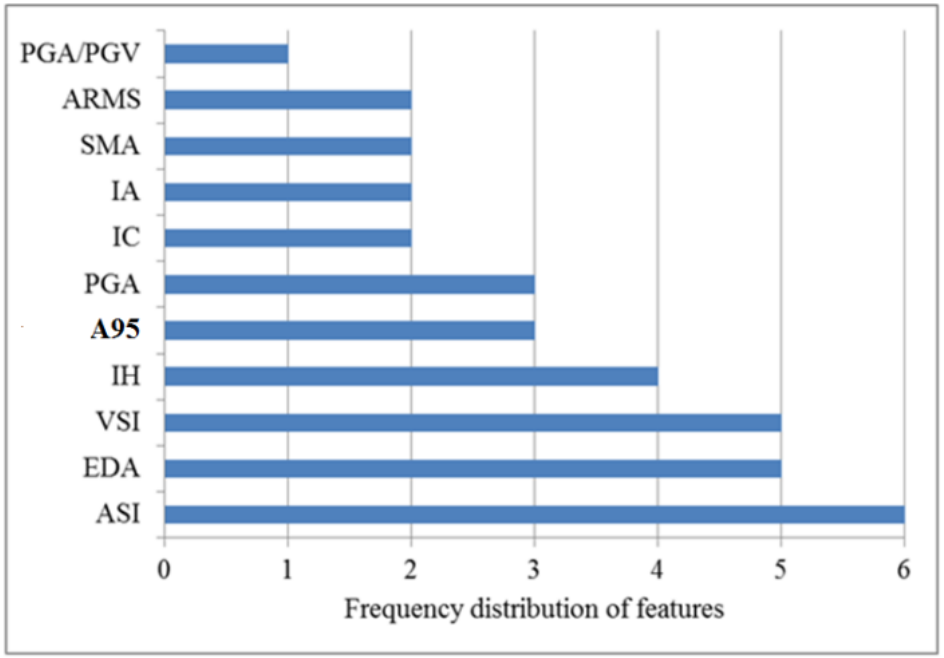
tests is determined. The results for the one-storey building clearly indicate that the ASI is the most frequently selected feature, followed by the EDA, VSI, IH, A95, PGA and others, as shown in Figure 7.21. For the two-storey building, the most frequently selected feature is the VSI, with no close competitors; the PGA, A95, ASI, IA and others are chosen at much smaller frequencies, as shown in Figure 7.22. The ROC analysis of the three most frequently selected features, i.e., the ASI, EDA, and VSI for the one-storey building and the VSI, PGA, and A95 for the two-storey building, also implies that the ASI produces better results than does the EDA (AUC of 0.965 vs. 0.954, respectively); both the ASI and EDA are better than the VSI (AUC=0.943) in terms of predicting the structural response of the one-storey building (Figure 7.21). Moreover, the superior performance of the VSI, with an AUC of 0.981, compared to AUCs of 0.962 and 0.957 for the PGA and A95, is clearly demonstrated by the ROC curve in Figure 7.22 for the two-storey building. The natural period of buildings possibly increased as a result of the stiffness degradation; however, this increase stopped within the acceleration and velocity ranges with respect to the one- and two-storey buildings, respectively (because a one-storey building has a shorter natural period than a two-storey building does, the increase in its natural period is less than that for a two-storey building). Therefore, the ASI is selected for the one-storey building, and the VSI is chosen for the two-storey building.

Table 7.32: Results of the classifier analysis using the optimum feature subsets chosen for each classifier and the one-storey building dataset (60 records); the wrapper method is used to obtain these values.

Classifiers	Feature set	TP rate	FP rate	Precision	ROC area	Error rate
1-NN	A95, RMSA, ASI, EDA, PGA, and VSI	0.981	0.021	0.981	0.98	1.95
Trees (Simple CART)	ASI, EDA, and IH	0.938	0.065	0.938	0.942	6.17
Logistic	ASI, IA, PGA, SMA, and VSI	0.958	0.047	0.958	0.982	4.22
MLP	A95, ASI, IC, PGA, and VSI	0.948	0.058	0.948	0.965	5.19
3-NN	A95, RMSA, ASI, EDA, IA, IH, and VSI	0.942	0.065	0.941	0.984	5.8
NB	ASI, EDA, IH, PGA/PGV, SMA, and VSI	0.938	0.06	0.939	0.982	6.16
Decision table	EDA, IC, and IH	0.942	0.065	0.941	0.979	5.84

Table 7.33: Results of the classifier analysis using the optimum feature subsets chosen for each classifier and the two-storey building dataset (60 records); the wrapper method is used to obtain these values.

Classifiers	Feature set	TP rate	FP rate	Precision	ROC area	Error rate
1-NN	A95, CAV, IA, and VSI	0.968	0.033	0.969	0.968	3.16
Trees (simple CART)	A95, IC, SED, VSI, and PGV/PGA	0.957	0.052	0.957	0.968	4.31
Logistic	ASI, RMSD, IH, PGA, and VSI	0.954	0.053	0.954	0.991	4.6
MLP	PGA and VSI	0.966	0.047	0.965	0.987	3.45
3-NN	A95, SMV, and VSI	0.96	0.05	0.96	0.972	4.02
NB	ASI and VSI	0.937	0.075	0.937	0.989	6.32
Decision table	EDA, IA, PGA, and VSI	0.954	0.062	0.954	0.983	4.6



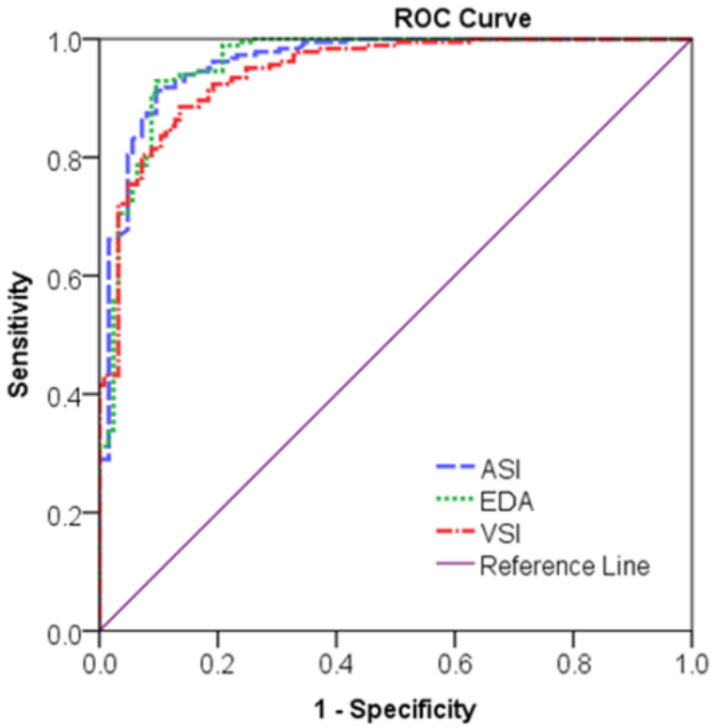
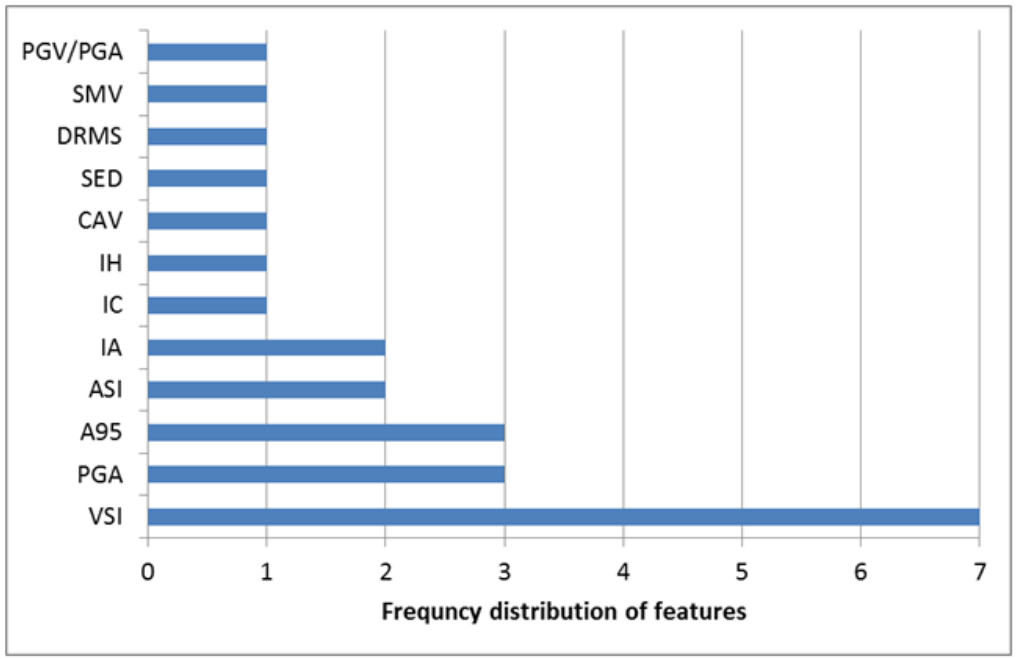


Figure 7.21: Frequency distribution of features chosen from the third dataset obtained by IDA using 60 records (one-storey building) for all classifiers (left); ROC curves for the three most frequently chosen ground-motion parameters for the collapse damage state of the studied one-storey building (right).



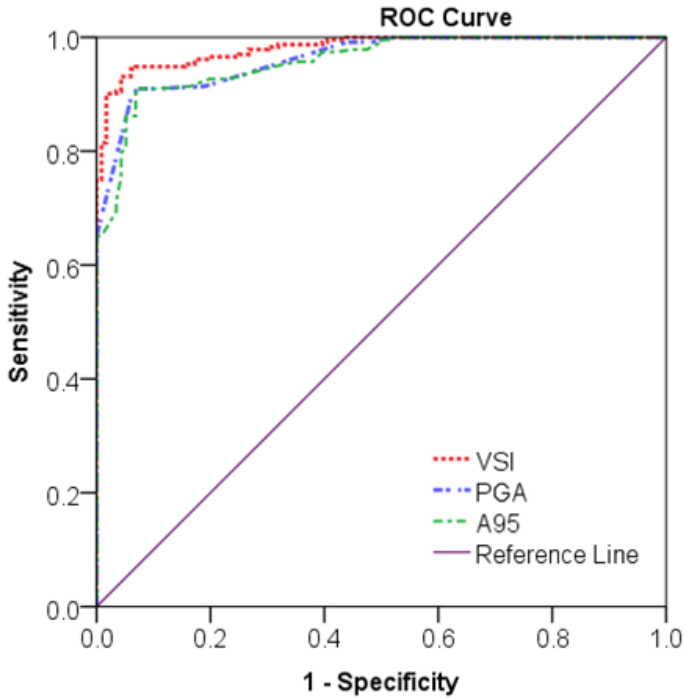


Figure 7.22: Frequency distribution of the features chosen from the third dataset obtained by IDA using 60 records (two-storey building) for all classifiers (left); ROC curves for the three most frequently chosen ground-motion parameters for the collapse damage state of the studied two-storey building (right).

7.2 Generation of Fragility Surfaces

As noted in the literature review, several studies (e.g., Cordova et al., 2001; Baker & Cornell, 2005; Luco et al., 2005; Kafali & Grigoriu, 2007; Rajeev et al., 2008; Sei'ichiro Fukushima, 2010; Seyedi et al., 2010; Gehl et al. 2011) have suggested the use of two ground-motion parameters for representing the seismic action in a region instead of a single IM (in fragility curves). In this particular case, the curves transfer to a surface, and the fragility curves are consequently called fragility surfaces. To verify the efficiency of using two parameters versus one IM and to ultimately derive fragility surfaces, different parameter pairs are selected in this section to measure the probability of failure and to derive fragility surfaces via a log-logistic regression function (equation 3.4). Considering the results of the wrapper and statistical regression methods (discussed in section 7.1.3), four IMs, namely, the ASI, VSI, PGA, and EDA, are selected to form pairs ASI+VSI, EDA+VSI, and PGA+VSI. Additionaly, two hybrid IMs, i.e., PGA/PGV and PVG^2/PGA are also selected to be combined with PGV, PGA and EDA parameters producing pairs of PGV+PGA/PGV, PGA+ PGV^2/PGA, and EDA+PGV^2/PGA, which have been noted in several studies discussed in the literature review.

To demonstrate how accurately the data points fit the statistical model used in for deriving fragility surfaces (i.e., log-logistic regression), the Nagelkerke R^2 (Nagelkerke, 1991) is used to represent a model's power of explanation. The Nagelkerke R^2 value is equivalent to the R^2 measure that is commonly used in normal regression models; it can be calculated via the following formula:

$$R^2 = \frac{1-\left\{\frac{L(M_{Intercept})}{L(M_{Full})}\right\}^{2/N'}}{1-L(M_{Intercept})^{2/N'}}, \qquad \text{Equation 7.1}$$

where M_{full} is the model with predictors, $M_{intercept}$ is the model without predictors, and L(M) is the conditional probability of the dependent variable given the independent variables. If N' observations are available in a dataset, then L(M) is the product of N' such probabilities. When the full model perfectly predicts the outcome with a likelihood of 1, the Nagelkerke R^2 value is 1.

Tables 7.34-7.37 list the percentages of correctly predicted damage states and the related Nagelkerke R^2 measures for the studied one- and two-storey buildings using various sets of earthquake records, including pairs of ground-motion parameters. The results clearly show that all of the considered IM pairs can predict the response of the

building with high accuracy, especially in the case of utilising a set of 7 records. Nevertheless, for predicting the structural response, the ASI-VSI pair provides a higher accuracy, i.e., the Nagelkerke R^2 value is closer to 1, than do the other pairs in most cases.

Table 7.34: Percentage of correct predictions and Nagelkerke R^2 values for the predicted structural response of the one-storey building in zone B using various sets of records and two ground-motion parameters.

Ground-motion intensity mesures		One-story (Zone B)											
		35 records		15 records-CMS		15 records-EC8		15 records-Jayaram et al.		15 records-Pro. Response spectrum		7 records	
IM1	IM2	Percentage Correct	Nagelkerke R^2	Percentage Correct	Nagelkerke R^2	Percentage Correct	Nagelkerke R^2	Percentage Correct	Nagelkerke R^2	Percentage Correct	Nagelkerke R^2	Percentage Correct	Nagelkerke R^2
PGA	PGV^2/PGA	92.7	0.833	95.3	0.897	93.4	0.910	91.9	0.833	88.7	0.810	96.4	0.934
PGA	ASI	92.1	0.829	92.2	0.906	93.4	0.904	95.2	0.868	91.9	0.877	96.4	0.945
PGA	VSI	93.9	0.857	96.9	0.963	95.1	0.902	91.9	0.869	91.9	0.836	100.0	1.000
EDA	PGV^2/PGA	92.7	0.842	93.8	0.878	93.4	0.904	91.9	0.817	90.3	0.858	96.4	0.874
EDA	ASI	92.7	0.819	93.8	0.879	95.1	0.930	95.2	0.858	95.2	0.915	100.0	1.000
EDA	VSI	93.3	0.871	93.8	0.920	93.4	0.907	91.9	0.863	93.5	0.916	92.9	0.888
ASI	VSI	94.5	0.880	93.8	0.916	95.1	0.904	98.4	0.946	95.2	0.921	100.0	1.000
PGV	PGA/PGV	92.7	0.830	95.3	0.878	93.4	0.905	91.9	0.832	88.7	0.808	100.0	1.000

Table 7.35: Percentage of correct predictions and Nagelkerke R^2 values for the predicted structural response of the two-storey building in zone B using various sets of records and two ground-motion parameters.

Ground-motion intensity mesures		Two-story (Zone B)											
		35 records		15 records-CMS		15 records-EC8		15 records-Jayaram et al.		15 records-Pro. Response spectrum		7 records	
IM1	IM2	Percentage Correct	Nagelkerke R^2	Percentage Correct	Nagelkerke R^2	Percentage Correct	Nagelkerke R^2	Percentage Correct	Nagelkerke R^2	Percentage Correct	Nagelkerke R^2	Percentage Correct	Nagelkerke R^2
PGA	PGV^2/PGA	93.5	0.877	94.3	0.843	94.4	0.925	93.2	0.880	94.3	0.875	96.9	0.924
PGA	ASI	91.4	0.809	94.3	0.857	95.8	0.909	93.2	0.813	92.9	0.841	96.9	0.935
PGA	VSI	96.2	0.915	94.3	0.905	100.0	1.000	100.0	1.000	97.1	0.948	100.0	1.000
EDA	PGV^2/PGA	92.5	0.854	92.9	0.852	93.0	0.902	93.2	0.864	94.3	0.845	100.0	1.000
EDA	ASI	88.2	0.777	94.3	0.855	94.4	0.893	91.8	0.801	90.0	0.828	100.0	1.000
EDA	VSI	94.1	0.905	94.3	0.902	100.0	1.000	100.0	1.000	97.1	0.948	100.0	1.000
ASI	VSI	96.8	0.930	95.7	0.912	98.6	0.953	98.6	0.957	97.1	0.953	100.0	1.000
PGV	PGA/PGV	93.5	0.876	94.3	0.841	94.4	0.924	93.2	0.880	94.3	0.875	100.0	1.000

Table 7.36: Percentage of correct predictions and Nagelkerke R^2 values for the predicted structural response of the one-storey building in zone A using various sets of records and two ground-motion parameters.

Ground-motion intensity mesures		One-story (Zone A)											
		35 records		15 records-CMS		15 records-EC8		15 records-Jayaram et al.		15 records-Pro. Response spectrum		7 records	
IM1	IM2	Percentage Correct	Nagelkerke R^2	Percentage Correct	Nagelkerke R^2	Percentage Correct	Nagelkerke R^2	Percentage Correct	Nagelkerke R^2	Percentage Correct	Nagelkerke R^2	Percentage Correct	Nagelkerke R^2
PGA	PGV^2/PGA	90.3	0.820	90.6	0.829	86.2	0.764	92.2	0.859	87.9	0.758	93.1	0.793
PGA	ASI	91.7	0.830	96.9	0.914	91.4	0.823	90.6	0.814	91.4	0.832	89.7	0.853
PGA	VSI	91.0	0.849	93.8	0.889	87.9	0.801	95.3	0.837	91.4	0.821	89.7	0.765
EDA	PGV^2/PGA	91.0	0.833	87.5	0.812	89.7	0.798	89.1	0.848	89.7	0.803	93.1	0.844
EDA	ASI	91.0	0.816	96.9	0.908	93.1	0.829	87.5	0.771	91.4	0.825	93.1	0.866
EDA	VSI	89.6	0.847	93.8	0.892	91.4	0.855	89.1	0.824	87.9	0.872	93.1	0.892
ASI	VSI	93.8	0.860	98.4	0.949	93.1	0.879	87.5	0.779	94.8	0.867	93.1	0.896
PGV	PGA/PGV	90.3	0.817	90.6	0.828	86.2	0.761	90.6	0.857	87.9	0.752	93.1	0.788

Table 7.37: Percentage of correct predictions and Nagelkerke R^2 values for the predicted structural response of the two-storey building in zone A using various sets of records and two ground-motion parameters.

Ground-motion intensity mesures		Two-story (Zone A)											
		35 records		15 records-CMS		15 records-EC8		15 records-Jayaram et al.		15 records-Pro. Response spectrum		7 records	
IM1	IM2	Percentage Correct	Nagelkerke R^2	Percentage Correct	Nagelkerke R^2	Percentage Correct	Nagelkerke R^2	Percentage Correct	Nagelkerke R^2	Percentage Correct	Nagelkerke R^2	Percentage Correct	Nagelkerke R^2
PGA	PGV^2/PGA	92.6	0.866	94.6	0.909	94.6	0.909	91.4	0.861	92.3	0.818	100.0	1.000
PGA	ASI	92.6	0.844	95.9	0.915	95.9	0.915	88.6	0.796	93.8	0.873	97.0	0.941
PGA	VSI	95.1	0.884	100.0	1.000	98.3	0.972	92.9	0.845	95.4	0.882	100.0	1.000
EDA	PGV^2/PGA	92.6	0.852	94.6	0.896	94.6	0.896	92.9	0.872	89.2	0.782	90.9	0.883
EDA	ASI	90.1	0.819	95.9	0.903	95.9	0.903	85.7	0.754	93.8	0.857	93.9	0.879
EDA	VSI	94.4	0.871	100.0	1.000	97.3	0.962	91.4	0.820	93.8	0.869	100.0	1.000
ASI	VSI	95.1	0.891	97.3	0.962	100.0	1.000	92.9	0.912	96.9	0.923	100.0	1.000
PGV	PGA/PGV	92.6	0.865	94.6	0.908	94.6	0.908	92.9	0.908	92.3	0.815	100.0	1.000

For the next step in this study, it is important to understand the effect of using two ground-motion parameters versus a single IM. To this end, the performance of the ASI-VSI pair in terms of predicting the structural response is examined in comparison with the use of a single parameter (i.e., the ASI or VSI) by identifying the -2LL measures that correspond to each of options based on log-logistic regression. Figures 7.23-7.28 present these values for the one-and two-storey buildings; various sets of earthquake records are considered. The results indicate that the ASI-VSI pair provides the best correlation with the damage states because it produces a smaller deviation in the predicted results compared with the two single parameters, i.e., the ASI and VSI. However, no substantial difference can be observed for sets of 7 records; it means that only one single parameter is capable of accurately predicting the damage state.

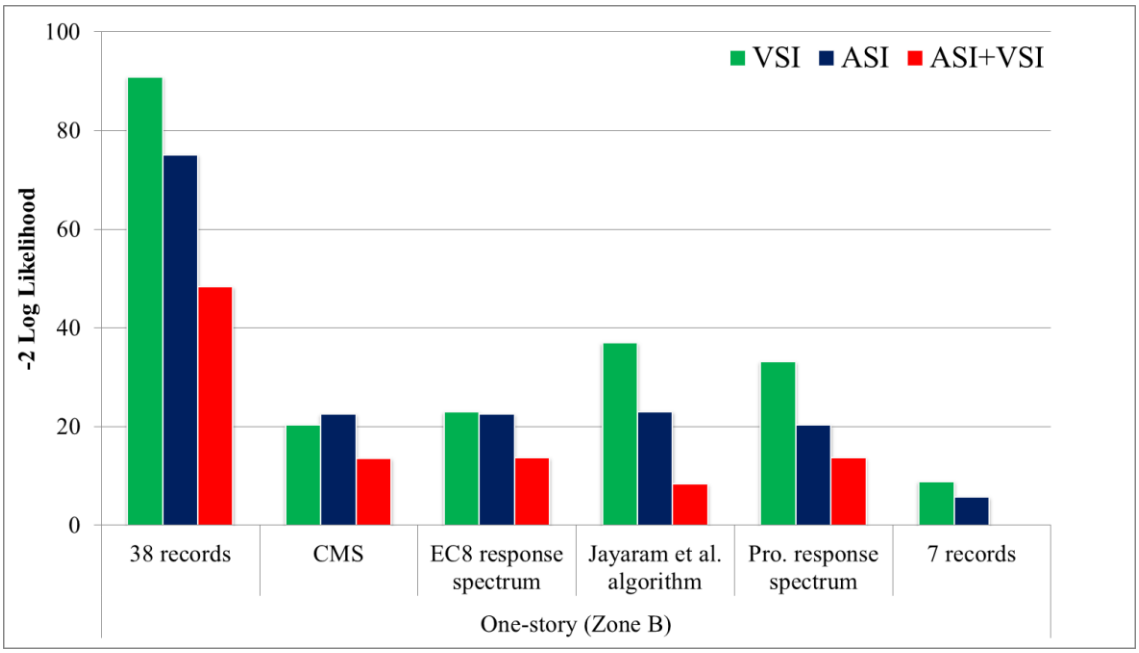


Figure 7.23: -2 log-likelihood (-2LL) measure demonstrating how poorly one IM and two IMs predict the response of the one-storey building using various sets of records (i.e., 38, 15 per each GMSM methods, and 7) in zone B.

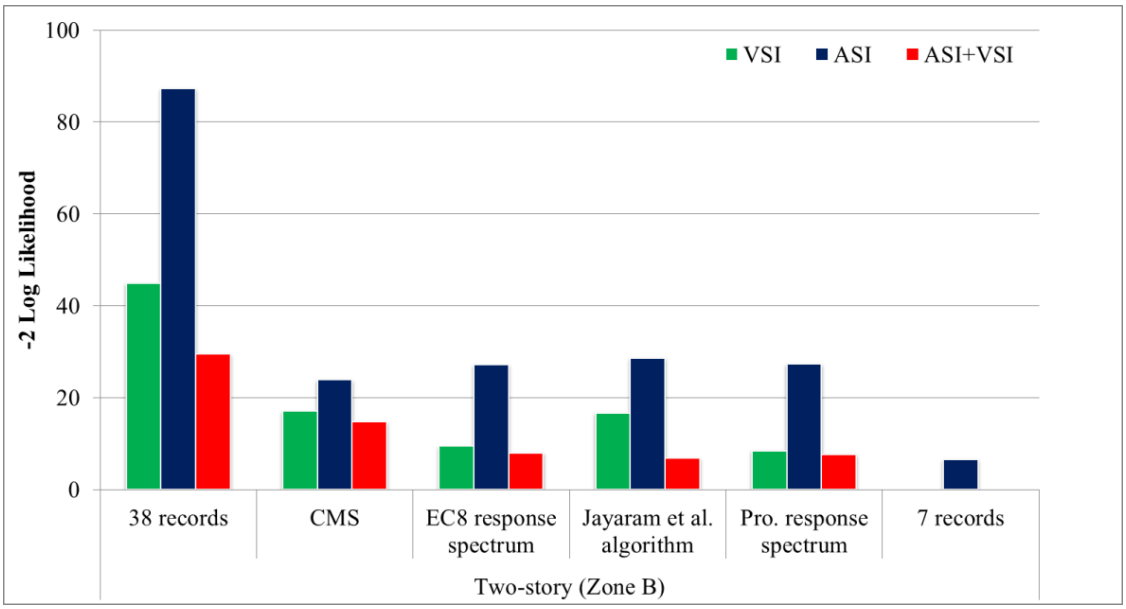


Figure 7.24: -2 log-likelihood (-2LL) measure demonstrating how poorly one IM and two IMs predict the response of the two-storey building using different sets of records (i.e., 38, 15 per each GMSM methods, and 7) in zone B.

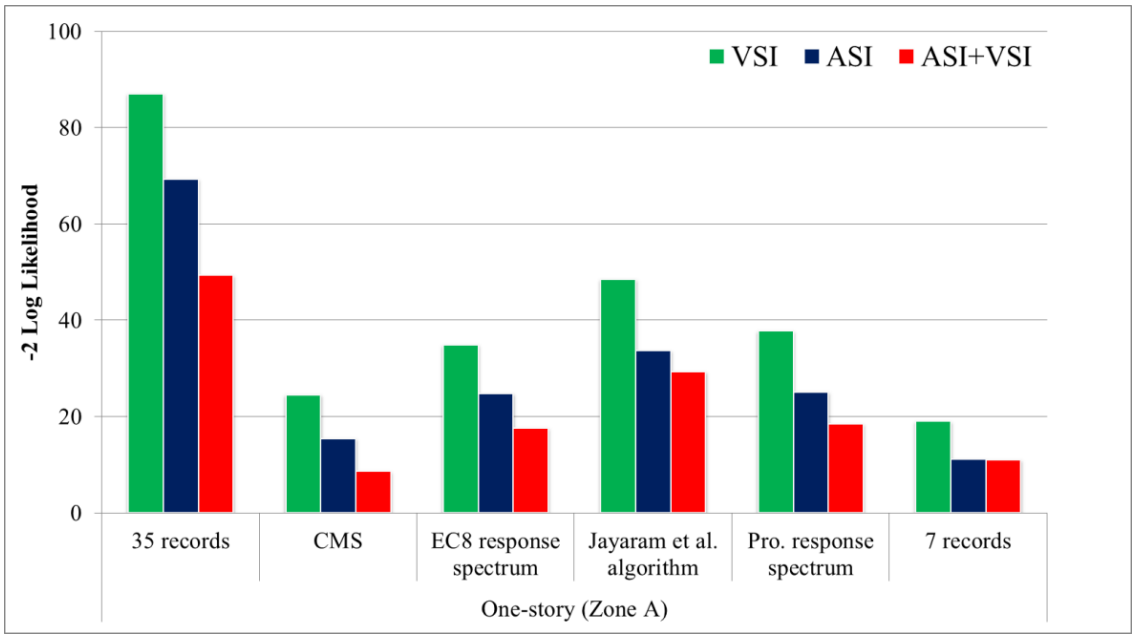


Figure 7.25: -2 log-likelihood (-2LL) measure demonstrating how poorly one IM and two IMs predict the response of the one-storey building using different sets of records (i.e., 38, 15 per each GMSM methods, and 7) in zone A.

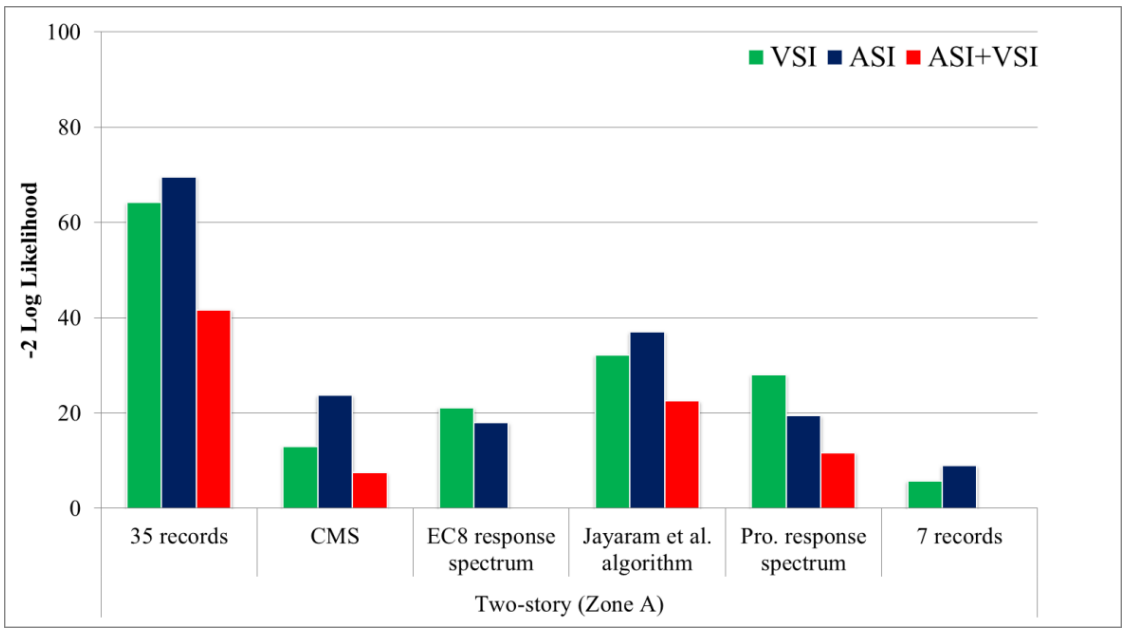


Figure 7.26: -2 log-likelihood (-2LL) measure demonstrating how poorly one IM and two IMs predict the response of the two-storey building using various sets of records (i.e., 38, 15 per each GMSM methods, and 7) in zone A.

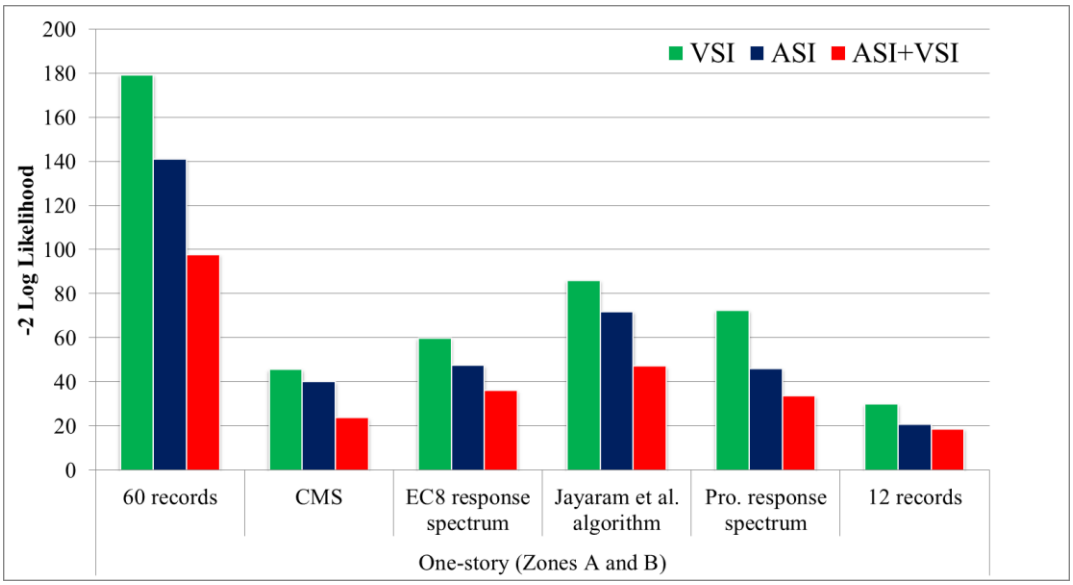


Figure 7.27: -2 log-likelihood (-2LL) measure demonstrating how poorly one IM and two IMs predict the response of the one-storey building using various combination of records (60, 25 per each GMSM methods, and 12) in zones A and B.

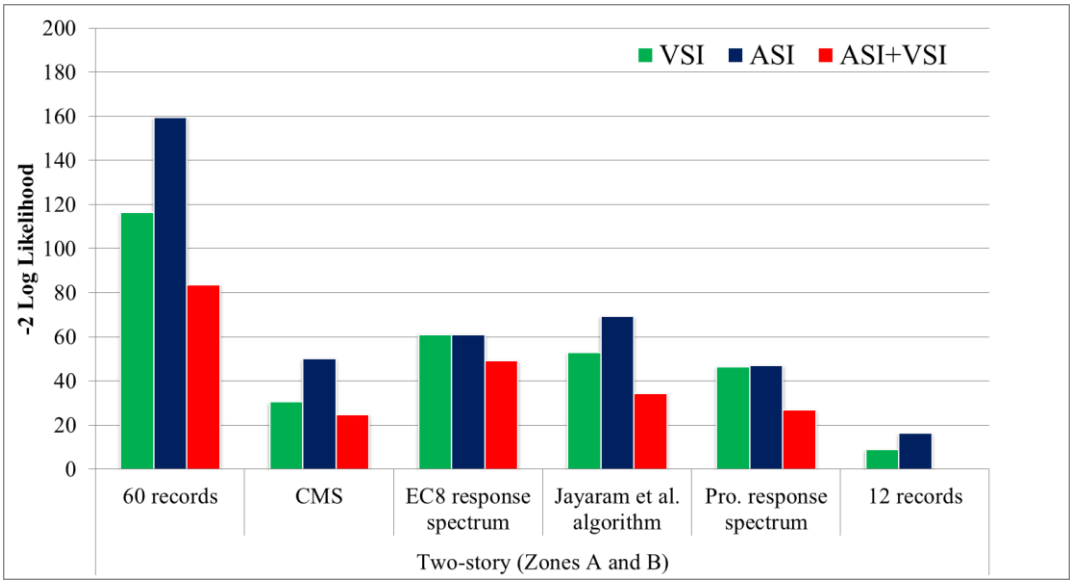


Figure 7.28: -2 log-likelihood (-2LL) measure demonstrating how poorly one IM and two IMs predict the response of the two-storey building using various combination of records (60, 25 per each GMSM methods, and 12) in zones A and B.

Based on the above analysis, it is now possible to create fragility surfaces for the tested buildings using the selected IM pair, i.e., the ASI-VSI pair. Using log-logistic regression, the fragility surfaces for the one- and two-storey buildings are developed and are depicted in Figures 7.29-7.30 for a set of 60 records. The scatter is pronounced when plotting single-parameter fragility curves (the ASI fragility curve) against fragility surface slices at different values of the second parameter, i.e., the VSI.

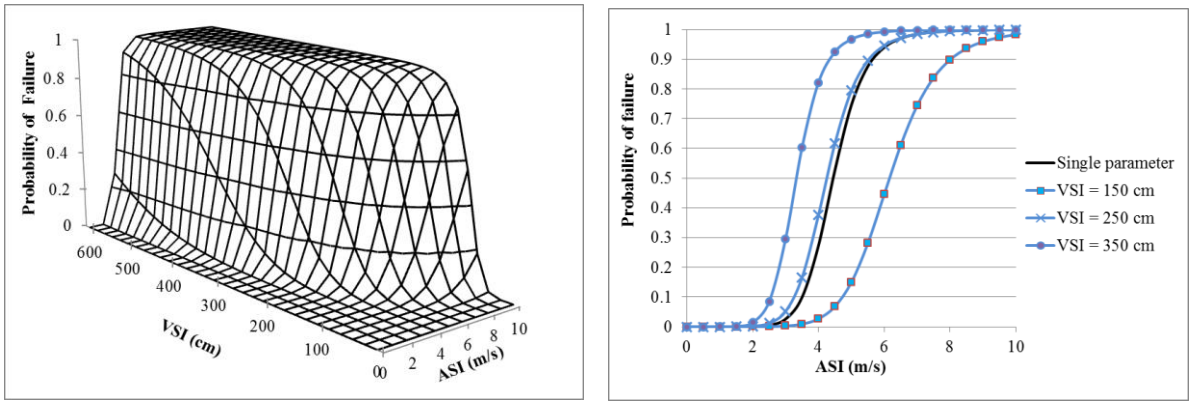


Figure 7.29: Fragility surface derived for the one-storey building and slices at different values of the second parameter (i.e., the VSI) using the set of 60 records.

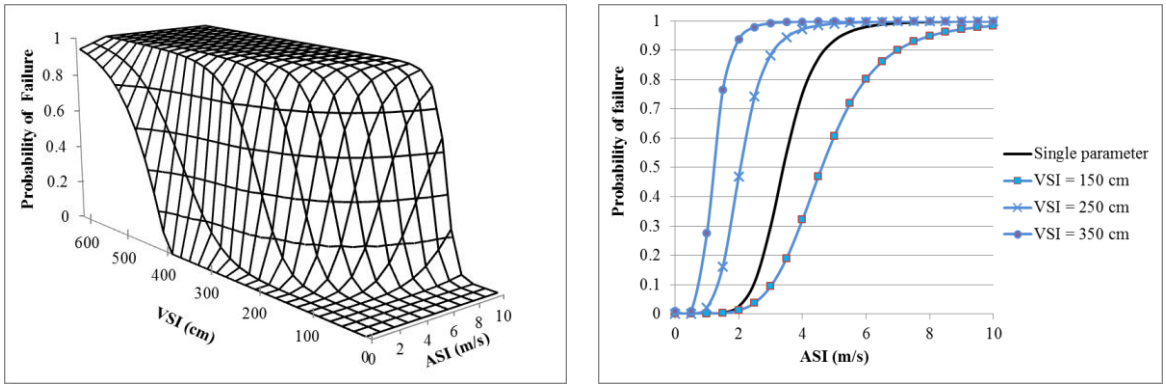


Figure 7.30: Fragility surface derived for the two-storey building and slices at different values of the second parameter (i.e., the VSI) using the set of 60 records.

CHAPTER 8. DISCUSSION OF THE RESULTS

In this chapter, the results of the structural analyses and the statistical analysis pertaining to the information obtained and presented in the previous chapters are discussed. Moreover, the reliability of the proposed technique for selecting a set of seven records corresponding to a set of seven different potential earthquake events is evaluated. The reliability of using the proposed KR response spectra for selecting time histories is also assessed. Finally, the application of machine learning in the wrapper-based approach for selecting the optimal IM for deriving fragility curves is discussed in detail.

8.1 Reliability of the Proposed Technique for the Selection of Seven Records and the Proposed Response Spectra for the Kurdistan Region

To select a set of seven records, the procedure presented in section 3.2 and four independent sets of fifteen ground motions are used for each seismic hazard zone of the KR by scaling the response spectra of the earthquake records (available in the PEER ground-motion database) to match the target spectra determined according to the four selected GMSM methods (Tables 5.13 and 5.14). The mean displacements in the x direction for the one- and two-storey masonry buildings in zone B are obtained by running several time history analyses for each set of records; Tables 8.1 and 8.2 introduce the calculated displacements for a node located at the middle of the roof for each set of records. As indicated in Table 8.2, the one- and two-storey buildings are displaced by approximately 0.7 and 2.0 cm, respectively, using records selected based on the proposed procedure and response spectrum. These values represent the approximate mean of the results obtained using the other methods.

Table 8.1: Selected set of seven records in zone B and the corresponding displacements of the test structures in the x direction.

No.	NGA record no.	Earthquake name	PGA (g)	x Displacement one-storey building (cm)	x Displacement two-storey building (cm)
1	126	Gazli, USSR	0.60	0.94	2.0
2	179	Imperial Valley-06	0.36	0.11	0.64
3	802	Loma Prieta	0.51	0.42	1.36
4	821	Erzican, Turkey	0.49	0.64	2.7
5	828	Cape Mendocino	0.59	0.72	2.41
6	1086	Northridge-01	0.60	0.87	2.44
7	1602	Duzce, Turkey	0.72	1.17	2.74
Mean			0.55	0.7	2.04

Table 8.2: Comparison of the mean responses for the tested buildings in zone B based on the selected records for each GMSM method.

Method	x displacement (cm)	
	One-storey building	Two-storey building
EC8 response spectrum	0.85	2.42
Proposed response spectrum	0.70	1.99
7 records	0.70	2.04
CMS	0.59	1.86
Jayaram et al. (2011) algorithm	0.39	1.76
38 records	0.64	2.08

The results of the time history analyses for the two buildings in zone A (one and two-storey structures) are provided in Tables 8.3 and 8.4. The proposed technique predicts a structural response (1.25 cm for the one-storey building and 3.22 cm for the two-storey building) that is approximately equal to the average of the remaining methods (particularly for the CMS and EC8 methods). However, the mean response of the buildings predicted using the sets of records obtained from the proposed response spectrum for zone A is closer to those produced using the records obtained by matching their response spectra to the EC8 target spectrum.

Based on a study of reinforced concrete buildings, Haselton et al. (2009) claimed that the estimated spectral shape is an interface between the different GMSM methods and the accurate prediction of their structural responses because nonlinear drift is significantly impacted by the spectral shape. Hence, they recommended using the CMS as an accurate method for selecting time histories. Considering the outcomes obtained from the time history analyses of both building types in this study, the CMS predicts the mean structural response fairly well compared with the results for a set of 35 or 38 records derived from a combination of all of the sets of records. Thus, with respect to the CMS method, the proposed response spectra for the KR tend to overpredict the mean structural response by 7 to 33%; notably, this range is approximately 30 to 44% for the EC8-code-based method. The proposed technique used to select sets of seven records can predict a structural response within a range of 10 to 25% of the CMS-related structural responses.

Although numerous methods have been proposed, there is no universal method for selecting and modifying an appropriate ground motion, as noted by Haselton et al. (2009). Therefore, a conservative prediction of the structural response is preferred because differences in the decision-making process for selection of a proper GMSM method can

lead to serious safety and economic challenges. Moreover, the estimation of the mean structural response can be conducted with an acceptable level of accuracy using the proposed technique for selecting sets of seven records and according to the proposed response spectra for the KR. The use of this approach is justified because the predicted response is not overestimated as much compared to the code-based (e.g., EC8) method and the CMS method. Furthermore, the proposed technique can be implemented without special concerns, such as selecting the best method or selecting an accurate record for a specific earthquake event, which is problematic for GMSM methods.

Table 8.3: Selected sets of seven records in zone A and the corresponding displacements of the test structures in the x direction: (a) option 1 and (b) option 2.

(a) Option 1							
No.	NGA record no.	Earthquake	PGA (g)	Mag.	EpiD (km)	x displacement (cm)	
						One storey	Two storey
1	126	Gazli, USSR	0.6	6.8	12.8	0.9	2.0
2	143	Tabas, Iran	0.8	7.4	55.2	1.8	4.6
3	568	San Salvador	0.9	5.8	7.9	1.5	4.6
4	779	Loma Prieta	1.0	6.9	18.5	2.1	4.3
5	828	Cape Mendocino	0.6	7.0	4.5	0.7	2.4
6	983	Northridge-01	0.6	6.7	13.0	1.0	2.6
7	1508	Chi-Chi, Taiwan	0.5	7.6	21.4	1.1	2.5
Mean			0.71	6.9	19.0	1.30	3.26
(b) Option 2							
No.	NGA record no.	Earthquake	PGA (g)	Mag.	EpiD (km)	x displacement (cm)	
						One storey	Two storey
1	126	Gazli, USSR	0.6	6.8	12.8	0.9	2.0
2	143	Tabas, Iran	0.8	7.4	55.2	1.8	4.6
3	779	Loma Prieta	1.0	6.9	18.5	2.1	4.3
4	828	Cape Mendocino	0.6	7.0	4.5	0.7	2.4
5	879	Landers	0.7	7.3	44.0	0.9	4.1
6	983	Northridge-01	0.6	6.7	13.0	1.0	2.6
7	1508	Chi-Chi, Taiwan	0.5	7.6	21.4	1.1	2.5
Mean			0.68	7.1	24.2	1.22	3.19

Table 8.4: Comparison of the mean responses for the tested buildings in zone A based on the selected records for each GMSM methods.

Method		x displacement (cm)	
		One storey building	Two storey building
EC8 response spectrum		1.47	3.63
Proposed response spectrum		1.36	3.47
7 records	Option 1	1.3	3.26
	Option 2	1.22	3.19
CMS		1.04	2.6
Jayaram et al. (2011) algorithm		0.98	2.8
35 records		1.08	2.88

The reliability of the results obtained using the proposed response spectra and the technique used to select seven earthquake records is further justified based on the fragility curves derived from various sets of records and the GMSM methods for buildings in zones A and B (see Figures 7.8-7.15). Having provided a well-selected IM, the fragility curves derived with a few records (i.e., minimum of 7 records) provide the same performance as those generated using many records (i.e., 60 records). Moreover, the fragility curves generated using the proposed-response-spectra-based records provide the same performance as do curves derived using the sets selected from the CMS, EC8, and Jayaram et al. methods when a correct IM is identified.

The proposed technique (the selection of seven records) is not an alternative to any of the available methods for selecting ground motions. Instead, it should be considered as an efficient technique and tool for determining the most accurate and appropriate ground motion from several ground motions selected using other methods for a specified site. Furthermore, the proposed procedure for selecting a set of seven records is applied to a specific case in this study. Therefore, the approach should be cautiously applied in different contexts, especially for different areas that may include different seismic characteristics and when using other GMSM methods.

8.2 Proposed Seismic Zone Factor Z for the KR

Formally, 50-year PGA values are usually implemented to zone a region for seismic design purposes. A seismic zone factor Z that represents the effective peak acceleration as the acceleration's decimal fraction due to gravity effects can be subsequeantly computed for each seismic zone based on the recommendations of ATC (1984), which is consistent with the philosophy of SEAOC (1988) and UBC (1991). A structure's minimum base shear is directly proportional to this factor (see Equation 5.1). Incorrectly defining this factor can greatly influence a structure's design, especially regarding its resistence to seismic loads.

The Iraqi seismic building code ultimately divides the country into four seismic zones with a maximum value of Z=0.09, which is found in northern KR (see Figure 5.14). However, based on the region's comprehensive seismological and geological charactersitics discussed in Chapter 5 and according to the seismic zonation map proposed in this study (Figure 5.25), the Z values provided by the Iraqi seismic building code for different regional seismic zones do not accurately reflect the severity of seismic actions within the study region. Figures 5.15 and 5.25 show that eastern KR is subject to the most seismic and hazardous activity, with a PGA exceeding 0.4g.

With a purpose of clarification, a comparison between seismic zonation maps of Iraq and Saudi Arabia is presented here. As mentioned by AL-Haddad et al. (1994) and as shown in Figure 8.1, the highest PGA for 10% probabilities of exceedance in 50 years for Saudi Arabia is approximately 0.2g. Four seismic zones are defined for the country (i.e., Saudi Arabia), in which the seismic zone factor Z ranges from 0.05 to 0.2, as shown in Figure 8.2 and Table 8.5. However, the Kurdistan region, which its highest PGA value exceeds two times the Saudi Arabia's one, is assigned four main seismic hazard zones with Z values ranging only from 0.0 to 0.09 (according to the Iraqi seismic building code).

To this end and based on the aformentioned facts, the use of the Z values listed in Table 8.6 is recommended for structural design and analysis in the study region.

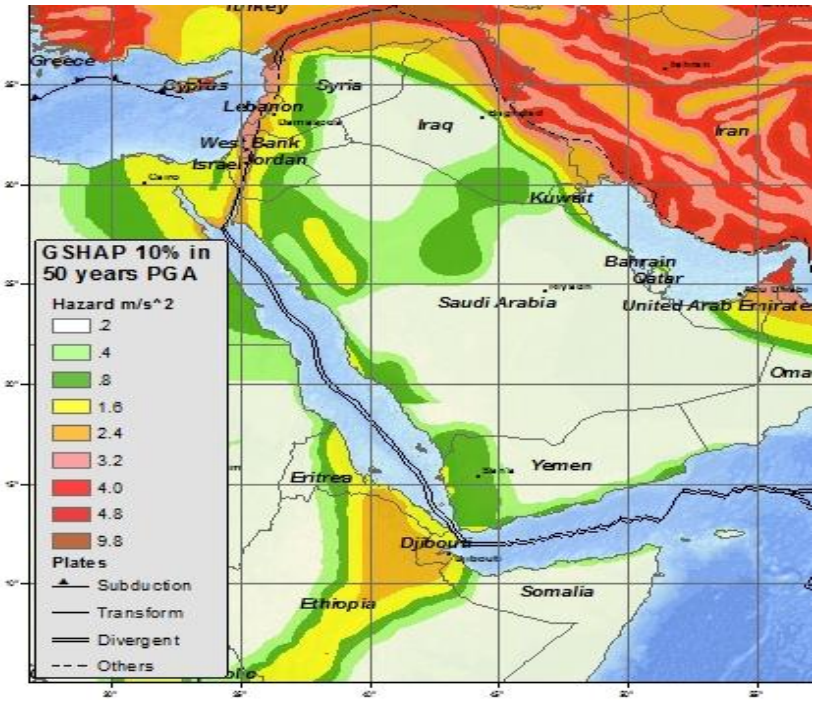


Figure 8.1: Peak ground acceleration (PGA in units of m/s^2) GSHAP map for rock with a 10% probability of exceedance in 50 years, 475-year return period (Grünthal et al., 1999); Saudi Arabia, Iraq, and surrounding areas are shown.

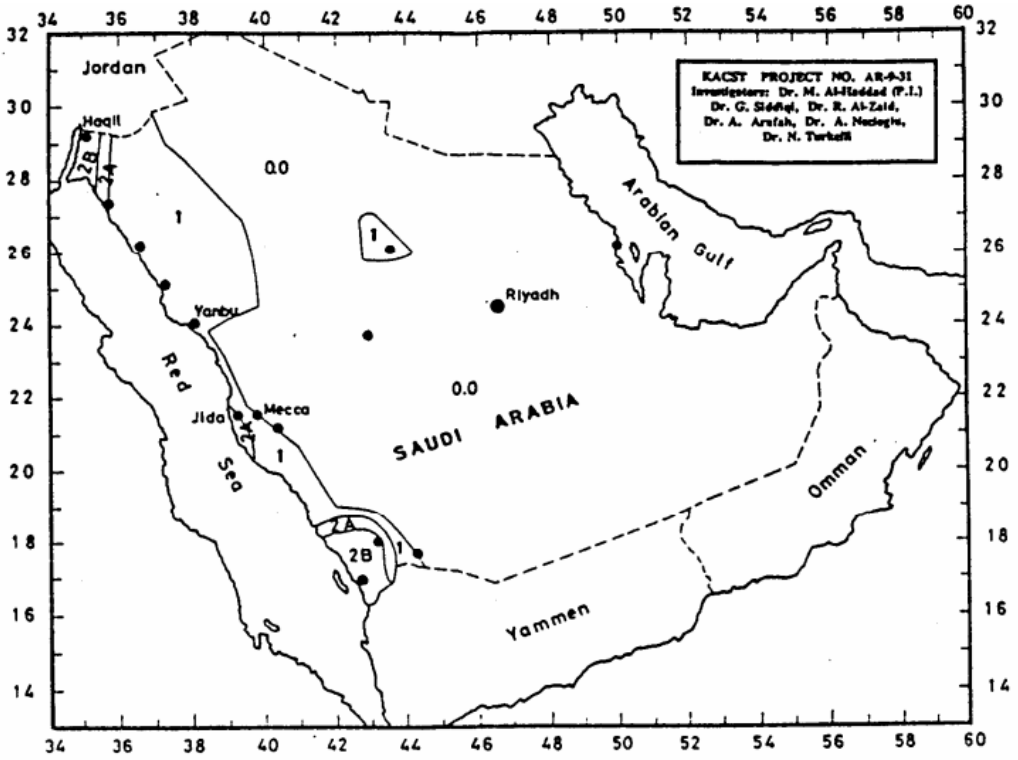


Figure 8.2: Seismic zonation map for Saudi Arabia (AL-Haddad et al., 1992; AL-Haddad et al., 1994).

Table 8.5: Seismic zone number, seismic zone factor Z, and corresponding PGA for Saudi Arabia (AL-Haddad et al., 1994).

Seismic zone number	Seismic zone factor Z	PGA (g) for 10% probability of exceedance in 50 years
0	0.05	<0.05
1	0.075	0.05 to 0.1
2A	0.15	0.1 to 0.15
2B	0.2	0.15 and above (the highest value is approximately 0.2g)

Table 8.6: Seismic zone number, seismic zone factor Z, and corresponding PGA for the Kurdistan region of Iraq based on this study.

Seismic zone (See Figure 5.16)	Seismic zone factor Z	PGA (g) for 10% probability of exceedance in 50 years
B	0.4	0.25-0.4
A	0.5	0.4 and above

8.3 Discussion of the Results Related to the Fragility Curves and Surfaces

To verify the assumption that the variability related to ground motions has a stronger impact on the structural response than do the variability of mechanical parameters, as discussed in section 7.1.1 for each of the considered PGA levels (i.e., 0.05g, 0.2g, and 0.8g), the total standard deviation associated with the maximum structural displacement is calculated as the square root of the sum of the squares of the standard deviations attributed to ground motion and mechanical parameter variability. A similar approach has been utilised in several studies for combining the uncertainties associated with capacity and demand to determine the total damage state variability (e.g., FEMA, 1999; RISK-UE, 2004; Rota et al., 2010). Figure 7.1 shows that the variability in mechanical properties is significantly smaller than the variability associated with different ground motions, especially for the tested buildings and selected ground motions in this study. Therefore, it is practical to neglect the variability in material parameters and consequently focus on different ground motions. Furthermore, this figure also shows that the standard deviation of

the displacement demand increases with increasing ground-motion severity, which is expected. This concept can be explained based on the increase in a building's nonlinear behaviour, which can modify the structural response. These results agree with Rota et al. (2010), who suggested that ignoring the variability in material parameters and focusing on ground-motion variability is a viable option. The results summarised in Figure 8.3 are obtained from Rota et al. (2010). This figure provides a comparison of the standard deviations obtained for three cases: a) 7 accelerograms using mean mechanical parameters; b) only1 accelerogram with stochastic parameters; and c) combining the two cases a and b. Their selected building prototype was a 3-storey masonry building in Benevento (southern Italy). Although similar conclusions can be drawn based on two different studies using different PGA levels and different building prototypes, further analyses with different buildings and ground-motion records are needed to confirm the results obtained in the two studies.

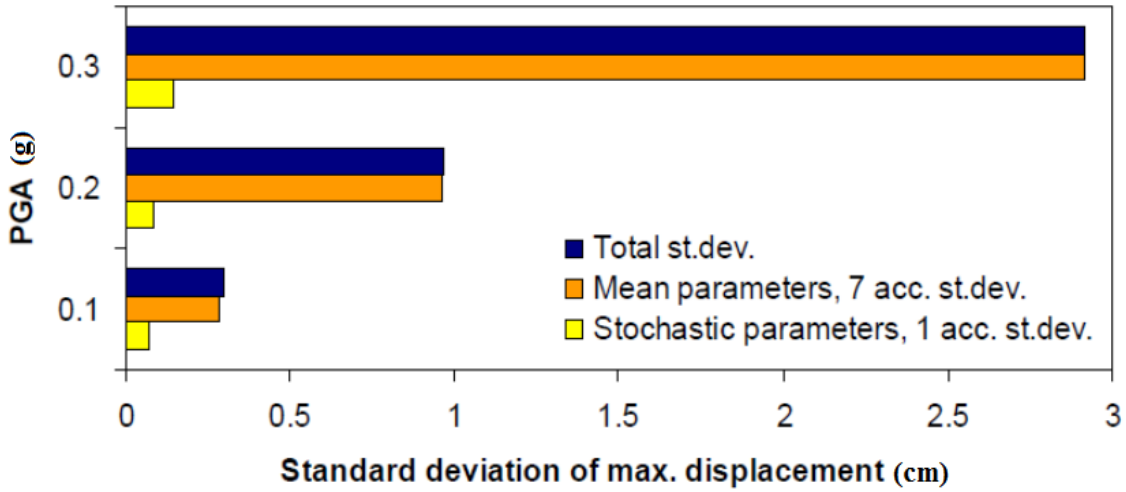


Figure 8.3: Comparison of the standard deviations obtained for: 7 accelerograms using mean mechanical parameters, only 1 accelerogram with stochastic parameters, and combining of two aforementioned cases (from Rota et al., 2010)

The results obtained in the current study confirm the results put forth by several researchers who identified the uncertainty in ground motions as the most significant source of uncertainty in the probabilistic assessment of structures (e.g., Dymiotis et al., 1999 and references therein; Pasticier et al., 2008). However, other studies, such as Parisi and Augenti (2012), drew the opposite conclusion, showing that the uncertainty propagation

from material properties to the seismic displacement capacity of an entire masonry structure (a two-storey URM building was selected to represent a typical residential buildings in southern Europe in their study) may not be small. As a result, they recommended using a set of values for each of the most uncertain parameters to perform seismic analyses and to determine the actual safety of a masonry structure. Furthermore, the authors also noted that both engineering judgment and expertise must continue to play a key role in assessing the structural safety of existing masonry buildings under seismic design loads. Additionally, other studies, such as Cattari et al. (2010) and Pagnini et al. (2011), found that the variability in the structural capacity is the main contributor to the global scatter in fragility curves via a comparison with the seismic demand and limit state definitions.

In addition, the fragility curves shown in Chapter 7 (e.g., Figures 7.8 and 7.9 or Figures 7.10 and 7.11) demonstrate that different IMs provide various levels of performance for different numbers of storeys in a typical URM building located in a seismic hazard zone. Notably, for buildings with the same number of storeys that are located in different seismic hazard zones, the selected IM in a given zone can also perform well in other zones. Thus, it can be concluded that a building's geometric variability, and more particularly the building's height, has a larger effect on the derived fragility curves when compared to the variability in ground motions. More specifically, the natural period of a structure, which is often related to its height in empirical equations used to predict natural periods, plays a significant role in selecting the best IMs. This relationship has been confirmed by numerous researchers; thus, it is currently suggested to classify ground-motion parameters based on their performance and applicability within specific period ranges and to select only those that occur in the same period range of a building under study prior to conducting any other procedures. For example, several studies, such as Fajfar et al. (1990), Riddell and Garcia (2001), Allen (2007), and Ye et al. (2011), have indicated that the acceleration-based, velocity-based, and displacement-based IMs perform better when applied to short-, medium-, and long-period structures, respectively.

Nevertheless, predicting the behaviour of structures subject to seismic actions before conducting a simulation analysis may be naturally complicated. For example, in the current case, both types of buildings were expected to have short natural periods, meaning that the acceleration-based IMs should have been better correlated with the damage index. While this was shown to be valid for the one-storey building, the velocity-based IMs performed better

when applied to the two-storey building. Therefore, the expected behaviour of a building under seismic loads is more important than its natural period when selecting the correct IM. This finding was also noted by Parisi and Augenti (2012), who indicated that engineering judgment plays a fundamental role in the seismic vulnerability assessment of a structure.

The aforementioned problem might be solved if two parameters are used in the fragility functions instead of a single IM. Furthermore, using two parameters can prevent the need for multiple fragility curves based on different IMs that are selected for different buildings with different numbers of storeys in the same area, which was noted earlier in this work. If only one parameter is used, two different fragility curves should be independently developed for one- and two-storey buildings using the ASI and VSI as the IMs, respectively. However, with two parameters, the damage probability of both buildings can be represented more accurately by deriving fragility surfaces with the same IMs, which is shown in Figures 7.29 and 7.30.

The fragility curves obtained for different sets of records (e.g., Figures 7.14 and 7.15) clearly indicate that when the correct IM is selected, the number of implemented records is irrelevant, even if the minimum number of records (7 records) is used. This finding was also noted by Gehl et al. (2013a), who stated that "a relatively small error is introduced into the final results by the limited number of analyses usually used" in developing fragility curves using a non-linear dynamic analysis method. However, this conclusion is only true if a correct IM and the GMSM method are utilised. As shown in Figure 7.14, the uncertainty in estimating the probability of collapse failure may decrease or increase depending on whether the ASI or EDA IMs are used, even if the same number of records is utilised. This finding is also shown in Figures 7.8-7.13 for all IMs. Therefore, even if a large number of records with the wrong IM are utilised, the results may not be the same as those expected when using a small number of records with the correct IM. Consequently, correctly selecting the IM is more important than the number of records when constructing fragility curves. However, the selection of an IM using a ground-motion selection method for a specific case does not imply that the same IM will be successfully selected for a different case (see Tables 7.8-7.11). Therefore, depending on the typology of a building (i.e., the number of storeys) and its location, the ground-motion selection methods may provide different optimal IMs. This finding may be a reason for the lack of a systematic ground-motion selection method to date. However, based on the obtained results, it can be concluded that if a correct IM is selected, it is irrelevant which ground-motion selection method is used for selecting the time histories.

In summary, to assess the relative importance of the various sources of uncertainty in the seismic vulnerability assessment of masonry buildings, auxiliary academic work is still required.

In short-period structures with distinct rigid behaviour, the structural response is closely associated with the applied ground motion. Thus, the peak structural response is scaled if the PGA is scaled (Dhakal et al., 2007). However, due to the loss of rigidity in the tested buildings used in this study and even though the PGA is used to scale the records in the IDA process, the ASI and VSI parameters provide better performance. The acceleration spectrum intensity (ASI), as discussed in the literature review, is the area under the acceleration response spectrum of a ground motion for a period from 0.1 to 0.5 s. This quantity was originally proposed by Von Thun et al. (1988) as a ground-motion intensity measure (IM) applicable to the seismic analysis of concrete dams and structures having fundamental periods of less than 0.5 s. Thus, it is not surprising that the ASI is the best IM for the tested one-storey building because the building can be considered as a concrete mass constructed primarily from solid concrete blocks and reinforced concrete materials. Furthermore, because of its consideration of the spectral periods' range, as indicated by Bradley (2010), the ASI can be considered as a useful ground-motion parameter for the simultaneous prediction of acceleration and displacement demands in individual structures and regions with prevalent short-period structures. Bradley (2010) asserted that the ASI provides a better predictability than do conventional IMs such as the elastic pseudospectral acceleration at a specific period. Similar to the ASI, the VSI can be defined as the area under the velocity spectrum intensity for periods between 0.1 and 2.5 s because many structures have fundamental periods between these values. Because the VSI represents several vibration frequencies, ground-motion characteristics and properties are intensively collected through this parameter. Thus, the VSI is a powerful IM for predicting the complex and nonlinear behaviours of a structure's response. Moreover, these parameters focus on the spectral acceleration (in terms of the ASI) and velocity (in terms of the VSI) over a wide range of periods. This range can encompass the possible increase in a structure's natural period due to progressive degradation, which explains why the ASI and VSI provide the best correlation with the defined damage states for the tested one- and two-storey URM buildings in this study. For the same reason, the two most commonly used contemporary intensity measures in earthquake engineering, namely, the spectral accelerations and spectral displacements at a building's natural period [i.e., S_a(T1) and

$S_d(T1)$], as mentioned by Song (2014), appear to be poorly correlated with the damage index of the buildings tested in this study. This was also one of the main conclusions of Deliverable 13's report (Pitilakis et al., 2011), which examined 49 ground-motion IMs to determine the most appropriate parameter for describing the fragility and vulnerability of historical masonry buildings. The acceleration spectral intensity (ASI) was found to be the best parameter in that study.

A central conclusion of this study is that certain ground-motion parameters are practical when selected based on a structure's typology. It can be argued that while some ground-motion parameters are computationally complex or typically unsuitable, there are others available that could be used practically, if selected properly with considering the structure's typology. These findings agree with Deliverable 13 (Pitilakis et al., 2011).

Furthermore, the results of the statistical regression approach agree with the findings obtained using the wrapper method because the ASI and VSI were found to be the best IMs for the one- and two-storey URM buildings in both methods, respectively. The wrapper method was used to classify a set of 20 IMs (obtained after reducing 36 IMs used in this study as mentioned in section 7.1.2) into different numbers of subsets containing various IMs. Moreover, the performance of each IM was evaluated with respect to the known response of the tested structures. The results illustrated that in addition to facilitating the detection of the correct IM, the wrapper method, which is based on the selected classifier function, is also able to predict structural responses with an accuracy comparable to those obtained by traditional regression functions such as the log-logistic regression used in this study. For example, considering the IMs selected by the 1-NN method provided in Tables 7.32 and 7.33, the error rate for classifying damage states using log-logistic regression was approximately 7 and 5% for the one- and two-storey building, respectively; comparatively, the 1-NN classifier had error rates of 1.95 and 3.16%, respectively, for these buildings. Based on the results presented herein, the maximum number of IMs selected within a subset for different algorithms was seven, and the minimum number of IMs was one. These results are encouraging for assessing the efficiency and sufficiency of IMs. Instead of testing all 36 IMs, there is an opportunity to use seven or fewer IMs based on the selected classifier function and the desired accuracy of the outcomes. By reducing the time spent testing IMs, this approach secures additional time for resolving other issues involved in the process of deriving fragility curves (e.g., variability in material properties, variability in damage states, and variability in regression

functions). Furthermore, the results of applying the wrapper feature selection method indicated that nearly all of the seven tested classifiers (NB, nearest neighbour (with K=1 and K=3), logistic, MLP, decision table, and decision tree) exhibited high accuracy, with an average error rate of 5% for the data considered. However, the most successful tested classifier algorithm was the 1-NN algorithm. The results also demonstrated that the feature selection may differ depending on the chosen algorithm. This point was clearly illustrated in Tables 7.12-7.19, 7.22-7.29, 7.32 and 7.33. For example, in Table 7.27, except for the VSI, the other parameters selected using the 1-NN, MLP and 3-NN classifiers are different, although all of the algorithms classify the data with an approximately equal error rate (1.35%) and prediction precision (0.987). This agrees with the findings of Kohavi and John (1997), who claimed that the feature set depends largely on the selected classifier algorithm and that it is difficult to find an optimal feature set for all classifiers.

The results of the wrapper-based approach using sets of 7 records applied to the studied one- and two-storey buildings (excluding those obtained for the one-storey building located in zone B and shown in Table 7.15) demonstrated that one IM is sufficient to predict the structural response with an acceptable range of accuracy, as shown in Tables 7.19, 7.25, and 7.29. This is evident from the -2 log-likelihood (-2LL) measure, which is depicted in Figures 7.24 and 7.25 to show how poorly the use of one and two IMs predicts the structural response. Therefore, it is not beneficial or pragmatic to use fragility surfaces with a limited number of records, namely 7 or fewer records, because a fragility curve with a well-selected IM may provide the same information as would a fragility surface. For example, the fragility surface for the one-storey building with 7 records in zone A (depicted in Figure 8.4) clearly indicates that one parameter is sufficient to obtain an accurate prediction of the damage states, i.e., no additional ground-motion parameter is needed.

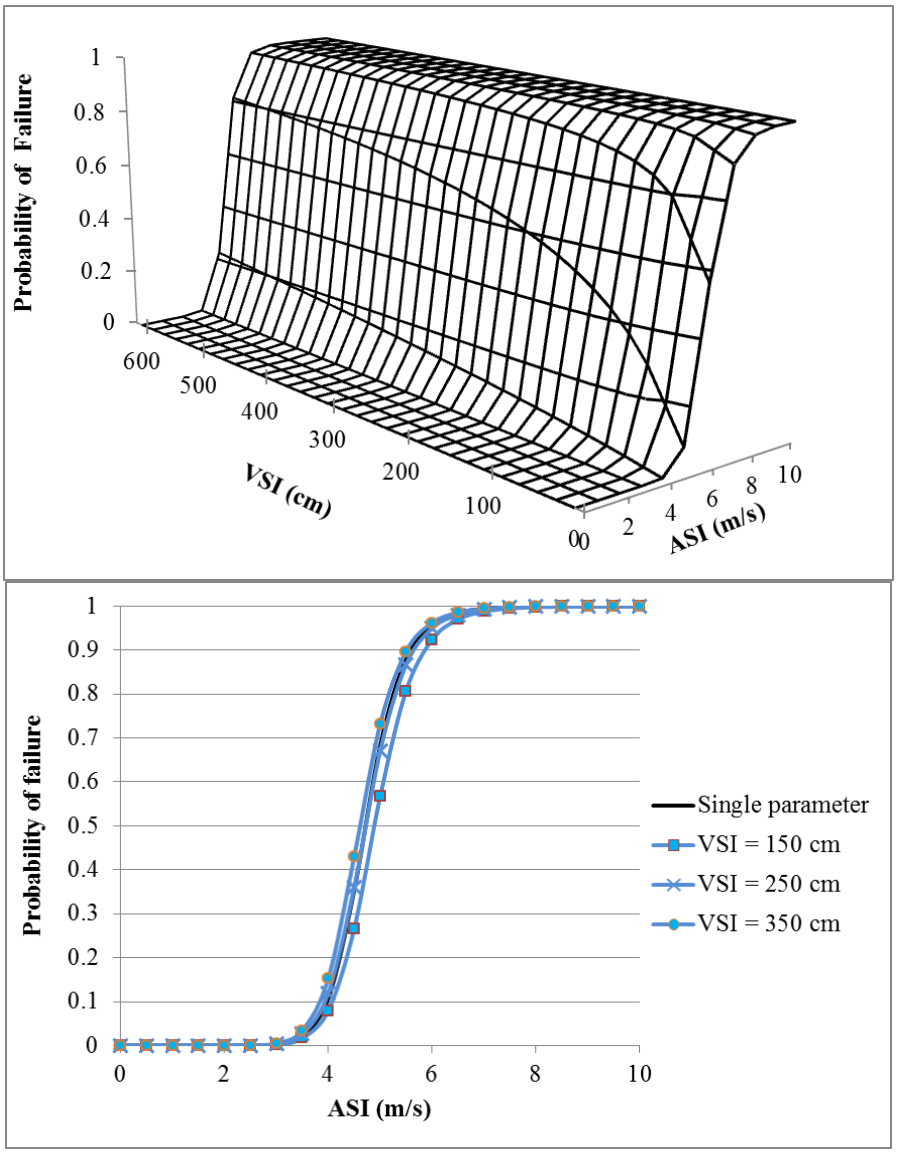


Figure 8.4: Fragility surface obtained for the one-storey building located in zone A using a set of 7 records.

The efficiency of using two parameters instead of one IM was clearly demonstrated by the fragility surfaces produced for the one- and two-storey buildings (Figures 7.29 and 7.30) using a set of 60 records. The scatter in the results was illustrated by plotting a single-parameter fragility curve (i.e., the ASI fragility curve) against slices of the fragility surface for different values of the second parameter (the VSI). These results indicated that fragility curves could accurately estimate the structural response only when the VSI was considered to be 180 and 250 cm for the two- and one-storey buildings, respectively. Otherwise, the results were likely to be underestimated or overestimated depending on the VSI level. For example, an earthquake record with ASI=3.47 m/s and VSI=266.29 cm

(e.g., the 1994 Northridge-01 earthquake: M_W=6.69, EpiD=13.39 km, and PGA=0.42g) may produce a collapse failure by the one-storey building by approximately of 21% vs. 79% of yield failure when two parameters are applied. However, for the same ASI but without considering the impact of the second parameter (VSI), the fragility curve exhibits probabilities of collapse and yield failure of 9 and 91%, respectively. The probability of failures for the same record applied to a two-storey building are 97 and 3% for collapse and yield, respectively, based on the fragility surface, whereas the fragility curve shows a probability of 53% for collapse failure and 47% for yield failure. Thus, the difference in the probability of failures for the considered one- and two-storey buildings ranges from 12 to 44%. This difference cannot be ignored, which means that the use of two parameters instead of a single parameter is highly recommended for representing the seismic action in a given region. Several recent studies, such as Omine et al. (2008), Seyedi et al. (2010), Gehl et al. 2011, Koutsourelakis (2010), and Gehl et al. (2013b), have supported this finding.

Finally, regarding the vulnerability of low-rise URM buildings in the KR to seismic loads, the summarised results based on the fragility surfaces shown in Figures 7.29 and 7.30 demonstrate that these types of buildings are highly vulnerable to seismic loads, suggesting that such structures must be strengthened to prevent structural failure.

Numerous resistant parts can be identified in masonry structures as a function of the considered load type. Nevertheless, the walls, the horizontal floors and the extent of connection between them are more important. In general, masonry buildings are complex three-dimensional structures. Due to their complexity, the analysis and design processes both require technical simplifications. The TREMURI software used in this study assumes that the global behaviour of a structure is managed through the in-plane response of its walls, implying that the out-of-plane mechanism is negligible. The latter is a local phenomenon of single facades or parts of them that often occurs due to the low level of connection between orthogonal walls and between walls and slabs. The out-of-plane mechanism can be avoided with the help of localised interventions. With respect to the URM buildings in the KR and those considered in this study, the connection between walls and slabs is achieved through RC ring beams that are monolithically cast with the RC slabs in addition to cement and mortar. Furthermore, a proper connection is also achievable at the intersection of corner walls via block toothing. Thus, it is a good assumption to neglect this failure mechanism. However, more recently, it has been noticed that the ring beam

above walls are neglected. In this case, a building is considered to not behave like a "rigid box" under seismic loads due to weakness in the connection between walls and the slab. The lack of connection and the poor bond between orthogonal walls, especially at the meeting points between the internal walls and the boundary walls, means that the structural response is not indicative of the global response. In this case, the behaviour of the walls is dominated by out-of-plane collapse. Because an earthquake induces shaking of an entire building, tying the buildings together to prevent them from being shaken apart is very important, although the connections must be strong.

The seismic forces affecting a building are inertial forces proportional to its weight. These lateral inertial forces are transferred by the floor slab to the walls, to the foundations, and ultimately to the soil system. Therefore, each of these structural elements (i.e., floor slabs, walls, and foundations) and the connections between them must be designed to transfer these inertial forces in the safest manner. From Newton's Second Law of Motion, the inertial force is the mass times the acceleration; its direction is opposite to that of the acceleration. Hence, more mass results in a greater inertial force. Therefore, lighter buildings are more capable of withstanding the shaking effects of earthquakes. In the KR, buildings are very heavy. There are unnecessary heavy loads imposed by the walls and floors, and their material density is very high. For example, these structures include: thick walls and floors; solid concrete blocks covered with thick layers of gypsum or cement mortar as the interior finishing component; and thick layers of grout plus stone or marble as a veneer. Typical reinforced concrete floors are heavy, with dead weight of 500-600 kg/m^2 (including finishing). Masonry veneer, which is an appealing architectural feature designed by owners, is very common in Kurdistan. The masonry veneers of stone placed on exterior walls are attached to the surfaces of solid concrete block walls. Although the veneers do not support any load, they physically add extra load to the building. Consequently, reducing the excessive dead weight by substituting existing heavy materials and securing lighter materials can assist in reducing excessive loads.

Moreover, nearly all URM buildings in the KR are designed to carry gravity loads; hence, these structures are not designed to resist horizontal loads. Their construction framework is traditional and conservative, meaning that they lack drawings and use concrete blocks with low-quality mortar. The poor quality of construction workmanship and the lack of skilled and knowledgeable workers, building inspectors, and construction supervisors are other critical problems. Solutions that rely on skilled human resources and

less material should be followed; although lower labour costs are not comparable to changes in material costs.

Finally, considering budget constraints and importance level of life safety, the following various types and levels of strengthening can be considered to improve existing URM buildings in the KR:

1 - Dead weight reduction by replacing existing heavy materials with lighter materials.

2 - Strengthening connections (e.g., tying of walls with steel ties, anchoring floors and roofs to walls, and reinforcing and/or stitching wall intersections).

3 - Integration of bracing and shear walls.

4 - Construction of reinforced cement or reinforced concrete jackets on one or both sides of walls to increase the shear resistance and lateral stiffness of walls.

5 - Grouting (filling voids with strong binding material, such as cement grout, to bond the loose masonry units together and to achieve a continuum in the masonry structure).

6 - Partial reconstruction of damaged or cracked masonry walls.

7 - Strengthening of the building's foundation (if necessary).

8 - Securing parapets, infill walls and veneers to the framing using ties.

The strengthening program in the Kurdistan region should be a gradual process that begins with the simplest solutions, such as securing walls, veneers, and parapets, and moving towards more complex and expensive approaches, such as the use of carbon fibre or reinforced concrete jacketing, to stimulate the authorities to allocate the appropriate funds without imposing a major burden on the nation's financial priorities.

CHAPTER 9. CONCLUSIONS

The central objective of this study was to investigate how existing URM buildings in the KR perform in response to future expected earthquakes. Using fragility analysis as an effective tool, several sources of uncertainty involved in the process were also studied. A framework for appropriately selecting the ground-motion parameters needed for fragility analysis was defined. During the course of this study, several seismic parameters were scrutinised based on the recorded overall structural damage in various seismic scenarios applied to both one- and two-storey URM buildings that were chosen to be representative of low-rise residential structures in the study region.

9.1 Summary of Research

Earthquakes, which are often considered to be the most devastating type of natural disaster, regularly impact the architectural environment in seismically exposed areas of the world and cause considerable social and economic losses. Approximately 1,685,000 officially reported deaths resulted from earthquakes in various countries around the world during the twentieth century (Coburn & Spence, 2002). From 2001 to 2011, earthquakes resulted in more than 780,000 deaths, representing nearly 60% of the entire natural-disaster-related mortality (Bartels & VanRooyen, 2012). Approximately 75% of the fatalities attributed to earthquakes in the 20th century were caused by the collapse of buildings (Coburn & Spence, 2002); URM buildings represent the greatest vulnerability and threat to humans (Grünthal, 1998). The substantial losses incurred due to destructive earthquakes has stimulated the need to assess the performance of existing buildings in response to potential future earthquake events to minimise the potential risks and prepare a proper emergency management, awareness, and rehabilitation plan.

In the Kurdistan region of Iraq, most existing buildings, which are typically URM buildings (approximately 87% of all buildings), are constructed to resist gravity loads and generally offer poor resistance to lateral seismic loads. Thus, the potential economic and social losses due to strong earthquake events can be extensive. Regardless of their location in a relatively active seismic zone, the collapse of structures during recent earthquakes, particularly in the countries around the KR (i.e., the 2011 Van earthquake in Turkey and the 2003 Bam earthquake in Iran) has raised concerns pertaining to the safety of existing

buildings in the region, including structures that will be constructed in the future. No research was found in the literature regarding the seismic vulnerability of this building type or other types of structures in the KR. Hence, a seismic vulnerability assessment of URM buildings in this region is the first step in developing seismic retrofitting and pre-disaster mitigation plans for the region. To this end, three main components must be considered and studied at the regional scale: seismic hazard (i.e., the intensity of the expected ground motion in a given region over a specified time period), exposure (i.e., the architectural environment), and vulnerability of exposure (i.e., an anticipated level of damage or loss caused by different seismic loading levels). The latter (e.g., vulnerability modelling), which is the central component of studies related to seismic vulnerability assessments of buildings, is typically represented by sets of fragility functions that explain the expected damage.

Considering the vulnerability of masonry as a construction material and its common use for architectural purposes in the KR, the main objective of this research was to examine the fragility of existing URM buildings in the KR with regard to the identified seismic hazard using an analytical approach. This goal was achieved via three levels of procedural implementations. First, an appropriate set of area-specific building stock databases was developed using information obtained from street surveys and a census in the region. Two typical URM buildings representing typical one- and two-storey residential buildings in urban areas were then assessed in this study. Consequently, for properly defining the seismic parameters required for structural analysis and due to the lack of KR-related data, including the out-dated Iraqi seismic building code, a seismic hazard zone map was designed based on a comprehensive study of all available information related to the area. Hazard zones ‘B’ and ‘A’, which represent areas with “high” to “very high” levels of seismic hazards, respectively, were defined for the region. The characteristics of each seismic hazard zone were identified, and a smoothed response spectrum corresponding to each zone was constructed based on recorded events in regions with similar seismic and tectonic characteristics. The sets of earthquake records were then selected to be compatible with the seismic characteristics of each seismic hazard zone. Then, incremental dynamic analysis was used to evaluate the nonlinear behaviour of the studied URM buildings and to estimate their performance under seismic loads in a more rigorous manner. The TREMURI software was implemented using macro-element theory. The fragility curves and surfaces were then developed using the statistical analysis results.

All of the aforementioned fields are formally uncertain; therefore, they are typically considered as the main sources of uncertainty involved in seismic fragility assessments of buildings. Thus, the accuracy of the information used in this process was crucial in reducing these uncertainties in this study. Three main sources of uncertainties were analysed in this study: the variability in the mechanical properties of the target buildings versus ground motions, the variability in the ground-motion intensity measures, and the variability in the number of records and the selection method. The literature includes only a few studies that have analysed the effects of the aforementioned uncertainties on the fragility analysis of URM buildings compared with other structural types, e.g., reinforced concrete and steel buildings.

The variability in material properties was determined by running approximately 1,100 time history analyses applied to the one-storey URM building in the x direction using a fixed accelerogram scaled to three PGA intensities (e.g., 0.05g, 0.2g, and 0.8g). For each intensity level, a set of variants was assigned with respect to the different material properties. A second set of analyses was performed by considering the variability due to ground motions and neglecting the variability induced by the material properties. The standard deviations based on the top displacement for the ground motion and material parameter variability were computed and compared.

Variability in the number of records (e.g., sets of 7, 15, and 35/38 records and sets of 12, 25 and 60 records) and the method of selecting these records (e.g., the CMS method, the algorithm of Jayaram et al., target spectra defined by EC8, and target spectra proposed for the KR) were considered for both one- and two-storey buildings located in seismic hazard zones A and B. In total, approximately 1,140 time history analyses were performed for both types of buildings (i.e., approximately 570 time histories were performed for each building in zones A and B).

The variability in the ground-motion intensity measure was evaluated using 36 different ground-motion parameters defined for each time history. Using a statistical regression procedure, the best IMs for deriving fragility curves were selected. Furthermore, to verify the efficiency of using two ground-motion parameters (i.e., fragility surface) instead of using a single parameter (i.e., fragility curve), as suggested by several studies (e.g., Baker & Cornell, 2005; Kafali & Grigoriu, 2007; Seyedi et al., 2010; Gehl et al., 2011; Gehl et al., 2013b), different pairs of parameters were compared based on their

ability to accurately predict the structural response. The log-logistic regression was used to fit the data and derive the fragility curves and surfaces.

The difficulty in developing the fragility curves was associated with identifying a suitable ground-motion intensity measure (IM), which is used to relate seismic hazards to structural response. Current techniques require considerable computational efforts because many regression analyses are necessary to select an optimal IM. Therefore, this study addressed the problem using machine learning models for IM selection. The wrapper method was used to classify a set of IMs into different subsets containing various IMs, and the performance of each IM was evaluated based on the known response of the tested structures. Therefore, the other main objective of this study was to investigate the feasibility of the machine learning tool (e.g., the wrapper method used in the research) for selecting the best IM for use as seismic input for obtaining fragility curves. The results of the wrapper method were compared with the results derived from a statistical regression method.

9.2 Significant Findings

Overall, the results discussed in this dissertation have provided several new contributions to our understanding of the main uncertainties involved in the process of fragility assessment for URM buildings. First, the uncertainty in the mechanical properties of materials is at least as important as the variability in ground motions, which means that it can be neglected during the development of fragility curves or surfaces. Considering the variability in the number of records as uncertainty, the results indicated that the use of IDA and a well-selected ground-motion IM allows fragility curves to be derived using a few records (i.e., minimum 7 records) with the same performance as when using many records (e.g., 60 records). This finding will be helpful in reducing the time spent collecting numerous records and conducting many time history analyses to obtain fragility curves. However, a proper selection of seven seismic inputs corresponding to the seismic and site conditions of the area where the structures are located may require extra efforts from researchers.

A procedure is outlined in the study to select sets of seven records, which is the recommended minimum based on various standards, e.g., Eurocode 8 (EC8), for estimating the average structural response, for each seismic hazard zone. The results of using this

technique showed that the estimation of the mean structural response and the development of corresponding fragility curves provide acceptable accuracy compared with several available ground-motion selection and modification methods (e.g., the CMS, the code-based selection of EC8, and the algorithm of Jayaram et al.). As various natural records for a single earthquake event become available, this technique for eliminating event-related bias could also be helpful, especially when a specific number of records are desired. Using this technique, several sets of real ground motions were selected based on numerous available ground-motion selection and modification methods. The records with substantial repetitions were identified and selected. Only one record was selected per event.

The results also indicated that, based on the number of records, the variability in the ground-motion selection methods may lead to the selection of different IMs. Increasing the number of records from 15 to 25 records mitigated the impact of the ground-motion selection methods. Furthermore, with regard to the number of storeys, different IMs were selected using the same ground-motion selection method when the number of storeys was increased from 1 to 2 within the same seismic hazard zone. These findings indicated that the effect of ground-motion selection methods on the fragility analysis results depends on the number of records and the number of storeys. However, this impact can be mitigated if a correct IM is selected. Generally, the selection of an IM using a ground-motion selection method in a specific case does not mean that the same IM will be selected by the same ground-motion selection method for a different case. Therefore, depending on the typology of buildings (e.g., the number of storeys), their locations, and the number of ground motions, the ground-motion selection methods may result in the selection of different optimal IMs. However, if a correct IM is selected, it is irrelevant which ground-motion selection method was used for selecting the time histories. In summary, the correct selection of an IM is more important than the other variabilities considered in this study for assessing the fragility of low-rise URM buildings.

The acceleration spectral intensity (ASI) and velocity spectral intensity (VSI) were found to be the best correlated parameters for the analysed one- and two-storey URM buildings, respectively. These parameters consider the spectral acceleration and velocity over a wide range of periods; notably, this range can account for increases in a building's natural period due to a loss in rigidity and progressive degradation. For the same reason, spectral accelerations and displacements near a building's natural period were found to be poorly correlated with the damage index (i.e., top displacement).

In this study, the proper selection of ground-motion parameters mainly depended on a structure's typology and its behaviour under seismic loads. Thus, prior knowledge of the structural behaviour under seismic loads is an important component to the correct selection of IMs. However, this task is complicated. As a result, a methodology was used in this study for that purpose and to select the most accurate IM. By clustering the variables (i.e., the ground-motion IM parameters) and subsequently applying the wrapper feature selection method and ROC analysis, the optimal IM was readily obtained. The maximum number of IMs selected within a subset for the different algorithms of the wrapper method was seven. This result is encouraging for assessing the performance of IMs. Because instead of testing all 36 IMs used in the study, it was possible to use seven or fewer IMs based on the selected classifier function and the desired accuracy of the results. By reducing the time spent testing IMs, this approach will provide additional time for considering other issues involved in the process of deriving fragility curves (e.g., variability in material properties, variability in damage states, and variability in regression functions). The results showed that the wrapper method was able to predict the structural responses and facilitate the detection of the correct IM. The prediction was achieved with an accuracy comparable to those obtained using traditional regression functions such as log-logistic regression. Finally, the feature set (here sets of ground-motion IMs) was found to depend largely on the classifier algorithm chosen in the wrapper-based approach, suggesting that it is not easy to find a feature set that is optimal for all classifiers.

Moreover, the significant difference in the results between using a single IM and two IMs cannot be ignored; the use of two parameters (i.e., fragility surfaces) instead of a single parameter (i.e., fragility curves) is advised when representing the seismic action of a given region. The results illustrated that fragility surfaces can provide a complete explanation of the impact of seismic actions on a structure by adding an additional layer of information and considering the epistemic uncertainty related to the second IM parameter. It was also found that it is not beneficial or practical to use fragility surfaces with a limited number of records, namely 7 or fewer records, because a fragility curve with a well-selected IM will provide the same information as the fragility surface.

Finally, the results indicated that the seismic safety of the investigated low-rise buildings in the region is questionable, implying that these structures must be strengthened and enhanced to prevent failure. Considering budget constraints and the importance of human safety, different types and levels of strengthening, such as reducing a structure's

dead weight, improving the connections, and grouting, were recommended to improve existing URM buildings in the KR. The strengthening program in the region should be implemented as a gradual process that begins with the simplest solutions, such as securing walls, veneers, and parapets, and moves towards more complicated and expensive approaches, such as the use of carbon fibre or reinforced concrete jacketing, to encourage authorities to allocate the appropriate funds without imposing a major burden on the nation's financial priorities. A seismic zonation map, response spectra, and seismic zone factor Z values designed and proposed in this study for the Kurdistan region can be used for analysing and designing common types of buildings in the territory. However, the preliminary preparation of a complete seismic building code specifically for the Kurdistan region is highly recommended as future work. Alternatively, updating the Iraqi seismic building code can be considered as an important solution.

9.3 Limitations

During the seismic fragility assessment of the URM buildings in the KR and the investigation of the importance of the seismic parameter selection, certain important caveats of the established methodology, e.g., the input earthquake record time histories and the software for non-linear dynamic analysis utilised in this study, should be identified.

Due to the absence of a database specific to the region, that includes the recorded earthquakes, natural time histories were gathered and selected over the world by considering their compatibility with the seismic signature of the study region. However, because seismic loading is complex, no single natural record that is similar to the one recorded in a different location can be found, especially when considering the significant affect of intensity, frequency content, and duration information on the structural response (elastic and inelastic). Therefore, it is necessary to be cautious when transferring results, thereby ensuring that the correct population of seismic records are selected. These records should correspond to the real site conditions under study.

The TREMURI model is commonly applied to the global analysis of newly designed and existing masonry structures for improving their seismic safety. The main assumption considered in this model is the proper prevention of potential local, mainly out-of-plane, failure modes in the masonry walls. However, depending on many factors (e.g., the connection between walls and the connection between slabs and walls), this prevention

may not be correctly obtained. Thus, the selection of a proper non-linear analysis program that considers the local failure modes in the masonry walls and a structure's global analysis is an important parameter for avoiding bias or inaccuracies in the results.

9.4 Recommendations for Future Research

Based on the aforementioned conclusions, future research should be extended in the following areas:

- It must be noted that this project focused on evaluating the best earthquake parameter based on the structural damage index for low-rise URM buildings in Kurdistan region of Iraq. Further research is required to extend the results obtained in this study by focusing on different types of buildings, layouts, and materials; different ground-motion selection methods; different seismic hazard zones; different numbers of storeys; and different numbers of records.
- Further research can be conducted to account for possible out-of-plane failure mechanisms of masonry walls that might occur if floors and walls are poorly connected.
- Experimental investigations should be performed to characterise the nonlinear deformation behaviour, drift thresholds and material strength parameters of concrete block masonry walls prevalent in the Kurdistan region. These investigations should also consider the effect of potential seismic retrofit schemes.
- Experimental and analytical studies require the identification of the most appropriate but least expensive methods of strengthening vulnerable buildings in the region. In addition, it is necessary to provide details of seismic resistance requirements for future regional construction practices.
- Future work must be focused on developing simplified vulnerability functions for social losses, such as the expected casualties assigned to the predicted damage states, primarily for the extensive and complete damage states.

- An updated database of experimental drift thresholds and capacity curves should be compiled for other buildings types in the Kurdistan region, including concrete and steel buildings, to provide future users with the option to apply site-related fragility functions.
- Furthermore, a major point of discussion during this study was the population of seismic events utilised for the obtained results. The use of natural accelerograms representative of the seismic characteristics of the study region and selected from across the world may introduce biases in the results because it is challenging to locate an earthquake record with identical characteristics but with a different origin. The use of a good sample of naturally occurring accelerograms specific to the study region, when they are available, will allow the results to approximate real conditions and reduce the bias in the results during identification of the best possible damage-describing earthquake parameter within a finite set of earthquake parameters.
- The proposed methodology, i.e., using the wrapper-based approach as an effective tool for selecting a seismic parameter that best correlates with the recorded structural damage, can be extended to include all available classification algorithms. By using different types of structures, materials, ground-motion records and parameters and applying the proposed methodology, there is distinct possibility to more easily identify and select the ground-motion parameter using appropriately designed software.
- Other feature selection methods, such as filter- and embedded-based approaches, can also be implemented in future studies for selecting the optimal ground-motion parameter required for determining fragility curves.

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APPENDIX

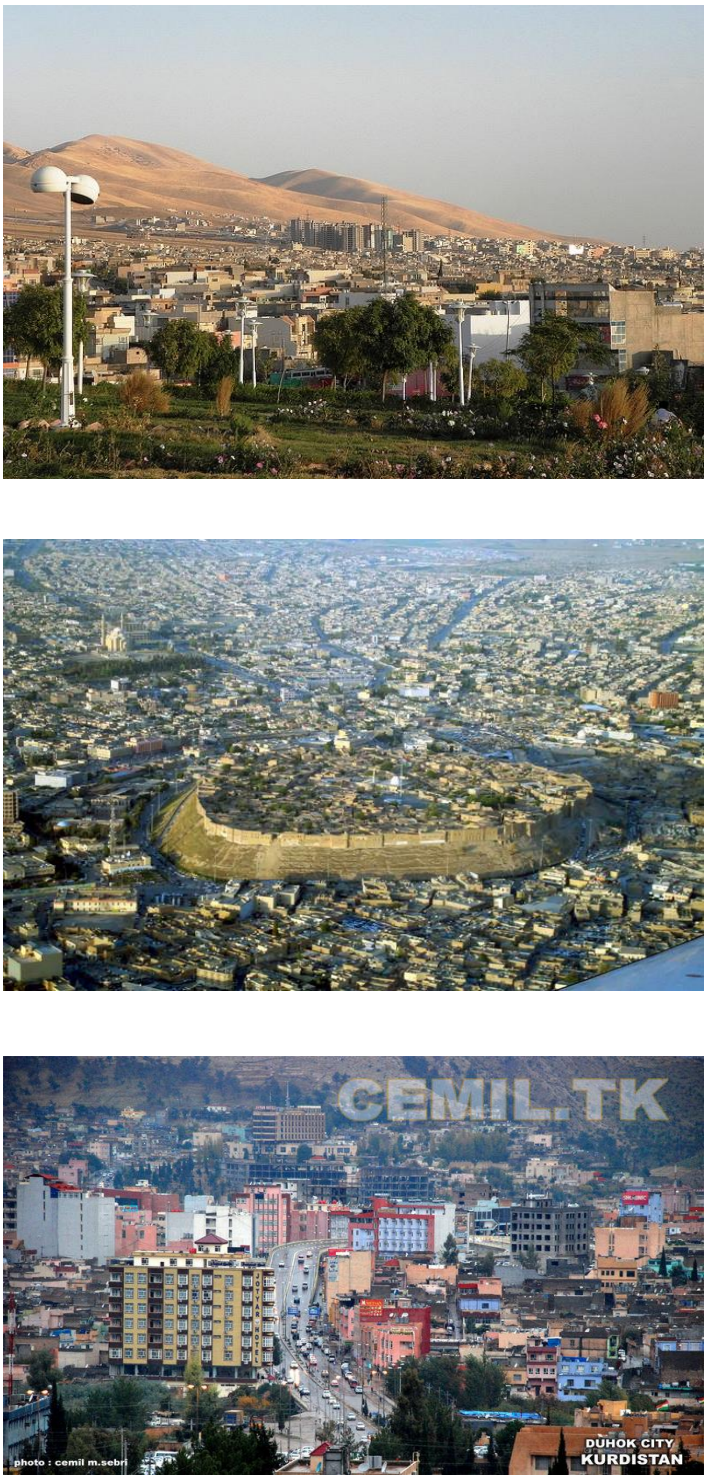


Figure A.1: From top to bottom: Sulaymaniya district centre (centre of Sulaymaniya Governorate); Erbil city, capital of the Iraqi Kurdistan Region; and Duhok district centre (centre of Duhok Governorate).

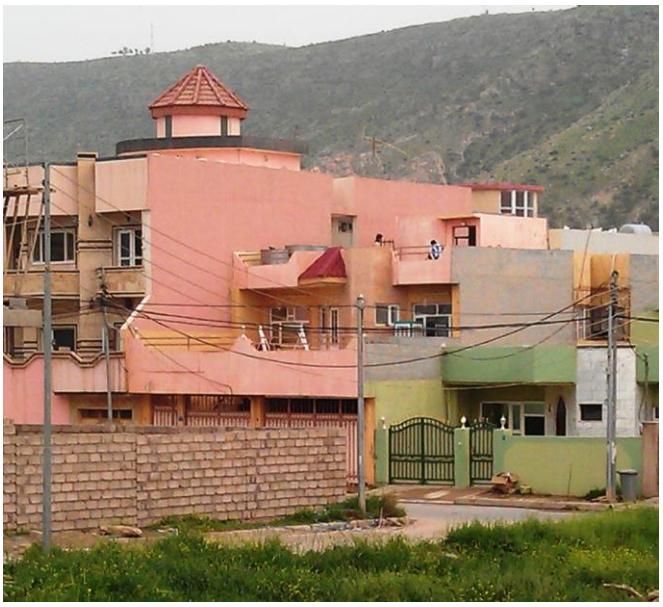


Figure A.2: Unreinforced masonry buildings with 1 to 3 storeys.



Figure A.3: Individual and attached unreinforced masonry buildings with one storey.



Figure A.4: Individual and attached unreinforced masonry buildings with two storeys.





Figure A.5: Dual system of buildings with three storeys.



Figure A.6: Mid- and high-rise reinforced concrete buildings.



Figure A.7: Mud clay, caravan and steel buildings.

Table A1: IDA of the one- and two-storey buildings using all sets of earthquake time histories.

NGA record no.	PGA (g)	One-storey building			Two-storey building		
		Max. x displacement (cm)	Damage state	Merged damage states	Max. x displacement (cm)	Damage state	Merged damage states
126	0.025	0.0065	No damage	None	0.0189	No damage	None
	0.05	0.0110	No damage	None	0.0335	No damage	None
	0.1	0.0215	No damage	None	0.0613	No damage	None
	0.2	0.0512	No damage	None	0.1637	Slight	Yield
	0.4	0.2790	Moderate	Yield	1.0343	Extensive	Yield
	0.6	0.9427	Very heavy	Collapse	1.9791	Very heavy	Collapse
	0.8	1.6400	Very heavy	Collapse	3.3052	Collapse	Collapse
	1	2.3800	Collapse	Collapse	3.9765	Collapse	Collapse
143	0.025	0.0050	No damage	None	0.0214	No damage	None
	0.05	0.0086	No damage	None	0.0365	No damage	None
	0.1	0.0160	No damage	None	0.0657	No damage	None
	0.2	0.0480	No damage	None	0.2781	Moderate	Yield
	0.4	0.3780	Moderate	Yield	0.9073	Extensive	Yield
	0.6	0.8500	Extensive	Yield	2.4018	Very heavy	Collapse
	0.8	1.8700	Collapse	Collapse	4.1526	Collapse	Collapse
	1	2.3400	Collapse	Collapse	5.7074	Collapse	Collapse
779	0.025	0.0034	No damage	None	0.0153	No damage	None
	0.05	0.0068	No damage	None	0.0238	No damage	None
	0.1	0.0110	No damage	None	0.0419	No damage	None
	0.2	0.0214	No damage	None	0.0876	No damage	None
	0.4	0.0837	Slight	Yield	0.6776	Extensive	Yield
	0.6	0.3300	Moderate	Yield	1.4878	Very heavy	Collapse
	0.8	0.8700	Extensive	Yield	4.1858	Collapse	Collapse
	1	2.3200	Collapse	Collapse	4.4184	Collapse	Collapse
828	0.025	0.0038	No damage	None	0.0202	No damage	None
	0.05	0.0070	No damage	None	0.0365	No damage	None
	0.1	0.0134	No damage	None	0.0677	No damage	None
	0.2	0.0290	No damage	None	0.1739	Slight	Yield
	0.4	0.2000	Moderate	Yield	0.8536	Extensive	Yield
	0.6	0.7100	Extensive	Yield	2.5449	Very heavy	Collapse
	0.8	1.5500	Very heavy	Collapse	4.5537	Collapse	Collapse
	1	2.8600	Collapse	Collapse	4.1536	Collapse	Collapse
879	0.025	0.0054	No damage	None	0.0198	No damage	None
	0.05	0.0098	No damage	None	0.0336	No damage	None
	0.1	0.0180	No damage	None	0.0617	No damage	None

	0.2	0.0549	Slight	Yield	0.1274	No damage	None
	0.4	0.1620	Moderate	Yield	0.5668	Moderate	Yield
	0.6	0.6320	Extensive	Yield	2.1315	Very heavy	Collapse
	0.8	1.4000	Very heavy	Collapse	4.0831	Collapse	Collapse
	1	3.0300	Collapse	Collapse	7.1089	Collapse	Collapse
983	0.025	0.0044	No damage	None	0.0148	No damage	None
	0.05	0.0060	No damage	None	0.0256	No damage	None
	0.1	0.0110	No damage	None	0.0518	No damage	None
	0.2	0.0280	No damage	None	0.1572	Slight	Yield
	0.4	0.3500	Moderate	Yield	1.3031	Extensive	Yield
	0.6	1.3800	Very heavy	Collapse	2.7626	Collapse	Collapse
	0.8	2.2300	Collapse	Collapse	3.9100	Collapse	Collapse
	1	2.4150	Collapse	Collapse	5.2107	Collapse	Collapse
1508	0.025	0.0036	No damage	None	0.0167	No damage	None
	0.05	0.0059	No damage	None	0.0280	No damage	None
	0.1	0.0110	No damage	None	0.0559	No damage	None
	0.2	0.0650	Slight	Yield	0.1750	Slight	Yield
	0.4	0.6110	Extensive	Yield	1.7101	Very heavy	Collapse
	0.6	1.4000	Very heavy	Collapse	3.3744	Collapse	Collapse
	0.8	2.2600	Collapse	Collapse	5.6544	Collapse	Collapse
	1	3.2000	Collapse	Collapse	7.3218	Collapse	Collapse
802	0.025	0.0045	No damage	None	0.0210	No damage	None
	0.05	0.0077	No damage	None	0.0377	No damage	None
	0.1	0.0142	No damage	None	0.0701	No damage	None
	0.2	0.0320	No damage	None	0.1641	Slight	Yield
	0.4	0.2160	Moderate	Yield	0.7157	Extensive	Yield
	0.6	0.5700	Extensive	Yield	1.8897	Very heavy	Collapse
	0.8	1.2967	Very heavy	Collapse	2.4584	Very heavy	Collapse
	1	1.8000	Collapse	Collapse	4.1135	Collapse	Collapse
825	0.025	0.0027	No damage	None	0.0134	No damage	None
	0.05	0.0063	No damage	None	0.0323	No damage	None
	0.1	0.0140	No damage	None	0.0701	No damage	None
	0.2	0.0346	No damage	None	0.1596	Slight	Yield
	0.4	0.1340	Moderate	Yield	0.5130	Moderate	Yield
	0.6	0.3400	Moderate	Yield	1.1757	Extensive	Yield
	0.8	0.6690	Extensive	Yield	1.9468	Very heavy	Collapse
	1	1.1000	Very heavy	Collapse	2.7059	Collapse	Collapse
963	0.025	0.0032	No damage	None	0.0118	No damage	None
	0.05	0.0070	No damage	None	0.0285	No damage	None
	0.1	0.0167	No damage	None	0.0633	No damage	None
	0.2	0.0400	No damage	None	0.2013	Slight	Yield

	0.4	0.4500	Extensive	Yield	1.0716	Extensive	Yield
	0.6	1.0800	Very heavy	Collapse	2.5665	Very heavy	Collapse
	0.8	1.7400	Collapse	Collapse	4.0315	Collapse	Collapse
	1	2.5400	Collapse	Collapse	4.7863	Collapse	Collapse
1004	0.025	0.0038	No damage	None	0.0196	No damage	None
	0.05	0.0068	No damage	None	0.0342	No damage	None
	0.1	0.0135	No damage	None	0.0655	No damage	None
	0.2	0.0313	No damage	None	0.1454	Slight	Yield
	0.4	0.1740	Moderate	Yield	0.8569	Extensive	Yield
	0.6	0.5000	Extensive	Yield	2.8409	Collapse	Collapse
	0.8	1.2900	Very heavy	Collapse	5.6121	Collapse	Collapse
	1	2.4000	Collapse	Collapse	8.7481	Collapse	Collapse
1085	0.025	0.0034	No damage	None	0.0106	No damage	None
	0.05	0.0053	No damage	None	0.0269	No damage	None
	0.1	0.0118	No damage	None	0.0599	No damage	None
	0.2	0.0256	No damage	None	0.1421	Slight	Yield
	0.4	0.2700	Moderate	Yield	1.1282	Extensive	Yield
	0.6	1.0000	Very heavy	Collapse	2.2014	Very heavy	Collapse
	0.8	1.4000	Very heavy	Collapse	3.0556	Collapse	Collapse
	1	2.7000	Collapse	Collapse	4.7972	Collapse	Collapse
1086	0.025	0.0035	No damage	None	0.0093	No damage	None
	0.05	0.0051	No damage	None	0.0241	No damage	None
	0.1	0.0115	No damage	None	0.0544	No damage	None
	0.2	0.0249	No damage	None	0.1274	No damage	None
	0.4	0.1850	Moderate	Yield	1.0621	Extensive	Yield
	0.6	0.8000	Extensive	Yield	2.5748	Very heavy	Collapse
	0.8	1.7300	Collapse	Collapse	3.6862	Collapse	Collapse
	1	3.2200	Collapse	Collapse	5.8653	Collapse	Collapse
1197	0.025	0.0040	No damage	None	0.0205	No damage	None
	0.05	0.0072	No damage	None	0.0355	No damage	None
	0.1	0.0132	No damage	None	0.0661	No damage	None
	0.2	0.0332	No damage	None	0.3585	Moderate	Yield
	0.4	0.7667	Extensive	Yield	1.4024	Extensive	Collapse
	0.6	1.2900	Very heavy	Collapse	3.0700	Collapse	Collapse
	0.8	2.6500	Collapse	Collapse	4.8452	Collapse	Collapse
	1	3.2670	Collapse	Collapse	5.9633	Collapse	Collapse
1507	0.025	0.0043	No damage	None	0.0225	No damage	None
	0.05	0.0077	No damage	None	0.0404	No damage	None
	0.1	0.0140	No damage	None	0.0735	No damage	None
	0.2	0.0538	Slight	Yield	0.3456	Moderate	Yield
	0.4	0.4280	Extensive	Yield	1.0087	Extensive	Yield
	0.6	0.9526	Very heavy	Collapse	2.6926	Collapse	Collapse

	0.8	1.7700	Collapse	Collapse	3.7818	Collapse	Collapse
	1	2.5280	Collapse	Collapse	4.3590	Collapse	Collapse
1511	0.025	0.0039	No damage	None	0.0204	No damage	None
	0.05	0.0067	No damage	None	0.0355	No damage	None
	0.1	0.0126	No damage	None	0.0685	No damage	None
	0.2	0.0710	Slight	Yield	0.3288	Moderate	Yield
	0.4	0.4847	Extensive	Yield	1.2787	Extensive	Yield
	0.6	1.2500	Very heavy	Collapse	3.5847	Collapse	Collapse
	0.8	2.9800	Collapse	Collapse	6.5198	Collapse	Collapse
	1	4.5200	Collapse	Collapse	9.2711	Collapse	Collapse
1512	0.025	0.0049	No damage	None	0.0270	No damage	None
	0.05	0.0090	No damage	None	0.0485	No damage	None
	0.1	0.0197	No damage	None	0.0945	No damage	None
	0.2	0.0890	Slight	Yield	0.4486	Moderate	Yield
	0.4	0.6120	Extensive	Yield	1.0939	Extensive	Yield
	0.6	1.1600	Very heavy	Collapse	2.1282	Very heavy	Collapse
	0.8	2.0200	Collapse	Collapse	3.3779	Collapse	Collapse
	1	2.6700	Collapse	Collapse	4.3346	Collapse	Collapse
1513	0.025	0.0045	No damage	None	0.0209	No damage	None
	0.05	0.0078	No damage	None	0.0365	No damage	None
	0.1	0.0146	No damage	None	0.0690	No damage	None
	0.2	0.0350	No damage	None	0.2000	Slight	Yield
	0.4	0.2960	Moderate	Yield	1.2413	Extensive	Yield
	0.6	0.8800	Very heavy	Collapse	2.4612	Very heavy	Collapse
	0.8	1.7280	Collapse	Collapse	3.6013	Collapse	Collapse
	1	2.0930	Collapse	Collapse	4.1713	Collapse	Collapse
568	0.025	0.0037	No damage	None	0.0197	No damage	None
	0.05	0.0066	No damage	None	0.0354	No damage	None
	0.1	0.0131	No damage	None	0.0675	No damage	None
	0.2	0.0308	No damage	None	0.1420	No damage	None
	0.4	0.0956	Slight	Yield	0.7633	Extensive	Yield
	0.6	0.3489	Moderate	Yield	2.2882	Very heavy	Collapse
	0.8	1.1656	Very heavy	Collapse	3.9766	Collapse	Collapse
	1	2.0700	Collapse	Collapse	5.3355	Collapse	Collapse
1111	0.025	0.0043	No damage	None	0.0207	No damage	None
	0.05	0.0075	No damage	None	0.0369	No damage	None
	0.1	0.0140	No damage	None	0.0708	No damage	None
	0.2	0.0327	No damage	None	0.2210	Slight	Yield
	0.4	0.5788	Extensive	Yield	1.4943	Very heavy	Collapse
	0.6	1.3060	Very heavy	Collapse	2.3049	Very heavy	Collapse
	0.8	1.6678	Very heavy	Collapse	3.0483	Collapse	Collapse

	1	2.0990	Collapse	Collapse	3.6548	Collapse	Collapse
1231	0.025	0.0036	No damage	None	0.0143	No damage	None
	0.05	0.0054	No damage	None	0.0246	No damage	None
	0.1	0.0123	No damage	None	0.0552	No damage	None
	0.2	0.0259	No damage	None	0.1389	No damage	None
	0.4	0.1709	Moderate	Yield	1.3316	Extensive	Yield
	0.6	1.0414	Very heavy	Collapse	2.6343	Very heavy	Collapse
	0.8	2.7500	Collapse	Collapse	5.0873	Collapse	Collapse
	1	2.7660	Collapse	Collapse	6.6582	Collapse	Collapse
1504	0.025	0.0041	No damage	None	0.0194	No damage	None
	0.05	0.0072	No damage	None	0.0342	No damage	None
	0.1	0.0132	No damage	None	0.0620	No damage	None
	0.2	0.0318	No damage	None	0.1314	No damage	None
	0.4	0.2388	Moderate	Yield	1.2298	Extensive	Yield
	0.6	0.9428	Very heavy	Collapse	5.7297	Collapse	Collapse
	0.8	3.7800	Collapse	Collapse	8.5219	Collapse	Collapse
	1	6.0300	Collapse	Collapse	12.0517	Collapse	Collapse
1509	0.025	0.0036	No damage	None	0.0178	No damage	None
	0.05	0.0060	No damage	None	0.0237	No damage	None
	0.1	0.0109	No damage	None	0.0546	No damage	None
	0.2	0.0270	No damage	None	0.1539	Slight	Yield
	0.4	0.4189	Extensive	Yield	1.8164	Very heavy	Collapse
	0.6	1.6586	Very heavy	Collapse	3.4607	Collapse	Collapse
	0.8	2.5000	Collapse	Collapse	4.8582	Collapse	Collapse
	1	3.1996	Collapse	Collapse	5.9728	Collapse	Collapse
139	0.025	0.0048	No damage	None	0.0246	No damage	None
	0.05	0.0093	No damage	None	0.0443	No damage	None
	0.1	0.0157	No damage	None	0.0966	No damage	None
	0.2	0.0588	Slight	Yield	0.2899	Moderate	Yield
	0.4	0.3689	Moderate	Yield	0.7306	Extensive	Yield
	0.6	0.5837	Extensive	Yield	1.3301	Extensive	Yield
	0.8	0.8742	Extensive	Yield	1.7825	Very heavy	Collapse
	1	1.4782	Very heavy	Collapse	2.3165	Very heavy	Collapse
178	0.025	0.0050	No damage	None	0.0239	No damage	None
	0.05	0.0087	No damage	None	0.0432	No damage	None
	0.1	0.0163	No damage	None	0.0827	No damage	None
	0.2	0.0480	No damage	None	0.4050	Moderate	Yield
	0.4	0.5864	Extensive	Yield	1.7830	Very heavy	Collapse
	0.6	1.5000	Very heavy	Collapse	3.4764	Collapse	Collapse
	0.8	2.5500	Collapse	Collapse	6.0180	Collapse	Collapse
	1	4.4200	Collapse	Collapse	6.7209	Collapse	Collapse

180	0.025	0.0050	No damage	None	0.0242	No damage	None
	0.05	0.0090	No damage	None	0.0439	No damage	None
	0.1	0.0180	No damage	None	0.0843	No damage	None
	0.2	0.0400	No damage	None	0.1815	Slight	Yield
	0.4	0.1857	Moderate	Yield	0.8333	Extensive	Yield
	0.6	0.6445	Extensive	Yield	1.8361	Very heavy	Collapse
	0.8	1.3727	Very heavy	Collapse	2.7227	Collapse	Collapse
	1	1.5340	Very heavy	Collapse	3.1751	Collapse	Collapse
183	0.025	0.0035	No damage	None	0.0154	No damage	None
	0.05	0.0063	No damage	None	0.0252	No damage	None
	0.1	0.0138	No damage	None	0.0562	No damage	None
	0.2	0.0305	No damage	None	0.1279	No damage	None
	0.4	0.1380	Moderate	Yield	0.4803	Moderate	Yield
	0.6	0.4480	Extensive	Yield	1.1607	Extensive	Yield
	0.8	1.0396	Very heavy	Collapse	1.9706	Very heavy	Collapse
	1	1.3360	Very heavy	Collapse	3.6449	Collapse	Collapse
1402	0.025	0.0052	No damage	None	0.0199	No damage	None
	0.05	0.0093	No damage	None	0.0329	No damage	None
	0.1	0.0175	No damage	None	0.0613	No damage	None
	0.2	0.0553	Slight	Yield	0.2292	Slight	Yield
	0.4	0.3450	Moderate	Yield	0.9188	Extensive	Yield
	0.6	0.8238	Extensive	Yield	1.5222	Very heavy	Collapse
	0.8	1.1786	Very heavy	Collapse	2.3984	Very heavy	Collapse
	1	1.6380	Very heavy	Collapse	3.1596	Collapse	Collapse
1503	0.025	0.0036	No damage	None	0.0126	No damage	None
	0.05	0.0050	No damage	None	0.0239	No damage	None
	0.1	0.0113	No damage	None	0.0539	No damage	None
	0.2	0.0256	No damage	None	0.1249	No damage	None
	0.4	0.1610	Moderate	Yield	0.6773	Extensive	Yield
	0.6	0.6119	Extensive	Yield	1.8616	Very heavy	Collapse
	0.8	1.3600	Very heavy	Collapse	3.8846	Collapse	Collapse
	1	2.3400	Collapse	Collapse	6.7972	Collapse	Collapse
1505	0.025	0.0036	No damage	None	0.0178	No damage	None
	0.05	0.0060	No damage	None	0.0301	No damage	None
	0.1	0.0118	No damage	None	0.0551	No damage	None
	0.2	0.0259	No damage	None	0.1446	Slight	Yield
	0.4	0.1806	Moderate	Yield	1.9579	Very heavy	Collapse
	0.6	1.1168	Very heavy	Collapse	9.6553	Collapse	Collapse
	0.8	8.1350	Collapse	Collapse	13.8564	Collapse	Collapse
	1	9.5830	Collapse	Collapse	18.5819	Collapse	Collapse
1524	0.025	0.0043	No damage	None	0.0230	No damage	None

	0.05	0.0076	No damage	None	0.0403	No damage	None
	0.1	0.0143	No damage	None	0.0776	No damage	None
	0.2	0.0460	No damage	None	0.2304	Slight	Yield
	0.4	0.3440	Moderate	Yield	1.7794	Very heavy	Collapse
	0.6	1.2520	Very heavy	Collapse	2.5772	Very heavy	Collapse
	0.8	1.8930	Collapse	Collapse	5.2532	Collapse	Collapse
	1	3.7440	Collapse	Collapse	7.1476	Collapse	Collapse
1549	0.025	0.0044	No damage	None	0.0199	No damage	None
	0.05	0.0075	No damage	None	0.0344	No damage	None
	0.1	0.0147	No damage	None	0.0648	No damage	None
	0.2	0.0360	No damage	None	0.1555	Slight	Yield
	0.4	0.1960	Moderate	Yield	0.4871	Moderate	Yield
	0.6	0.4615	Extensive	Yield	1.2918	Extensive	Yield
	0.8	0.9060	Very heavy	Collapse	1.7586	Very heavy	Collapse
	1	1.4170	Very heavy	Collapse	2.4467	Very heavy	Collapse
1596	0.025	0.0044	No damage	None	0.0197	No damage	None
	0.05	0.0075	No damage	None	0.0350	No damage	None
	0.1	0.0140	No damage	None	0.0611	No damage	None
	0.2	0.0335	No damage	None	0.1443	Slight	Yield
	0.4	0.1834	Moderate	Yield	0.5080	Moderate	Yield
	0.6	0.4690	Extensive	Yield	1.2862	Extensive	Yield
	0.8	0.9317	Very heavy	Collapse	2.0581	Very heavy	Collapse
	1	1.4350	Very heavy	Collapse	2.4445	Very heavy	Collapse
1633	0.025	0.0051	No damage	None	0.0198	No damage	None
	0.05	0.0087	No damage	None	0.0357	No damage	None
	0.1	0.0156	No damage	None	0.0670	No damage	None
	0.2	0.0525	Slight	Yield	0.2366	Slight	Yield
	0.4	0.3090	Moderate	Yield	0.8526	Extensive	Yield
	0.6	0.7240	Extensive	Yield	1.4897	Very heavy	Collapse
	0.8	1.0700	Very heavy	Collapse	2.2096	Very heavy	Collapse
	1	1.5000	Very heavy	Collapse	3.7043	Collapse	Collapse
179	0.025	0.0044	No damage	None	0.0187	No damage	None
	0.05	0.0073	No damage	None	0.0327	No damage	None
	0.1	0.0143	No damage	None	0.0625	No damage	None
	0.2	0.0337	No damage	None	0.1370	No damage	None
	0.4	0.1724	Moderate	Yield	0.9054	Extensive	Yield
	0.6	0.6099	Extensive	Yield	5.6646	Collapse	Collapse
	0.8	2.0022	Collapse	Collapse	9.9294	Collapse	Collapse
	1	9.1170	Collapse	Collapse	16.2144	Collapse	Collapse
821	0.025	0.0037	No damage	None	0.0192	No damage	None

	0.05	0.0070	No damage	None	0.0351	No damage	None
	0.1	0.0130	No damage	None	0.0649	No damage	None
	0.2	0.0316	No damage	None	0.1905	Slight	Yield
	0.4	0.2890	Moderate	Yield	1.4518	Very heavy	Collapse
	0.6	1.0170	Very heavy	Collapse	4.0135	Collapse	Collapse
	0.8	2.0330	Collapse	Collapse	4.5051	Collapse	Collapse
	1	3.8210	Collapse	Collapse	5.3769	Collapse	Collapse
1602	0.025	0.0037	No damage	None	0.0165	No damage	None
	0.05	0.0059	No damage	None	0.0249	No damage	None
	0.1	0.0127	No damage	None	0.0464	No damage	None
	0.2	0.0281	No damage	None	0.1280	No damage	None
	0.4	0.1650	Moderate	Yield	0.8899	Extensive	Yield
	0.6	0.7611	Extensive	Yield	1.9899	Very heavy	Collapse
	0.8	1.3490	Very heavy	Collapse	3.1633	Collapse	Collapse
	1	1.8250	Collapse	Collapse	4.6233	Collapse	Collapse
181	0.025	0.0047	No damage	None	0.0162	No damage	None
	0.05	0.0086	No damage	None	0.0273	No damage	None
	0.1	0.0163	No damage	None	0.0493	No damage	None
	0.2	0.0550	Slight	Yield	0.1255	No damage	None
	0.4	0.3524	Moderate	Yield	1.1525	Extensive	Yield
	0.6	1.7200	Collapse	Collapse	3.3148	Collapse	Collapse
	0.8	3.5400	Collapse	Collapse	4.4154	Collapse	Collapse
	1	4.8125	Collapse	Collapse	7.4206	Collapse	Collapse
803	0.025	0.0039	No damage	None	0.0166	No damage	None
	0.05	0.0068	No damage	None	0.0299	No damage	None
	0.1	0.0124	No damage	None	0.0544	No damage	None
	0.2	0.0284	No damage	None	0.2095	Slight	Yield
	0.4	0.4299	Extensive	Yield	1.4273	Very heavy	Collapse
	0.6	1.4466	Very heavy	Collapse	4.6961	Collapse	Collapse
	0.8	4.1900	Collapse	Collapse	6.3380	Collapse	Collapse
	1	7.0522	Collapse	Collapse	8.6051	Collapse	Collapse
174	0.025	0.0052	No damage	None	0.0219	No damage	None
	0.05	0.0079	No damage	None	0.0397	No damage	None
	0.1	0.0156	No damage	None	0.0747	No damage	None
	0.2	0.0637	Slight	Yield	0.3450	Moderate	Yield
	0.4	0.3789	Moderate	Yield	1.0335	Extensive	Yield
	0.6	0.9650	Very heavy	Collapse	1.9546	Very heavy	Collapse
	0.8	1.4080	Very heavy	Collapse	2.9918	Collapse	Collapse
	1	2.0960	Collapse	Collapse	4.0940	Collapse	Collapse
729	0.025	0.0045	No damage	None	0.0213	No damage	None
	0.05	0.0078	No damage	None	0.0411	No damage	None
	0.1	0.0144	No damage	None	0.0786	No damage	None

	0.2	0.0366	No damage	None	0.2722	Moderate	Yield
	0.4	0.5736	Extensive	Yield	2.9320	Collapse	Collapse
	0.6	2.4257	Collapse	Collapse	5.1550	Collapse	Collapse
	0.8	4.4300	Collapse	Collapse	6.4263	Collapse	Collapse
	1	4.3000	Collapse	Collapse	8.5095	Collapse	Collapse
1148	0.025	0.0045	No damage	None	0.0233	No damage	None
	0.05	0.0079	No damage	None	0.0417	No damage	None
	0.1	0.0170	No damage	None	0.0768	No damage	None
	0.2	0.0345	No damage	None	0.2889	Moderate	Yield
	0.4	0.3870	Extensive	Yield	0.6858	Extensive	Yield
	0.6	0.6020	Extensive	Yield	1.0867	Extensive	Yield
	0.8	0.7660	Extensive	Yield	1.4798	Very heavy	Collapse
	1	1.0315	Very heavy	Collapse	2.2006	Very heavy	Collapse
1244	0.025	0.0041	No damage	None	0.0193	No damage	None
	0.05	0.0070	No damage	None	0.0331	No damage	None
	0.1	0.0133	No damage	None	0.0613	No damage	None
	0.2	0.0309	No damage	None	0.1597	Slight	Yield
	0.4	0.3240	Moderate	Yield	1.4136	Very heavy	Collapse
	0.6	1.1022	Very heavy	Collapse	3.5563	Collapse	Collapse
	0.8	3.4280	Collapse	Collapse	6.5228	Collapse	Collapse
	1	4.8500	Collapse	Collapse	8.9955	Collapse	Collapse
1485	0.025	0.0045	No damage	None	0.0211	No damage	None
	0.05	0.0078	No damage	None	0.0368	No damage	None
	0.1	0.0147	No damage	None	0.0684	No damage	None
	0.2	0.0328	No damage	None	0.1721	Slight	Yield
	0.4	0.2719	Moderate	Yield	1.2665	Extensive	Yield
	0.6	0.9670	Very heavy	Collapse	2.5193	Very heavy	Collapse
	0.8	1.6260	Very heavy	Collapse	2.8844	Collapse	Collapse
	1	1.7158	Collapse	Collapse	3.6466	Collapse	Collapse
1489	0.025	0.0044	No damage	None	0.0229	No damage	None
	0.05	0.0077	No damage	None	0.0404	No damage	None
	0.1	0.0144	No damage	None	0.0760	No damage	None
	0.2	0.0348	No damage	None	0.1916	Slight	Yield
	0.4	0.2200	Moderate	Yield	1.1558	Extensive	Yield
	0.6	1.0580	Very heavy	Collapse	3.1471	Collapse	Collapse
	0.8	1.9670	Collapse	Collapse	4.5613	Collapse	Collapse
	1	3.2090	Collapse	Collapse	6.7950	Collapse	Collapse
1492	0.025	0.0038	No damage	None	0.0184	No damage	None
	0.05	0.0066	No damage	None	0.0314	No damage	None
	0.1	0.0121	No damage	None	0.0576	No damage	None
	0.2	0.0278	No damage	None	0.1944	Slight	Yield
	0.4	0.2270	Moderate	Yield	5.0865	Collapse	Collapse

	0.6	6.4700	Collapse	Collapse	11.6002	Collapse	Collapse
	0.8	8.1000	Collapse	Collapse	16.9657	Collapse	Collapse
	1	13.1700	Collapse	Collapse	24.9798	Collapse	Collapse
1510	0.025	0.0041	No damage	None	0.0197	No damage	None
	0.05	0.0071	No damage	None	0.0341	No damage	None
	0.1	0.0132	No damage	None	0.0629	No damage	None
	0.2	0.0313	No damage	None	0.1939	Slight	Yield
	0.4	0.3359	Moderate	Yield	2.6683	Collapse	Collapse
	0.6	1.6900	Very heavy	Collapse	8.0722	Collapse	Collapse
	0.8	7.4275	Collapse	Collapse	14.7473	Collapse	Collapse
	1	9.6160	Collapse	Collapse	18.4094	Collapse	Collapse
169	0.025	0.0046	No damage	None	0.0223	No damage	None
	0.05	0.0076	No damage	None	0.0405	No damage	None
	0.1	0.0140	No damage	None	0.0766	No damage	None
	0.2	0.2097	Moderate	Yield	0.2856	Moderate	Yield
	0.4	0.7743	Extensive	Yield	1.6497	Very heavy	Collapse
	0.6	1.5640	Very heavy	Collapse	2.7859	Collapse	Collapse
	0.8	2.2245	Collapse	Collapse	3.6366	Collapse	Collapse
	1	3.0589	Collapse	Collapse	5.3746	Collapse	Collapse
182	0.025	0.0036	No damage	None	0.0182	No damage	None
	0.05	0.0068	No damage	None	0.0331	No damage	None
	0.1	0.0124	No damage	None	0.0606	No damage	None
	0.2	0.0291	No damage	None	0.1937	Slight	Yield
	0.4	0.2360	Moderate	Yield	2.0055	Very heavy	Collapse
	0.6	1.6350	Very heavy	Collapse	4.0778	Collapse	Collapse
	0.8	3.8070	Collapse	Collapse	8.0666	Collapse	Collapse
	1	7.4420	Collapse	Collapse	11.9338	Collapse	Collapse
184	0.025	0.0049	No damage	None	0.0243	No damage	None
	0.05	0.0080	No damage	None	0.0437	No damage	None
	0.1	0.0146	No damage	None	0.0950	No damage	None
	0.2	0.0680	Slight	Yield	0.3041	Moderate	Yield
	0.4	0.4860	Extensive	Yield	1.2026	Extensive	Yield
	0.6	0.8910	Very heavy	Collapse	2.8344	Collapse	Collapse
	0.8	1.9780	Collapse	Collapse	7.7985	Collapse	Collapse
	1	5.8110	Collapse	Collapse	10.7691	Collapse	Collapse
723	0.025	0.0045	No damage	None	0.0192	No damage	None
	0.05	0.0067	No damage	None	0.0289	No damage	None
	0.1	0.0122	No damage	None	0.0610	No damage	None
	0.2	0.0272	No damage	None	0.1903	Slight	Yield
	0.4	0.8366	Extensive	Yield	3.5429	Collapse	Collapse
	0.6	3.6130	Collapse	Collapse	6.2465	Collapse	Collapse
	0.8	4.7438	Collapse	Collapse	10.1350	Collapse	Collapse
	1	7.0388	Collapse	Collapse	12.3013	Collapse	Collapse

959	0.025	0.0040	No damage	None	0.0218	No damage	None
	0.05	0.0073	No damage	None	0.0323	No damage	None
	0.1	0.0170	No damage	None	0.0692	No damage	None
	0.2	0.0520	Slight	Yield	0.2455	Slight	Yield
	0.4	0.5656	Extensive	Yield	1.1398	Extensive	Yield
	0.6	1.1178	Very heavy	Collapse	2.7305	Collapse	Collapse
	0.8	1.9360	Collapse	Collapse	4.0619	Collapse	Collapse
	1	2.8670	Collapse	Collapse	5.1494	Collapse	Collapse
1048	0.025	0.0051	No damage	None	0.0217	No damage	None
	0.05	0.0081	No damage	None	0.0394	No damage	None
	0.1	0.0112	No damage	None	0.0740	No damage	None
	0.2	0.0386	No damage	None	0.1936	Slight	Yield
	0.4	0.2340	Moderate	Yield	1.0768	Extensive	Yield
	0.6	1.2300	Very heavy	Collapse	2.5333	Very heavy	Collapse
	0.8	1.8816	Collapse	Collapse	3.6902	Collapse	Collapse
	1	2.2760	Collapse	Collapse	4.8106	Collapse	Collapse
953	0.025	0.0043	No damage	None	0.0167	No damage	None
	0.05	0.0064	No damage	None	0.0291	No damage	None
	0.1	0.0107	No damage	None	0.0527	No damage	None
	0.2	0.0273	No damage	None	0.1863	Slight	Yield
	0.4	0.7900	Extensive	Yield	2.4070	Very heavy	Collapse
	0.6	1.8359	Collapse	Collapse	4.4995	Collapse	Collapse
	0.8	2.8726	Collapse	Collapse	6.7000	Collapse	Collapse
	1	3.9215	Collapse	Collapse	8.4454	Collapse	Collapse
1013	0.025	0.0040	No damage	None	0.0162	No damage	None
	0.05	0.0057	No damage	None	0.0280	No damage	None
	0.1	0.0100	No damage	None	0.0502	No damage	None
	0.2	0.0229	No damage	None	0.1366	No damage	None
	0.4	0.2598	Moderate	Yield	1.6514	Very heavy	Collapse
	0.6	1.1110	Very heavy	Collapse	2.9183	Collapse	Collapse
	0.8	1.9477	Collapse	Collapse	4.2058	Collapse	Collapse
	1	2.9830	Collapse	Collapse	6.4421	Collapse	Collapse
1044	0.025	0.0044	No damage	None	0.0213	No damage	None
	0.05	0.0076	No damage	None	0.0387	No damage	None
	0.1	0.0139	No damage	None	0.0729	No damage	None
	0.2	0.0368	No damage	None	0.3627	Moderate	Yield
	0.4	0.6020	Extensive	Yield	1.3684	Extensive	Yield
	0.6	1.4765	Very heavy	Collapse	2.8623	Collapse	Collapse
	0.8	1.9780	Collapse	Collapse	4.5752	Collapse	Collapse
	1	3.0956	Collapse	Collapse	5.7360	Collapse	Collapse
1119	0.025	0.0041	No damage	None	0.0205	No damage	None
	0.05	0.0072	No damage	None	0.0366	No damage	None

	0.1	0.0144	No damage	None	0.0698	No damage	None
	0.2	0.0342	No damage	None	0.1607	Slight	Yield
	0.4	0.1940	Moderate	Yield	1.1296	Extensive	Yield
	0.6	0.7000	Extensive	Yield	2.5266	Very heavy	Collapse
	0.8	1.6200	Very heavy	Collapse	4.1425	Collapse	Collapse
	1	2.5600	Collapse	Collapse	6.5548	Collapse	Collapse
1605	0.025	0.0036	No damage	None	0.0186	No damage	None
	0.05	0.0069	No damage	None	0.0340	No damage	None
	0.1	0.0128	No damage	None	0.0625	No damage	None
	0.2	0.0353	No damage	None	0.4124	Moderate	Yield
	0.4	0.8169	Extensive	Yield	2.4285	Very heavy	Collapse
	0.6	1.7240	Collapse	Collapse	4.0054	Collapse	Collapse
	0.8	3.6750	Collapse	Collapse	6.7260	Collapse	Collapse
	1	4.7426	Collapse	Collapse	9.6594	Collapse	Collapse
1546	0.025	0.0044	No damage	None	0.0214	No damage	None
	0.05	0.0075	No damage	None	0.0386	No damage	None
	0.1	0.0140	No damage	None	0.0718	No damage	None
	0.2	0.0400	No damage	None	0.4111	Moderate	Yield
	0.4	0.7400	Extensive	Yield	1.3490	Extensive	Yield
	0.6	1.1040	Very heavy	Collapse	3.7472	Collapse	Collapse
	0.8	3.4420	Collapse	Collapse	5.7650	Collapse	Collapse
	1	5.4800	Collapse	Collapse	8.7782	Collapse	Collapse
1063	0.025	0.0037	No damage	None	0.0186	No damage	None
	0.05	0.0068	No damage	None	0.0341	No damage	None
	0.1	0.0134	No damage	None	0.0626	No damage	None
	0.2	0.0308	No damage	None	0.1588	Slight	Yield
	0.4	0.1977	Moderate	Yield	2.0031	Very heavy	Collapse
	0.6	1.4440	Very heavy	Collapse	7.0216	Collapse	Collapse
	0.8	4.9200	Collapse	Collapse	9.6059	Collapse	Collapse
	1	6.6340	Collapse	Collapse	10.8667	Collapse	Collapse

Table A2: Dependency check for the residuals of the one-storey building response on the ground-motion magnitude using 60 unscaled records and the ASI as the ground-motion parameter.

Model Summary[b]

Model	R	R Squared	Adjusted R Squared	Std. Error of the Estimate
1	.093[a]	.009	-.009	.37714

a. Predictors: (Constant), Magnitude

b. Dependent Variable: Residual of response

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	.070	1	.070	.495	.485[b]
	Residual	8.107	57	.142		
	Total	8.178	58			

a. Dependent Variable: Residual of response

b. Predictors: (Constant), Magnitude

Table A3: Dependency check for the residuals of the one-storey building response on the ground-motion source distance using 60 unscaled records and the ASI as the ground-motion parameter.

Model Summary[b]

Model	R	R Squared	Adjusted R Squared	Std. Error of the Estimate
1	.171[a]	.029	.012	0.37318

a. Predictors: (Constant), EpiD

b. Dependent Variable: Residual of response

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	0.240	1	0.240	1.722	.195[b]
	Residual	7.938	57	0.139		
	Total	8.178	58			

a. Dependent Variable: Residual of response

b. Predictors: (Constant), EpiD

Table A4: Dependency check for the residuals of the two-storey building response on the ground-motion magnitude using 60 unscaled records and the ASI as the ground-motion parameter.

Model Summary[b]

Model	R	R Squared	Adjusted R Squared	Std. Error of the Estimate
1	.108[a]	.012	-.006	1.23052

a. Predictors: (Constant), Magnitude

b. Dependent Variable: Residual of response

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	1.023	1	1.023	.676	.414[b]
	Residual	86.309	57	1.514		
	Total	87.332	58			

a. Dependent Variable: Residual of response

b. Predictors: (Constant), Magnitude

Table A5: Dependency check for the residuals of the two-storey building response on the ground-motion source distance using 60 unscaled records and the ASI as the ground-motion parameter.

Model Summary[b]

Model	R	R Squared	Adjusted R Squared	Std. Error of the Estimate
1	.018[a]	.000	-.017	1.23759

a. Predictors: (Constant), EpiD

b. Dependent Variable: Residual of response

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	0.029	1	0.029	.019	.892[b]
	Residual	87.303	57	1.532		
	Total	87.332	58			

a. Dependent Variable: Residual of response

b. Predictors: (Constant), EpiD

Table A6: Dependency check for the residuals of the one-storey building response on the ground-motion magnitude using 60 unscaled records and the VSI as the ground-motion parameter.

Model Summary[b]

Model	R	R Squared	Adjusted R Squared	Std. Error of the Estimate
1	.068[a]	.005	-.013	.37095

a. Predictors: (Constant), Magnitude

b. Dependent Variable: Residual of response

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	.036	1	.036	.264	.610[b]
	Residual	7.844	57	.138		
	Total	7.880	58			

a. Dependent Variable: Residual of response

b. Predictors: (Constant), Magnitude

Table A7: Dependency check for the residuals of the one-storey building response on the ground-motion source distance using 60 unscaled records and the VSI as the ground-motion parameter.

Model Summary[b]

Model	R	R Squared	Adjusted R Squared	Std. Error of the Estimate
1	.111[a]	.012	-.005	0.36951

a. Predictors: (Constant), EpiD

b. Dependent Variable: Residual of response

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	0.097	1	0.097	.710	.403[b]
	Residual	7.783	57	0.137		
	Total	7.880	58			

a. Dependent Variable: Residual of response

b. Predictors: (Constant), EpiD

Table A8: Dependency check for the residuals of the two-storey building response on the ground-motion magnitude using 60 unscaled records and the VSI as the ground-motion parameter.

Model Summary[b]

Model	R	R Squared	Adjusted R Squared	Std. Error of the Estimate
1	.085[a]	.007	-.010	.89825

a. Predictors: (Constant), Magnitude

b. Dependent Variable: Residual of response

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	.336	1	.336	.417	.521[b]
	Residual	45.990	57	.807		
	Total	46.327	58			

a. Dependent Variable: Residual of response

b. Predictors: (Constant), Magnitude

Table A9: Dependency check for the residuals of the two-storey building response on the ground-motion source distance using 60 unscaled records and the VSI as the ground-motion parameter.

Model Summary[b]

Model	R	R Squared	Adjusted R Squared	Std. Error of the Estimate
1	.160[a]	.026	.009	0.88989

a. Predictors: (Constant), EpiD

b. Dependent Variable: Residual of response

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	1.188	1	1.188	1.500	.226[b]
	Residual	45.139	57	0.792		
	Total	46.327	58			

a. Dependent Variable: Residual of response

b. Predictors: (Constant), EpiD

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Research Ethics Review Checklist

University of Portsmouth

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Postgraduate Research Student (PGRS) Information				**Student ID:**	605287
PGRS Name:	ABDULHAMEED A. YASEEN				
Department:	SCES	**First Supervisor:**	Dr Nikos Nanos		
Start Date: (or progression date for Prof Doc students)		**Oct. 2011**			
Study Mode and Route:	Part-time ☐ Full-time ☒	MPhil ☐ PhD ☒		MD ☐ Professional Doctorate ☐	

Title of Thesis:	Seismic Fragility Assessment of Masonry Buildings in the Kurdistan Region
Thesis Word Count: (excluding ancillary data)	53000

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UKRIO Finished Research Checklist:
(If you would like to know more about the checklist, please see your Faculty or Departmental Ethics Committee rep or see the online version of the full checklist at: http://www.ukrio.org/what-we-do/code-of-practice-for-research/)

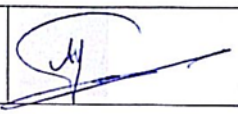
a) Have all of your research and findings been reported accurately, honestly and within a reasonable time frame?	YES ☒ NO ☐
b) Have all contributions to knowledge been acknowledged?	YES ☒ NO ☐
c) Have you complied with all agreements relating to intellectual property, publication and authorship?	YES ☒ NO ☐
d) Has your research data been retained in a secure and accessible form and will it remain so for the required duration?	YES ☒ NO ☐
e) Does your research comply with all legal, ethical, and contractual requirements?	YES ☒ NO ☐

Candidate Statement:

I have considered the ethical dimensions of the above named research project, and have successfully obtained the necessary ethical approval(s)

Ethical review number(s) from Faculty Ethics Committee (or from NRES/SCREC):	Student number: 510664 Certificate code: 5AB4-B9DA-3EAB-B98E-7184-98BB-EA90-3AFC

If you have *not* submitted your work for ethical review, and/or you have answered 'No' to one or more of questions a) to e), please explain below why this is so:

Signed *(PGRS)*:		**Date:** 16/10/2015

UPR16 – August 2015